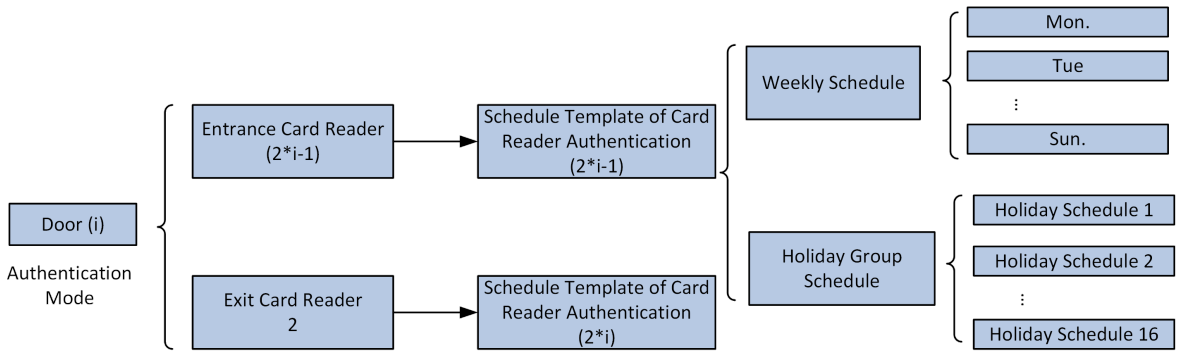
1 Authentication Schedule Management

1.1 Introduction to the Function

By calling the following APIs, the entrance&exit time schedule and authentication mode can be applied to the card reader. An access control terminal is controlled by two card readers. For example: access control terminal 1 is controlled by entrance card reader 1 and exit card reader 2; access control terminal 2 is controlled by entrance card reader 3 and exit card reader 4, etc. Card reader n is corresponding to schedule template n.

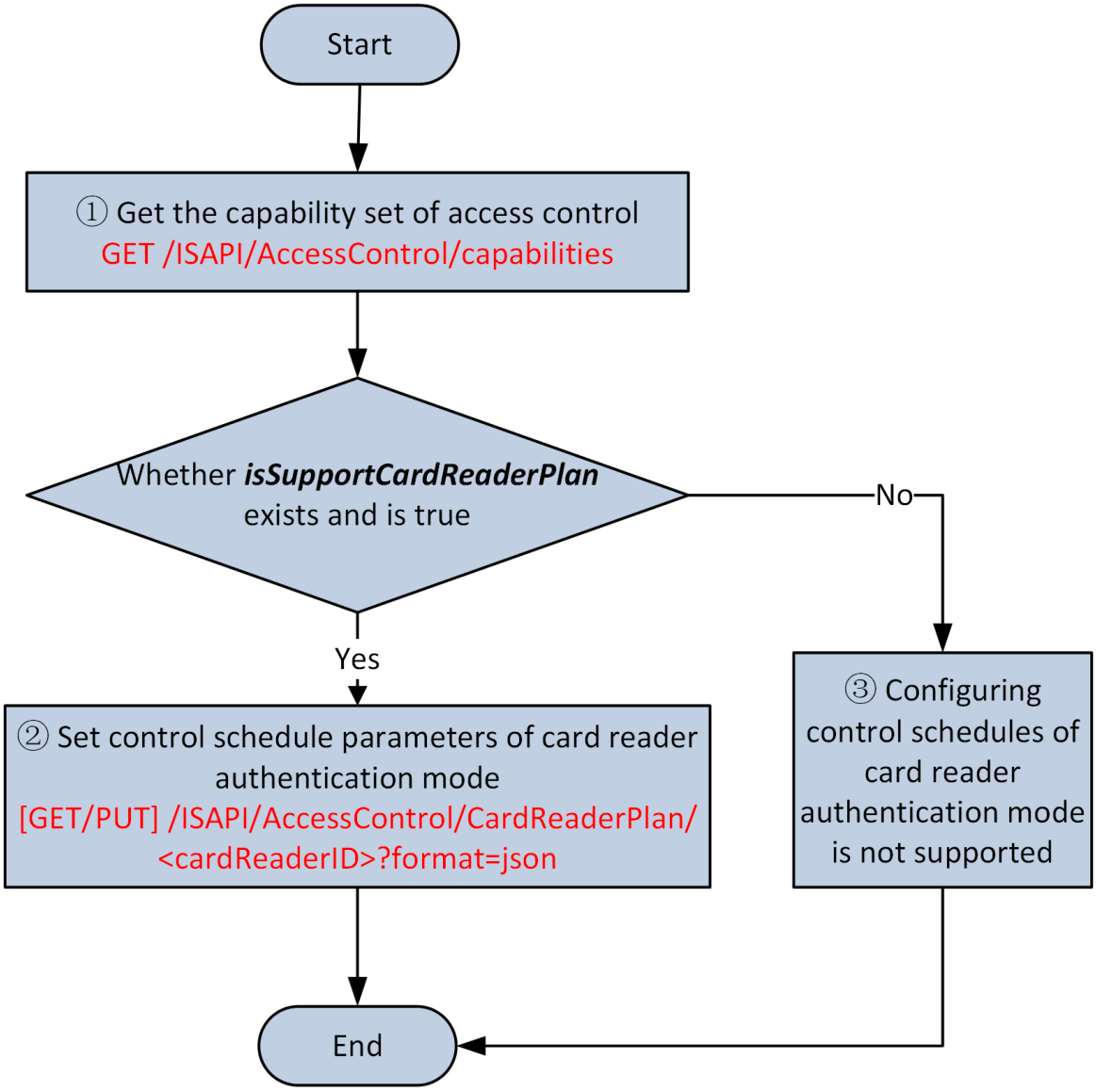
1 weekly schedule and 4 holiday groups can be added in each schedule template. The priority of holiday schedule is higher than that of weekly schedule. A weekly schedule can be configured by date of a week and 8 different time periods of a day. 16 holiday schedules can be added to a holiday group schedule. Each holiday schedule has its start and end date, and the time period is same in the range (8 time periods can be added). The access control can follow the schedule template to manage the time of person's permissions.

For Person Management of Person and Credential Management, the priority of the authentication method for person is higher than that of the authentication schedule. If the authentication method (the node is userVerifyMode) is applied to a person, the person can access according to the authentication method for person.



1.2 API Calling Flow

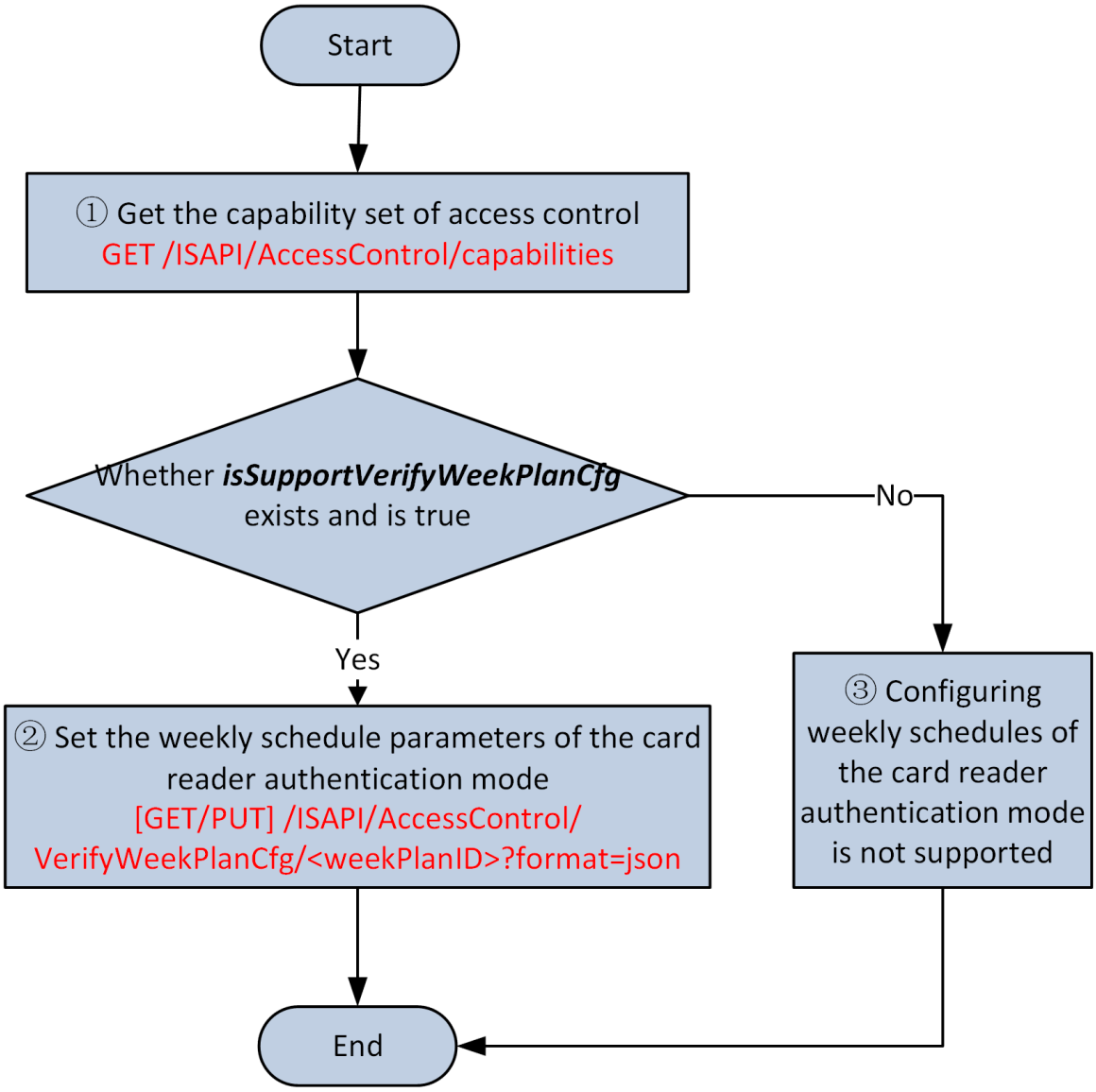
1.2.1 Card Reader's Authentication Schedule Configuration



The API calling flow is as follows:

1. Check whether the device supports configuring control schedules of card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardReaderPlan is returned and its value is "true", it indicates that the device supports configuring acontrol schedules of card reader authentication mode (and the device also supports configuring schedule templates of the card reader authentication mode).
2. Set control schedule parameters of card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/CardReaderPlan/<cardReaderID>?format=json`; the value of cardReaderID should be the same as the value of templateNo.
3. If the node isSupportCardReaderPlan is returned and its value is "false", it indicates that the device does not support configuring control schedules of card reader authentication mode.

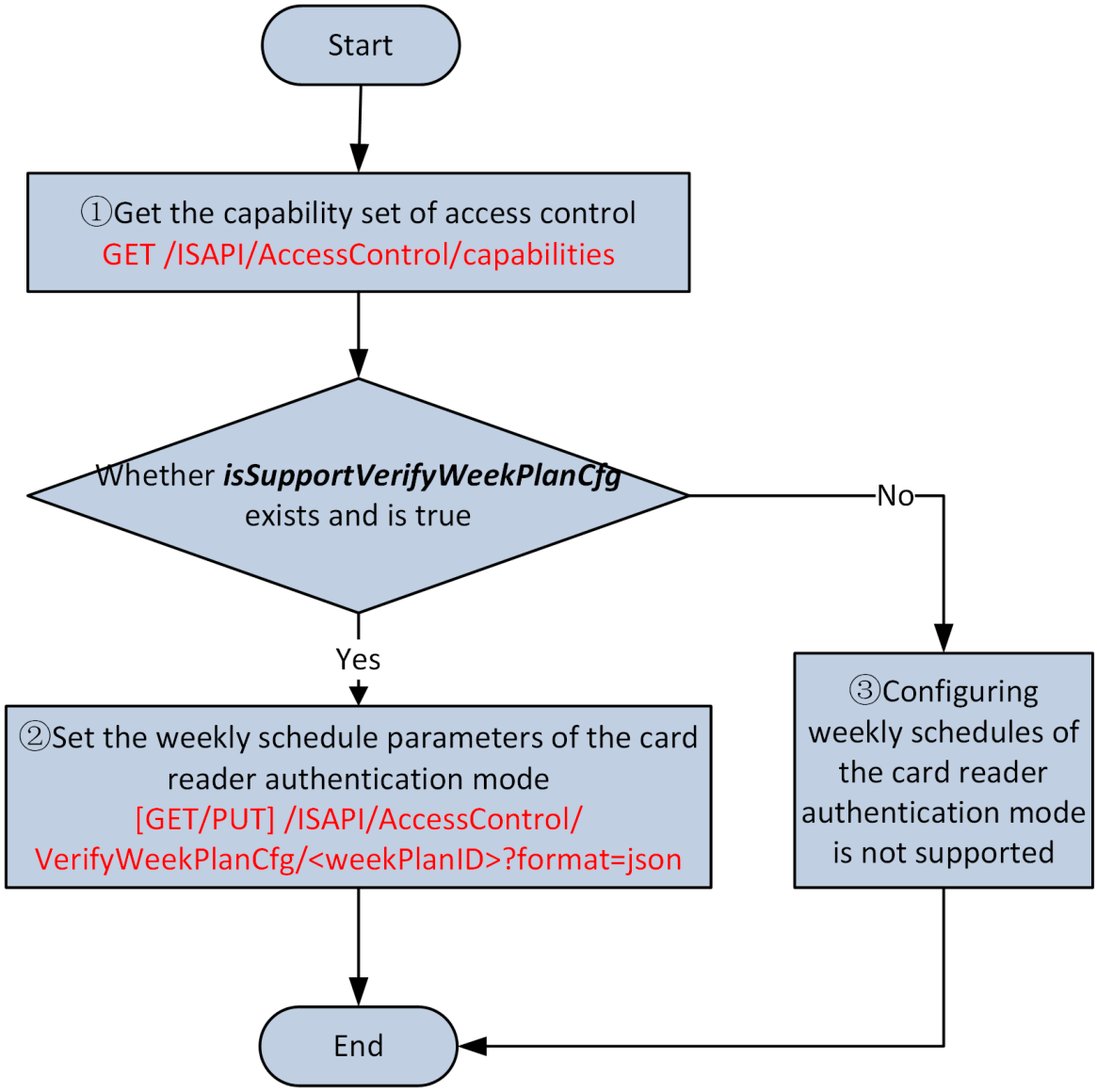
1.2.2 Authentication Schedule Template Configuration



The API calling flow is as follows:

1. Check whether the device supports configuring schedule templates of the card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportVerifyPlanTemplate is returned and its value is "true", it indicates that the device supports configuring schedule templates of the card reader authentication mode (and the device also supports configuring weekly schedules of the card reader authentication mode).
2. Set the schedule template parameters of the card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyPlanTemplate/<planTemplateID>?format=json`.
3. If the node isSupportVerifyPlanTemplate is returned and its value is "false", it indicates that the device does not support configuring schedule templates of the card reader authentication mode.

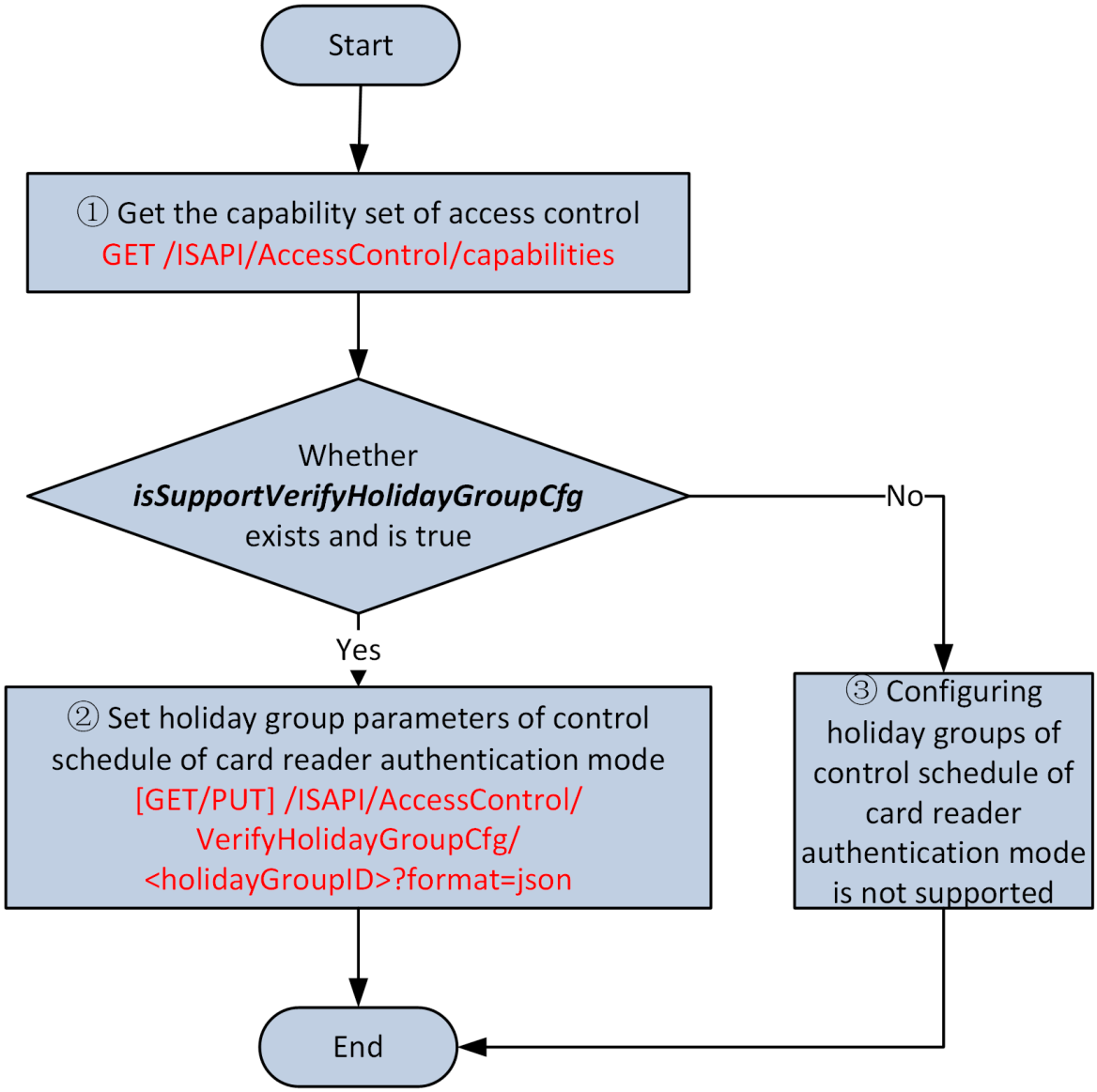
1.2.3 Weekly Authentication Schedule Template Configuration



The API calling flow is as follows:

1. Check whether the device supports configuring weekly schedules of the card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardRightWeekPlanCfg is returned and its value is "true", it indicates that the device supports configuring weekly schedules of the card reader authentication mode.
2. Set the weekly schedule parameters of the card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyWeekPlanCfg/<weekPlanID>?format=json`.
3. If the node isSupportVerifyWeekPlanCfg is returned and its value is "false", it indicates that the device does not support configuring weekly schedules of the card reader authentication mode.

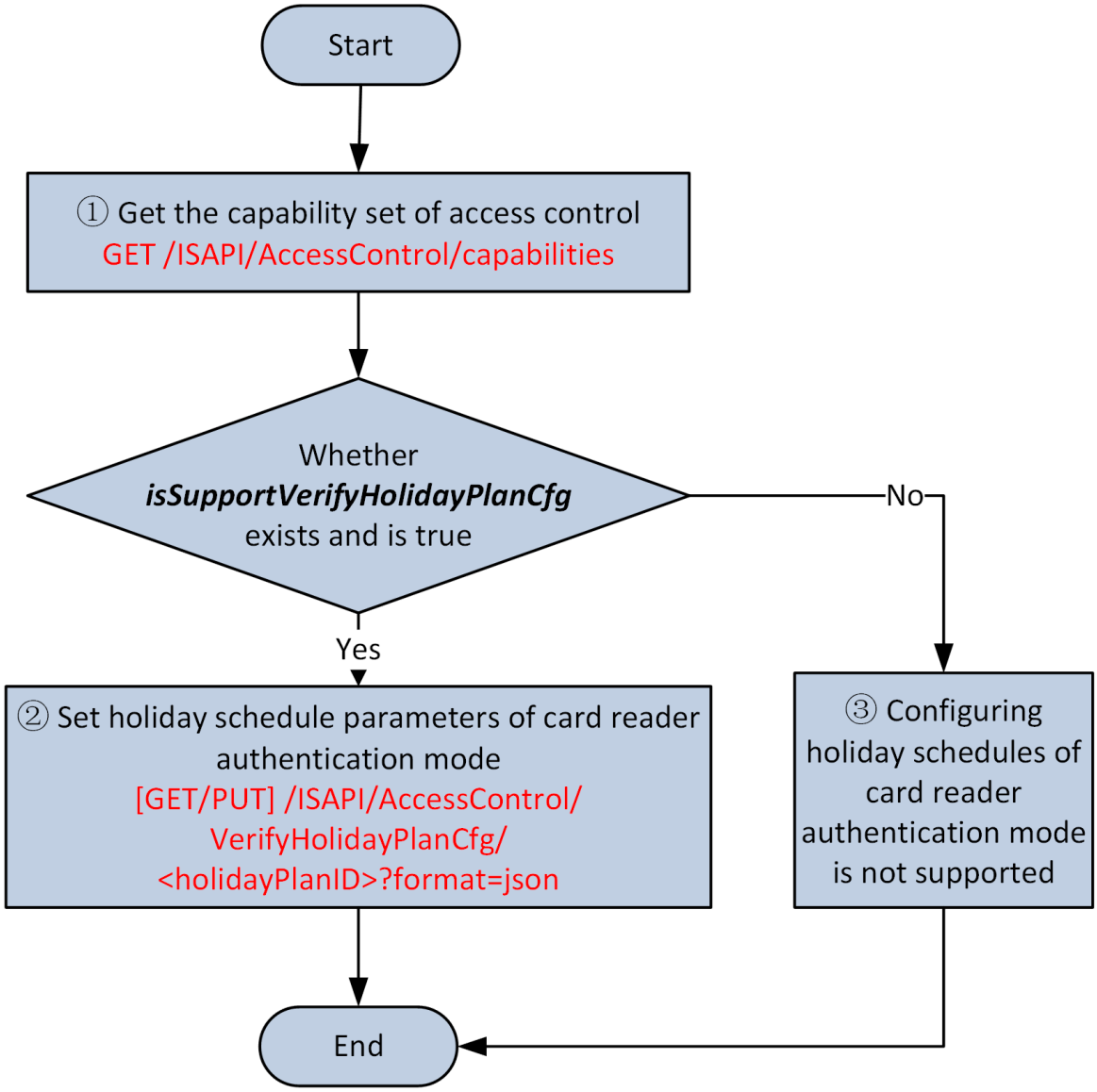
1.2.4 Holiday Authentication Group Configuration



The API calling flow is as follows:

1. Check whether the device supports configuring holiday groups of control schedule of card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportVerifyHolidayGroupCfg is returned and its value is "true", it indicates that the device supports configuring holiday groups of control schedule of card reader authentication mode (and the device also supports configuring holiday schedules of card reader authentication mode).
2. Set holiday group parameters of control schedule of card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyHolidayGroupCfg/<holidayGroupID>?format=json`.
3. If the node isSupportVerifyHolidayGroupCfg is returned and its value is "false", it indicates that the device does not support configuring holiday groups of control schedule of card reader authentication mode.

1.2.5 Holiday Authentication Schedule Configuration



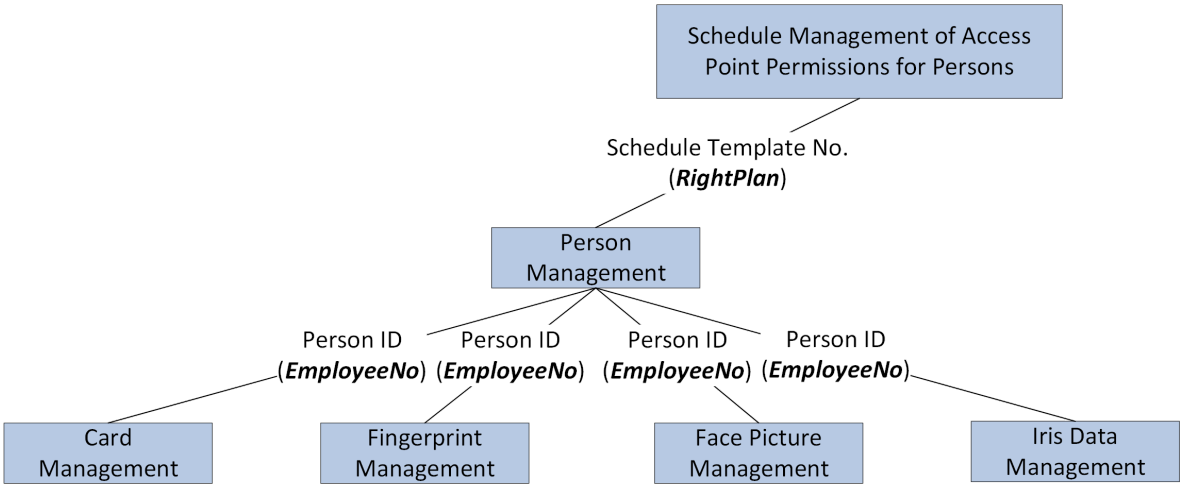
The API calling flow is as follows:

1. Check whether the device supports configuring holiday schedules of card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportVerifyHolidayPlanCfg is returned and its value is "true", it indicates that the device supports configuring holiday schedules of card reader authentication mode.
2. Set holiday schedule parameters of card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyHolidayPlanCfg/<holidayPlanID>?format=json`.
3. If the node isSupportVerifyHolidayPlanCfg is returned and its value is "false", it indicates that the device does not support configuring holiday schedules of card reader authentication mode.

2 Person and Credential Management

2.1 Introduction to the Function

The person and credential management function is person-based, and is for managing persons, credentials (cards, fingerprints, face pictures, and iris data), and permission schedules which control the permissions for persons to enter and exit the controlled areas. Its architecture is shown below.



This document mainly introduces the calling flows for person management and credential management (card, fingerprint, face picture, iris data management). For details about the calling flow for permission schedule management, refer to the “Management of Permission Schedules for Persons and Access Points”.

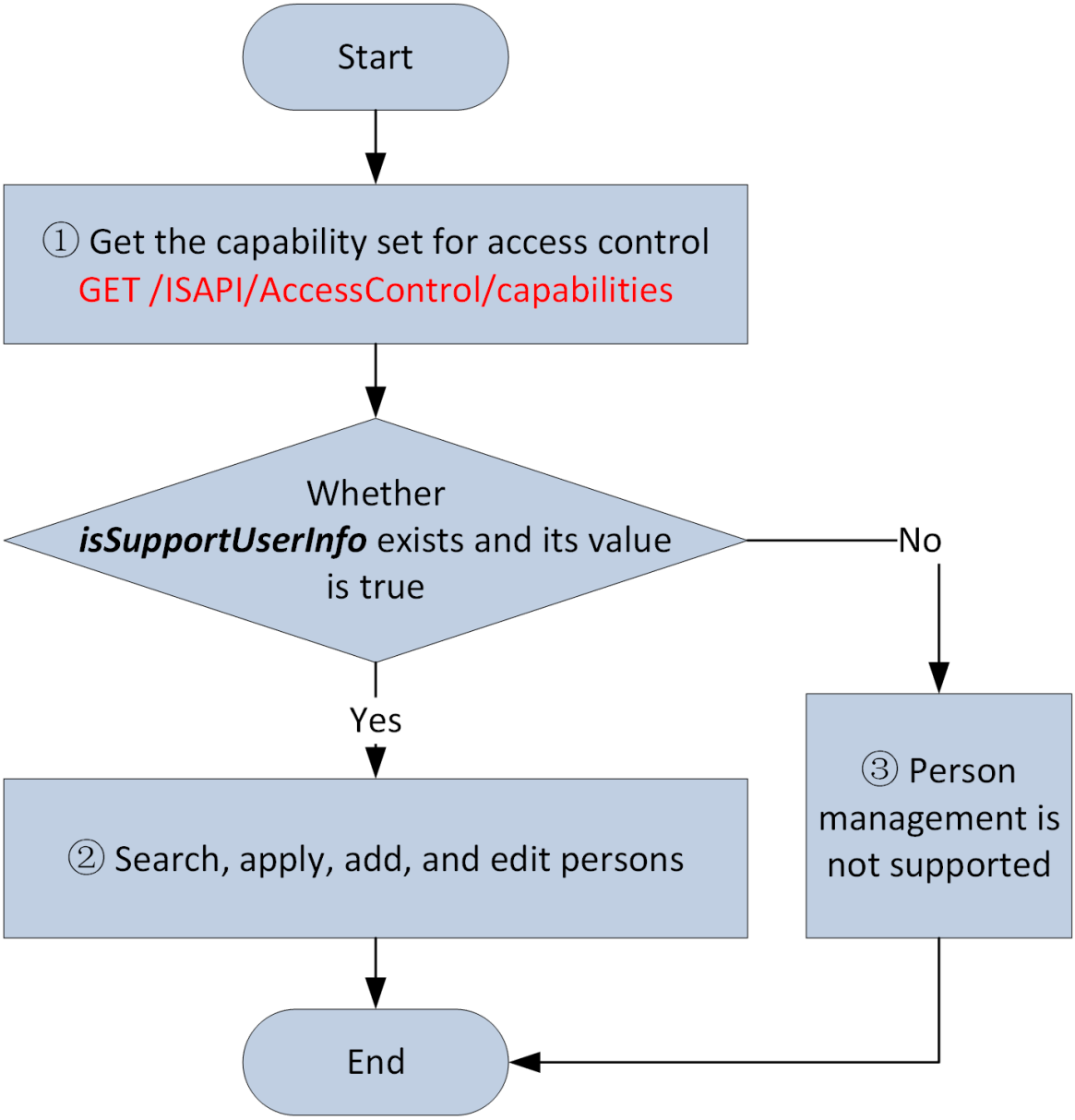
3 Person Management

3.1 Introduction to the Function

Person management includes searching, applying, adding, editing, and deleting persons.

3.2 API Calling Flow

3.2.1 Check Whether the Device Supports Person Management



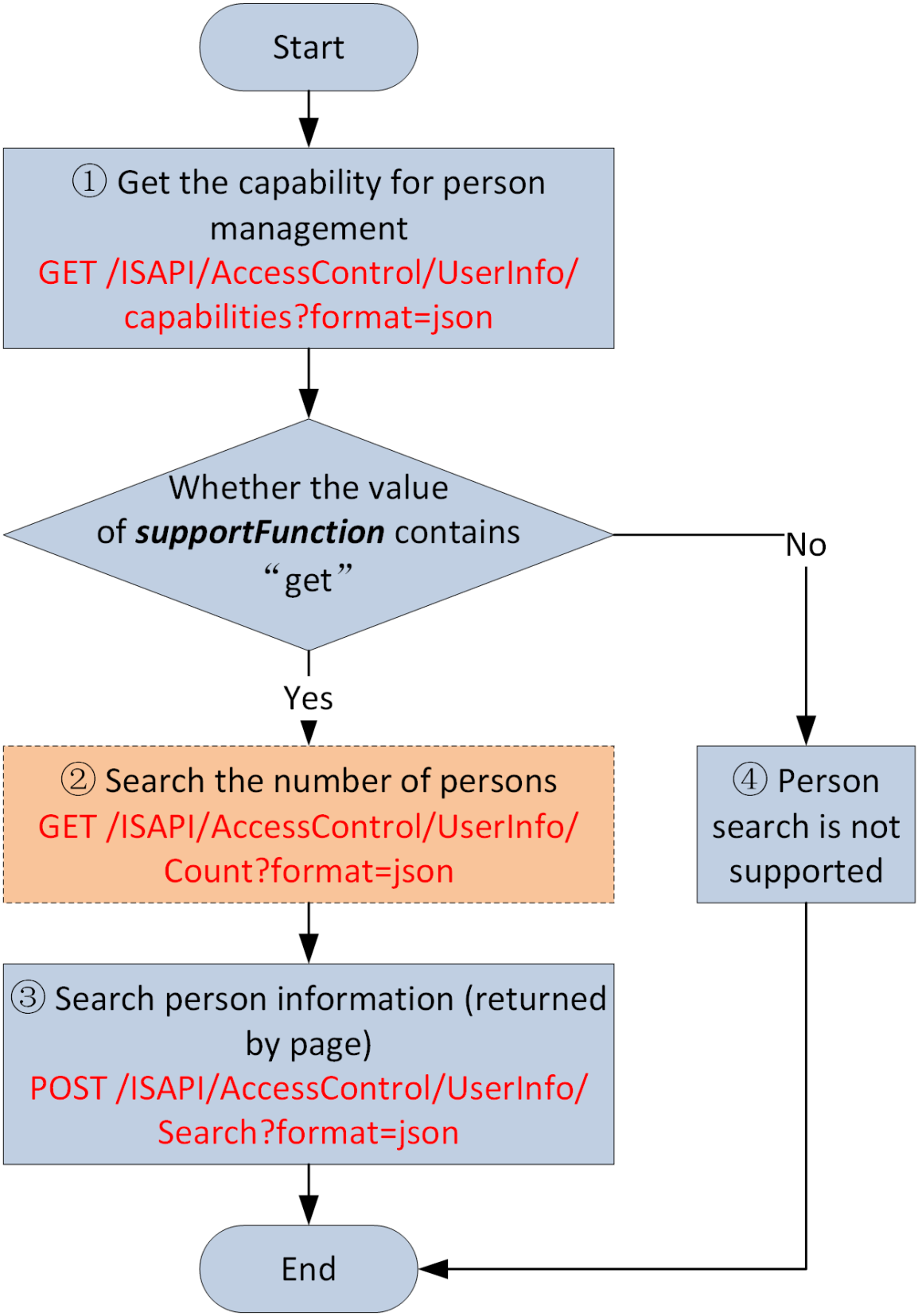
Before calling the API for person management, make sure that the device supports person management.

1. Check whether the device supports person management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportUserInfo is returned and its value is true, it indicates that the device supports person management.
2. Search, apply, add, and edit persons.
3. If the node isSupportUserInfo is returned and its value is false, it indicates that the device does not support person management.

Note:

The person ID (EmployeeNo) is the unique identifier for person and credential management. After calling `GET /ISAPI/AccessControl/capabilities`, through the child nodes of EmployeeNoInfo which are employeeNo, characterType, and isSupportCompress, the maximum string length and character types of the person ID supported by the device can be checked. Generally, devices support up to 32 bytes and any type of characters. But for access controllers and distribution-type access control devices, check through the child nodes mentioned above.

3.2.2 Person Search



The person search function is for searching the number of persons and person information added to the device.

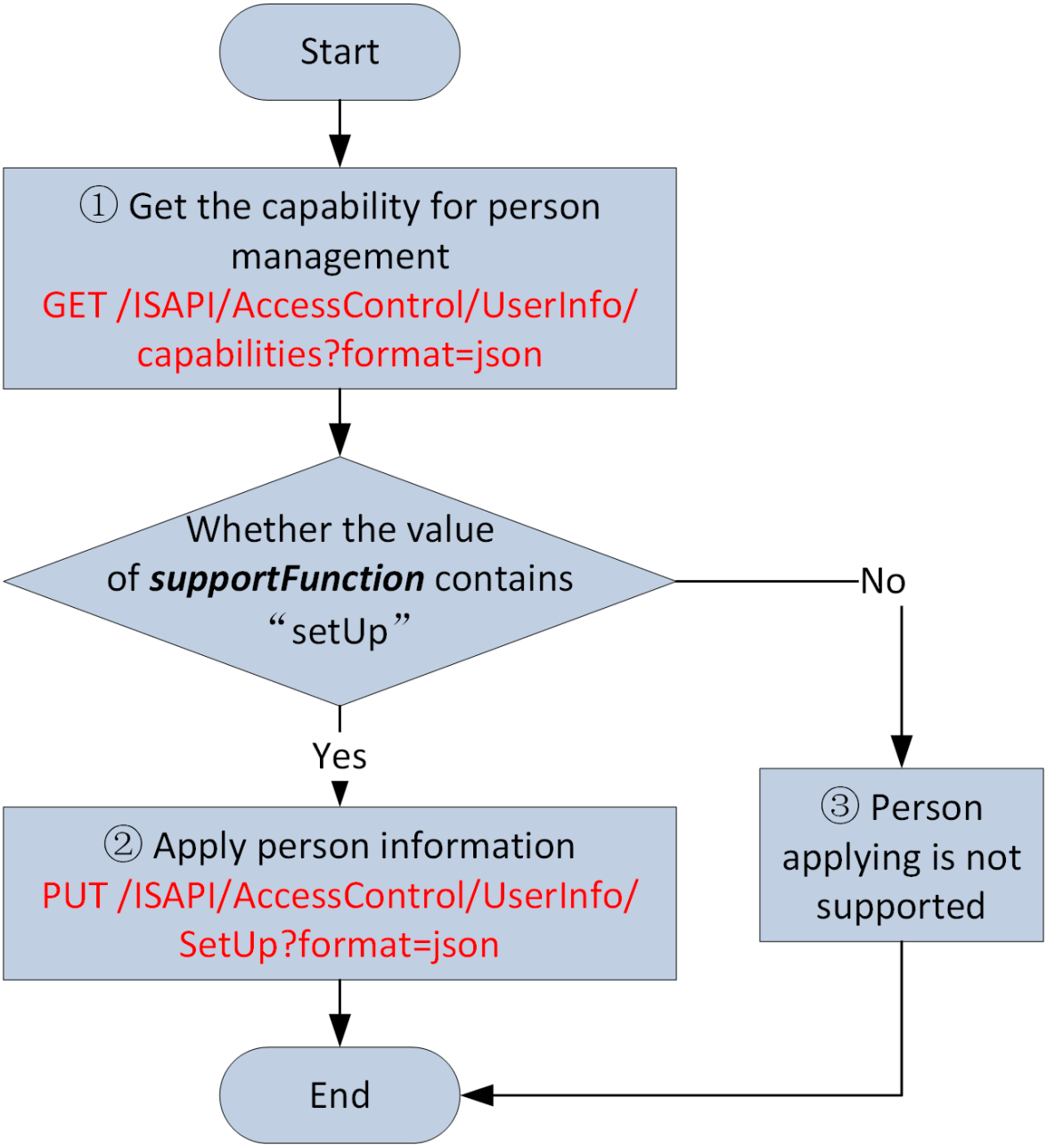
1. Check whether the device supports person search: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains "get", it indicates that the device supports person search.
2. Search the number of persons: `GET /ISAPI/AccessControl/UserInfo/Count?format=json`; the returned value of the node userNumber is the number of the persons added to the device.

3. Search person information: `POST /ISAPI/AccessControl/UserInfo/Search?format=json`; the person information is returned by page.
4. If the value of the node supportFunction does not contain “get”, it indicates that the device does not support person search.

Note:

The value of the node maxRecordNum returned by calling `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json` is the maximum number of persons supported by the device.

3.2.3 Person Applying



Person information can be applied to the device via the person applying function. If the person has been added to the device, the person information will be edited; if the person has not been added to the device, the person information will be applied to the device.

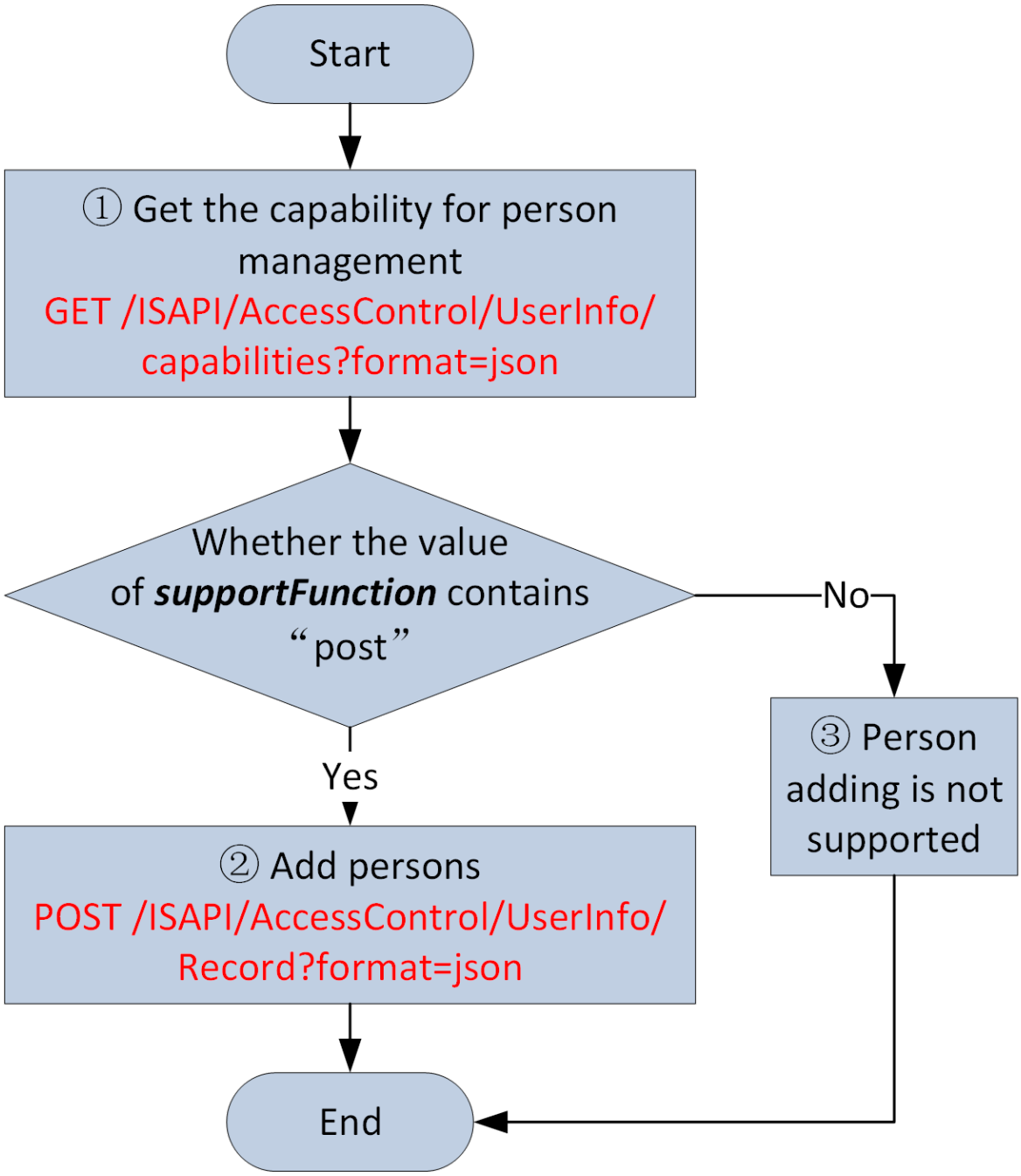
1. Check whether the device supports person applying: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains “setUp”, it indicates that the device supports person applying.
2. Apply person information: `PUT /ISAPI/AccessControl/UserInfo/SetUp?format=json`.

3. If the value of the node supportFunction does not contain setUp, it indicates that the device does not support person applying.

Note:

Check whether the person has been added to the device via the node employeeNo returned after calling the API for person applying.

3.2.4 Person Adding



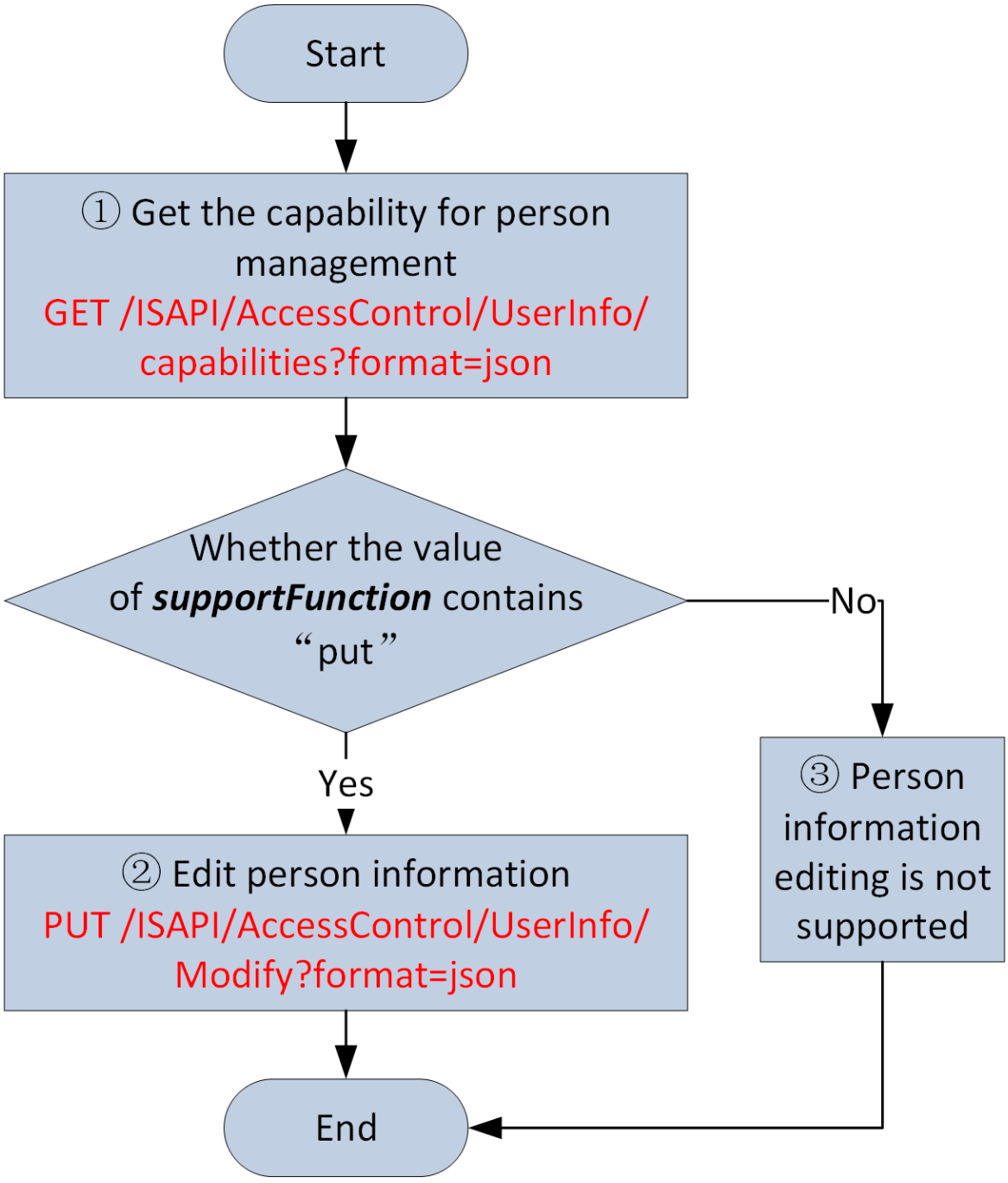
Person can be added to the device via the person adding function. If the person has been added to the device, the device will report an error; if the person has not been added to the device, the person will be added to the device.

1. Check whether the device supports person adding: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports person adding.
2. Add persons: `POST /ISAPI/AccessControl/UserInfo/Record?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support person adding.

Note:

Check whether the person has been added to the device via the node employeeNo returned after calling the API for person adding.

3.2.5 Person Information Editing



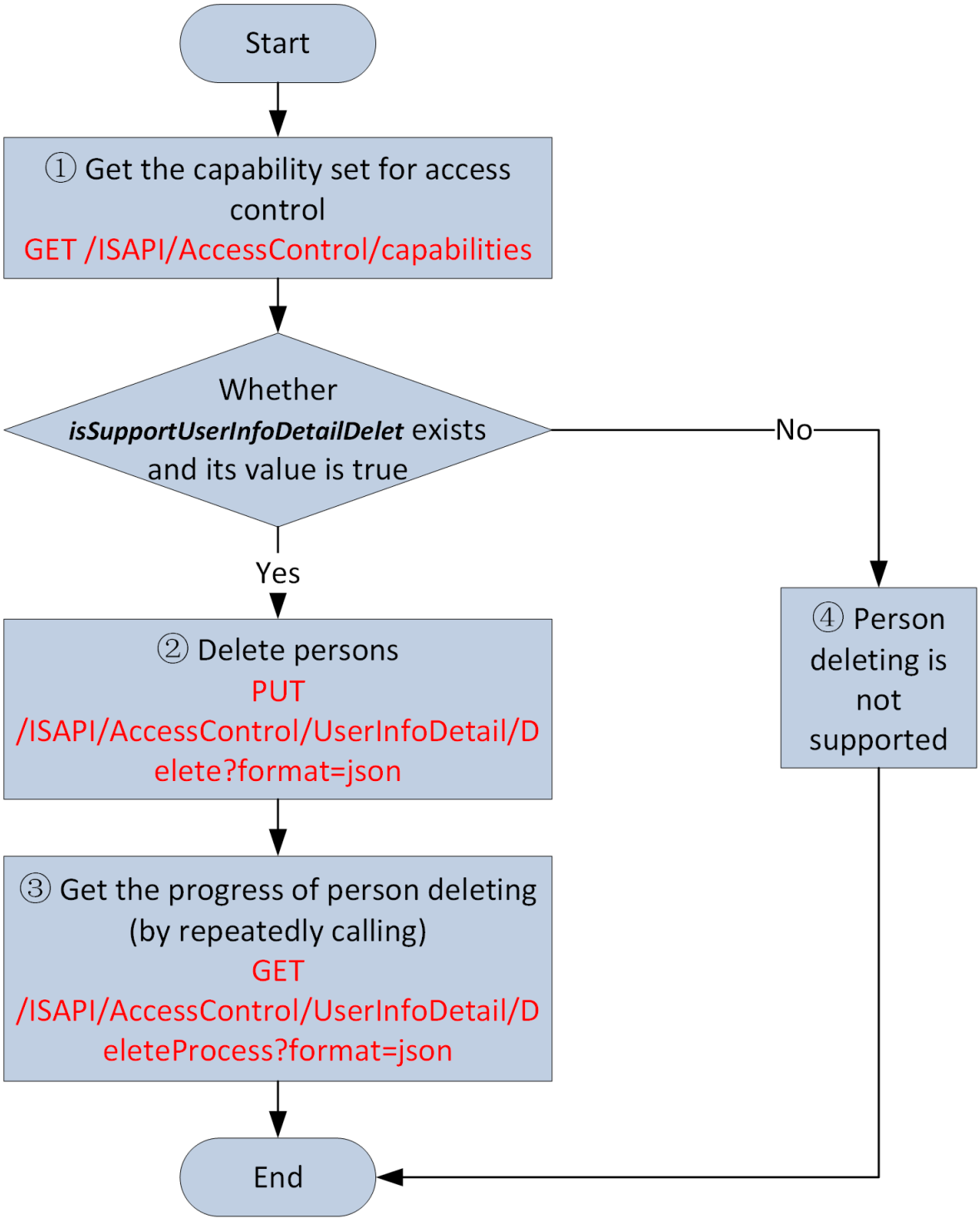
Person information added to the device can be edited via the person information editing function. If the person has been added to the device, the person information will be edited; if the person has not been added to the device, the device will report an error.

1. Check whether the device supports person information editing: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports person information editing.
2. Edit Person Information: `PUT /ISAPI/AccessControl/UserInfo/Modify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support person information editing.

Note:

Check whether the person has been added to the device via the node employeeNo returned after calling the API for person information editing.

3.2.6 Person Deleting



The person added to the device can be deleted via the person deleting function. The device will not report an error if the person to be deleted is not added to the device.

1. Check whether the device supports person deleting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportUserInfoDetailDelete is returned and its value is "true", it indicates that the device supports person deleting.
2. Delete persons: `PUT /ISAPI/AccessControl/UserInfoDetail/Delete?format=json`; if calling succeeded, it indicates that the device has started to execute person deleting, but it does not indicate that the device has deleted the person.

3. Get the progress of deleting person information: `GET /ISAPI/AccessControl/UserInfoDetail/DeleteProcess`; repeatedly call this API to get the progress of person deleting.
4. If the node isSupportUserInfoDetailDelete is returned and its value is "false", it indicates that the device does not support person deleting.

Note:

When the person is deleted, the information on the credentials (the card, fingerprint, face picture, and iris data) linked via the person ID will also be deleted.

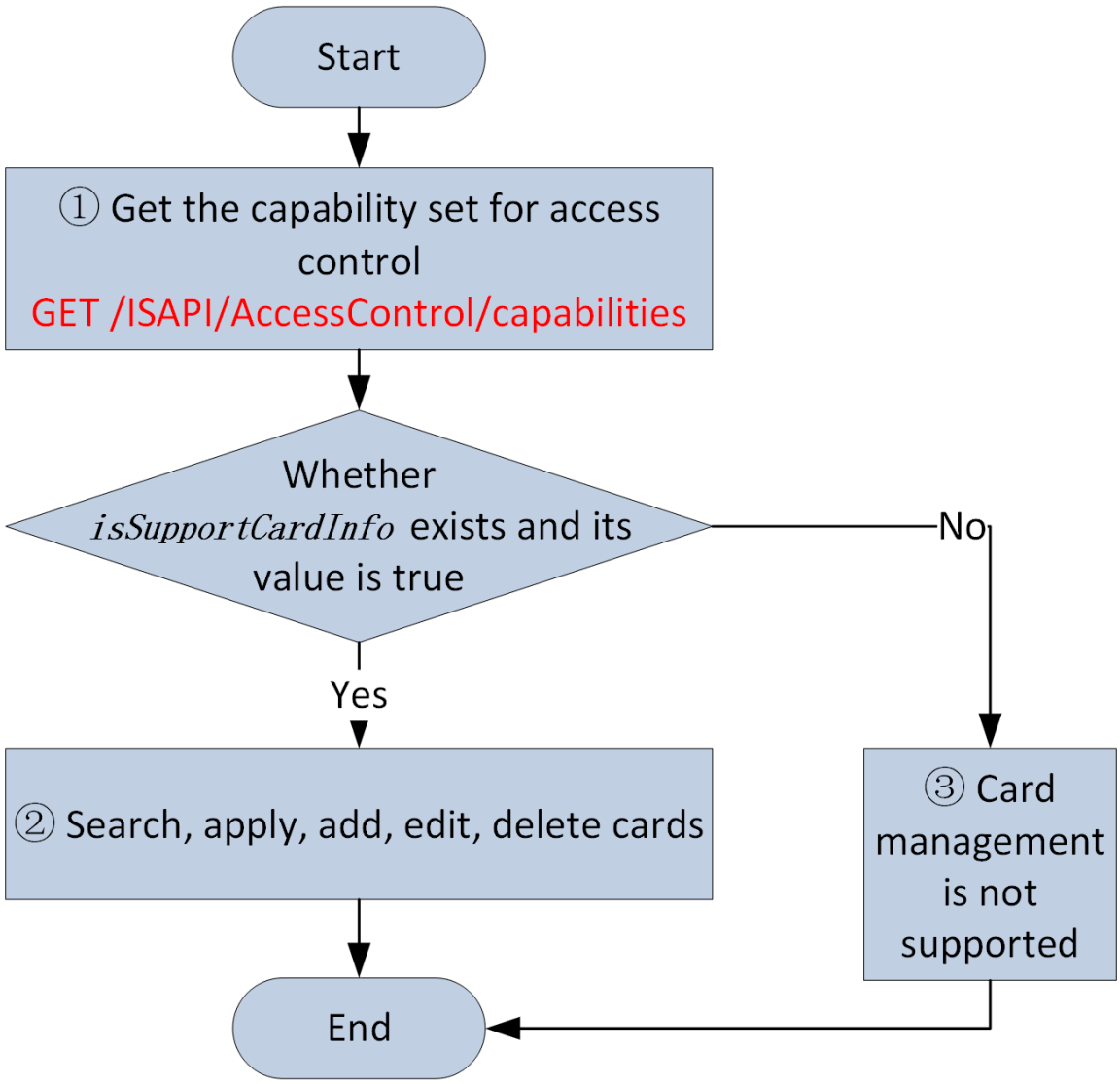
4 Card Management

4.1 Introduction to the Function

Card management includes searching, applying, adding, editing, deleting, and collecting cards.

4.2 API Calling Flow

4.2.1 Check Whether the Device Supports Card Management



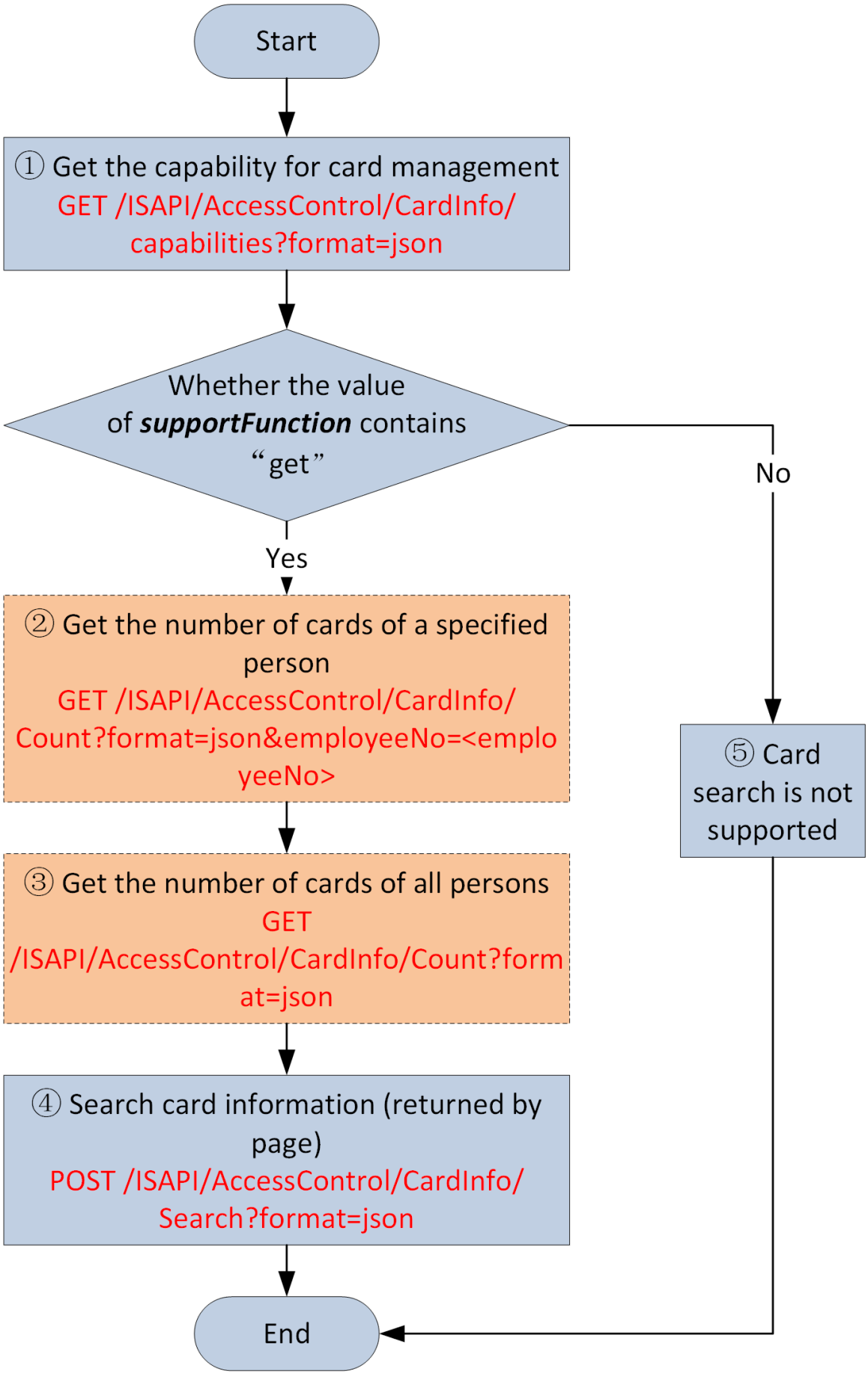
Before calling the API for card management, make sure that the device supports card management.

1. Check whether the device supports card management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardInfo is returned and its value is “true”, it indicates that the device supports card management.
2. Search, apply, add, edit, and delete cards.
3. If the node isSupportCardInfo is returned and its value is “false”, it indicates that the device does not support card management.

Note:

- Before applying, adding, or editing cards on the device, make sure that the related person information linked to the person ID has been applied to the device.
- The value of the node numberPerPerson returned by calling `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json` is the maximum number of cards supported per person. If the value returned is 255, it indicates that the number of cards per person is unlimited. If the node is not returned, it indicates that the maximum number of cards can be applied is 5.
- Manage cards of different card number lengths by calling `[GET/PUT] /ISAPI/AccessControl/CardVerificationRule?format=json`.

4.2.2 Card Search



The card search function is for searching the number of cards and card information applied to the device.

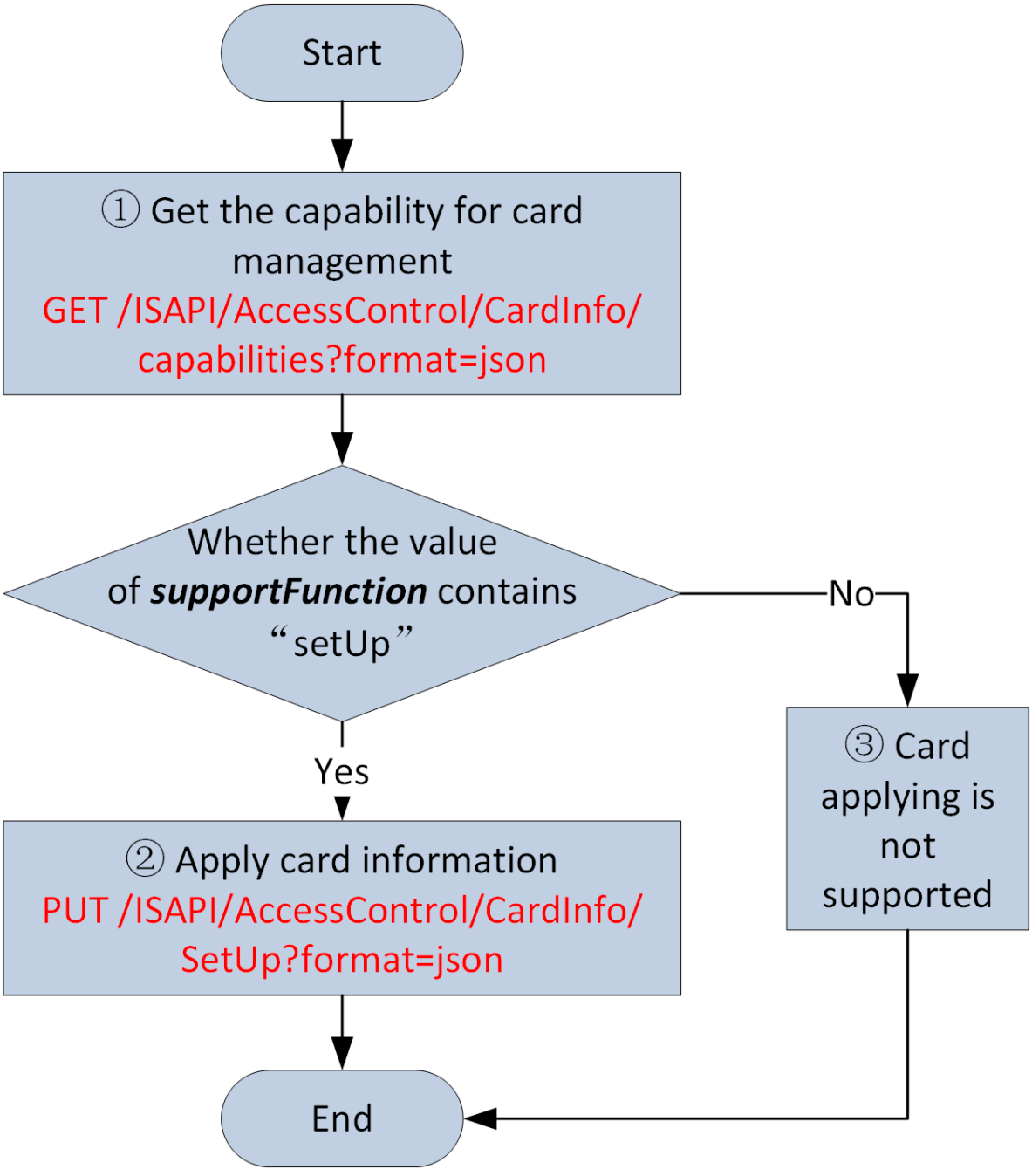
1. Check whether the device supports card search: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "get", it indicates that the device supports card search.

2. Search the number of specified persons' cards: `GET /ISAPI/AccessControl/CardInfo/Count?format=json&employeeNo=<employeeNo>`; the returned value of the node cardNumber is the number of the cards added to the specified persons.
3. Search the number of all persons' cards: `GET /ISAPI/AccessControl/CardInfo/Count?format=json`; the returned value of the node cardNumber is the number of the cards added to all persons.
4. Search card information: `POST /ISAPI/AccessControl/CardInfo/Search?format=json`; the card information is returned by page.
5. If the value of the node supportFunction does not contain "get", it indicates that the device does not support card search.

Note:

The value of the node maxRecordNum returned by calling `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json` is the maximum number of cards supported by the device.

4.2.3 Card Applying



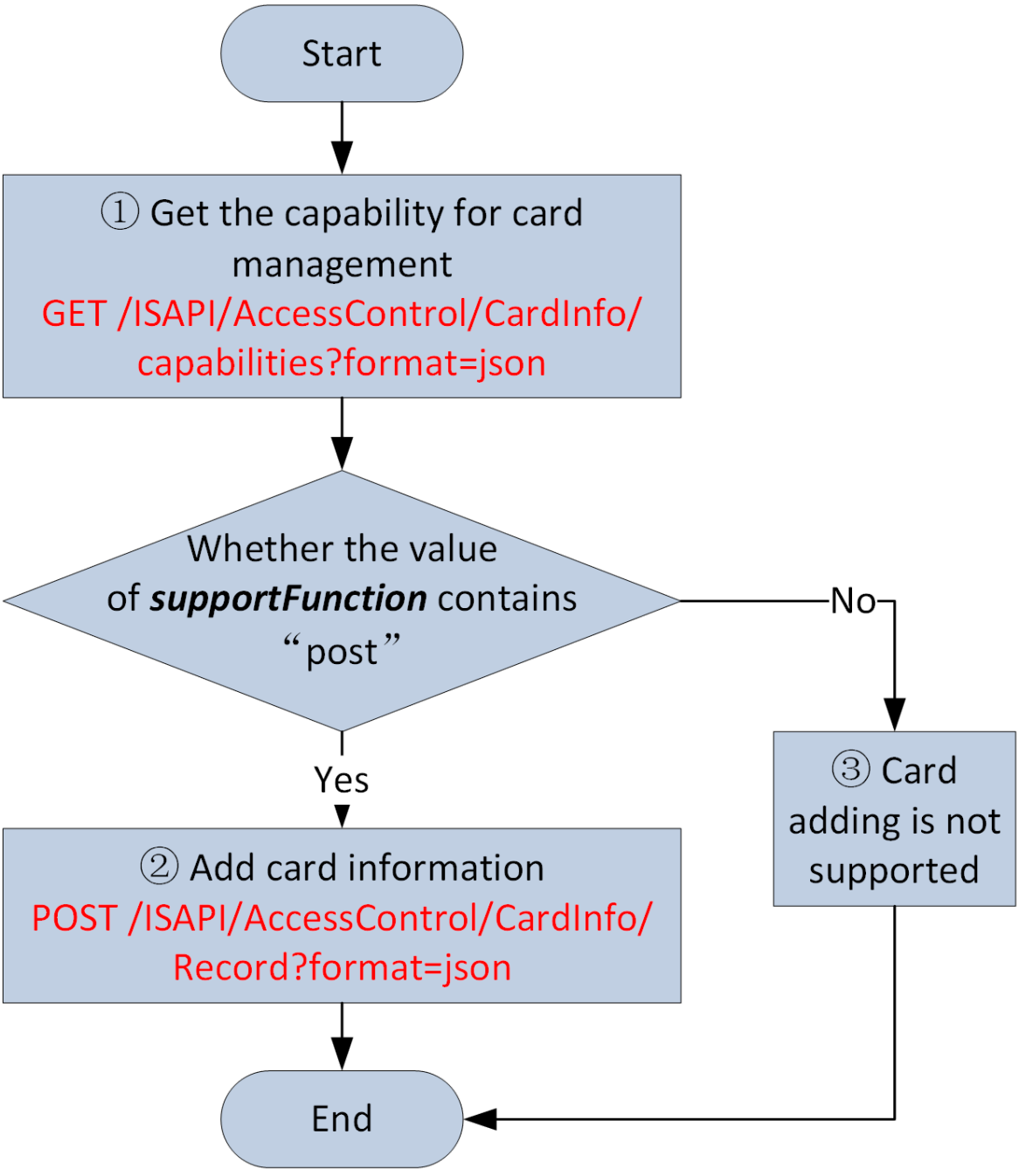
Card information can be applied to the device via the card applying function. If the card has been added to the device, the card information will be edited; if the card has not been added to the device, the card information will be added to the device.

1. Check whether the device supports card applying: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "setUp", it indicates that the device supports card applying.
2. Apply card information: `PUT /ISAPI/AccessControl/CardInfo/SetUp?format=json`.
3. If the value of the node supportFunction does not contain "setUp", it indicates that the device does not support card applying.

Note:

Check whether the card has been added to the device via the node cardNo returned after calling the API for card applying.

4.2.4 Card Adding



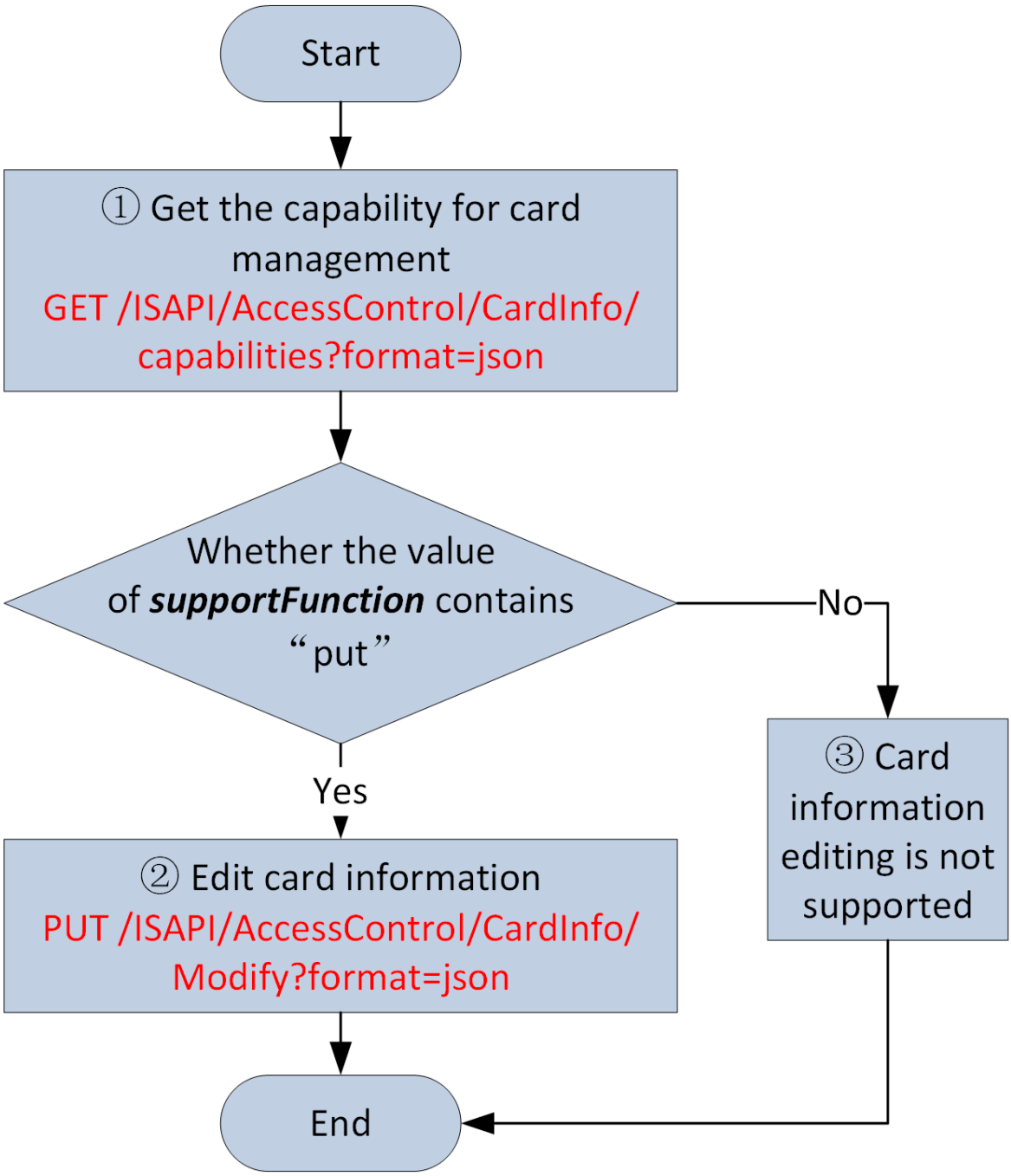
Card information can be added to the device via the card adding function. If the card has been added to the device, the device will report an error; if the card has not been added to the device, the card information will be added to the device.

1. Check whether the device supports card adding: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports card adding.
2. Add card information: `POST /ISAPI/AccessControl/CardInfo/Record?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support card adding.

Note:

Check whether the card has been added to the device via the node cardNo returned after calling the API for card adding.

4.2.5 Card Information Editing



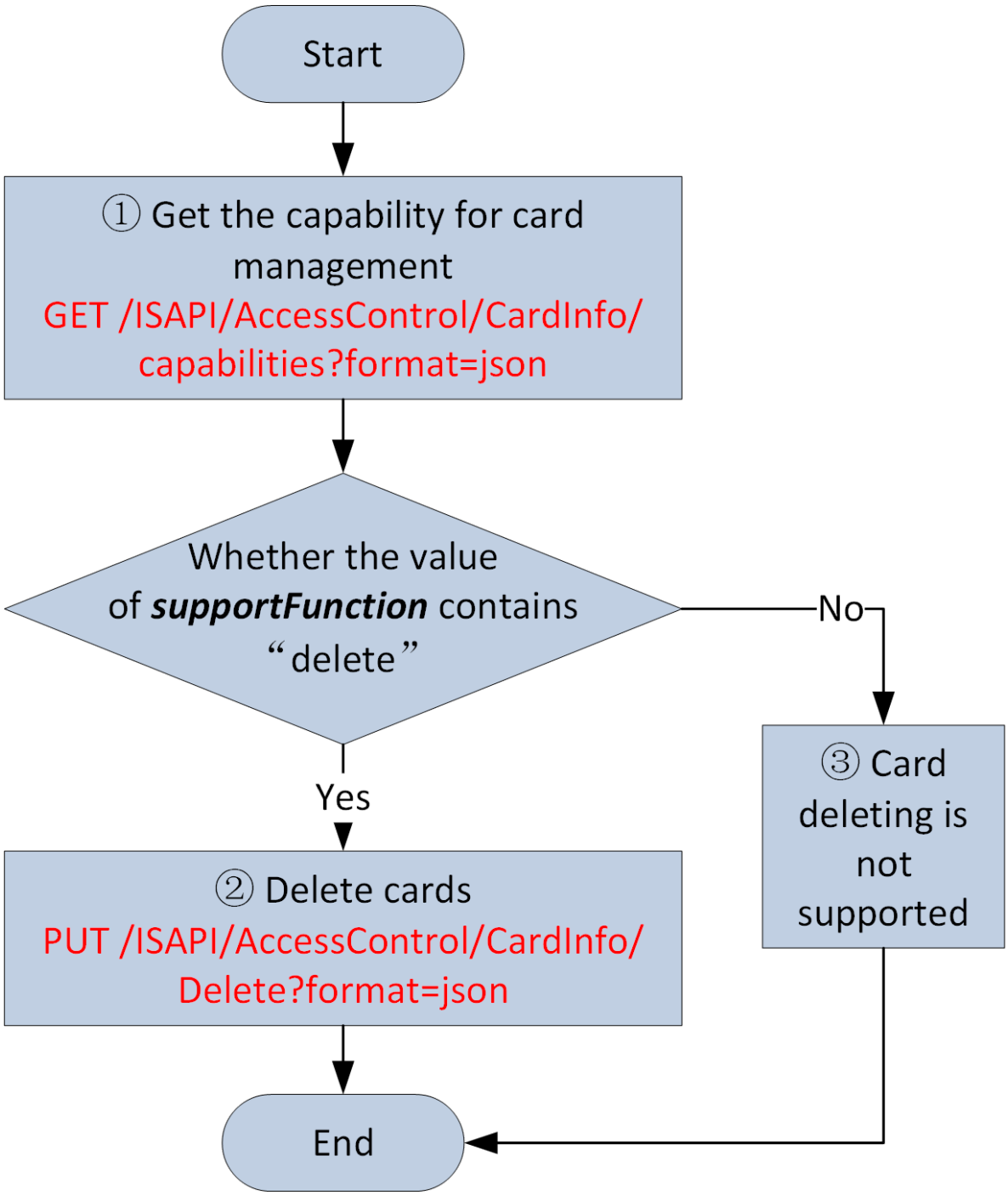
Card information on the device can be edited via the card information editing function. If the card has been added to the device, the card information will be edited; if the card has not been added to the device, the device will report an error.

1. Check whether the device supports card information editing: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports card information editing.
2. Edit card information: `PUT /ISAPI/AccessControl/CardInfo/Modify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support card information editing.

Note:

Check whether the card has been added to the device via the node cardNo returned after calling the API for card information editing.

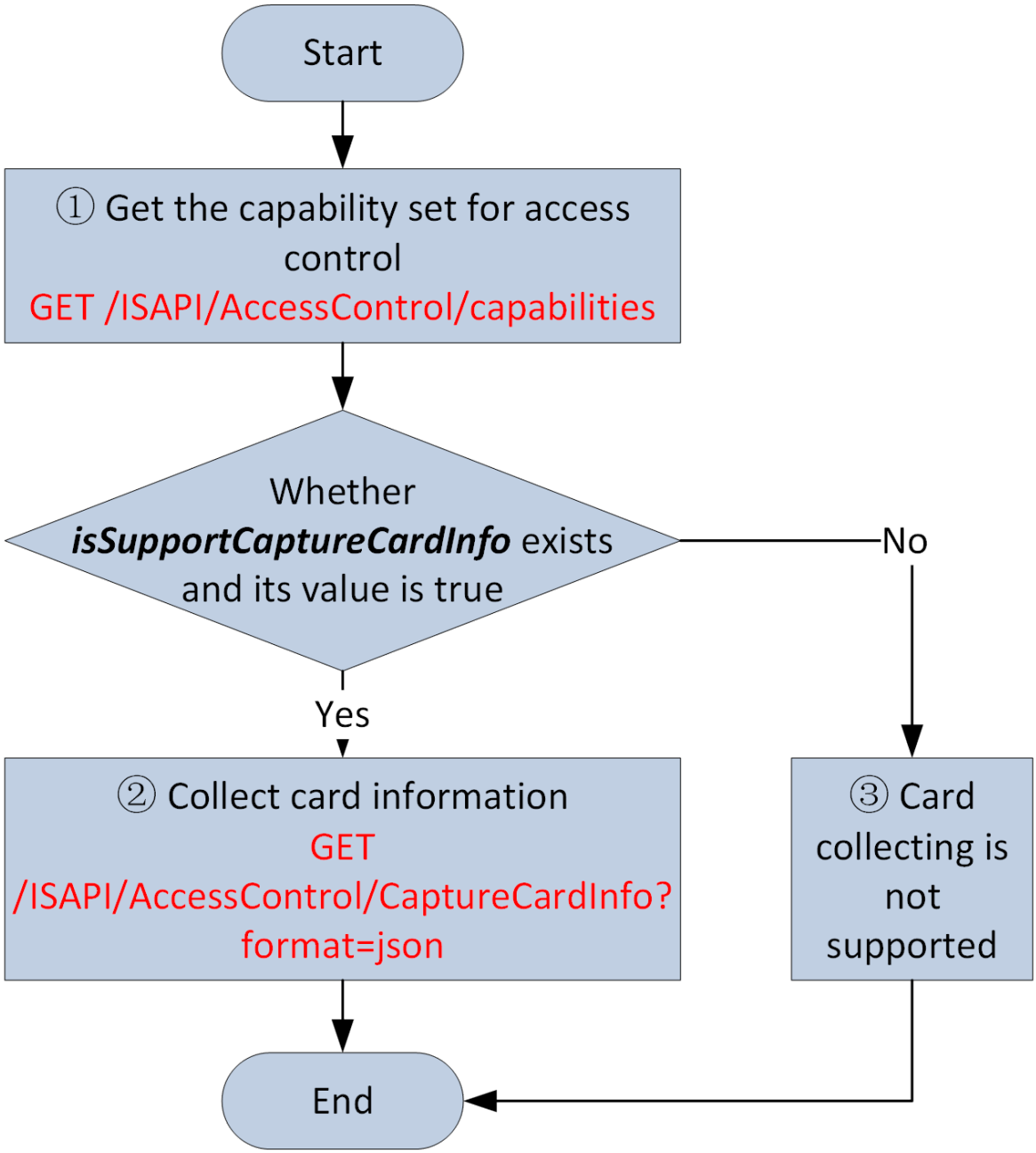
4.2.6 Card Deleting



The card information on the device can be deleted via the card deleting function. The device will not report an error if the card information to be deleted is not added to the device.

1. Check whether the device supports card deleting: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "delete", it indicates that the device supports card deleting.
2. Delete cards: `PUT /ISAPI/AccessControl/CardInfo/Delete?format=json`; if calling succeeded, it indicates that the device has deleted the cards.
3. If the value of the node supportFunction does not contain "delete", it indicates that the device does not support card deleting.

4.2.7 Card Collecting



The card collecting function is for collecting the card No., card type, etc.

1. Check whether the device supports card collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureCardInfo is returned and its value is "true", it indicates that the device supports card collecting.
2. Collect card information: `GET /ISAPI/AccessControl/CaptureCardInfo?format=json`.

3. If the node isSupportCaptureCardInfo is returned and its value is “false”, it indicates that the device does not support card collecting.

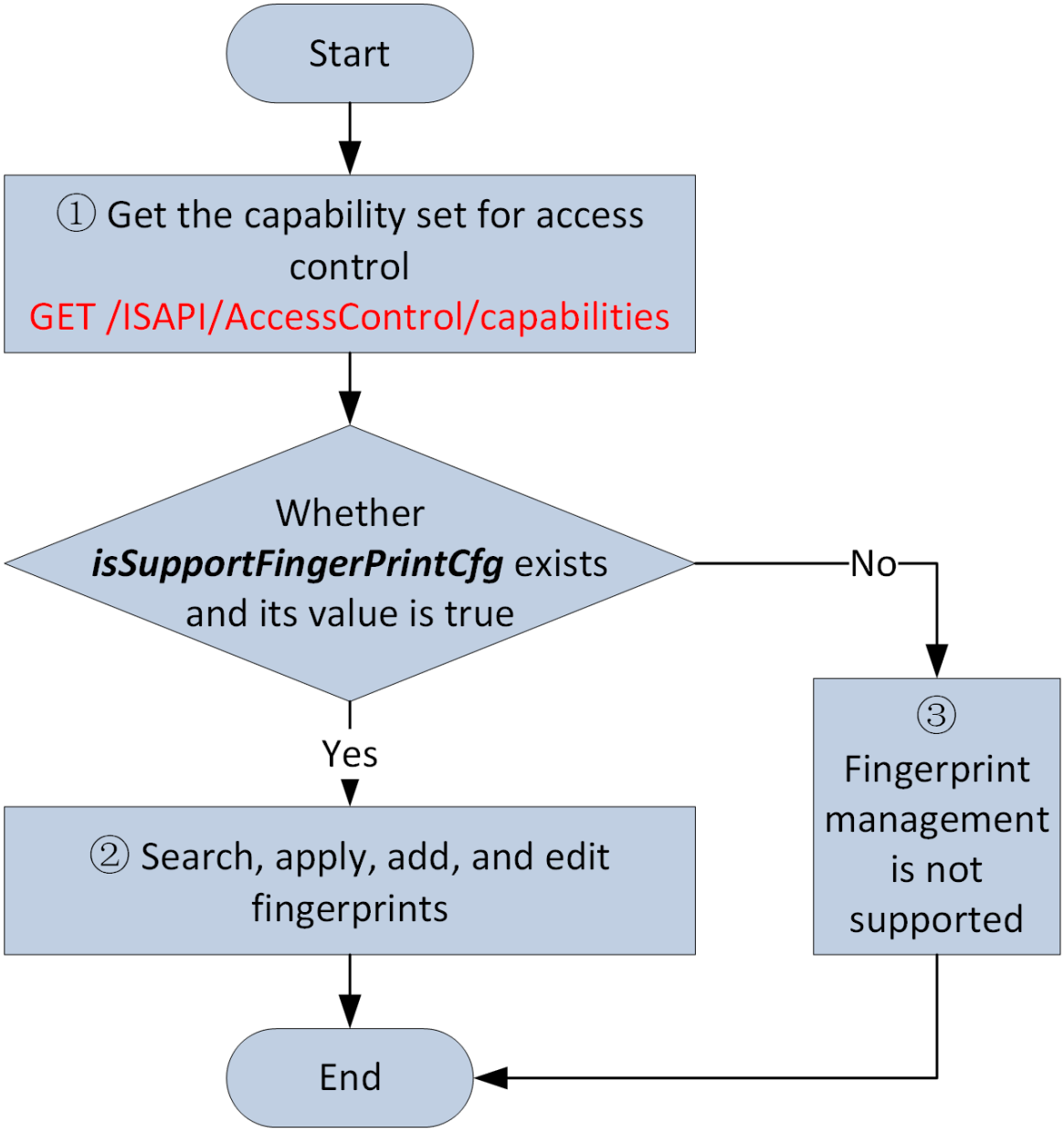
5 Fingerprint Management

5.1 Introduction to the Function

Fingerprint management includes searching, applying, deleting, and collecting fingerprints.

5.2 API Calling Flow

5.2.1 Check Whether the Device Supports Fingerprint Management



Before calling the API for fingerprint management, make sure that the device supports fingerprint management.

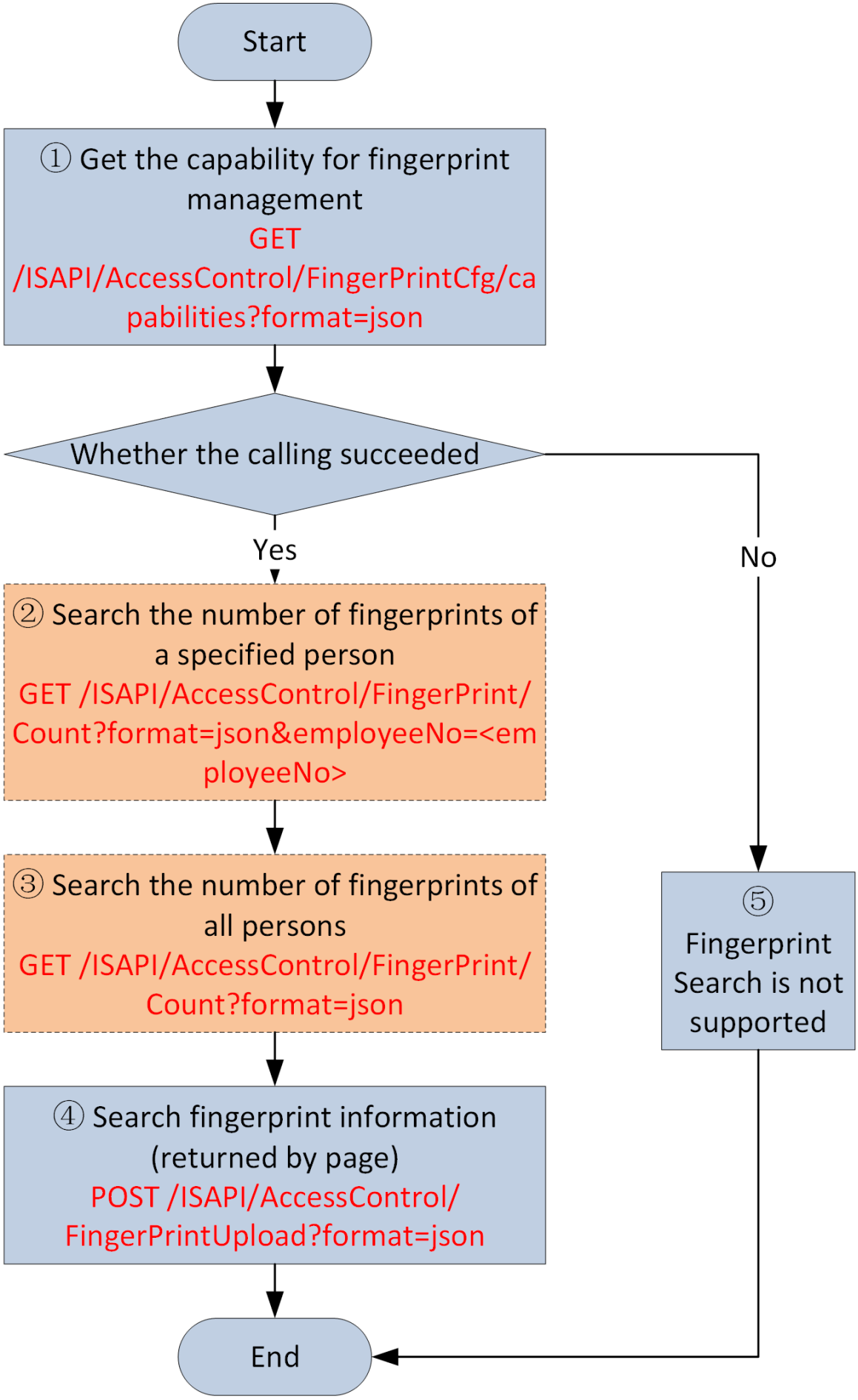
1. Check whether the device supports fingerprint management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportFingerPrintCfg is returned and its value is “true”, it indicates that the device supports fingerprint management.
2. Search, apply, add, and edit fingerprints.

3. If the node isSupportFingerPrintCfg is returned and its value is “false”, it indicates that the device does not support fingerprint management.

Note:

- Before applying the fingerprint information to the device, make sure that the related person information linked to the person ID has been applied to the device.
- The maximum number of fingerprints that can be applied to the device per person is 10 (the 10 fingerprints of a person).

5.2.2 Fingerprint Search



The fingerprint search function is for searching the number of fingerprints and fingerprint information added to the device.

1. Check whether the device supports fingerprint search: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if calling succeeded,

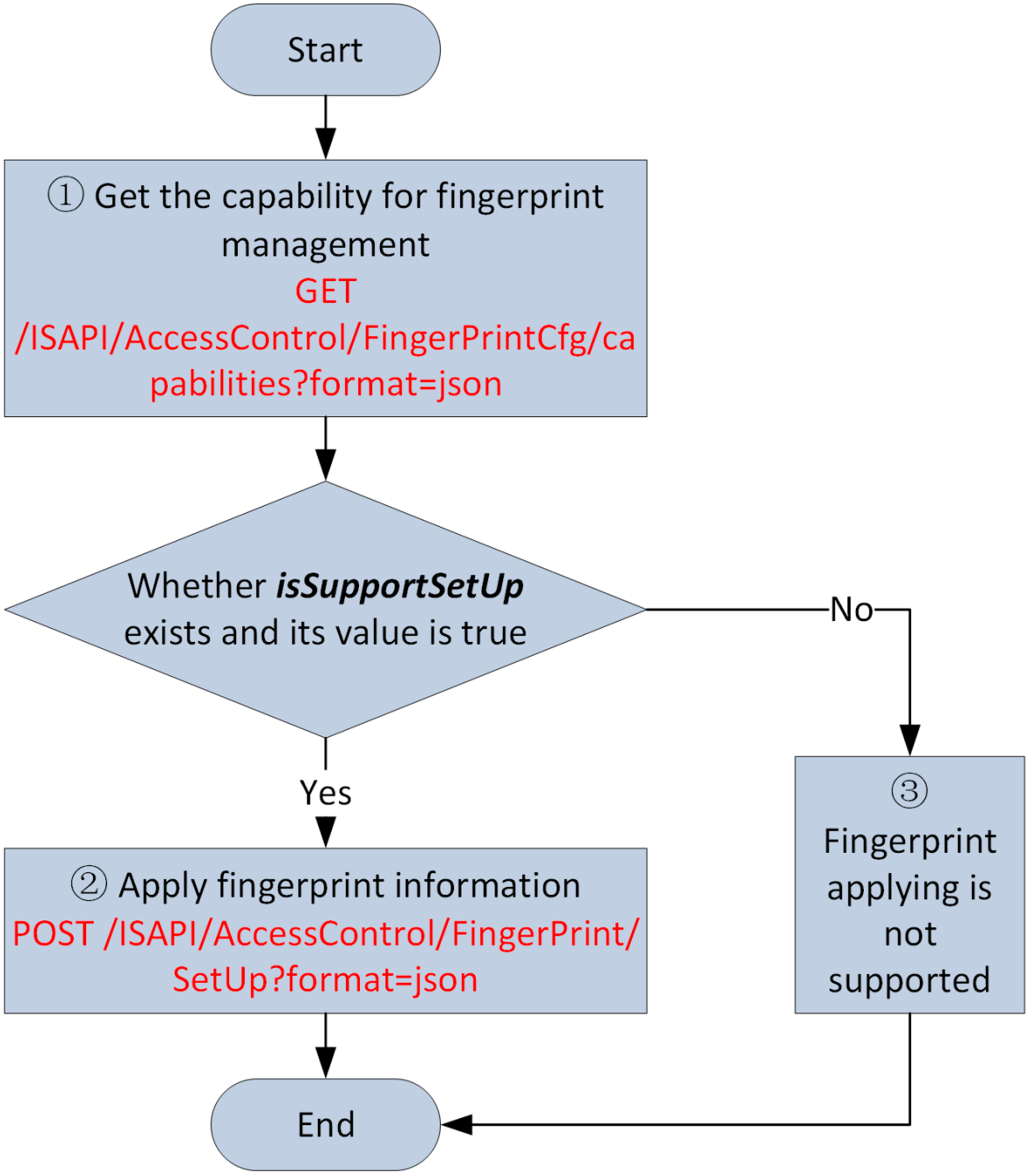
it indicates that the device supports fingerprint search.

2. Search the number of the specified persons' fingerprints: `GET /ISAPI/AccessControl/FingerPrint/Count?format=json&employeeNo=<employeeNo>`; the returned value of the node numberOfFP is the number of the added fingerprints of the specified persons.
3. Search the number of fingerprints of all persons: `GET /ISAPI/AccessControl/FingerPrint/Count?format=json`; the returned value of the node numberOfFP is the number of the added fingerprints of all persons.
4. Search fingerprint information: `POST /ISAPI/AccessControl/FingerPrintUpload?format=json`; the fingerprint information is returned by page. If the value of the child node status of the node FingerPrintInfo is "NoFP", it indicates that all fingerprint information are returned.
5. If calling failed, it indicates that the device does not support fingerprint search.

Note:

- The value of the node fingerPrintCapacity returned by calling `GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json` is the maximum number of fingerprints supported by the card reader.
- The value of the node fingerPrintNum returned by calling `GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json` is the number of fingerprints added to the card reader.

5.2.3 Fingerprint Applying



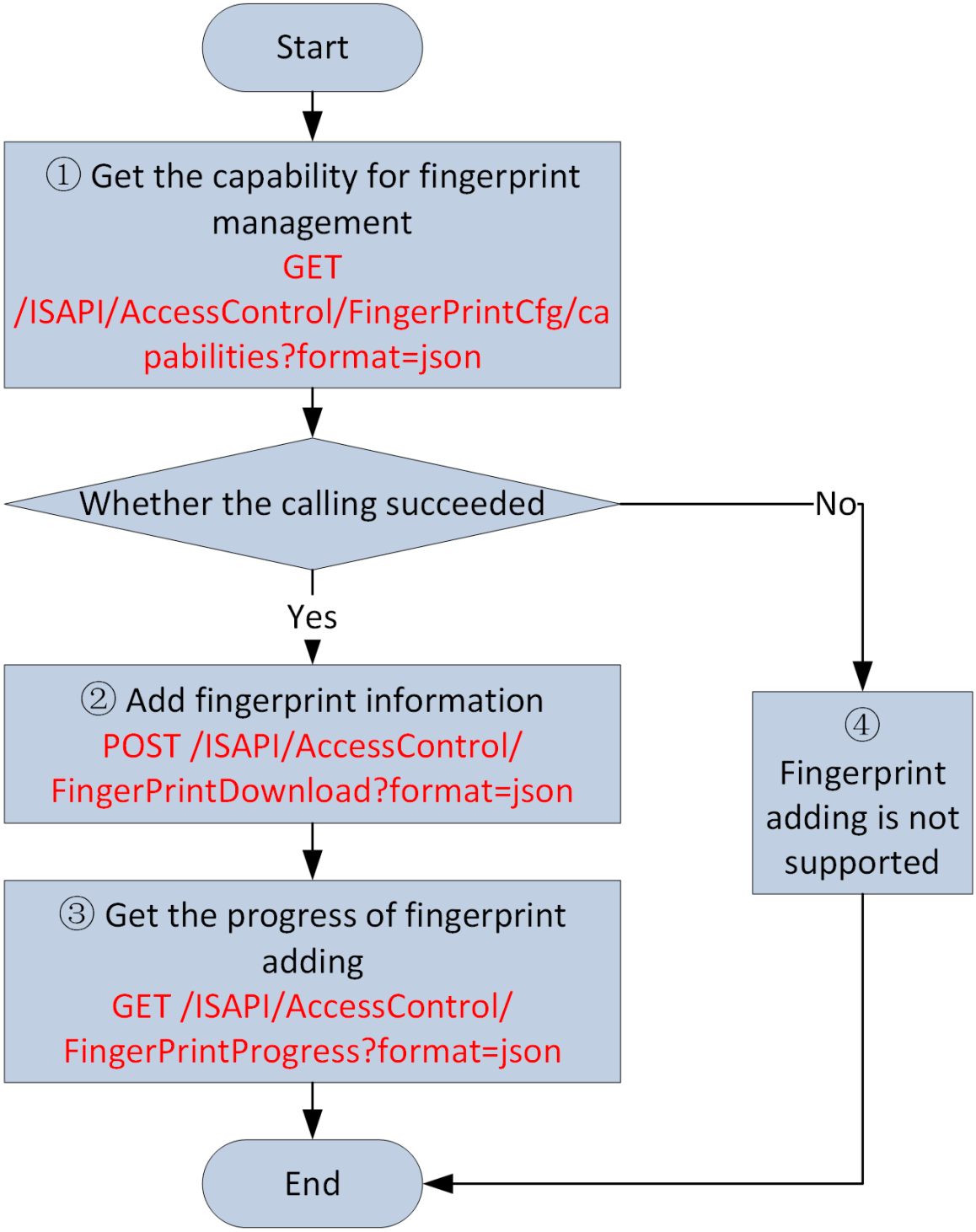
Fingerprint information can be applied to the device via the fingerprint applying function. If the fingerprint has been added to the device, the fingerprint information will be edited; if the fingerprint has not been added to the device, the fingerprint will be added to the device.

1. Check whether the device supports fingerprint applying: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if the node isSupportSetUp is returned and its value is "true", it indicates that the device supports fingerprint applying.
2. Apply fingerprint information: `POST /ISAPI/AccessControl/FingerPrint/SetUp?format=json`.
3. If the node isSupportSetUp is returned and its value is false, it indicates that the device does not support fingerprint applying.

Note:

Check whether the fingerprint has been added to the device via the nodes employeeNo and fingerPrintID returned after calling the API for fingerprint applying.

5.2.4 Fingerprint Adding



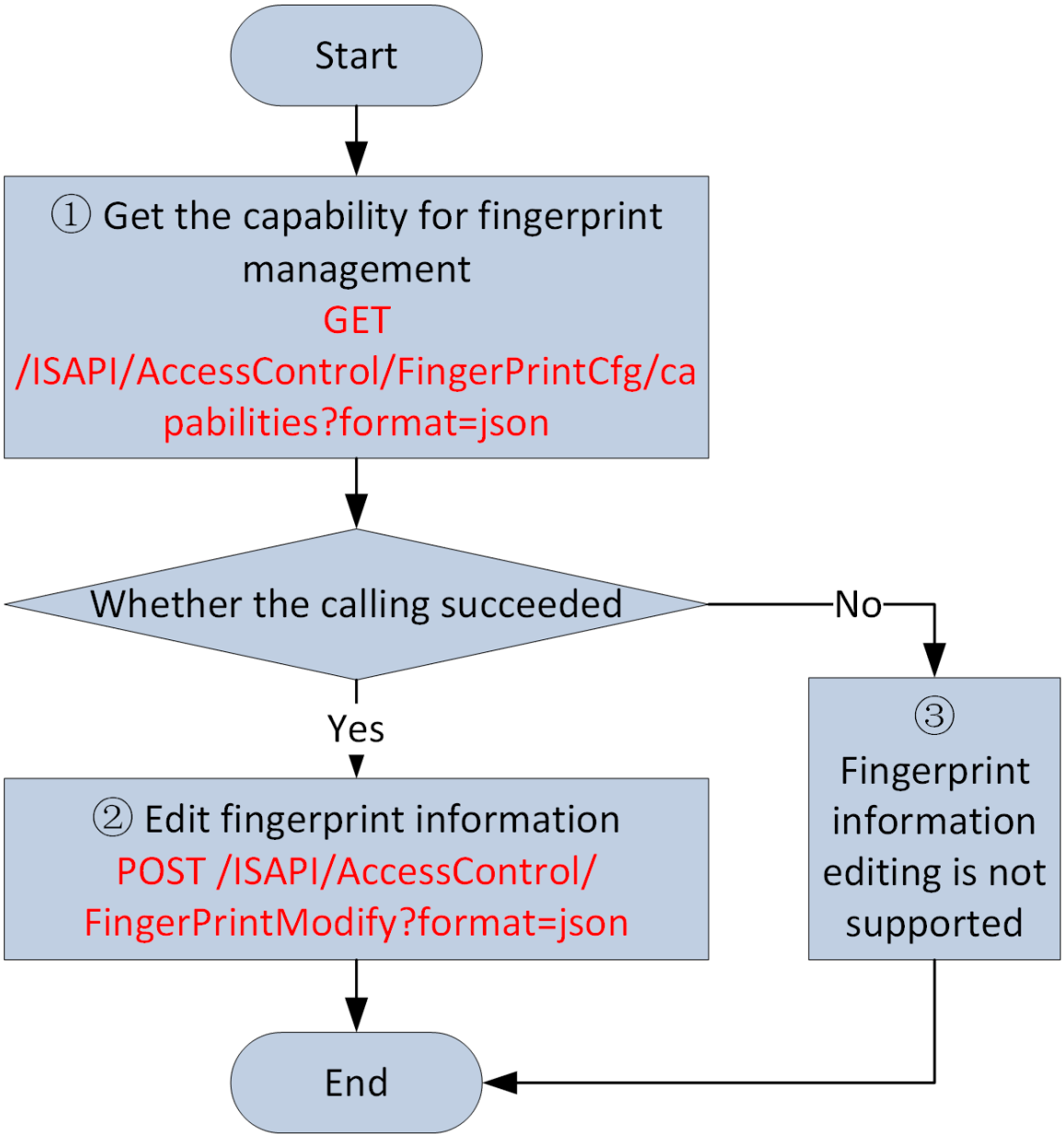
Fingerprint information can be added to the device via the fingerprint adding function. If the fingerprint has been added to the device, the device will report an error; if the fingerprint has not been added to the device, the fingerprint will be added to the device.

1. Check whether the device supports fingerprint adding: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if calling succeeded, it indicates that the device supports fingerprint adding.
2. Add fingerprint information: `POST /ISAPI/AccessControl/FingerPrintDownload?format=json`; if calling succeeded, it indicates that the device has started to execute fingerprint adding, but it does not indicate that the device has added the fingerprint.
3. Get the progress of fingerprint adding: `GET /ISAPI/AccessControl/FingerPrintProgress?format=json`; repeatedly call this API to get the progress of fingerprint adding.
4. If calling failed, it indicates that the device does not support fingerprint adding.

Note:

Check whether the fingerprint has been added to the device via the nodes employeeNo and fingerPrintID returned after calling the API for fingerprint adding.

5.2.5 Fingerprint Information Editing



The fingerprint information on the device can be edited via the fingerprint information editing function. If the fingerprint has been added to the device, the fingerprint information will be edited; if the fingerprint has not been added to the device, the device will report an error.

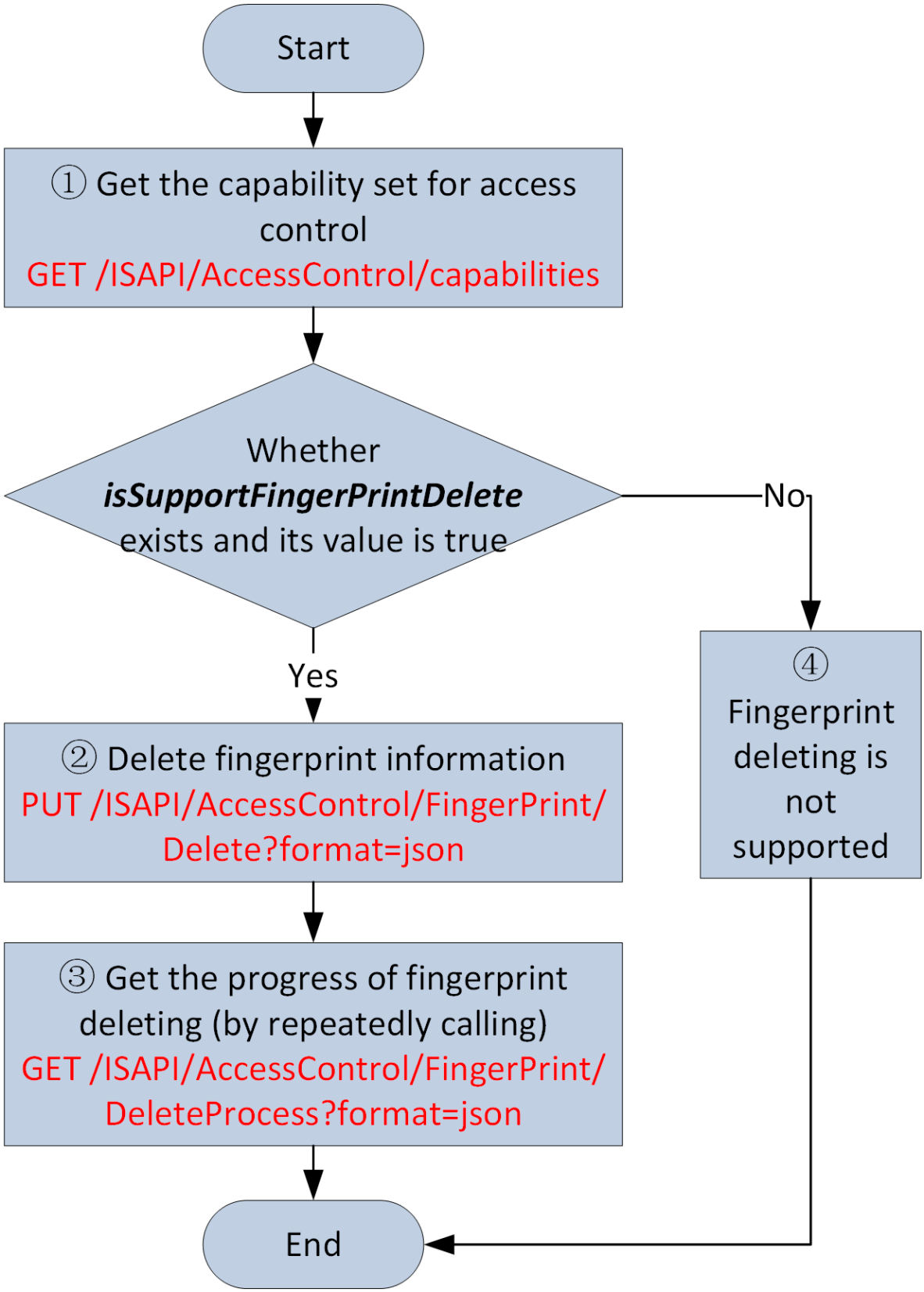
1. Check whether the device supports fingerprint information editing: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if calling succeeded, it indicates that the device supports fingerprint information editing.
2. Edit fingerprint information: `POST /ISAPI/AccessControl/FingerPrintModify?format=json`.
3. If calling failed, it indicates that the device does not support fingerprint information editing.

Note:

- Check whether the fingerprint has been added to the device via the nodes employeeNo and fingerPrintID returned after calling the API for fingerprint information editing.

- When the fingerprint information is edited, only the fingerprint parameters will be edited; the fingerprint data will not be edited.

5.2.6 Fingerprint Deleting



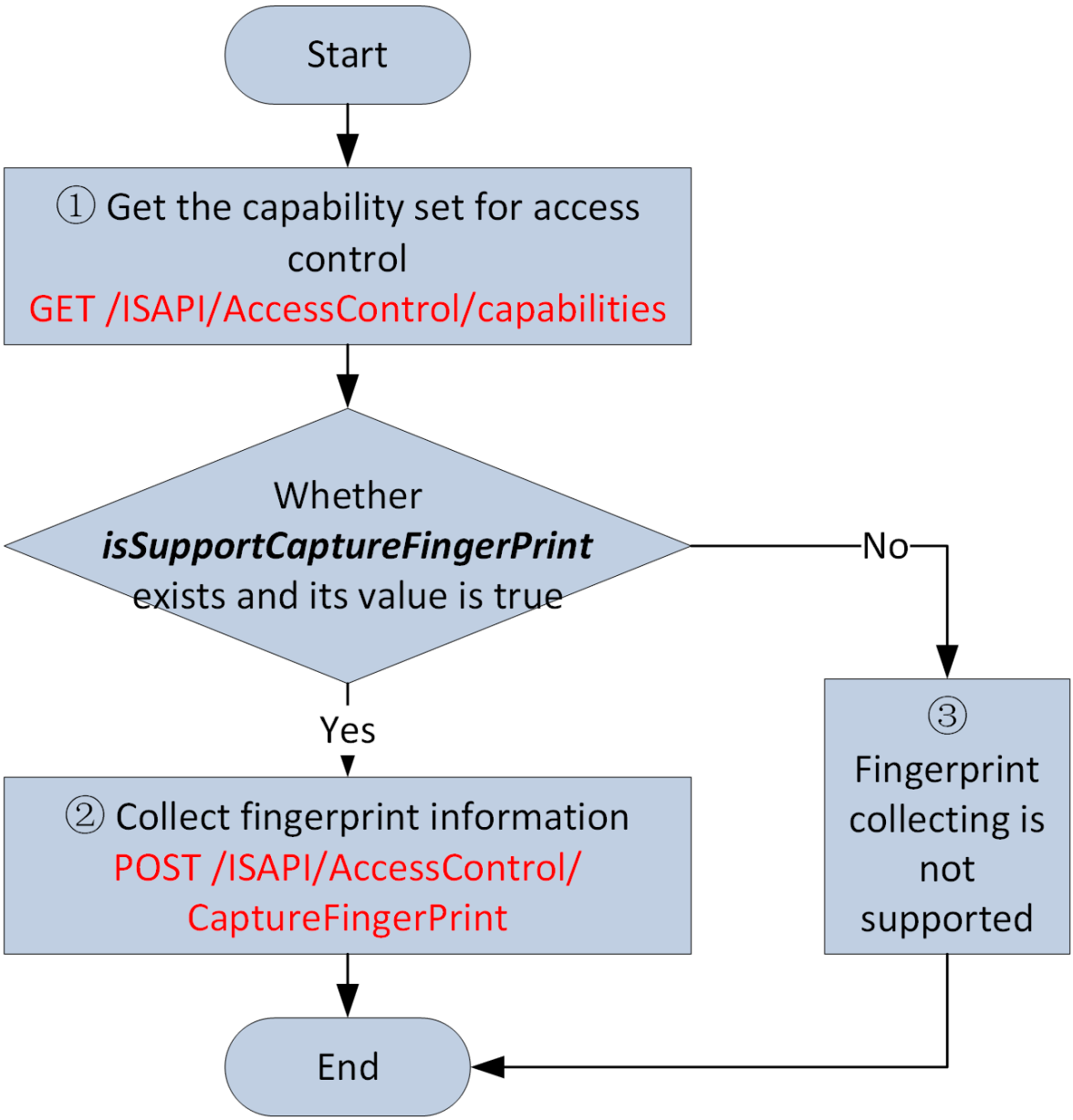
The fingerprint information on the device can be deleted via the fingerprint deleting function. The device will not report an error if the fingerprint information to be deleted is not added to the device.

1. Check whether the device supports fingerprint deleting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportFingerPrintDelete is returned

and its value is “true”, it indicates that the device supports fingerprint deleting.

2. Delete fingerprint information: `PUT /ISAPI/AccessControl/FingerPrint/Delete?format=json`; if calling succeeded, it indicates that the device has started to execute fingerprint deleting, but it does not indicate that the device has deleted the fingerprints.
3. Get the progress of fingerprint deleting: `GET /ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json`; repeatedly call this API to get the progress of fingerprint deleting.
4. If the node isSupportFingerPrintDelete is returned and its value is “false”, it indicates that the device does not support fingerprint deleting.

5.2.7 Fingerprint Collecting



The fingerprint collecting function is for collecting the fingerprint data, fingerprint quality, etc.

1. Check whether the device supports fingerprint collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureFingerPrint is returned and its value is “true”, it indicates that the device supports fingerprint collecting.
2. Collect fingerprint information: `POST /ISAPI/AccessControl/CaptureFingerPrint`.
3. If the node isSupportCaptureFingerPrint is returned and its value is “false”, it indicates that the device does not support fingerprint collecting.

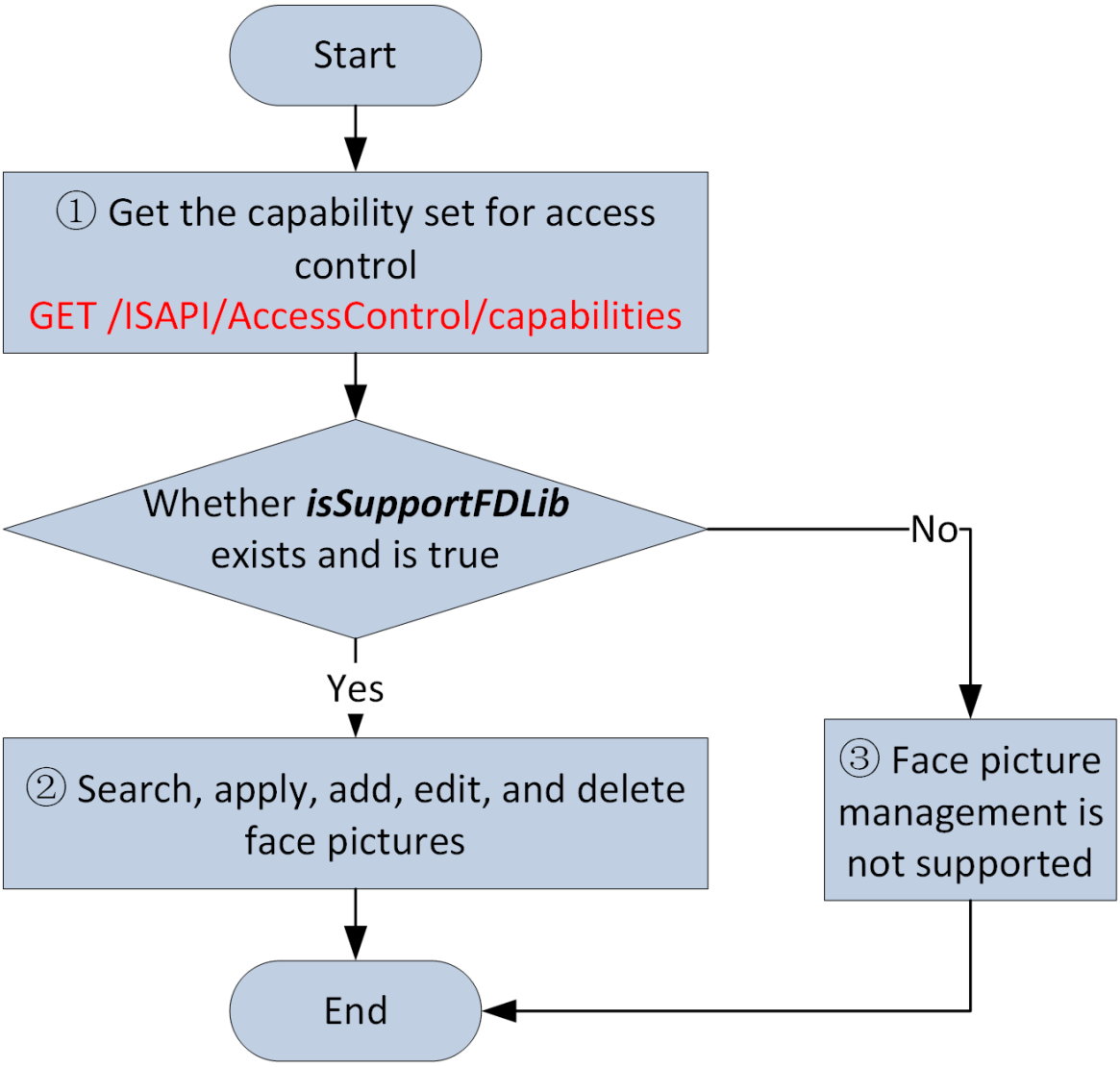
6 Face Picture Management

6.1 Introduction to the Function

Face picture management includes searching, applying, adding, editing, deleting, and collecting face pictures.

6.2 API Calling Flow

6.2.1 Check Whether the Device Supports Face Picture Management



Before calling the API for face picture management, make sure that the device supports face picture management.

1. Check whether the device supports face picture management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportFDLib is returned and its value is "true", it indicates that the device supports face picture management.
2. Search, apply, add, edit, and delete face pictures.
3. If the node isSupportFDLib is returned and its value is "false", it indicates that the device does not support face picture management.

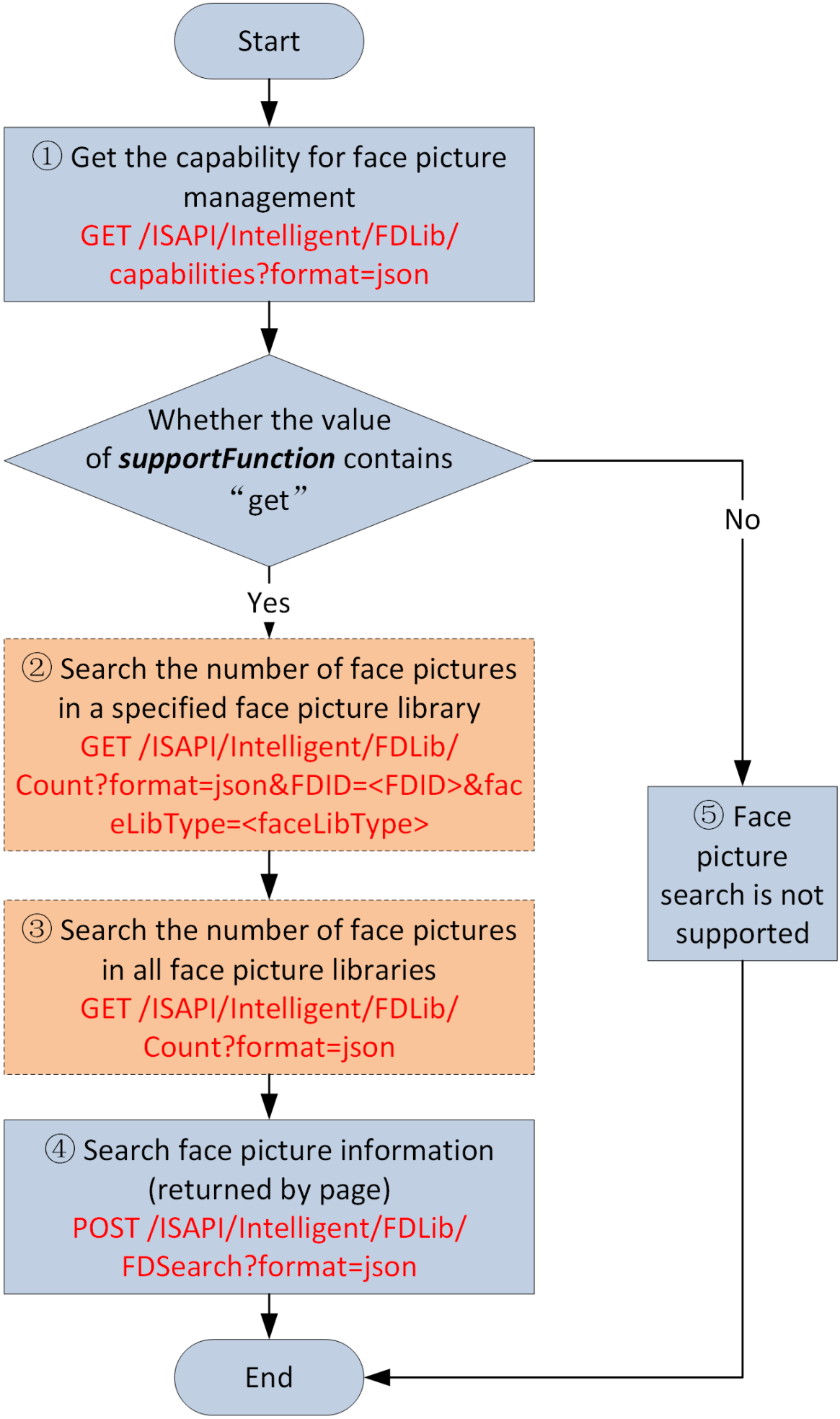
Note:

- Before applying, adding, or editing face picture information on the device, make sure that the related person information linked to the person ID has been applied to the device, and

make sure that the device has its face picture library by calling `GET /ISAPI/Intelligent/FDLib?format=json` (if the device has no face picture library, then create the face picture library by calling `POST /ISAPI/Intelligent/FDLib?format=json`), and the ID of the library of face pictures captured in visible light (FDID) is 1.

- If the value of the node mode returned by calling `GET /ISAPI/AccessControl/FaceRecognizeMode/capabilities?format=json` contains "deepMode", it indicates that the device supports the deep mode, which compares face pictures captured in infrared light. For devices which support the deep mode, if the face picture library ID (FDID) is 2, face pictures captured in infrared light will be applied to the face picture library and be used for face picture comparison; if the face picture library ID (FDID) is 1, face pictures captured in visible light will be applied to the face picture library and be displayed on the device.
- Switch between the deep mode and normal mode: `[GET/PUT] /ISAPI/AccessControl/FaceRecognizeMode?format=json`; the modes can be switched via the node mode.

6.2.2 Face Picture Search



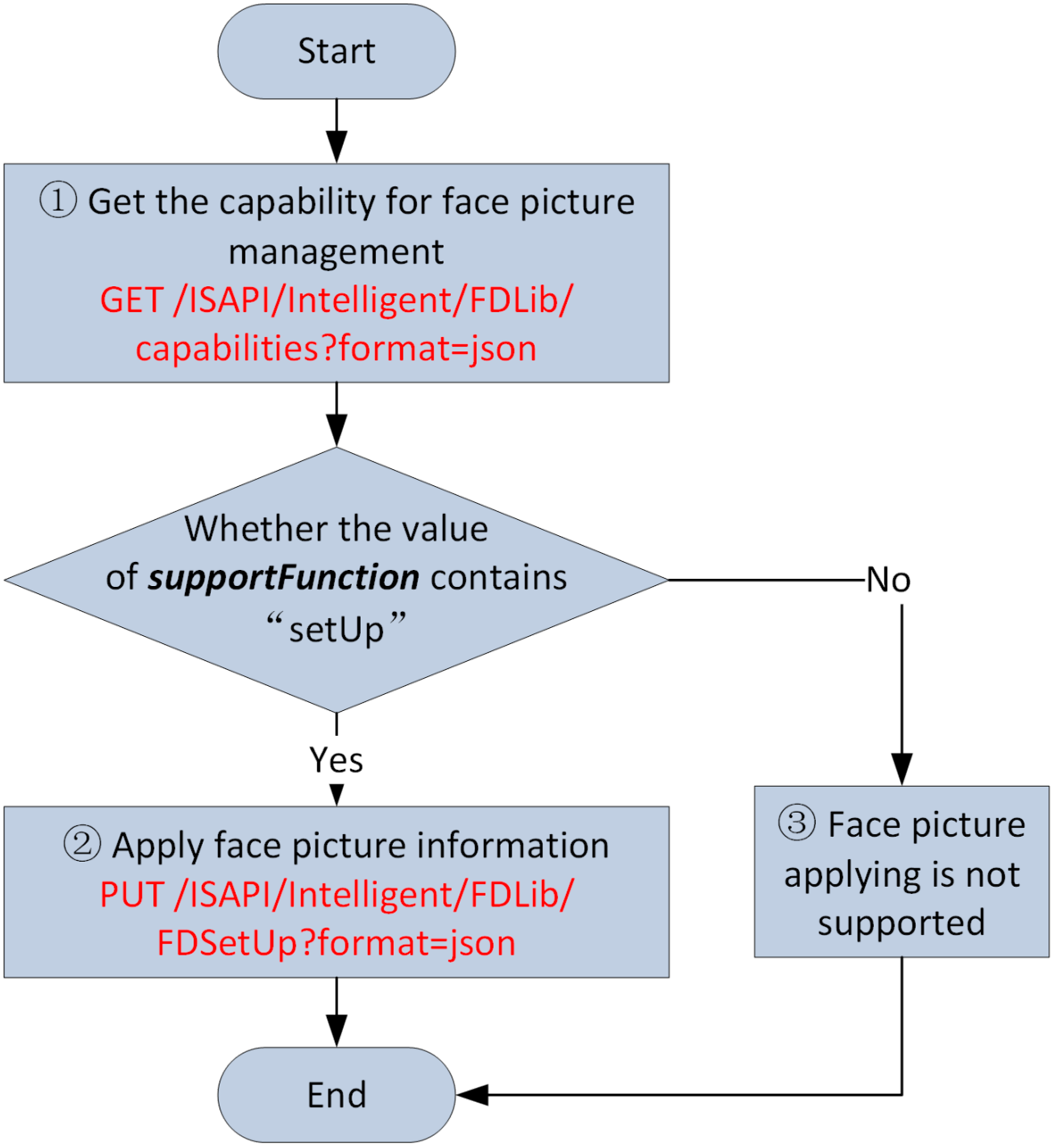
The face picture search function is for searching the number of face pictures and face picture information added to the device.

1. Check whether the device supports face picture search: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains “get”, it indicates that the device supports face picture search.
2. Search the number of face pictures in the specified face picture libraries: `GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>`; the returned value of the node recordDataNumber is the number of the added face pictures of the specified face picture libraries.
3. Search the number of face pictures in all face picture libraries: `GET /ISAPI/Intelligent/FDLib/Count?format=json`; the returned value of the node recordDataNumber is the number of face pictures in all face picture libraries.
4. Search face picture information: `POST /ISAPI/Intelligent/FDLib/FDSearch?format=json`; the face picture information is returned by page.
5. If the value of the node supportFunction does not contain “get”, it indicates that the device does not support face picture search.

Note:

The value of the node FDRecordDataMaxNum returned by calling `GET /ISAPI/Intelligent/FDLib/capabilities?format=json` is the maximum number of face pictures supported by the device.

6.2.3 Face Picture Applying



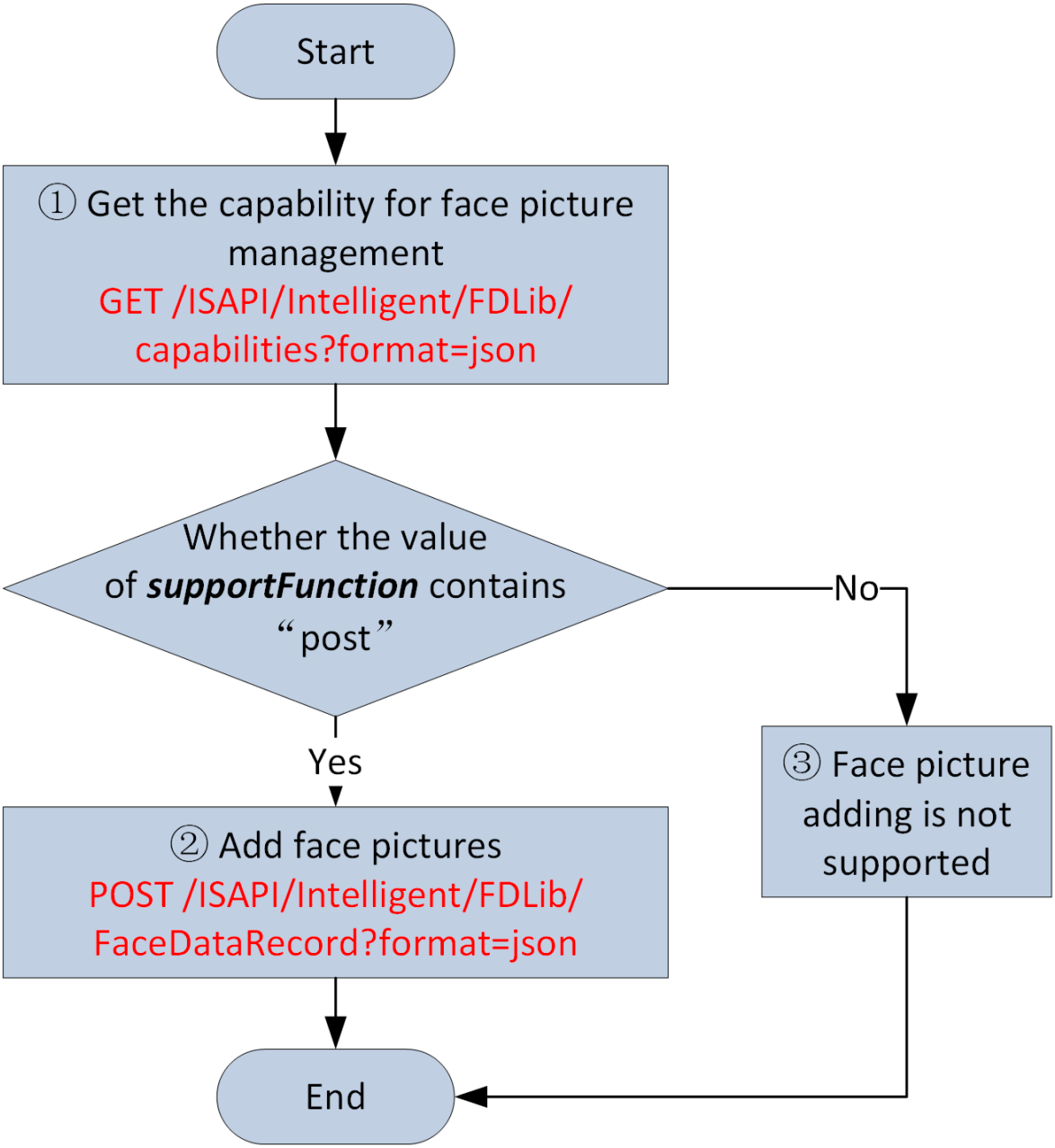
Face picture information can be applied to the device via the face picture applying function. If the face picture has been added to the device, the face picture information will be edited; if the face picture has not been added to the device, the face picture information will be added to the device.

1. Check whether the device supports face picture applying: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "setUp", it indicates that the device supports face picture applying.
2. Apply face picture information: `PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json`.
3. If the value of the node supportFunction does not contain "setUp", it indicates that the device does not support face picture applying.

Note:

Check whether the face picture has been added to the device via the node FPID returned by calling the API for face picture applying, and link the face picture to the person information via the node FPID in face picture management and the node employeeNo in person management.

6.2.4 Face Picture Adding



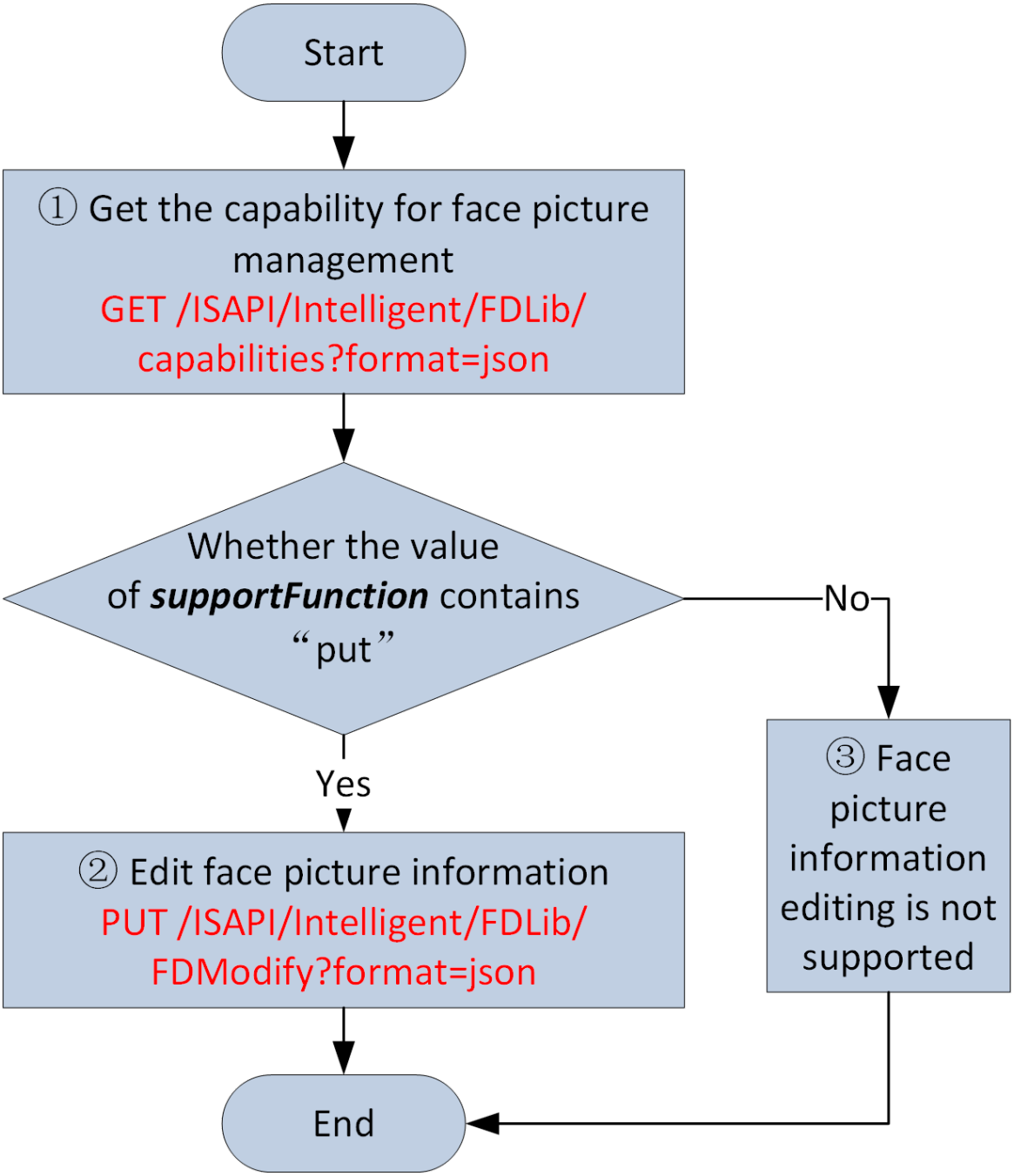
Face picture information can be added to the device via the face picture adding function. If the face picture has been added to the device, the device will report an error; if the face picture has not been added to the device, the face picture information will be added to the device.

1. Check whether the device supports face picture adding: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports face picture adding.
2. Add face picture information: `POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support face picture adding.

Note:

Check whether the face picture has been added to the device via the node FPID returned by calling the API for face picture adding, and link the face picture to the person information via the node FPID in face picture management and the node employeeNo in person management.

6.2.5 Face Picture Information Editing



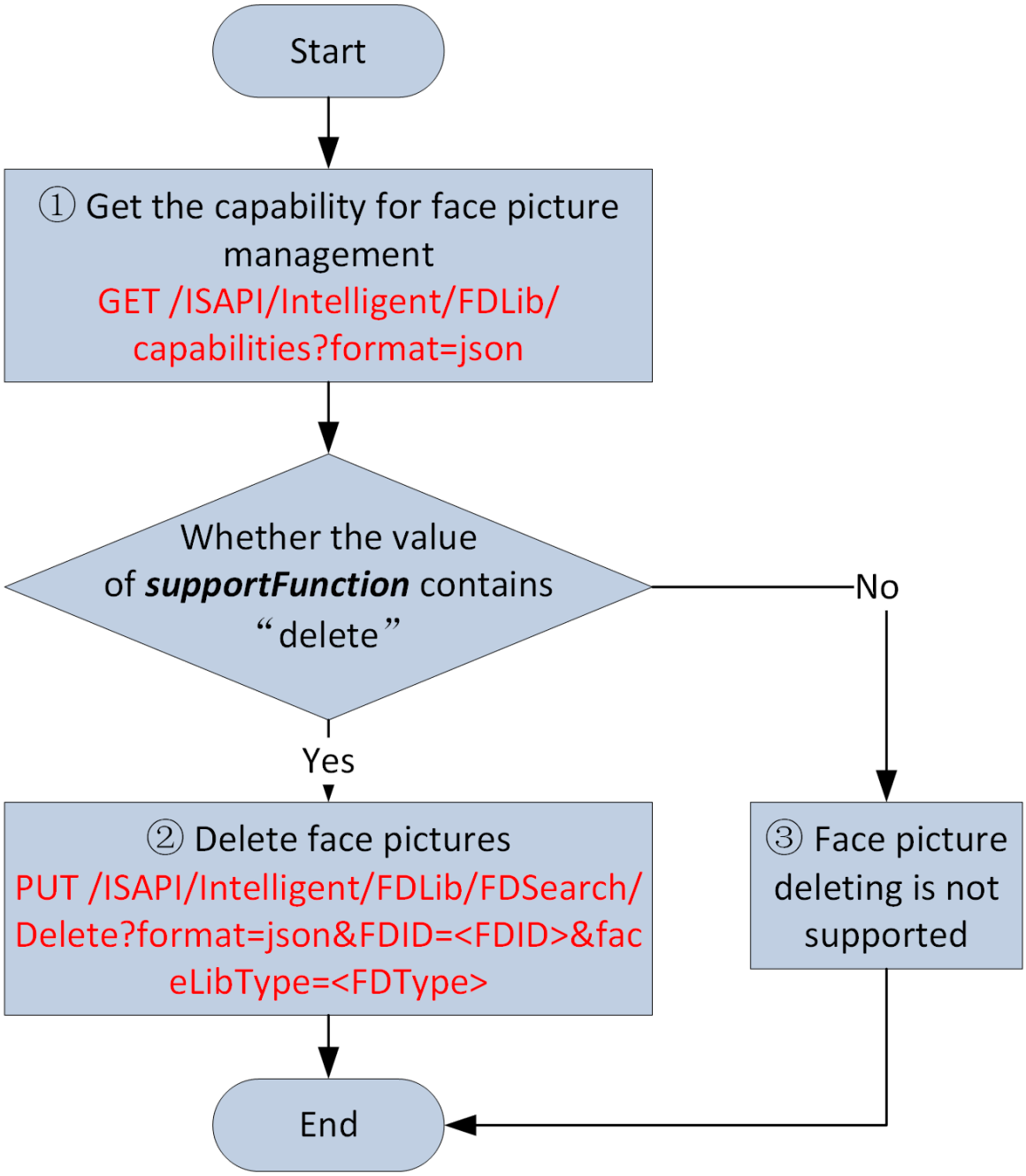
Face picture information on the device can be edited via the face picture information editing function. If the face picture has been added to the device, the face picture information will be edited; if the face picture has not been added to the device, the device will report an error.

1. Check whether the device supports face picture information editing: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports face picture information editing.
2. Edit face picture information: `PUT /ISAPI/Intelligent/FDLib/FDModify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support face picture information editing.

Note:

Check whether the face picture has been added to the device via the node FPID returned by calling the API for face picture information editing, and link the face picture to the person information via the node FPID in face picture management and the node employeeNo in person management.

6.2.6 Face Picture Deleting



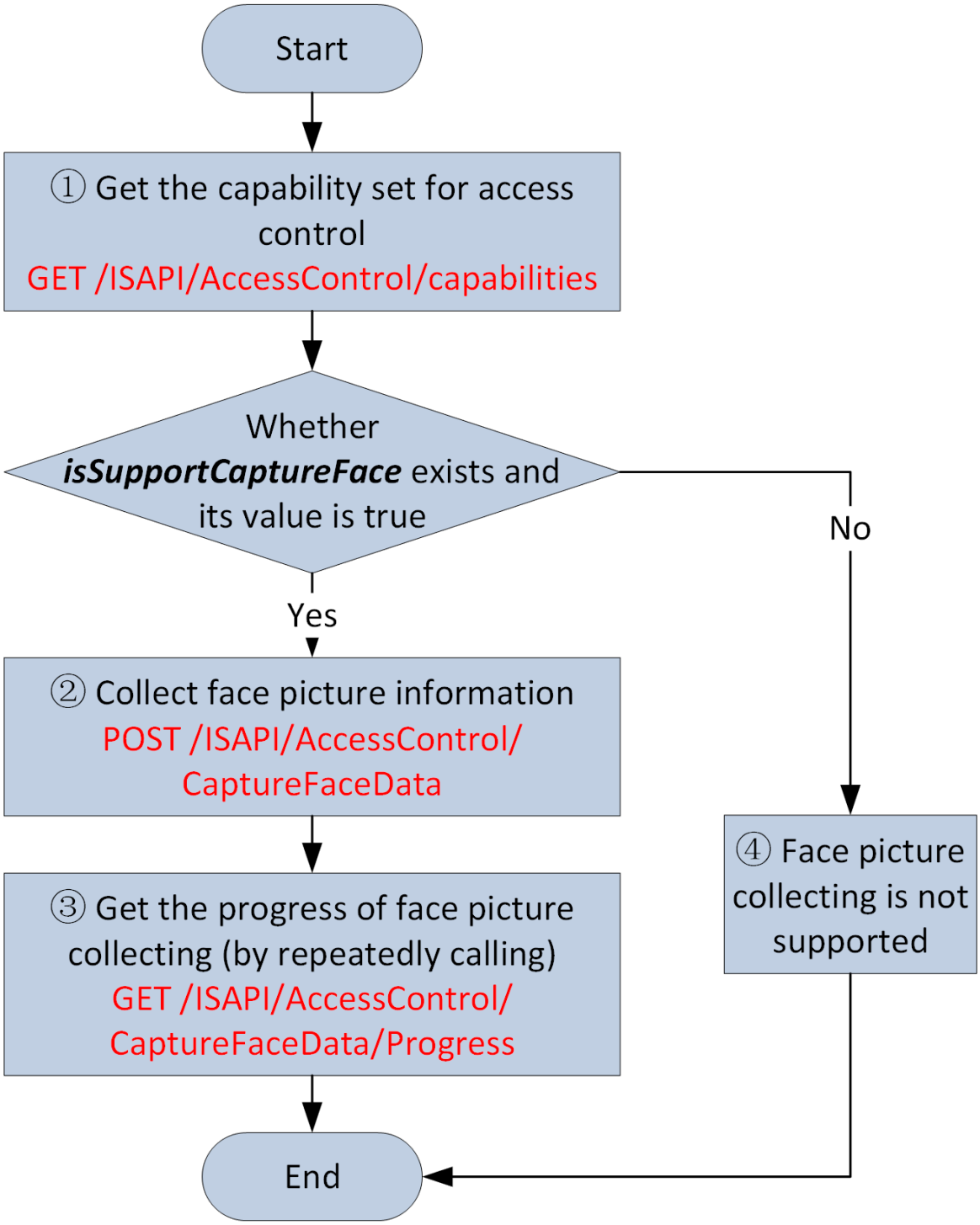
The face picture information on the device can be deleted via the face picture deleting function. The device will not report an error if the face picture to be deleted is not added to the device.

1. Check whether the device supports face picture deleting: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "delete", it indicates that the device supports face picture deleting.
2. Delete face pictures: `PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>`; if calling succeeded, it indicates that the device has deleted the face pictures.
3. If the value of the node supportFunction does not contain "delete", it indicates that the device does not support face picture deleting.

Note:

All the face picture libraries and the face picture information in the libraries on the device can be deleted by calling `DELETE /ISAPI/Intelligent/FDLib?format=json`.

6.2.7 Face Picture Collecting



Face picture data, face picture quality grades, etc., can be collected via the face picture collecting function.

1. Check whether the device supports face picture collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureFace is returned and its value is "true", it indicates that the device supports face picture (captured in visible light) collecting. If the node isSupportCaptureInfraredFace is returned and its value is "true", it indicates that the device supports face picture (captured in infrared light) collecting.
2. Collect face picture information: `POST /ISAPI/AccessControl/CaptureFaceData`.

- If the node captureProgress is returned, and the value is 100, it indicates that the face picture has been collected, and the binary data or URL of the collected face picture will be parsed.

- If the node captureProgress is returned and the value is 0, it indicates that the face picture has not been collected, and you need to get the progress of face picture collecting.

3. Get the progress of face picture collecting: `GET /ISAPI/AccessControl/CaptureFaceData/Progress`; repeatedly call this API to get the progress of face picture collecting.

- Repeatedly call this API until the node captureProgress is returned and its value is 100, which indicates that the face picture has been collected and the binary data and URL of the face picture will be parsed.
- If the value of the node captureProgress is 0 and the value of the isCurRequestOver is true, which indicates that the face picture collecting failed, stop calling the API.

4. If the node isSupportCaptureFace is returned and its value is “false”, it indicates that the device does not support face picture collecting.

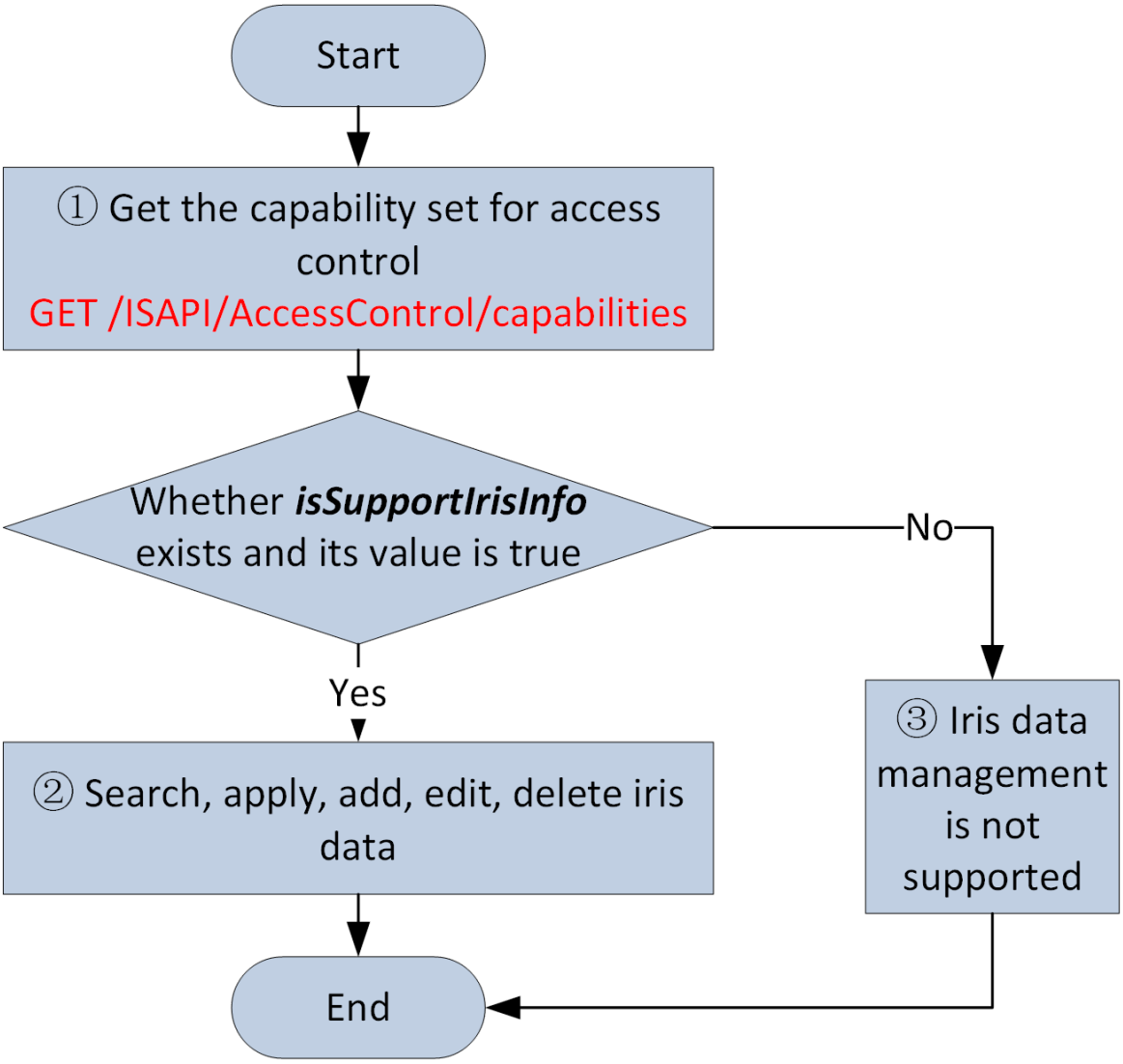
7 Iris Data Management

7.1 Introduction to the Function

Iris data management includes searching, applying, adding, editing, deleting, and collecting iris data.

7.2 API Calling Flow

7.2.1 Check Whether the Device Supports Iris Data Management



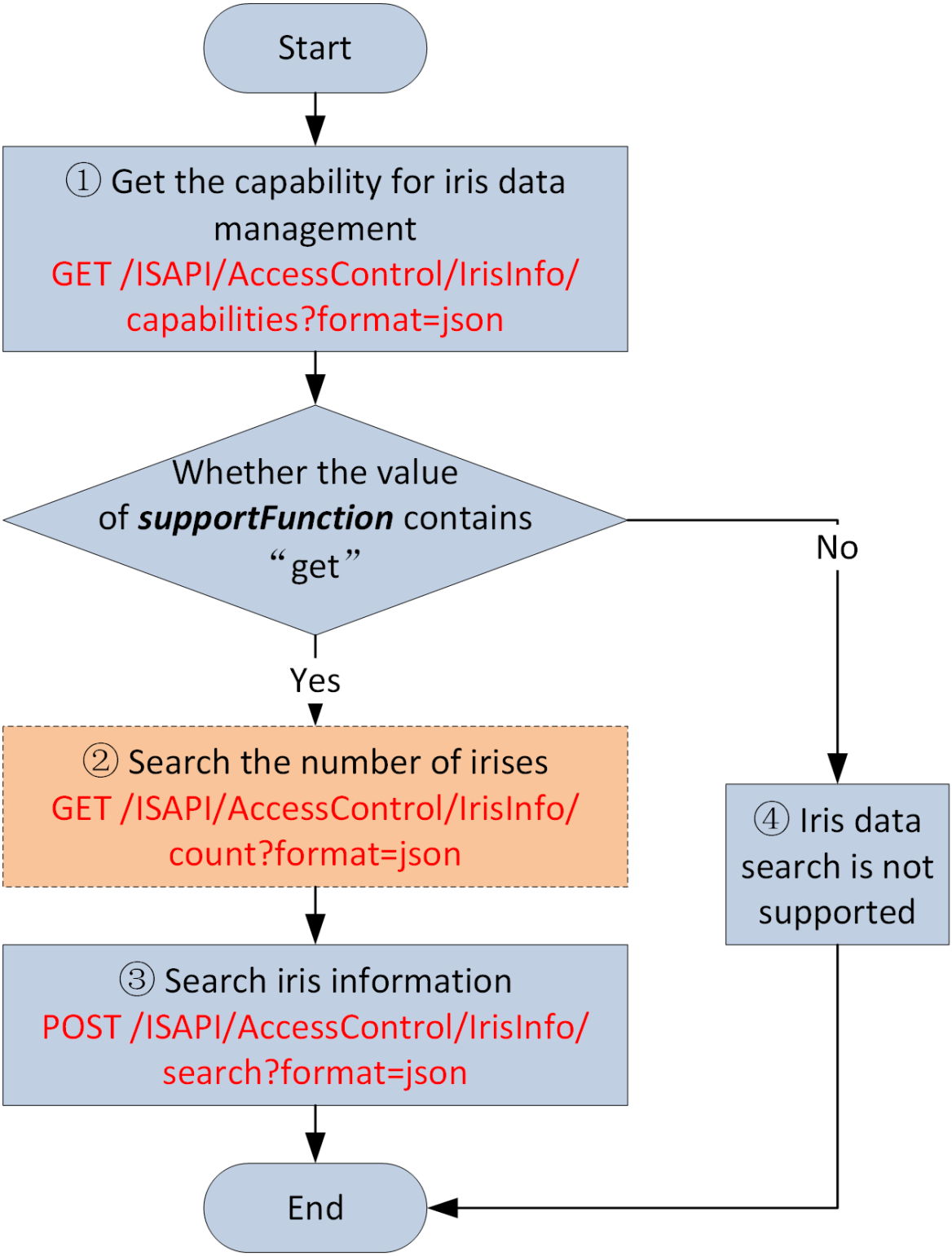
Before calling the API for iris data management, make sure that the device supports iris data management:

1. Check whether the device supports iris data management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportIrisInfo is returned and its value is true, it indicates that the device supports iris data management.
2. Search, apply, add, edit, and delete iris data.
3. If the node isSupportIrisInfo is returned and its value is false, it indicates that the device does not support iris data management.

Note:

- Before applying, adding, and editing the iris data on the device, make sure that the related person information linked to the person ID has been applied to the device.
- Data for up to two irises of each person (two eyes of each person) can be applied to the device.

7.2.2 Iris Data Search



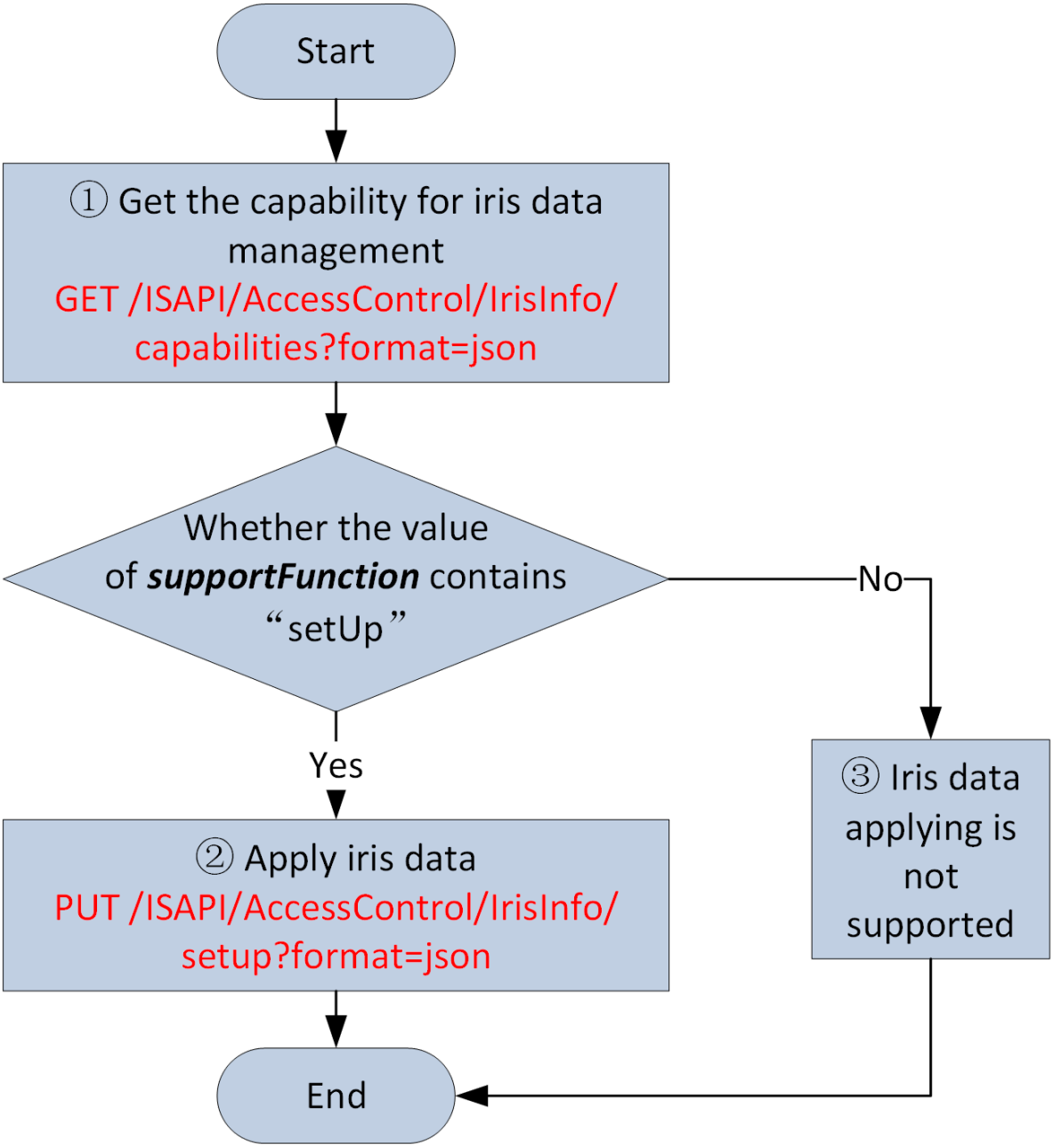
The iris data search function is for searching the number of irises and iris information added to the device.

1. Check whether the device supports iris data search: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the node supportFunction is returned, and the value contains "get", it indicates that the device supports iris data search.
2. Search the number of irises: `GET /ISAPI/AccessControl/IrisInfo/count?format=json`; the value of the node IrisNumber is the number of added irises on the device.
3. Search iris information: `POST /ISAPI/AccessControl/IrisInfo/search?format=json`; the searched iris information will be returned by page.
4. If the value of the node supportFunction does not contain "get", it indicates that the device does not support iris data search.

Note:

The value of the node maxRecordNum returned by calling `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json` is the maximum number of irises supported by the device.

7.2.3 Iris Data Applying



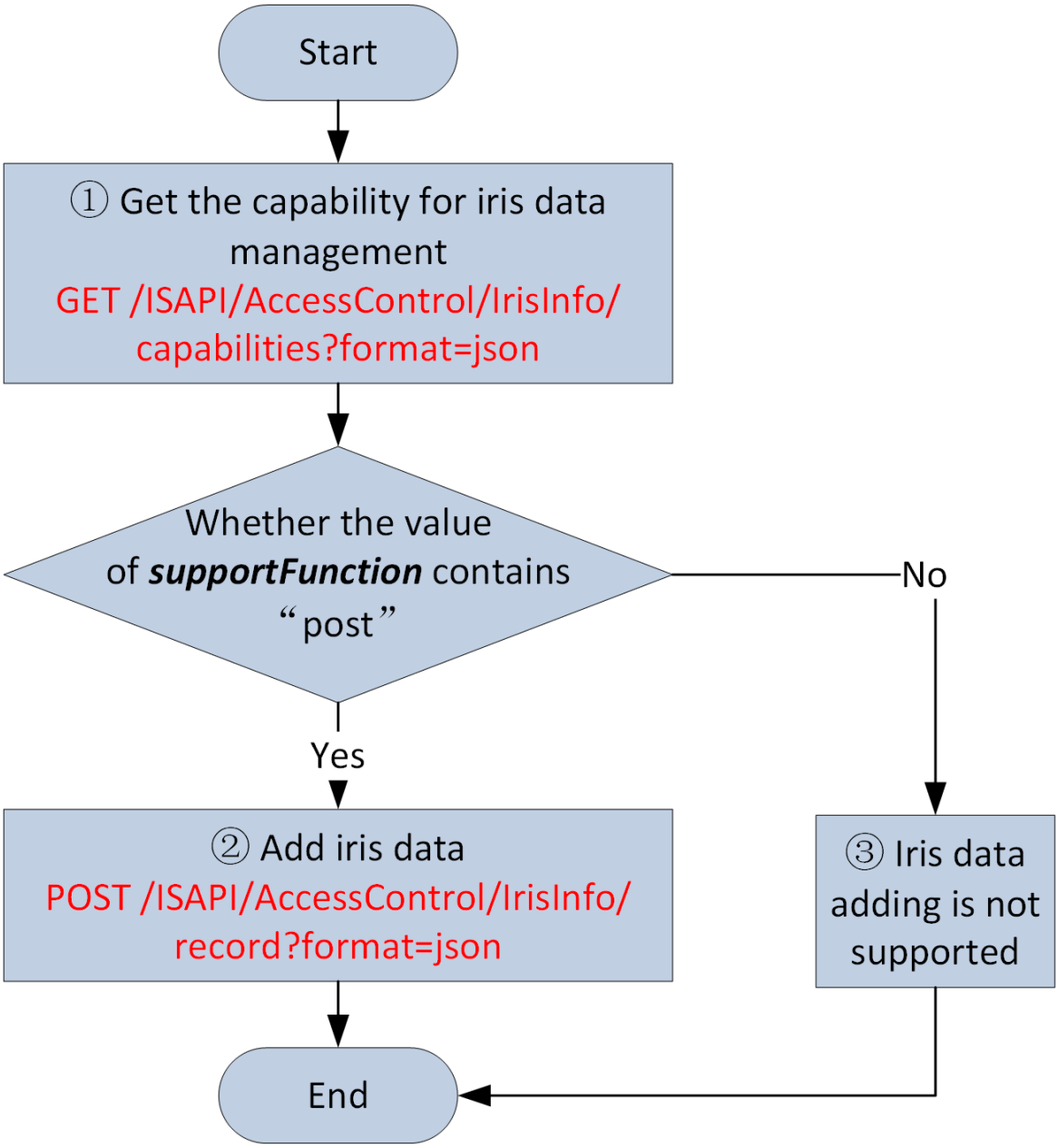
The iris data applying function is for applying iris information to the device. If the iris data has already been applied to the device, the information about the iris will be edited; if the iris data has not been applied to the device, the iris information will be added to the device.

1. Check whether the device supports iris data applying: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the node supportFunction is returned and the value contains "setUp", it indicates that the device supports iris data applying.
2. Iris Data Applying: `PUT /ISAPI/AccessControl/IrisInfo/setup?format=json`.
3. If the value of the node supportFunction does not contain "setUp", it indicates that the device does not support iris data applying.

Note:

Check whether the iris data have been applied to the device via the nodes employeeNo and id returned by calling the API for iris data applying.

7.2.4 Iris Data Adding



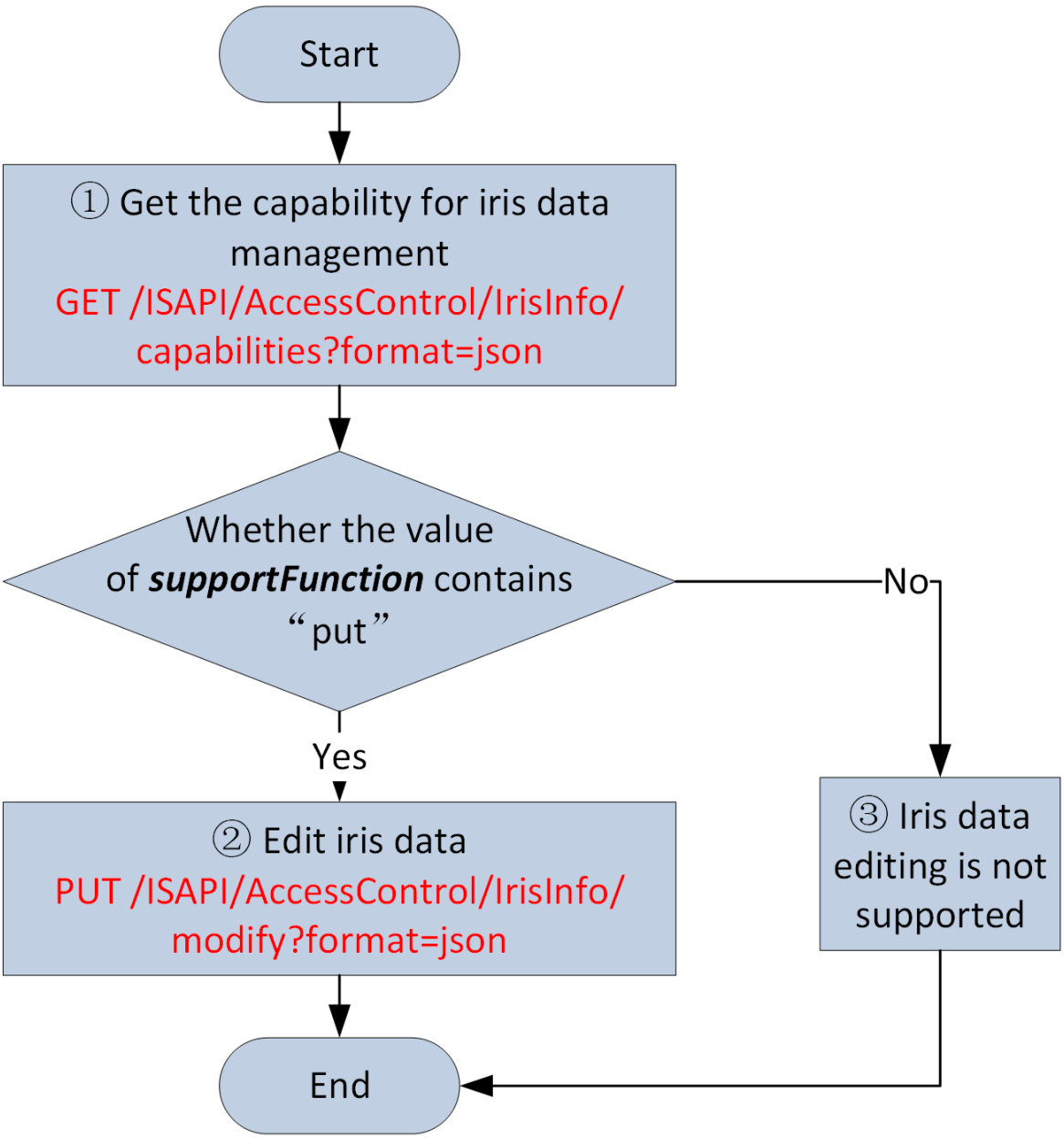
The iris data adding function is for adding the iris data to the device. If the iris data have already been added to the device, the device will report an error; if the iris data have not been added to the device, the iris data will be added to the device.

1. Check whether the device supports iris data adding: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports iris data applying.
2. Add iris data: `POST /ISAPI/AccessControl/IrisInfo/record?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support iris data applying.

Note:

Check whether the iris data have been applied to the device via the nodes employeeNo and id returned by calling the API for iris data adding.

7.2.5 Iris Data Editing



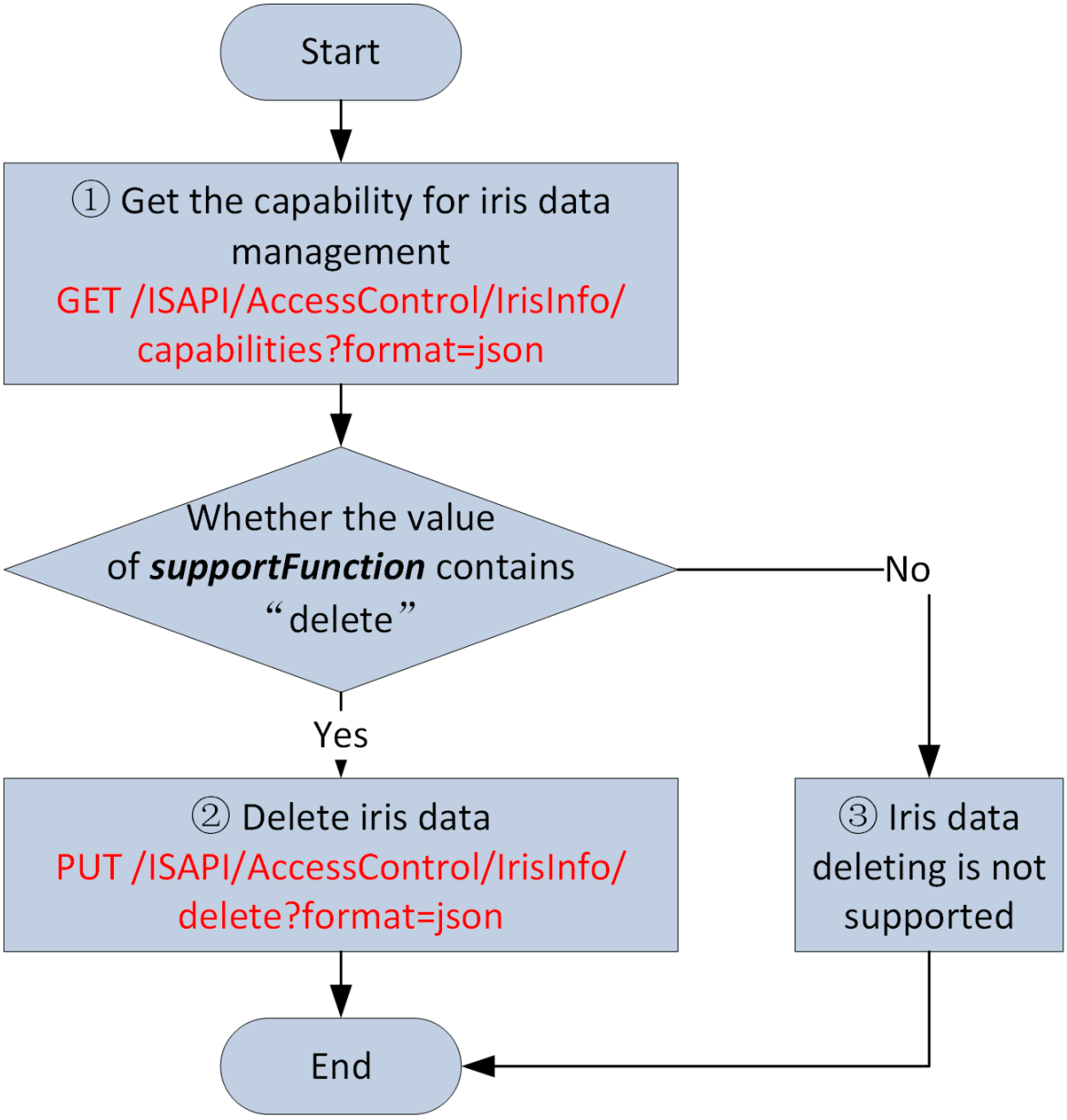
The iris data editing function is for editing the applied iris information on the device. If the iris data have already been added to the device, the iris information will be edited; if the iris data have not been added to the device, the device will report an error.

1. Check whether the device supports iris data editing: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports iris data editing.
2. Edit iris information: `PUT /ISAPI/AccessControl/IrisInfo/modify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support iris data editing.

Note:

Check whether the iris data have been applied to the device via the nodes employeeNo and id returned by calling the API for iris data editing.

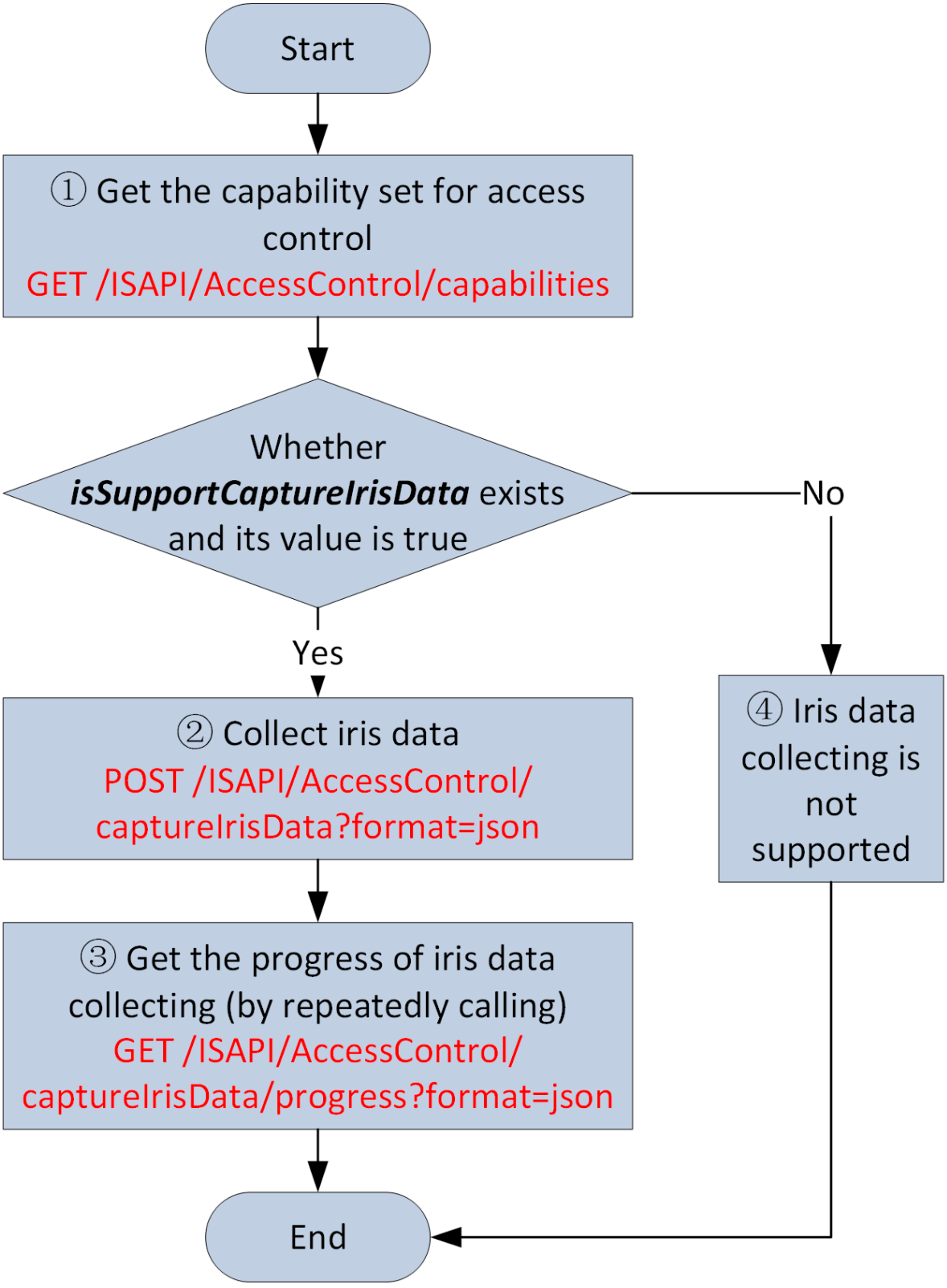
7.2.6 Iris Data Deleting



The iris data deleting function is for deleting the applied iris information on the device. If the iris data to be deleted have not been applied to the device, the device will not report an error.

1. Check whether the device supports iris data deleting: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the value of the node supportFunction contains "delete", it indicates that the device supports iris data deleting.
2. Delete iris information: `PUT /ISAPI/AccessControl/IrisInfo/delete?format=json`; if calling succeeded, it indicates that the iris information has been deleted.
3. If the value of the node supportFunction does not contain "delete", it indicates that the device does not support iris data deleting.

7.2.7 Iris Data Collecting



The iris data collecting function is for collecting iris data and information.

1. Check whether the device supports iris data collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureIrisData is returned and its value is "true", it indicates that the device supports iris data collecting.
2. Collect iris information: `POST /ISAPI/AccessControl/captureIrisData?format=json`; if calling succeeded, it indicates that the device has started to execute the collection.
3. Get the progress of iris data collecting: `GET /ISAPI/AccessControl/captureIrisData/progress?format=json`; repeatedly call this API until the value of captureProgress is returned and is 100, which indicates that the collecting completed.

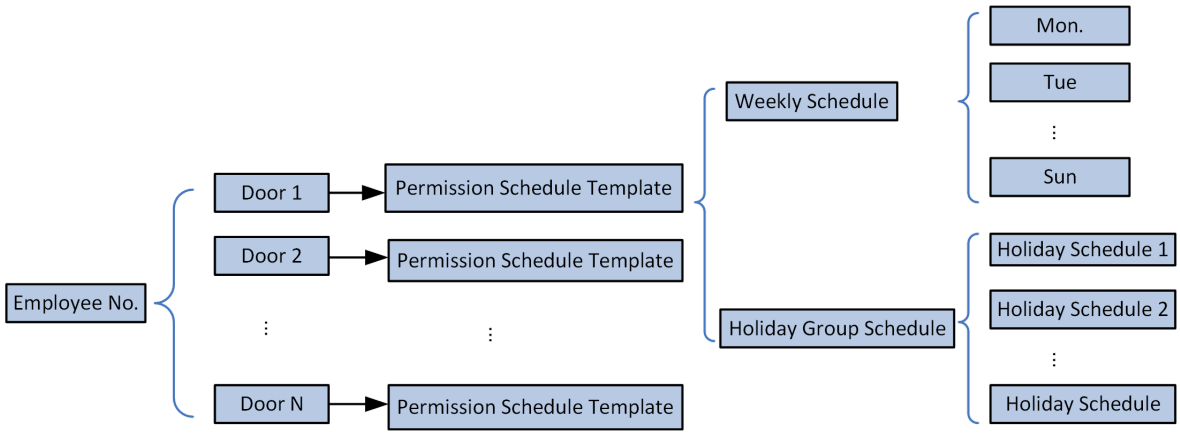
4. If the node isSupportCaptureIrisData is returned and its value is “false”, it indicates that the device does not support iris data collecting.

8 Schedules Management of Persons' Access Permission

8.1 Introduction to the Function

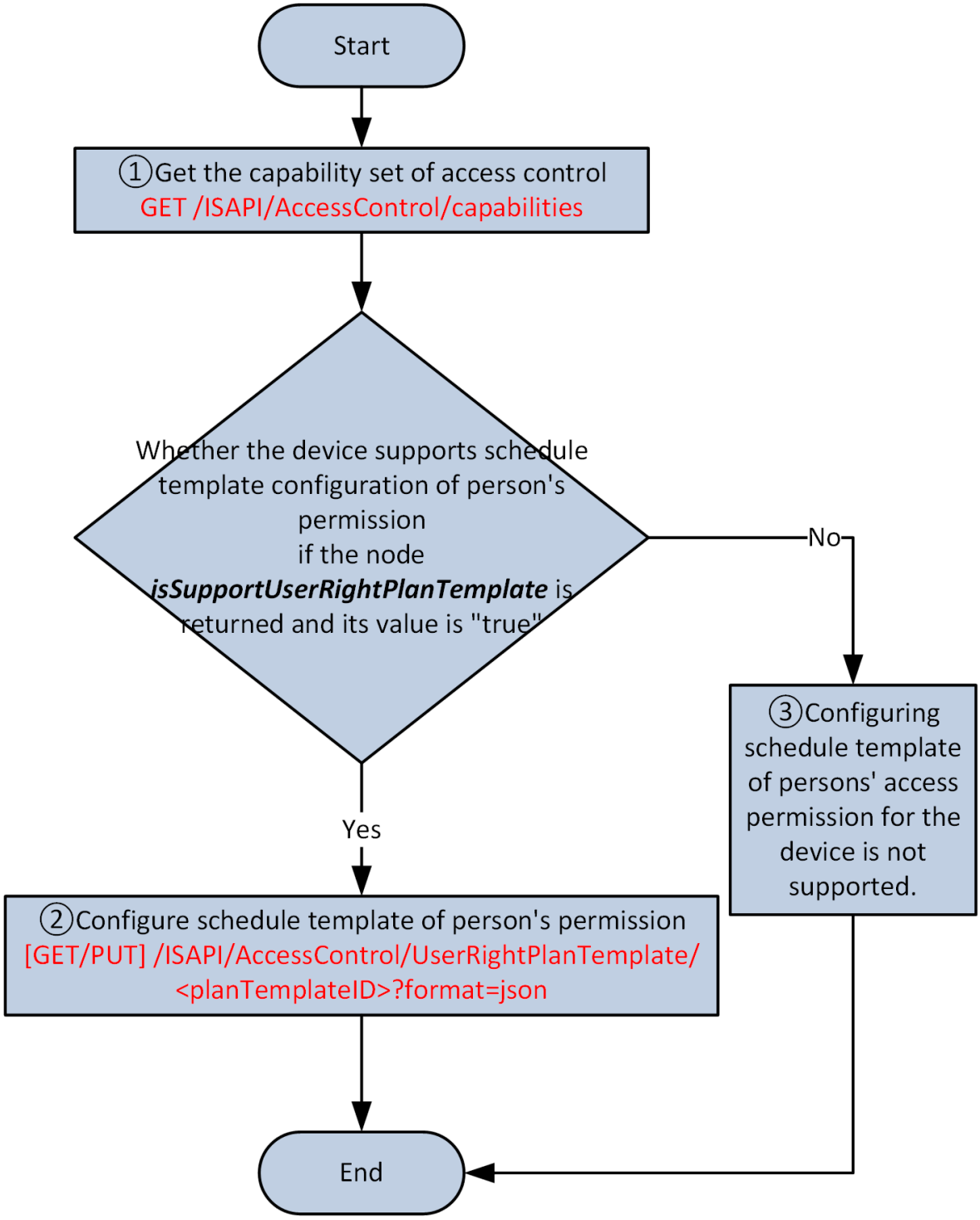
It is required to connect to door permission and schedule template of access permission related to each door before applying permissions to persons. For applying permissions to persons, see calling flow of *Person Management* of *Person and Credential Management.* Configuring schedules of persons' access permission is required, or the related persons cannot access.

1 weekly schedule and 4 holiday groups can be added in each schedule template. The priority of holiday schedule is higher than that of weekly schedule. Weekly schedule can be configured by day of a week and 8 different time period of a day. 16 holiday schedules can be added to a holiday group schedule. Each holiday schedule has its start and end day, and the time period is same in the holiday range (8 time periods can be added). The access control can follow the schedule template to manage person's permissions by time.



8.2 API Calling Flow

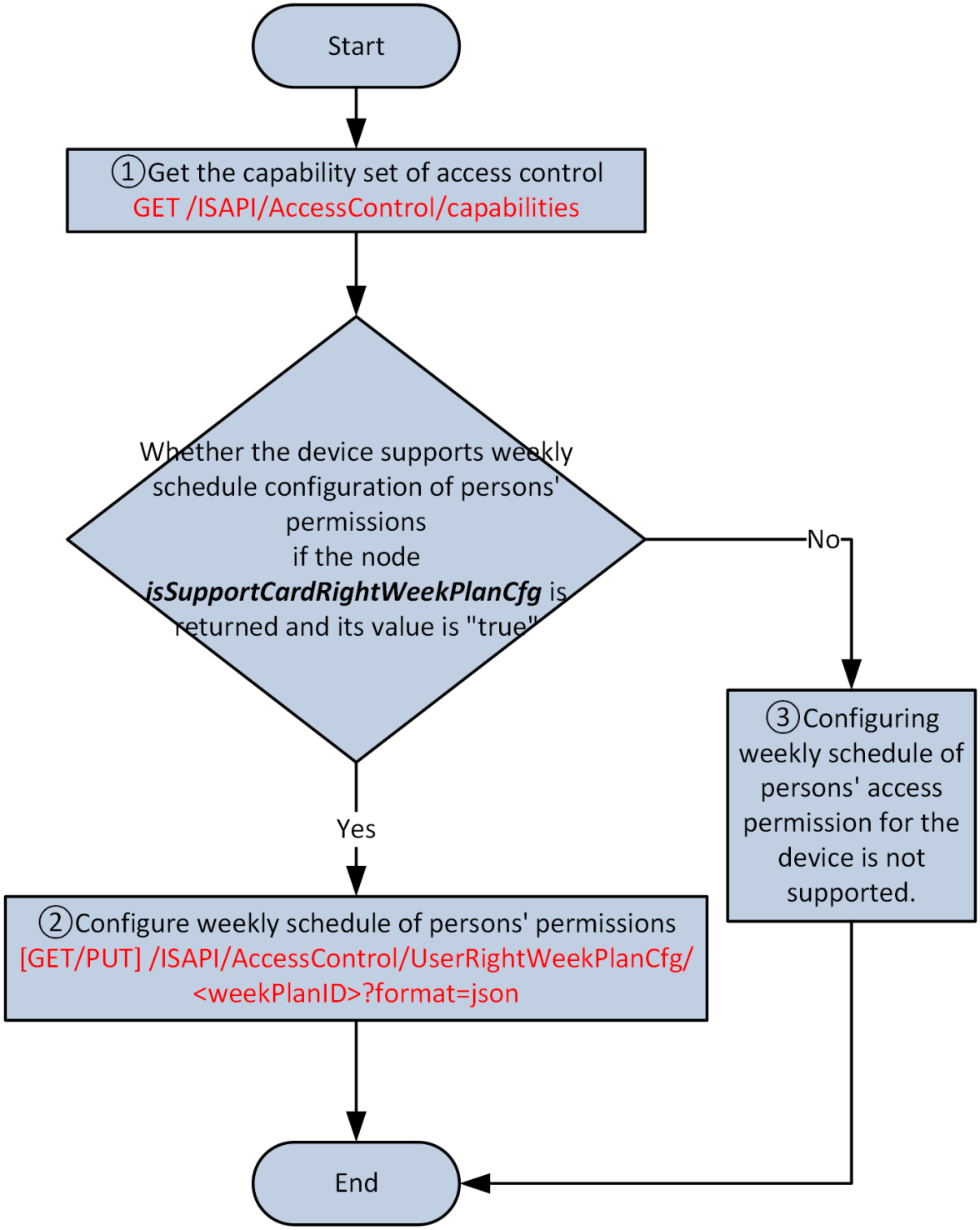
8.2.1 Schedule Template of Persons' Access Permission



Calling Flow:

1. Check whether the device supports schedule template configuration of person's permission: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportUserRightPlanTemplate is returned and its value is "true", it indicates that the device supports schedule template configuration of person's permission (if it supports, it also supports weekly schedule configuration of persons' permission).
2. Schedule template configuration of persons' permission: `[GET/PUT] /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json`.
3. Configuring schedule template of persons' access permission control for the device is not supported.

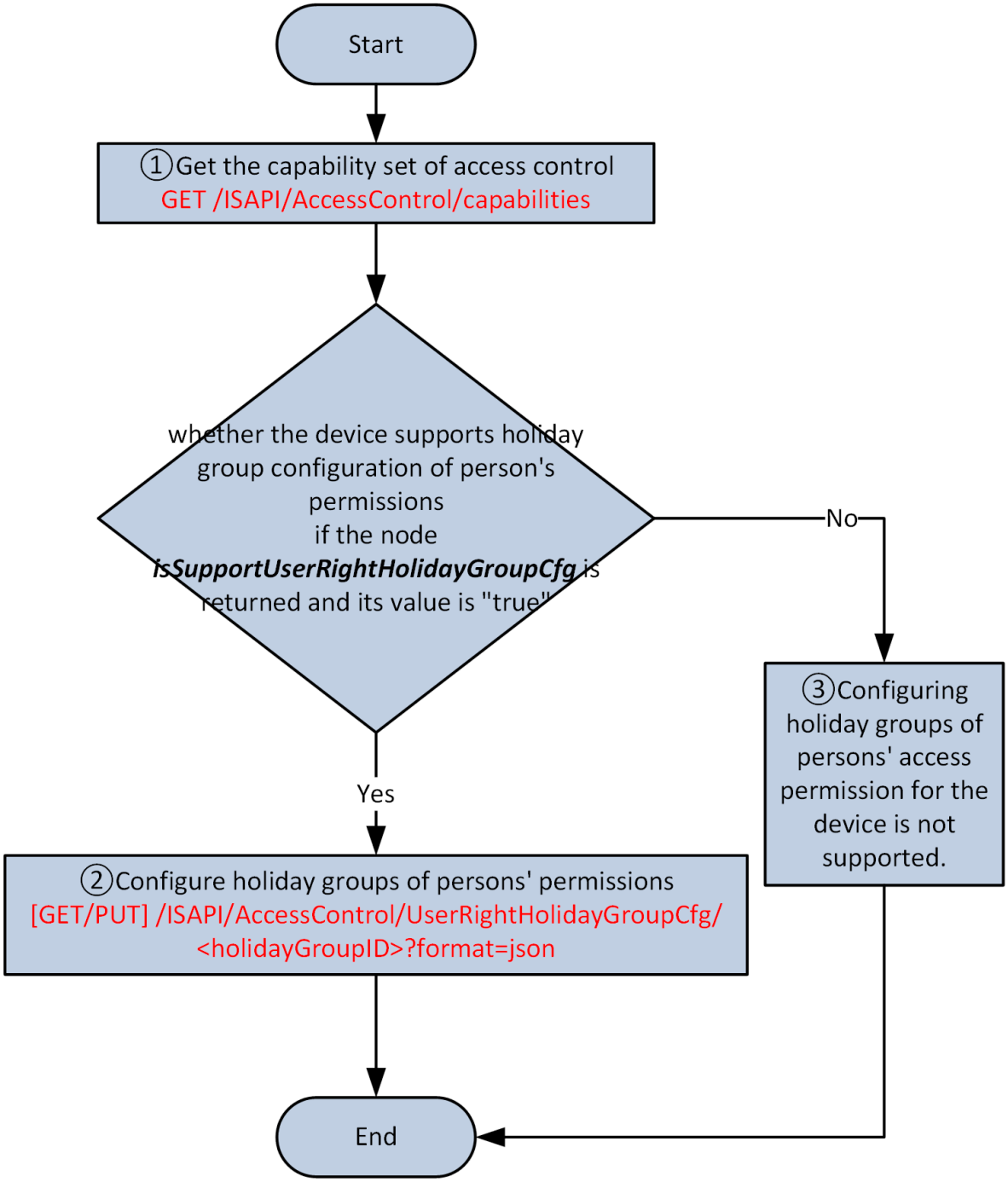
8.2.2 Weekly Schedule of Persons' Access Permission



Calling Flow:

1. Check whether the device supports weekly schedule configuration of persons' permissions: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardRightWeekPlanCfg is returned and its value is "true", it indicates that the device supports weekly schedule configuration of person's permissions.
2. Weekly schedule configuration of persons' permissions: `[GET/PUT] /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json`.
3. Configuring weekly schedule of persons' access permission control for the device is not supported.

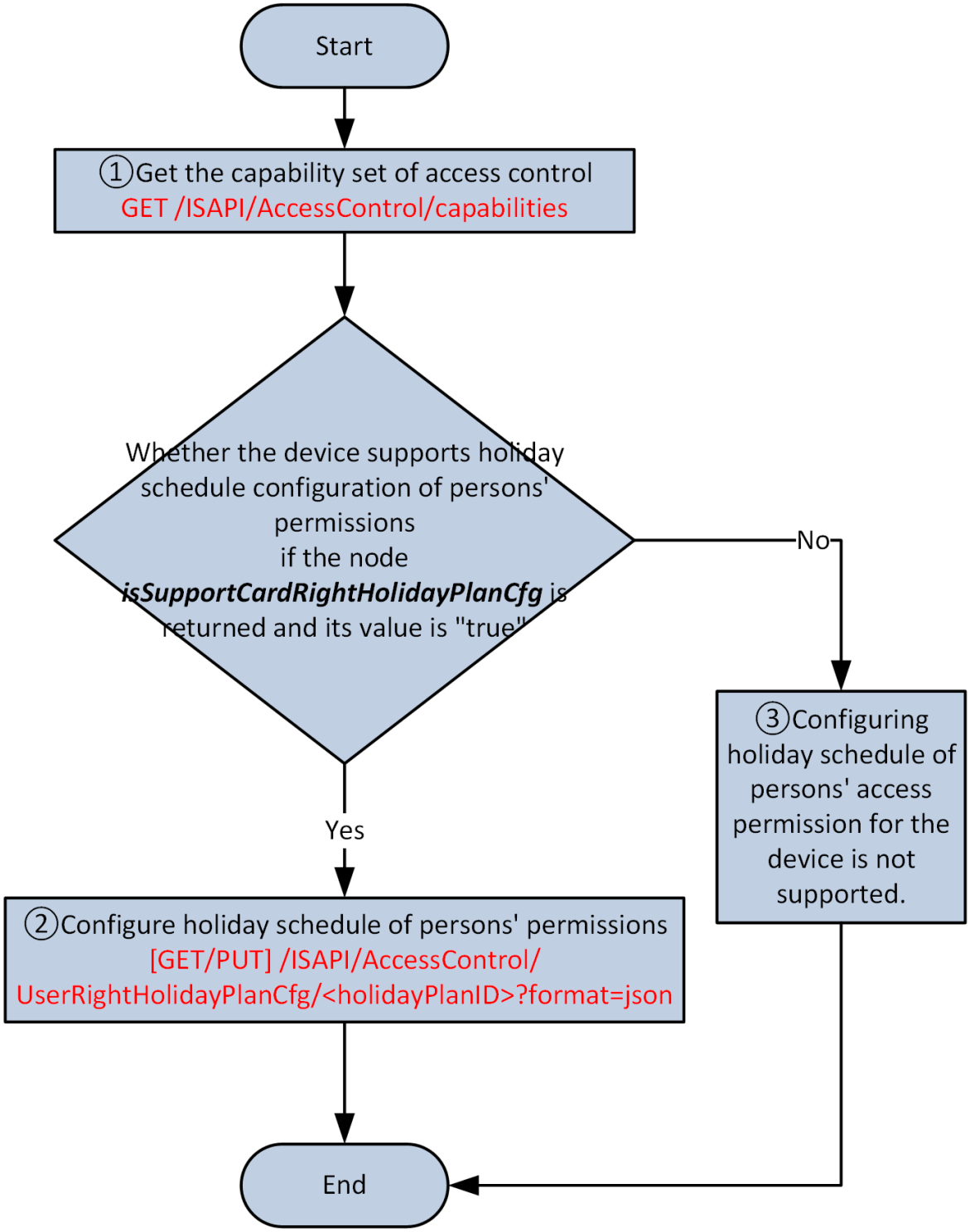
8.2.3 Holiday Groups of Persons' Access Permission



Calling Flow:

1. Check whether the device supports holiday group configuration of person's permissions: `GET /ISAPI/AccessControl/capabilities`; if the nodeisSupportUserRightHolidayGroupCfg is returned and its value is "true", it indicates that the device supports holiday group configuration of person's permission (if it supports, it also supports holiday schedule configuration of persons' permissions).
2. Holiday group configuration of persons' permissions: `[GET/PUT] /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json`.
3. Configuring holiday groups of persons' access permission control for the device is not supported.

8.2.4 Holiday Schedule of Persons' Access Permission



Calling Flow:

1. Check whether the device supports holiday schedule configuration of persons' permissions: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardRightHolidayPlanCfg is returned and its value is "true", it indicates that the device supports holiday schedule configuration of person's permissions.
2. Holiday schedule configuration of persons' permissions: `[GET/PUT] /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json`.
3. Configuring holiday schedule of persons' access permission control for the device is not supported.

9 Store and Search for Access Control Event

9.1 Introduction to the Function

The device supports configuring storage parameters of access control event, searching for access control events, and searching for the amount of access control events. There are three storage modes: deleting old events periodically, deleting old events by specified time and overwriting.

9.2 API Calling Flow

9.2.1 Configure Storage Parameters of Access Control Events

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventStorageCfg is returned and its value is "true", it indicates that the device supports configuring the storage parameters of access control events.
2. Get the configuration capability of storing access control events: `GET /ISAPI/AccessControl/AcsEvent/StorageCfg/capabilities?format=json`.
3. Get and set storage parameters of access control events: `GET|PUT /ISAPI/AccessControl/AcsEvent/StorageCfg?format=json`.

9.2.2 Search for Access Control Events

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAcsEvent is returned and its value is "true", it indicates that the device supports searching for access control events.
2. Get the capability of searching for access control events: `GET /ISAPI/AccessControl/AcsEvent/capabilities?format=json`.
3. Search for access control events: `POST /ISAPI/AccessControl/AcsEvent?format=json`.

9.2.3 Get Total Number of Access Control Events

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAcsEventTotalNum is returned and its value is "true", it indicates that the device supports getting the total number of access control events.
2. Get the capability of getting total number of access control events by specific conditions: `GET /ISAPI/AccessControl/AcsEventTotalNum/capabilities?format=json`.
3. Get the total number of access control events by specific conditions: `POST /ISAPI/AccessControl/AcsEventTotalNum?format=json`.

10 Event and Card Linkage Parameters

10.1 Introduction to the Function

The device supports linking specific actions when access control events triggered. The linkage actions include event, card No., MAC address, and employee No. Take event for example, if a person authenticated by the device, the door will open for the person.

10.2 API Calling Flow

10.2.1 Configure Parameters of Event and Card Linkage

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardLinkageCfg is returned and its value is "true", it indicates that the device supports configuring the parameters of event and card linkage.
2. Get the configuration capability of the event and card linkage: `GET /ISAPI/AccessControl/EventCardLinkageCfg/capabilities?format=json`.

3. Get and set the parameters of event and card linkage: `GET|PUT /ISAPI/AccessControl/EventCardLinkageCfg/<ACEID>?format=json`.

10.2.2 Search for Parameters of Event and Card Linkage

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardLinkageCfgSearch is returned and its value is "true", it indicates that the device supports searching for the parameters of event and card linkage.
2. Get the capability of searching for parameters of event and card linkage: `GET /ISAPI/AccessControl/EventCardLinkageCfg/search/capabilities?format=json`.
3. Search for parameters of event and card linkage: `POST /ISAPI/AccessControl/EventCardLinkageCfg/search?format=json`.

10.2.3 Delete Parameters of Specific Event and Card Linkage

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardLinkageCfgDelete is returned and its value is "true", it indicates that the device supports deleting parameters of event and card linkage.
2. Delete parameters of specific event and card linkage: `PUT /ISAPI/AccessControl/EventCardLinkageCfgDelete?format=json`.

10.2.4 Get List of Event and Card Linkage ID

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportOfflineCapture is returned and its value is "true", it indicates that the device supports offline data collection.
2. Get the capability of the list of event and card linkage ID: `GET /ISAPI/AccessControl/EventCardNoList/capabilities?format=json`.
3. Get the list of event and card linkage ID: `GET /ISAPI/AccessControl/EventCardNoList?format=json`.

10.2.5 Optimize Event

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node iisSupportEventOptimizationCfg is returned and its value is "true", it indicates that the device supports event optimization.
2. Get the configuration capability of event optimization: `GET /ISAPI/AccessControl/EventOptimizationCfg/capabilities?format=json`.
3. Get the event optimization configuration parameters: `GET|PUT /ISAPI/AccessControl/EventOptimizationCfg?format=json`.

11 Arming Information

11.1 Introduction to the Function

The device supports getting the arming information, such as the armed device IP and port, arming type, and protocol type.

11.2 API Calling Flow

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportDeployInfo is returned and its value is "true", it indicates that the device supports getting arming information.

2. Get arming information capability: `GET /ISAPI/AccessControl/DeployInfo/capabilities`.
3. Get arming information: `GET /ISAPI/AccessControl/DeployInfo`.

12 Device Hardware Asset Management

12.1 Introduction to the Function

Hardware assets: the host assets (CPU, memory, and HDD) and peripheral assets (hardware connected to the host including camera, sensor, and USB flash drive).
Typical application: financial industry, where the head office needs to regularly count the security device assets of each branch to collect the number, operation status, and other basic information of deployed host/storage/peripheral cameras.
Extended application: management of industry-related service applications for assets customized by the industry platform (e.g., information and software assets, person assets, and service assets) based on the hardware asset integration.

12.2 API Calling Flow

1. Get the search capability of the device hardware asset data via `GET /ISAPI/System/deviceInfo/capabilities`: when `isSupportSearchHardwareAssets` is true, it indicates the device supports searching for asset data.
2. Search for device hardware asset data via `GET /ISAPI/System/deviceInfo/ExportDeviceAssets?format=json` to get the information of hardware assets on device including host assets, connected sub-device assets, HDD assets, etc.
3. Export the device hardware asset information in binary data in Excel format.

13 API Reference

13.1 Authentication Schedule Management

13.1.1 Get the holiday group configuration capability of the control schedule of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyHolidayGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "VerifyHolidayGroupCfg": {
    /*ro, opt, object*/
        "groupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
```

```
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:whether to enable: "true"-
enable,"false"-disable*/
        "groupName": {
        /*ro, opt, object, length of holiday group name*/
            "@min":  1,
            /*ro, opt, int, the minimum length*/
            "@max":  32
            /*ro, opt, int, the maximum length*/
        },
        "holidayPlanNo": {
        /*ro, opt, object, holiday group schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

13.1.2 Get the schedule template configuration capability of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyPlanTemplate/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "VerifyPlanTemplate": {
    /*ro, opt, object*/
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (yes), false (no)*/
        "templateName": {
        /*ro, opt, object, template name length*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "weekPlanNo": {
        /*ro, opt, object, weekly schedule No.*/
```

```
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "holidayGroupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        }
    }
}
```

13.1.3 Get the control schedule configuration parameters of the card reader authentication mode

URL

GET /ISAPI/AccessControl/CardReaderPlan/<cardReaderID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "CardReaderPlan": {
    /*ro, req, object, schedule template structure*/
        "templateNo":  1
        /*ro, req, int, schedule template number, desc:0-cancel linking the
template to the schedule and restore to the default status (normal status)*/
    }
}
```

13.1.4 Set control schedule parameters of card reader authentication mode

URL

PUT /ISAPI/AccessControl/CardReaderPlan/<cardReaderID>?format=json

Query Parameter

Request Message

```
{
    "CardReaderPlan": {
    /*req, object*/
        "templateNo":  1
        /*req, int, schedule template No., desc:0-cancel linking the template to
the schedule and restore to the default status (normal status)*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.1.5 Get the control schedule configuration capability of the card reader authentication mode

URL

GET /ISAPI/AccessControl/CardReaderPlan/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardReaderPlan": {
    /*ro, opt, object, card reader No. node*/
        "cardReaderNo": {
        /*ro, opt, object, card reader No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  4
            /*ro, opt, int*/
        },
        "templateNo": {
        /*ro, opt, object, schedule template No. node*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16,
            /*ro, opt, int*/
            "@opt": [65535, 65534, 65533]
            /*ro, opt, array, subType:int*/
        },
        "verifyMode": {
```

```
        /*ro, opt, object, authentication mode, dep:and,
{$.CardReaderPlan.templateNo.@opt[*],eq,65535}*/
            "@opt":
 "cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpO
rCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNo
AndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,fac
eOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,cardOrFpOrPw,fa
ceOrPw,employeeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrI
ris,faceOrCardOrPwOrIris,sleep,invalid"
            /*ro, opt, string, desc:"cardAndPw" (card + password), "card"
(card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw"
(fingerprint + password), "fpOrCard" (fingerprint or card), "fpAndCard"
(fingerprint + card), "fpAndCardAndPw" (fingerprint + card + password),
"faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp"
(face + fingerprint), "faceAndPw" (face + password), "faceAndCard" (face +
card), "face" (face), "employeeNoAndPw" (employee No. + password), "fpOrPw"
(fingerprint or password), "employeeNoAndFp" (employee No. + fingerprint),
"employeeNoAndFpAndPw" (employee No. + fingerprint + password),
"faceAndFpAndCard" (face + fingerprint + card), "faceAndPwAndFp" (face +
password + fingerprint), "employeeNoAndFace" (employee No. + face),
"faceOrfaceAndCard" (face or face + card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "faceOrPw" (face or password),
"employeeNoAndFaceAndPw" (employee No. + face + card), "cardOrFaceOrFaceAndCard"
(card or face or face + card), "iris", "faceOrFpOrCardOrPwOrIris"(face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or  card
or password or iris), "sleep", "invalid"*/
        }
    }
}
```

13.1.6 Set holiday schedule parameters of card reader authentication mode

URL

PUT /ISAPI/AccessControl/VerifyHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Request Message

```
{
    "VerifyHolidayPlanCfg": {
    /*opt, object, holiday schedule parameters of the card reader authentication
mode*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "beginDate":  "1970-01-01",
        /*req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-02",
        /*req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*req, int, time period No., range:[1,8]*/
```

```
                "enable":  true,
                /*req, bool, whether to enable, desc:"true"-enable, "false"-
disable*/
                "verifyMode":  "cardAndPw",
                /*req, enum, authentication mode, subType:string,
desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password),
"fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
                "TimeSegment": {
                /*opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "10:00:00"
                    /*req, time, end time of the time period, desc:device local
time*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.1.7 Get holiday schedule parameters of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "VerifyHolidayPlanCfg": {
    /*ro, opt, object, holiday schedule parameters of the card reader
authentication mode*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
        "beginDate":  "1970-01-01",
        /*ro, req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-02",
        /*ro, req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable the holiday schedule,
desc:true (enable), false (disable)*/
                "verifyMode":  "cardAndPw",
```

```
                /*ro, req, enum, authentication mode, subType:string,
desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password),
"fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
                "TimeSegment": {
                /*ro, opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time of the time period, desc:device
local time*/
                }
            }
        ]
    }
}
```

13.1.8 Get the holiday schedule configuration capability of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyHolidayPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
```

```
    "VerifyHolidayPlanCfg": {
    /*ro, opt, object, holiday schedule of card reader authentication mode*/
        "planNo": {
        /*ro, opt, object, holiday schedule template No.*/
            "@min":  1,
            /*ro, opt, int, maximum value*/
            "@max":  16
            /*ro, opt, int, minimum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (enable), false
(disable)*/
        "beginDate":  "1970-01-01",
        /*ro, opt, date, start date of the holiday (device local time), desc:
(device local time)*/
        "endDate":  "1970-01-02",
        /*ro, opt, date, end date of the holiday (device local time), desc:
(device local time)*/
        "HolidayPlanCfg": {
        /*ro, opt, object, holiday schedule parameters*/
            "maxSize":  8,
            /*ro, opt, int, maximum value*/
            "id": {
            /*ro, opt, object, time period No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  8
                /*ro, opt, int, maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable, desc:true (enable), false
(disable)*/
            "verifyMode": {
            /*ro, opt, object, authentication mode*/
                "@opt":
 "cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpO
rCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNo
AndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,fac
eOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw,employ
eeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCar
dOrPwOrIris,sleep,invalid,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,cardOrFpOrIris
OrPw,cardOrIrisOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw
,faceAndIrisAndPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw,cardAndVp,face
AndVp,fpAndVp,pwAndVp"
```

```
                /*ro, opt, string, authentication mode, desc:"cardAndPw"
(card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time, desc:(device local time)*/
                "endTime":  "10:00:00",
                /*ro, opt, time, end time, desc:(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour",
"minute", "second". If this node is not returned, the default time accuracy is
"minute"*/
            }
        },
        "purePwdVerifyEnable":  true
        /*ro, opt, bool*/
    }
}
```

13.1.9 Get the week schedule configuration parameters of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "VerifyWeekPlanCfg": {
    /*ro, opt, object, the week schedule configuration parameters of the card reader authentication mode*/
        "enable":  true,
        /*ro, req, bool, whether to enable the week schedule configuration parameters of the card reader authentication mode, desc:true (enable), false (disable)*/
        "WeekPlanCfg": [
        /*ro, req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable week schedule*/
                "verifyMode":  "cardAndPw",
                /*ro, req, enum, authentication mode, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw" (employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password), "cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
                "TimeSegment": {
                /*ro, req, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time, desc:device local time*/
                }
            }
        ]
    }
}
```

13.1.10 Set the week schedule parameters of the card reader authentication mode

URL

PUT /ISAPI/AccessControl/VerifyWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Request Message

```
{
    "VerifyWeekPlanCfg": {
    /*opt, object, the week schedule configuration parameters of the card reader authentication mode*/
        "enable": true,
        /*req, bool, whether to enable, desc:"true" (enable), "false" (disable)*/
        "WeekPlanCfg": [
        /*req, array, week schedule parameters, subType:object*/
            {
                "week": "Monday",
                /*req, enum, days of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id": 1,
                /*req, int, time period No., range:[1,8]*/
                "enable": true,
                /*req, bool, whether to enable week schedule*/
                "verifyMode": "cardAndPw",
                /*req, enum, authentication mode, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw" (employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password), "cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
                "TimeSegment": {
                /*req, object, time*/
```

```
                    "beginTime":  "00:00:00",
                    /*req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "10:00:00"
                    /*req, time, end time of the time period, desc:device local
time*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.1.11 Get the weekly schedule configuration capability of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyWeekPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "VerifyWeekPlanCfg": {
    /*ro, opt, object, weekly schedule parameters of the card reader
authentication mode*/
        "planNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  16
```

```
            /*ro, opt, int, maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:"true"-enable, "false"-
disable*/
        "WeekPlanCfg": {
        /*ro, opt, object, week schedule parameters*/
            "maxSize":  56,
            /*ro, opt, int, maximum value*/
            "week": {
            /*ro, opt, object, day of a week*/
                "@opt":
 "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"
                /*ro, opt, string, day of a week, desc:"Monday", "Tuesday",
"Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
            },
            "id": {
            /*ro, opt, object, weekly schedule No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  8
                /*ro, opt, int, maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable, desc:true (enable), false
(disable)*/
            "verifyMode": {
            /*ro, opt, object, authentication mode*/
                "@opt":
 "cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpO
rCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNo
AndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,fac
eOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw,employ
eeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCar
dOrPwOrIris,sleep,invalid,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,cardOrFpOrIris
OrPw,cardOrIrisOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw
,faceAndIrisAndPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw,cardAndVp,face
AndVp,fpAndVp,pwAndVp"
```

```
                /*ro, opt, string, authentication mode, desc:"cardAndPw"
(card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris), "sleep",
"invalid"*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time, desc:start time of the time period
(device local time)*/
                "endTime":  "10:00:00",
                /*ro, opt, time, end time, desc:end time of the time period
(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:If this
node is not returned, it indicates that the time accuracy is "minute". "hour",
"minute", "second"*/
            }
        },
        "purePwdVerifyEnable":  true
        /*ro, opt, bool*/
    }
}
```

13.1.12 Set holiday group parameters of control schedule of card reader authentication mode

URL

PUT /ISAPI/AccessControl/VerifyHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Request Message

```
{
    "VerifyHolidayGroupCfg": {
    /*opt, object, holiday group parameters of control schedule of card reader
authentication mode*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "groupName":  "test",
        /*req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*opt, string, holiday group schedule No., desc:holiday group schedule
No.*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.1.13 Set the schedule template parameters of the card reader authentication mode

URL

PUT /ISAPI/AccessControl/VerifyPlanTemplate/<planTemplateID>?format=json

Query Parameter

Request Message

```
{
    "VerifyPlanTemplate": {
    /*opt, object*/
        "enable":  true,
        /*req, bool, whether to enable, desc:true (yes), false (no)*/
        "templateName":  "test",
        /*req, string, template name*/
        "weekPlanNo":  1,
        /*req, int, week schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*req, string, holiday group No., desc:holiday group No.*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.1.14 Get the schedule template parameters of card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyPlanTemplate/<planTemplateID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "VerifyPlanTemplate": {
    /*ro, opt, object, the schedule template parameters of card reader
authentication mode*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-
disable*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, week schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*ro, req, string, holiday group No., desc:holiday group No.*/
    }
}
```

13.1.15 Get the holiday group configuration parameters of the control schedule of the card reader authentication mode

URL

GET /ISAPI/AccessControl/VerifyHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "VerifyHolidayGroupCfg": {
    /*ro, opt, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (yes), false-no
(default)*/
        "groupName":  "test",
        /*ro, req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*ro, opt, string, holiday group schedule No., desc:holiday group
schedule No.*/
    }
}
```

13.2 Access Control System Management

13.2.1 Get the functional capability of access control

URL

GET /ISAPI/AccessControl/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AccessControl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the functional capability of access control,
attr:version{req, string, protocolVersion}-->
  <isSupportWiegandCfg>
    <!--ro, opt, bool, whether it supports Wiegand configuration, desc:true
(support), this node is not returned (not support)-->true
  </isSupportWiegandCfg>
  <isSupportModuleStatus>
    <!--ro, opt, bool, whether it supports getting the status of secure door
control unit, desc:true (support), this node is not returned (not support)--
>true
  </isSupportModuleStatus>
  <isSupportSNAPConfig>
    <!--ro, opt, bool, whether it supports getting capture linkage parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportSNAPConfig>
  <LocalController>
    <!--ro, opt, object-->
    <isSupportLocalControllerManage>
      <!--ro, opt, bool, whether it supports distributed access controller
management, desc:true (support), this node is not returned (not support)-->true
    </isSupportLocalControllerManage>
    <isSupportLocalControllerControl>
      <!--ro, opt, bool, whether it supports distributed access controller
control, desc:true (support), this node is not returned (not support)-->true
    </isSupportLocalControllerControl>
  </LocalController>
  <isSupportUSBManage>
    <!--ro, opt, bool, whether it supports USB management of access control
device, desc:true (support), this node is not returned (not support)-->true
  </isSupportUSBManage>
  <isSupportLadderControlRelay>
    <!--ro, opt, bool, whether it supports setting elevator controller relay,
desc:true (support), this node is not returned (not support)-->true
  </isSupportLadderControlRelay>
  <isSupportIdentityTerminal>
    <!--ro, opt, bool, whether it supports face recognition terminal
configuration, desc:true (support), this node is not returned (not support)--
>true
  </isSupportIdentityTerminal>
  <isSupportDepartmentParam>
    <!--ro, opt, bool, whether it supports setting department parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportDepartmentParam>
  <isSupportSchedulePlan>
    <!--ro, opt, bool, whether it supports setting shift schedule, desc:true
(support), this node is not returned (not support)-->true
  </isSupportSchedulePlan>
  <isSupportAttendanceRule>
    <!--ro, opt, bool, Attendance Rule Configuration, desc:true (support), this
node is not returned (not support)-->true
```

```
  </isSupportAttendanceRule>
  <isSupportOrdinaryClass>
    <!--ro, opt, bool, whether it supports setting normal shift parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportOrdinaryClass>
  <isSupportWorkingClass>
    <!--ro, opt, bool, whether it supports setting man-hour shift parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportWorkingClass>
  <isSupportAttendanceHolidayGroup>
    <!--ro, opt, bool, whether it supports setting holiday group for time and
attendance, desc:true (support), this node is not returned (not support)-->true
  </isSupportAttendanceHolidayGroup>
  <isSupportAttendanceHolidayPlan>
    <!--ro, opt, bool, whether it supports setting holiday schedule for time and
attendance, desc:true (support), this node is not returned (not support)-->true
  </isSupportAttendanceHolidayPlan>
  <isSupportWiegandRuleCfg>
    <!--ro, opt, bool, whether it supports setting Wiegand rule, desc:true
(support), this node is not returned (not support)-->true
  </isSupportWiegandRuleCfg>
  <isSupportM1CardEncryptCfg>
    <!--ro, opt, bool, configuration capability of the M1 card encryption
verification, desc:true (support), this node is not returned (not support)--
>true
  </isSupportM1CardEncryptCfg>
  <isSupportDeployInfo>
    <!--ro, opt, bool, whether it supports getting arming information-->true
  </isSupportDeployInfo>
  <isSupportSubmarineBack>
    <!--ro, opt, bool, whether it supports specifying anti-passing back server,
desc:true (support), this node is not returned (not support)-->true
  </isSupportSubmarineBack>
  <isSupportSubmarineBackHostInfo>
    <!--ro, opt, bool, whether it supports setting access controllers with anti-
passing back enabled, desc:true (support), this node is not returned (not
support)-->true
  </isSupportSubmarineBackHostInfo>
  <isSupportStartReaderInfo>
    <!--ro, opt, bool, whether it supports setting first card reader, desc:true
(support), this node is not returned (not support)-->true
  </isSupportStartReaderInfo>
  <isSupportSubmarineBackReader>
    <!--ro, opt, bool, whether it supports setting anti-passing back card
reader, desc:true (support), this node is not returned (not support)-->true
  </isSupportSubmarineBackReader>
  <isSupportServerDevice>
    <!--ro, opt, bool, whether it supports setting anti-passing back server
information, desc:true (support), this node is not returned (not support)-->true
  </isSupportServerDevice>
  <isSupportReaderAcrossHost>
    <!--ro, opt, bool, whether it supports enabling cross-controller anti-
passing back function of card reader, desc:true (support), this node is not
returned (not support)-->true
  </isSupportReaderAcrossHost>
  <isSupportClearCardRecord>
```

```
    <!--ro, opt, bool, whether it supports clearing card swiping records in
anti-passing back server, desc:true (support), this node is not returned (not
support)-->true
  </isSupportClearCardRecord>
  <isSupportSubmarineBackMode>
    <!--ro, opt, bool, whether it supports setting anti-passing back mode,
desc:true (support), this node is not returned (not support)-->true
  </isSupportSubmarineBackMode>
  <isSupportClearSubmarineBack>
    <!--ro, opt, bool, whether it supports clearing cross-controller anti-
passing back information, desc:true (support), this node is not returned (not
support)-->true
  </isSupportClearSubmarineBack>
  <isSupportSupplementLight>
    <!--ro, opt, bool-->true
  </isSupportSupplementLight>
  <isSupportFaceCompareCond>
    <!--ro, opt, bool, whether it supports configuring restriction condition
parameters of face picture comparison-->true
  </isSupportFaceCompareCond>
  <isSupportRightControllerAudio>
    <!--ro, opt, bool-->true
  </isSupportRightControllerAudio>
  <isSupportChannelControllerCfg>
    <!--ro, opt, bool-->true
  </isSupportChannelControllerCfg>
  <isSupportGateDialAndInfo>
    <!--ro, opt, bool-->true
  </isSupportGateDialAndInfo>
  <isSupportGateStatus>
    <!--ro, opt, bool-->true
  </isSupportGateStatus>
  <isSupportGateIRStatus>
    <!--ro, opt, bool-->true
  </isSupportGateIRStatus>
  <isSupportGateRelatedPartsStatus>
    <!--ro, opt, bool-->true
  </isSupportGateRelatedPartsStatus>
  <isSupportChannelControllerAlarmLinkage>
    <!--ro, opt, bool-->true
  </isSupportChannelControllerAlarmLinkage>
  <isSupportChannelControllerAlarmOut>
    <!--ro, opt, bool-->true
  </isSupportChannelControllerAlarmOut>
  <isSupportChannelControllerAlarmOutControl>
    <!--ro, opt, bool-->true
  </isSupportChannelControllerAlarmOutControl>
  <isSupportChannelControllerTypeCfg>
    <!--ro, opt, bool-->true
  </isSupportChannelControllerTypeCfg>
  <isSupportRemoteCtrllerModeCfg>
    <!--ro, opt, bool-->true
  </isSupportRemoteCtrllerModeCfg>
  <isSupportRemoteControlDoor>
    <!--ro, opt, bool, whether it supports remote door,elevator,and lock contro,
desc:true (support), this node is not returned (not support)-->true
  </isSupportRemoteControlDoor>
  <isSupportUserInfo>
```

```
    <!--ro, opt, bool, whether it supports person management based on person,
desc:true (support), this node is not returned (not support)-->true
  </isSupportUserInfo>
  <EmployeeNoInfo>
    <!--ro, opt, object, employee No. (person ID) information, desc:this node is
valid only when the isSupportUserInfo is "true"-->
    <employeeNo min="1" max="10">
      <!--ro, opt, int, employee No. (person ID), attr:min{req, int},max{req,
int}-->1
    </employeeNo>
    <characterType opt="any,number">
      <!--ro, opt, enum, employee No. (person) ID type, subType:string,
attr:opt{req, string}, desc:"any"-any characters (default), "number"-digits
(from 0 to 9), only one value can be returned-->any
    </characterType>
    <isSupportCompress>
      <!--ro, opt, bool, whether it supports compressing employee No. (person
ID) for storage, desc:true (support), this node is not returned (not support)--
>true
    </isSupportCompress>
  </EmployeeNoInfo>
  <isSupportCardInfo>
    <!--ro, opt, bool, whether it supports card management based on person,
desc:true (support), this node is not returned (not support)-->true
  </isSupportCardInfo>
  <isSupportFDLib>
    <!--ro, opt, bool, whether it supports face picture library management--
>true
  </isSupportFDLib>
  <isSupportUserInfoDetailDelete>
    <!--ro, opt, bool, whether it supports deleting person information and
permission, desc:true (support), this node is not returned (not support)-->true
  </isSupportUserInfoDetailDelete>
  <isSupportAuthCodeInfo>
    <!--ro, opt, bool, whether it supports authentication password management,
desc:true (support), this node is not returned (not support)-->true
  </isSupportAuthCodeInfo>
  <isSupportFingerPrintCfg>
    <!--ro, opt, bool, whether it supports configuring fingerprint parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportFingerPrintCfg>
  <isSupportFingerPrintDelete>
    <!--ro, opt, bool, whether it supports deleting fingerprints, desc:true
(support), this node is not returned (not support)-->true
  </isSupportFingerPrintDelete>
  <isSupportCaptureFingerPrint>
    <!--ro, opt, bool, whether it supports collecting fingerprint information,
desc:true (support), this node is not returned (not support)-->true
  </isSupportCaptureFingerPrint>
  <isSupportDoorStatusWeekPlanCfg>
    <!--ro, opt, bool, whether it supports configuring door control week
schedule, desc:true (support), this node is not returned (not support)-->true
  </isSupportDoorStatusWeekPlanCfg>
  <isSupportVerifyWeekPlanCfg>
    <!--ro, opt, bool, whether it supports configuring week schedule of the card
reader authentication mode, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVerifyWeekPlanCfg>
```

```
  <isSupportCardRightWeekPlanCfg>
    <!--ro, opt, bool, whether it supports configuring week schedule of the
access permission control, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCardRightWeekPlanCfg>
  <isSupportDoorStatusHolidayPlanCfg>
    <!--ro, opt, bool, whether it supports configuring door control holiday
schedule, desc:true (support), this node is not returned (not support)-->true
  </isSupportDoorStatusHolidayPlanCfg>
  <isSupportVerifyHolidayPlanCfg>
    <!--ro, opt, bool, whether it supports configuring holiday schedule of the
card reader authentication mode, desc:true (support), this node is not returned
(not support)-->true
  </isSupportVerifyHolidayPlanCfg>
  <isSupportCardRightHolidayPlanCfg>
    <!--ro, opt, bool, whether it supports configuring holiday schedule of the
access permission control, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCardRightHolidayPlanCfg>
  <isSupportDoorStatusHolidayGroupCfg>
    <!--ro, opt, bool, whether it supports configuring holiday group of the door
control schedule, desc:true (support), this node is not returned (not support)--
>true
  </isSupportDoorStatusHolidayGroupCfg>
  <isSupportVerifyHolidayGroupCfg>
    <!--ro, opt, bool, whether it supports configuring holiday group of the
control schedule of the card reader authentication mode, desc:true (support),
this node is not returned (not support)-->true
  </isSupportVerifyHolidayGroupCfg>
  <isSupportUserRightHolidayGroupCfg>
    <!--ro, opt, bool, whether it supports configuring holiday group of the
access permission control schedule, desc:true (support), this node is not
returned (not support)-->true
  </isSupportUserRightHolidayGroupCfg>
  <isSupportDoorStatusPlanTemplate>
    <!--ro, opt, bool, whether it supports configuring door control schedule
template, desc:true (support), this node is not returned (not support)-->true
  </isSupportDoorStatusPlanTemplate>
  <isSupportVerifyPlanTemplate>
    <!--ro, opt, bool, whether it supports configuring schedule template of the
card reader authentication mode, desc:true (support), this node is not returned
(not support)-->true
  </isSupportVerifyPlanTemplate>
  <isSupportUserRightPlanTemplate>
    <!--ro, opt, bool, whether it supports configuring schedule template of the
access permission control, desc:true (support), this node is not returned (not
support)-->true
  </isSupportUserRightPlanTemplate>
  <isSupportDoorStatusPlan>
    <!--ro, opt, bool, whether it supports configuring door control schedule,
desc:true (support), this node is not returned (not support)-->true
  </isSupportDoorStatusPlan>
  <isSupportCardReaderPlan>
    <!--ro, opt, bool, whether it supports configuring control schedule of the
card reader authentication mode, desc:true (support), this node is not returned
(not support)-->true
  </isSupportCardReaderPlan>
  <isSupportClearPlansCfg>
```

```
    <!--ro, opt, bool, whether it supports clearing the access control schedule
parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportClearPlansCfg>
  <isSupportRemoteControlBuzzer>
    <!--ro, opt, bool, whether it supports remotely controlling the buzzer of
the card reader, desc:true (support), this node is not returned (not support)--
>true
  </isSupportRemoteControlBuzzer>
  <isSupportEventCardNoList>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportEventCardNoList>
  <isSupportEventCardLinkageCfg>
    <!--ro, opt, bool, whether it supports configuring event and card linkage
parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportEventCardLinkageCfg>
  <isSupportClearEventCardLinkageCfg>
    <!--ro, opt, bool, whether it supports clearing event and card linkage
parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportClearEventCardLinkageCfg>
  <isSupportAcsEvent>
    <!--ro, opt, bool, whether it supports searching for access control events,
desc:true (support), this node is not returned (not support)-->true
  </isSupportAcsEvent>
  <isSupportAcsEventTotalNum>
    <!--ro, opt, bool, whether it supports getting total number of access
control events by specific conditions, desc:true (support), this node is not
returned (not support)-->true
  </isSupportAcsEventTotalNum>
  <isSupportEventOptimizationCfg>
    <!--ro, opt, bool, whether it supports configuring event optimization,
desc:true (support), this node is not returned (not support)-->true
  </isSupportEventOptimizationCfg>
  <isSupportAcsWorkStatus>
    <!--ro, opt, bool, whether it supports getting working status of the access
control device, desc:true (support), this node is not returned (not support)--
>true
  </isSupportAcsWorkStatus>
  <isSupportDoorCfg>
    <!--ro, opt, bool, whether it supports configuring door parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportDoorCfg>
  <isSupportCardReaderCfg>
    <!--ro, opt, bool, whether it supports configuring card reader parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportCardReaderCfg>
  <isSupportAcsCfg>
    <!--ro, opt, bool, whether it supports configuring parameters of access
control device, desc:true (support), this node is not returned (not support)--
>true
  </isSupportAcsCfg>
  <isSupportRemoteCheck>
    <!--ro, opt, bool, whether it supports verifying access control events
remotely, desc:true (support), this node is not returned (not support)-->true
  </isSupportRemoteCheck>
  <isSupportMaskDetection>
    <!--ro, opt, bool, whether it supports mask detection, desc:true (support),
this node is not returned (not support)-->true
```

```
  </isSupportMaskDetection>
  <isSupportGroupCfg>
    <!--ro, opt, bool, whether it supports configuring group parameters:,
desc:true (support), this node is not returned (not support)-->true
  </isSupportGroupCfg>
  <isSupportClearGroupCfg>
    <!--ro, opt, bool, whether it supports clearing group parameters, desc:true
(support), this node is not returned (not support)-->true
  </isSupportClearGroupCfg>
  <isSupportMultiCardCfg>
    <!--ro, opt, bool, whether it supports configuring multiple authentication
mode, desc:true (support), this node is not returned (not support)-->true
  </isSupportMultiCardCfg>
  <isSupportMultiDoorInterLockCfg>
    <!--ro, opt, bool, whether it supports configuring multi-door interlocking
parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportMultiDoorInterLockCfg>
  <isSupportAntiSneakCfg>
    <!--ro, opt, bool, whether it supports configuring anti-passing back
parameters in the device, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAntiSneakCfg>
  <isSupportCardReaderAntiSneakCfg>
    <!--ro, opt, bool, whether it supports configuring anti-passing back
parameters for the card reader in the device, desc:true (support), this node is
not returned (not support)-->true
  </isSupportCardReaderAntiSneakCfg>
  <isSupportClearAntiSneakCfg>
    <!--ro, opt, bool, whether it supports clearing anti-passing back
parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportClearAntiSneakCfg>
  <isSupportClearAntiSneak>
    <!--ro, opt, bool, whether it supports clearing anti-passing back records in
the device, desc:true (support), this node is not returned (not support)-->true
  </isSupportClearAntiSneak>
  <isSupportSmsRelativeParam>
    <!--ro, opt, bool, whether it supports configuring message function,
desc:true (support), this node is not returned (not support)-->true
  </isSupportSmsRelativeParam>
  <isSupportPhoneDoorRightCfg>
    <!--ro, opt, bool, whether it supports configuring the door permission
linked to the mobile phone number: "true"-yes,this node is not returned-no,
desc:true (support), this node is not returned (not support)-->true
  </isSupportPhoneDoorRightCfg>
  <isSupportOSDPStatus>
    <!--ro, opt, bool, whether it supports searching for OSDP card reader
status, desc:true (support), this node is not returned (not support)-->true
  </isSupportOSDPStatus>
  <isSupportOSDPModify>
    <!--ro, opt, bool, whether it supports editing OSDP card reader ID,
desc:true (support), this node is not returned (not support)-->true
  </isSupportOSDPModify>
  <isSupportRemoteControlPWChcek>
    <!--ro, opt, bool, whether it supports verifying the password for remote
door control, desc:true (support), this node is not returned (not support)--
>true
  </isSupportRemoteControlPWChcek>
  <isSupportRemoteControlPWCfg>
```

```
    <!--ro, opt, bool, whether it supports configuring the password for remote
door contro, desc:true (support), this node is not returned (not support)-->true
  </isSupportRemoteControlPWCfg>
  <isSupportAttendanceStatusModeCfg>
    <!--ro, opt, bool, whether it supports configuring attendance mode,
desc:true (support), this node is not returned (not support)-->true
  </isSupportAttendanceStatusModeCfg>
  <isSupportAttendanceStatusRuleCfg>
    <!--ro, opt, bool, whether it supports configuring attendance status and
rule, desc:true (support), this node is not returned (not support)-->true
  </isSupportAttendanceStatusRuleCfg>
  <isSupportLogModeCfg>
    <!--ro, opt, bool, whether it supports configuring log mode, desc:true
(support), this node is not returned (not support)-->true
  </isSupportLogModeCfg>
  <isSupportQRCodeEncryption>
    <!--ro, opt, bool-->true
  </isSupportQRCodeEncryption>
  <isSupportQRCodeInfo>
    <!--ro, opt, bool-->true
  </isSupportQRCodeInfo>
  <isSupportBluetoothEncryptionInfo>
    <!--ro, opt, bool-->true
  </isSupportBluetoothEncryptionInfo>
  <isSupportBluetoothEncryptionVersion>
    <!--ro, opt, bool-->true
  </isSupportBluetoothEncryptionVersion>
  <FactoryReset>
    <!--ro, opt, object-->
    <isSupportFactoryReset>
      <!--ro, opt, bool, whether it supports restoring to default settings by
condition-->true
    </isSupportFactoryReset>
    <mode opt="full,basic,part">
      <!--ro, opt, enum, conditions for restoring to default settings,
subType:string, attr:opt{req, string}, desc:conditions for restoring to default
settings-->full
    </mode>
  </FactoryReset>
  <isSupportCaptureFace>
    <!--ro, opt, bool, whether it supports collecting face pictures, desc:true
(support), this node is not returned (not support)-->true
  </isSupportCaptureFace>
  <isSupportCaptureInfraredFace>
    <!--ro, opt, bool, whether it supports collecting infrared face pictures,
desc:true (support), this node is not returned (not support)-->true
  </isSupportCaptureInfraredFace>
  <isSupportFaceRecognizeMode>
    <!--ro, opt, bool, whether it supports configuring facial recognition mode,
desc:true (support), this node is not returned (not support)-->true
  </isSupportFaceRecognizeMode>
  <isSupportCaptureCardInfo>
    <!--ro, opt, bool, whether it supports collecting card information,
desc:true (support), this node is not returned (not support)-->true
  </isSupportCaptureCardInfo>
  <isSupportCaptureIDInfo>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
```

```
  </isSupportCaptureIDInfo>
  <isSupportCaptureRule>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCaptureRule>
  <isSupportCapturePresetParam>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCapturePresetParam>
  <isSupportOfflineCapture>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportOfflineCapture>
  <isSupportCardOperations>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCardOperations>
  <isSupportNFCCfg>
    <!--ro, opt, bool, configuration capability of enabling NFC (Near-Field
Communication) function, desc:true (support), this node is not returned (not
support)-->true
  </isSupportNFCCfg>
  <isSupportRFCardCfg>
    <!--ro, opt, bool, whether it supports enabling or disabling RF card
recognition, desc:true (support), this node is not returned (not support)-->true
  </isSupportRFCardCfg>
  <isSupportTTSText>
    <!--ro, opt, bool, whether it supports configuring the text of the audio
prompt, desc:true (support), this node is not returned (not support)-->true
  </isSupportTTSText>
  <isSupportIDBlackListCfg>
    <!--ro, opt, bool, whether it supports applying ID card blocklist, desc:true
(support), this node is not returned (not support)-->true
  </isSupportIDBlackListCfg>
  <isSupportUserInfoAndRight>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportUserInfoAndRight>
  <isSupportUserDataImport>
    <!--ro, opt, bool, whether it supports importing person permission data--
>true
  </isSupportUserDataImport>
  <isSupportUserDataExport>
    <!--ro, opt, bool, whether it supports exporting person permission data--
>true
  </isSupportUserDataExport>
  <isSupportMaintenanceDataExport>
    <!--ro, opt, bool, whether it supports exporting maintenance data-->true
  </isSupportMaintenanceDataExport>
  <isSupportLockTypeCfg>
    <!--ro, opt, bool, whether it supports configuring door lock status when the
device is powered off-->true
  </isSupportLockTypeCfg>
  <isSupportSafetyHelmetDetection>
    <!--ro, opt, bool, whether it supports configuring hard hat detection,
desc:true (support), this node is not returned (not support)-->true
  </isSupportSafetyHelmetDetection>
  <isSupportKeyCfgAttendance>
```

```
    <!--ro, opt, bool, whether it supports configuring parameters of attendance
check by pressing the key, desc:true (support), this node is not returned (not
support)-->true
  </isSupportKeyCfgAttendance>
  <isSupportAsyncImportDatas>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAsyncImportDatas>
  <isSupportAsyncImportPic>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAsyncImportPic>
  <isSupportIDBlackListClear>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportIDBlackListClear>
  <isSupportIDBlackListParamCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportIDBlackListParamCfg>
  <isSupportIDBlackListTemplate>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportIDBlackListTemplate>
  <isSupportCertificateCompareCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCertificateCompareCfg>
  <isSupportVisitorRightListCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVisitorRightListCfg>
  <isSupportVisitorRegisterInfoCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVisitorRegisterInfoCfg>
  <isSupportVisitorReceiptInfoCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVisitorReceiptInfoCfg>
  <isSupportVisitorParamCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVisitorParamCfg>
  <isSupportVisitorAdvanceParamCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVisitorAdvanceParamCfg>
  <isSupportVisitorRegisterAudio>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportVisitorRegisterAudio>
  <isSupportFingerDataEncryption>
    <!--ro, opt, bool-->true
  </isSupportFingerDataEncryption>
  <isSupportAttendanceWeekPlan>
```

```
    <!--ro, opt, bool, whether it supports configuring parameters of the week
attendance schedule, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAttendanceWeekPlan>
  <isSupportClearAttendancePlan>
    <!--ro, opt, bool, whether it supports clearing the week attendance
schedule, desc:true (support), this node is not returned (not support)-->true
  </isSupportClearAttendancePlan>
  <isSupportAttendanceStatusPlan>
    <!--ro, opt, bool-->true
  </isSupportAttendanceStatusPlan>
  <isSupportAttendanceMode>
    <!--ro, opt, bool, whether it supports configuring the attendance mode--
>true
  </isSupportAttendanceMode>
  <isSupportAttendancePlanTemplate>
    <!--ro, opt, bool, whether it supports configuring the attendance schedule
template, desc:true (support), this node is not returned (not support)-->true
  </isSupportAttendancePlanTemplate>
  <isSupportAttendancePlanTemplateList>
    <!--ro, opt, bool, whether it supports getting the list of attendance
schedule templates, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAttendancePlanTemplateList>
  <isSupportClearPictureCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportClearPictureCfg>
  <isSupportEventStorageCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportEventStorageCfg>
  <isSupportCardVerificationRule>
    <!--ro, opt, bool, whether it supports configuring card No. authentication
mode, desc:true (support), this node is not returned (not support)-->true
  </isSupportCardVerificationRule>
  <isSupportFaceTemperatureEvent>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportFaceTemperatureEvent>
  <isSupportQRCodeEvent>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportQRCodeEvent>
  <isSupportIDCardInfoEvent>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportIDCardInfoEvent>
  <isSupportTTSTextHolidayPlan>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportTTSTextHolidayPlan>
  <isSupportConferencePlan>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportConferencePlan>
  <isSupportDeleteConferencePlan>
```

```
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportDeleteConferencePlan>
  <isSupportConferenceBasic>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportConferenceBasic>
  <isSupportPersonInfoExtendName>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportPersonInfoExtendName>
  <isSupportCardDisplayCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCardDisplayCfg>
  <isSupportBluetooth>
    <!--ro, opt, bool, whether it supports bluetooth configuration-->true
  </isSupportBluetooth>
  <isSupportFaceCompareParamUpload>
    <!--ro, opt, bool-->true
  </isSupportFaceCompareParamUpload>
  <isSupportTemperatureMeasureCfg>
    <!--ro, opt, bool, whether it supports configuring temperature measurement
parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportTemperatureMeasureCfg>
  <isSupportTemperatureMeasureAreaCfg>
    <!--ro, opt, bool, whether it supports configuring parameters of the
temperature measurement area, desc:true (support), this node is not returned
(not support)-->true
  </isSupportTemperatureMeasureAreaCfg>
  <isSupportTemperatureMeasureAreaCalibrationCfg>
    <!--ro, opt, bool, whether it supports configuring calibration parameters of
the temperature measurement area, desc:true (support), this node is not returned
(not support)-->true
  </isSupportTemperatureMeasureAreaCalibrationCfg>
  <isSupportBlackObjectCfg>
    <!--ro, opt, bool, whether it supports configuring black body parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportBlackObjectCfg>
  <isSupportHealthCodeCfg>
    <!--ro, opt, bool, whether it supports configuring health code parameters,
desc:true (support), this node is not returned (not support)-->true
  </isSupportHealthCodeCfg>
  <isSupportShowHealthCodeCfg>
    <!--ro, opt, bool, whether it supports configuring display parameters of the
health code, desc:true (support), this node is not returned (not support)-->true
  </isSupportShowHealthCodeCfg>
  <isSupportLOGOCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportLOGOCfg>
  <isSupportCustomInfoCfg>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCustomInfoCfg>
  <isSupportReaderList>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
```

```
  </isSupportReaderList>
  <isSupportFingerprintRecognition>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportFingerprintRecognition>
  <isSupportFaceRegionReader>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportFaceRegionReader>
  <isSupportFaceRecognitionReader>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportFaceRecognitionReader>
  <isSupportMaskDetectionReader>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportMaskDetectionReader>
  <isSupportAntiPassbackResetRules>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAntiPassbackResetRules>
  <isSupportAntiPassbackTimeRange>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAntiPassbackTimeRange>
  <isSupportAcrossHostInterlockServer>
    <!--ro, opt, bool-->true
  </isSupportAcrossHostInterlockServer>
  <isSupportAcrossHostInterlockHostInfo>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAcrossHostInterlockHostInfo>
  <isSupportAcrossHostInterlockGroup>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAcrossHostInterlockGroup>
  <isSupportAcrossHostInterlockDoor>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportAcrossHostInterlockDoor>
  <isSupportAcrossHostInterlockClient>
    <!--ro, opt, bool-->true
  </isSupportAcrossHostInterlockClient>
  <isSupportWiegandBindReader>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportWiegandBindReader>
  <isSupportUserAndRightShow>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportUserAndRightShow>
  <isSupportDoorMagneticDefiniteRule>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportDoorMagneticDefiniteRule>
  <isSupportTTSTextSearchHolidayPlan>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
```

```
  </isSupportTTSTextSearchHolidayPlan>
  <isSupportAddCustomAudio>
    <!--ro, opt, bool, whether it supports importing custom audio, desc:related
URI: /ISAPI/AccessControl/customAudio/addCustomAudio?format=json-->true
  </isSupportAddCustomAudio>
  <isSupportDeleteCustomAudio>
    <!--ro, opt, bool, whether it supports deleting custom audio, desc:related
URI: /ISAPI/AccessControl/customAudio/deleteCustomAudio?format=json-->true
  </isSupportDeleteCustomAudio>
  <isSupportSearchCustomAudio>
    <!--ro, opt, bool, whether it supports searching for custom audio,
desc:related URI: /ISAPI/AccessControl/customAudio/searchCustomAudioStatus?
format=json-->true
  </isSupportSearchCustomAudio>
  <isSupportIrisPreview>
    <!--ro, opt, bool-->true
  </isSupportIrisPreview>
  <isSupportIrisCalibration>
    <!--ro, opt, bool-->true
  </isSupportIrisCalibration>
  <isSupportIrisCalibrationParam>
    <!--ro, opt, bool-->true
  </isSupportIrisCalibrationParam>
  <isSupportCaptureIrisData>
    <!--ro, opt, bool-->true
  </isSupportCaptureIrisData>
  <isSupportIrisInfo>
    <!--ro, opt, bool-->true
  </isSupportIrisInfo>
  <isSupportIrisCfg>
    <!--ro, opt, bool-->true
  </isSupportIrisCfg>
  <isSupportIrisRecognition>
    <!--ro, opt, bool-->true
  </isSupportIrisRecognition>
  <isSupportImageFlip>
    <!--ro, opt, bool-->true
  </isSupportImageFlip>
  <isSupportUncertificateCompareCfg>
    <!--ro, opt, bool-->true
  </isSupportUncertificateCompareCfg>
  <isSupportCertificateCaptureEvent>
    <!--ro, opt, bool-->true
  </isSupportCertificateCaptureEvent>
  <isSupportUncertificateCompareEvent>
    <!--ro, opt, bool-->true
  </isSupportUncertificateCompareEvent>
  <isSupportCustomAudio>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCustomAudio>
  <isSupportEventCardLinkageCfgDelete>
    <!--ro, opt, bool-->true
  </isSupportEventCardLinkageCfgDelete>
  <isSupportCustomAudioBatchDelete>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCustomAudioBatchDelete>
```

```
<isSupportEventCardLinkageCfgSearch>
  <!--ro, opt, bool-->true
</isSupportEventCardLinkageCfgSearch>
<isSupportCustomAudioPlay>
  <!--ro, opt, bool-->true
</isSupportCustomAudioPlay>
<isSupportCustomAudioStop>
  <!--ro, opt, bool-->true
</isSupportCustomAudioStop>
<isSupportTTSAudio>
  <!--ro, opt, bool-->true
</isSupportTTSAudio>
<isSupportBoxInfoDelete>
  <!--ro, opt, bool-->true
</isSupportBoxInfoDelete>
<LocalAttendanceCap>
  <!--ro, opt, object-->
  <isSupportGroupSearch>
    <!--ro, opt, bool-->true
  </isSupportGroupSearch>
  <isSupportGroup>
    <!--ro, opt, bool-->true
  </isSupportGroup>
  <isSupportRule>
    <!--ro, opt, bool-->true
  </isSupportRule>
  <isSupportShiftSearch>
    <!--ro, opt, bool-->true
  </isSupportShiftSearch>
  <isSupportShift>
    <!--ro, opt, bool-->true
  </isSupportShift>
  <isSupportHolidayPlanSearch>
    <!--ro, opt, bool-->true
  </isSupportHolidayPlanSearch>
  <isSupportHolidayPlan>
    <!--ro, opt, bool-->true
  </isSupportHolidayPlan>
  <isSupportHolidayGroupSearch>
    <!--ro, opt, bool-->true
  </isSupportHolidayGroupSearch>
  <isSupportHolidayGroup>
    <!--ro, opt, bool-->true
  </isSupportHolidayGroup>
  <isSupportWeekPlanSearch>
    <!--ro, opt, bool-->true
  </isSupportWeekPlanSearch>
  <isSupportWeekPlan>
    <!--ro, opt, bool-->true
  </isSupportWeekPlan>
  <isSupportPlanTemplateSearch>
    <!--ro, opt, bool-->true
  </isSupportPlanTemplateSearch>
  <isSupportPlanTemplate>
    <!--ro, opt, bool-->true
  </isSupportPlanTemplate>
  <isSupportGroupShiftSearch>
    <!--ro, opt, bool-->true
```

```
    </isSupportGroupShiftSearch>
    <isSupportGroupShift>
      <!--ro, opt, bool-->true
    </isSupportGroupShift>
    <isSupportPersonalShiftNum>
      <!--ro, opt, bool-->true
    </isSupportPersonalShiftNum>
    <isSupportRecordStorage>
      <!--ro, opt, bool-->true
    </isSupportRecordStorage>
    <isSupportImportTemplate>
      <!--ro, opt, bool-->true
    </isSupportImportTemplate>
    <isSupportExportTemplate>
      <!--ro, opt, bool-->true
    </isSupportExportTemplate>
    <isSupportExportReport>
      <!--ro, opt, bool-->true
    </isSupportExportReport>
    <isSupportPersonalStatisticalData>
      <!--ro, opt, bool-->true
    </isSupportPersonalStatisticalData>
    <isSupportPersonalRecordOverview>
      <!--ro, opt, bool-->true
    </isSupportPersonalRecordOverview>
    <isSupportPersonalAttendanceDetails>
      <!--ro, opt, bool-->true
    </isSupportPersonalAttendanceDetails>
    <isSupportSearchSummarySheet>
      <!--ro, opt, bool-->true
    </isSupportSearchSummarySheet>
    <isSupportSearchAnomalySheet>
      <!--ro, opt, bool-->true
    </isSupportSearchAnomalySheet>
    <isSupportSearchCardSheet>
      <!--ro, opt, bool-->true
    </isSupportSearchCardSheet>
    <isSupportSearchRecordSheet>
      <!--ro, opt, bool-->true
    </isSupportSearchRecordSheet>
    <isSupportGroupStatisticalData>
      <!--ro, opt, bool-->true
    </isSupportGroupStatisticalData>
    <isSupportUserBasicData>
      <!--ro, opt, bool-->true
    </isSupportUserBasicData>
    <isSupportGroupAttendanceDetails>
      <!--ro, opt, bool-->true
    </isSupportGroupAttendanceDetails>
    <isSupportSearchPersonShiftOverview>
      <!--ro, opt, bool-->true
    </isSupportSearchPersonShiftOverview>
    <isSupportEmployeeRecordOverview>
      <!--ro, opt, bool-->true
    </isSupportEmployeeRecordOverview>
    <dataTimeRange>
      <!--ro, opt, int, range:[0,60], step:1, unit:d-->1
    </dataTimeRange>
```

```
  </LocalAttendanceCap>
  <isSupportCardInterceptionRule>
    <!--ro, opt, bool, desc:true (support), this node is not returned (not
support)-->true
  </isSupportCardInterceptionRule>
  <isSupportAuthorizationCodeCfg>
    <!--ro, opt, bool-->true
  </isSupportAuthorizationCodeCfg>
  <isSupportMaintenanceDataExportCap>
    <!--ro, opt, bool-->true
  </isSupportMaintenanceDataExportCap>
  <isSupportPeopleCounting>
    <!--ro, opt, bool-->true
  </isSupportPeopleCounting>
  <isSupportClearPeopleCounting>
    <!--ro, opt, bool-->true
  </isSupportClearPeopleCounting>
  <isSupportGateTime>
    <!--ro, opt, bool-->true
  </isSupportGateTime>
  <isSupportVerificationCustomAudioFile>
    <!--ro, opt, bool-->true
  </isSupportVerificationCustomAudioFile>
  <isSupportIRMode>
    <!--ro, opt, bool-->true
  </isSupportIRMode>
  <isSupportIRInductionOpen>
    <!--ro, opt, bool-->true
  </isSupportIRInductionOpen>
  <isSupportIRAntiPinch>
    <!--ro, opt, bool-->true
  </isSupportIRAntiPinch>
  <isSupportMaintenanceDataProgress>
    <!--ro, opt, bool-->true
  </isSupportMaintenanceDataProgress>
  <isSupportUserDataProgress>
    <!--ro, opt, bool-->true
  </isSupportUserDataProgress>
  <isSupportRemoteControlPermission>
    <!--ro, opt, bool-->true
  </isSupportRemoteControlPermission>
  <isSupportChannelLampEffect>
    <!--ro, opt, bool-->true
  </isSupportChannelLampEffect>
  <isSupportAddVisitorPicture>
    <!--ro, opt, bool-->true
  </isSupportAddVisitorPicture>
  <isSupportPersonnelChannelPicture>
    <!--ro, opt, bool-->true
  </isSupportPersonnelChannelPicture>
  <isSupportCustomPrompt>
    <!--ro, opt, bool-->true
  </isSupportCustomPrompt>
  <isSupportCardRecognitionReader>
    <!--ro, opt, bool-->true
  </isSupportCardRecognitionReader>
  <isSupportCardInfoCollection>
    <!--ro, opt, bool-->true
```

```
  </isSupportCardInfoCollection>
  <isSupportReleaseCardInfoCollectionResult>
    <!--ro, opt, bool-->true
  </isSupportReleaseCardInfoCollectionResult>
  <isSupportHealthInfoResult>
    <!--ro, opt, bool-->true
  </isSupportHealthInfoResult>
  <isSupportESDParamCfg>
    <!--ro, opt, bool-->true
  </isSupportESDParamCfg>
  <isSupportAlarmIOLinkageCfg>
    <!--ro, opt, bool-->true
  </isSupportAlarmIOLinkageCfg>
  <isSupportUserAndRightFreeze>
    <!--ro, opt, bool-->true
  </isSupportUserAndRightFreeze>
  <isSupportThermopileTemperatureMeasureCalibration>
    <!--ro, opt, bool-->true
  </isSupportThermopileTemperatureMeasureCalibration>
  <isSupportThermopileTOFCalibration>
    <!--ro, opt, bool-->true
  </isSupportThermopileTOFCalibration>
  <isSupportClearUserAuthenticationTimes>
    <!--ro, opt, bool-->true
  </isSupportClearUserAuthenticationTimes>
  <personnelChannelNum>
    <!--ro, opt, int, step:1-->true
  </personnelChannelNum>
  <isSupportSearchModelFailRecord>
    <!--ro, opt, bool-->1
  </isSupportSearchModelFailRecord>
</AccessControl>
```

13.3 Door Status Control

13.3.1 Remotely control the door or elevator

URL

PUT /ISAPI/AccessControl/RemoteControl/door/<doorID>

Query Parameter

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RemoteControlDoor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <cmd>
    <!--req, enum, command, subType:string, desc:"open" (open the door), "close"
(close the door (controlled)), "alwaysOpen" (remain unlocked (free)),
"alwaysClose" (remain open (disabled)), "visitorCallLadder" (call elevator
(visitor)), "householdCallLadder" (call elevator (resident))-->open
  </cmd>
  <password>
    <!--opt, string, range:[8,16]-->test
```

```
  </password>
  <employeeNo>
    <!--wo, opt, string, employee No., range:[0,32], desc:employee No.-->test
  </employeeNo>
  <channelNo>
    <!--opt, int, range:[0,256], step:1-->1
  </channelNo>
  <controlType>
    <!--opt, enum, subType:string-->monitor
  </controlType>
</RemoteControlDoor>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string,
protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device
Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK"
(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML
Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes
the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

13.3.2 Get the capability set of remote door status control

URL

GET /ISAPI/AccessControl/RemoteControl/door/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<RemoteControlDoor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, remote door status control, attr:version{req, float,
protocolVersion}-->
  <doorNo min="1" max="10">
    <!--ro, opt, int, range of the door No., attr:min{req, int},max{req, int}--
>1
  </doorNo>
  <cmd
opt="open,close,alwaysOpen,alwaysClose,visitorCallLadder,householdCallLadder,res
ume">
    <!--ro, req, enum, command, subType:string, attr:opt{req, string},
desc:"open" (open the door), "close" (close the door (controlled)), "alwaysOpen"
(remain unlocked (free)), "alwaysClose" (remain locked (disabled)),
"visitorCallLadder" (call elevator (visitor)), "householdCallLadder" (call
elevator (resident))-->open
  </cmd>
  <password min="8" max="16">
    <!--ro, opt, string, door opening password, range:[8,16], attr:min{req, int,
range:[8,16]},max{req, int, range:[8,16]}-->test
  </password>
  <employeeNo min="0" max="32">
    <!--ro, opt, string, employee No., range:[0,32], attr:min{req, int, range:
[0,32]},max{req, int, range:[0,32]}-->test
  </employeeNo>
  <channelNo min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </channelNo>
  <controlType opt="monitor,calling">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->monitor
  </controlType>
</RemoteControlDoor>
```

13.4 Access Point Status Schedule

13.4.1 Get the configuration parameters of the door control holiday schedule

URL

GET /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "DoorStatusHolidayPlanCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
        "beginDate":  "2017-10-01",
        /*ro, req, date, start date of the holiday*/
```

```
        "endDate":  "2017-10-08",
        /*ro, req, date, end date of the holiday*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
                "doorStatus":  "remainClosed",
                /*ro, req, enum, door status, subType:string, desc:“remainOpen”-
remain open (access without authentication), “remainClosed”-remain closed
(access is not allowed), “normal”-access by authentication, "sleep", "invalid",
“induction”, “barrierFree”*/
                "TimeSegment": {
                /*ro, opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time of the time period, desc:device
local time*/
                }
            }
        ]
    }
}
```

13.4.2 Get the configuration capability of the door control holiday schedule

URL

GET /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusHolidayPlanCfg": {
    /*ro, opt, object*/
        "planNo": {
        /*ro, opt, object, holiday schedule No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable: "true"-enable,"false"-disable,
desc:"true" (enable), "false" (disable)*/
```

```
        "beginDate":  "1970-01-01",
        /*ro, opt, date, start date of the holiday*/
        "endDate":  "1971-01-01",
        /*ro, opt, date, end date of the holiday*/
        "HolidayPlanCfg": {
        /*ro, opt, object*/
            "maxSize":  8,
            /*ro, opt, int*/
            "id": {
            /*ro, opt, object, time period No.*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  8
                /*ro, opt, int*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable: "true"-enable,"false"-disable,
desc:"true" (enable), "false" (disable)*/
            "doorStatus": {
            /*ro, opt, object, door status*/
                "@opt":
 "remainOpen,remainClosed,normal,sleep,invlid,induction,barrierFree"
                /*ro, opt, string, desc:"remainOpen" (remain open (access
without authentication)), "remainClosed" (remain closed (access is not
allowed)), "normal" (access by authentication), "sleep", "invalid"*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time of the time period (device local
time), desc:device local time*/
                "endTime":  "00:00:00",
                /*ro, opt, time, end time of the time period (device local
time), desc:device local time*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour",
"minute", "second"; if this node is not returned, the default time accuracy is
"minute"*/
            }
        }
    }
}
```

13.4.3 Get the configuration capability of the door control schedule template

URL

GET /ISAPI/AccessControl/DoorStatusPlanTemplate/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusPlanTemplate": {
    /*ro, opt, object, schedule template*/
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value of schedule template No.*/
            "@max":  16
            /*ro, opt, int, the maximum value of schedule template No.*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (enable), false
(disable)*/
        "templateName": {
        /*ro, opt, object, template name length*/
            "@min":  1,
            /*ro, opt, int, the minimum value of template name length*/
            "@max":  32
            /*ro, opt, int, the maximum value of template name length*/
        },
        "weekPlanNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value of weekly schedule No.*/
            "@max":  16
            /*ro, opt, int, the maximum value of weekly schedule No.*/
        },
        "holidayGroupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value of holiday group No.*/
            "@max":  16
            /*ro, opt, int, the maximum value of holiday group No.*/
        }
    }
}
```

13.4.4 Set parameters of door control weekly schedule

URL

PUT /ISAPI/AccessControl/DoorStatusWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Request Message

```
{
    "DoorStatusWeekPlanCfg": {
    /*opt, object, weekly schedule of door control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "WeekPlanCfg": [
        /*req, array, weekly schedule parameters, subType:object*/
            {
                "week":  "Monday",
```

```
                /*req, enum, days of the week, subType:string, desc:"Monday",
"Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*req, bool, whether to enable*/
                "doorStatus":  "remainClosed",
                /*req, enum, door control schedule, subType:string,
desc:"remainOpen"-remain open (access without authentication), "remainClosed"-
remain closed (access is not allowed), "normal"-access by authentication,
"sleep", "invalid"*/
                "TimeSegment": {
                /*req, object, time*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*req, time, end time, desc:device local time*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.4.5 Get the configuration parameters of the door control week schedule

URL

GET /ISAPI/AccessControl/DoorStatusWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "DoorStatusWeekPlanCfg": {
    /*ro, opt, object, door control week schedule*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
        "WeekPlanCfg": [
        /*ro, req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*ro, req, enum, days of the week, subType:string,
desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday",
"Sunday"*/
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable: "true"-enable, "false"-
disable*/
                "doorStatus":  "remainClosed",
                /*ro, req, enum, door status, subType:string, desc:"remainOpen"-
remain open (access without authentication), "remainClosed"-remain closed
(access is not allowed), "normal"-access by authentication, "sleep","invalid",
“induction”, “barrierFree”*/
                "TimeSegment": {
                /*ro, req, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time of the time period, desc:device
local time*/
                }
            }
        ]
    }
}
```

13.4.6 Get the configuration capability of the door control week schedule

URL

GET /ISAPI/AccessControl/DoorStatusWeekPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusWeekPlanCfg": {
    /*ro, opt, object*/
        "planNo": {
        /*ro, opt, object, week schedule No.*/
            "@min":  1,
```

```
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable: "true"-enable,"false"-disable,
desc:"true" (enable), "false" (disable)*/
        "WeekPlanCfg": {
        /*ro, opt, object, week schedule parameters*/
            "maxSize":  56,
            /*ro, opt, int*/
            "week": {
            /*ro, opt, object*/
                "@opt":
 "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"
                /*ro, opt, string, desc:"Monday", "Tuesday", "Wednesday",
"Thursday", "Friday", "Saturday", "Sunday"*/
            },
            "id": {
            /*ro, opt, object, weekly schedule No.*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  8
                /*ro, opt, int*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable: "true"-enable,"false"-disable,
desc:"true" (enable), "false" (disable)*/
            "doorStatus": {
            /*ro, opt, object, door status*/
                "@opt":
 "remainOpen,remainClosed,normal,sleep,invalid,induction,barrierFree"
                /*ro, opt, string, desc:"remainOpen" (remain open (access
without authentication)), "remainClosed" (remain closed (access is not
allowed)), "normal" (access by authentication), "sleep", "invalid"*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time of the time period (device local
time), desc:device local time*/
                "endTime":  "10:00:00",
                /*ro, opt, time, end time of the time period (device local
time), desc:device local time*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour",
"minute", "second"; if this node is not returned, the default time accuracy is
"minute"*/
            }
        }
    }
}
```

13.4.7 Get the configuration capability of the door control schedule

URL

GET /ISAPI/AccessControl/DoorStatusPlan/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusPlan": {
    /*ro, opt, object*/
        "doorNo": {
        /*ro, opt, object, door No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  4
            /*ro, opt, int*/
        },
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        }
    }
}
```

13.4.8 Set holiday group parameters of door control schedule

URL

PUT /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Request Message

```
{
    "DoorStatusHolidayGroupCfg": {
    /*opt, object*/
        "enable":  true,
        /*req, bool, whether to enable, desc:whether to enable*/
        "groupName":  "test",
        /*req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*opt, string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.4.9 Get the holiday group configuration parameters of the door control schedule

URL

GET /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "DoorStatusHolidayGroupCfg": {
    /*ro, opt, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable: "true"-enable, "false"-disable*/
        "groupName":  "test",
        /*ro, req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*ro, req, string*/
    }
}
```

13.4.10 Get parameters of door control schedule template

URL

GET /ISAPI/AccessControl/DoorStatusPlanTemplate/<planTemplateID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "DoorStatusPlanTemplate": {
    /*ro, opt, object, door control schedule template*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-
disable*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, weekly schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*ro, req, string, holiday group No.*/
    }
}
```

13.4.11 Set parameters of door control holiday schedule

URL

PUT /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Request Message

```
{
    "DoorStatusHolidayPlanCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-
disable*/
        "beginDate":  "2017-10-01",
        /*ro, req, date, start date of the holiday*/
        "endDate":  "2017-10-08",
        /*ro, req, date, end data of the holiday*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-
disable*/
                "doorStatus":  "remainClosed",
                /*ro, req, enum, door status, subType:string, desc:"remainOpen"-
remain open (access without authentication), "remainClosed"-remain closed
(access is not allowed), "normal"-access by authentication, "sleep", "invalid"*/
                "TimeSegment": {
                /*ro, opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time, desc:device local time*/
                }
            }
        ]
    }
```

```
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.4.12 Set parameters of door control schedule template

URL

PUT /ISAPI/AccessControl/DoorStatusPlanTemplate/<planTemplateID>?format=json

Query Parameter

Request Message

```
{
    "DoorStatusPlanTemplate": {
    /*ro, opt, object, door control schedule template*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-
disable*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, weekly schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*ro, req, string, holiday group No., desc:holiday group No.*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.4.13 Get the configuration capability of door status parameters of holiday group

URL

GET /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusHolidayGroupCfg": {
    /*ro, opt, object*/
        "groupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (enable)*/
        "groupName": {
        /*ro, opt, object, length of holiday group name*/
            "@min":  1,
            /*ro, opt, int, the minimum length*/
            "@max":  32
            /*ro, opt, int, the maximum length*/
        },
        "holidayPlanNo": {
        /*ro, opt, object, holiday group plan No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
```

```
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

13.4.14 Set parameters of door control schedule

URL

PUT /ISAPI/AccessControl/DoorStatusPlan/<doorID>?format=json

Query Parameter

Request Message

```
{
    "DoorStatusPlan": {
    /*req, object*/
        "templateNo":  1
        /*req, int, schedule template No., desc:0-cancel linking the template
with the schedule and restore to the default status (normal status)*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.4.15 Get the configuration parameters of the door control schedule

URL

GET /ISAPI/AccessControl/DoorStatusPlan/<doorID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "DoorStatusPlan": {
    /*ro, req, object*/
        "templateNo":  1
        /*ro, req, int, schedule template No., desc:0-cancel linking the
template with the schedule and restore to the default status (normal status)*/
    }
}
```

13.4.16 Clear access control schedule configuration parameters

URL

PUT /ISAPI/AccessControl/ClearPlansCfg?format=json

Query Parameter

None

Request Message

```
{
    "ClearPlansCfg": {
    /*opt, object*/
        "ClearFlags": {
        /*opt, object*/
            "doorStatusWeekPlan":  true,
            /*opt, bool, whether to clear the week schedule of the door
control*/
            "cardReaderWeekPlan":  true,
            /*opt, bool, whether to clear the week schedule of the card reader
authentication mode control*/
            "userRightWeekPlan":  true,
            /*opt, bool, whether to clear the week schedule of the access
permission control*/
            "doorStatusHolidayPlan":  true,
            /*opt, bool, whether to clear the holiday schedule of the door
control*/
            "cardReaderHolidayPlan":  true,
            /*opt, bool, whether to clear the holiday schedule of the card
reader authentication mode control*/
            "userRightHolidayPlan":  true,
            /*opt, bool, whether to clear the holiday schedule of the access
permission control*/
            "doorStatusHolidayGroup":  true,
            /*opt, bool, whether to clear the holiday group of the door
control*/
            "cardReaderHolidayGroup":  true,
            /*opt, bool, whether to clear the holiday group of the card reader
authentication mode control*/
            "userRightHolidayGroup":  true,
            /*opt, bool, whether to clear the holiday group of the access
permission control*/
            "doorStatusTemplate":  true,
            /*opt, bool, whether to clear the schedule template of the door
control*/
            "cardReaderTemplate":  true,
```

```
            /*opt, bool, whether to clear the control schedule template of card
reader authentication mode*/
            "userRightTemplate":  true
            /*opt, bool, whether to clear the schedule template of access
permission control*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded);
it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when
the value of statusCode is not 1*/
}
```

13.4.17 Get the capability of clearing access control schedule configuration.

URL

GET /ISAPI/AccessControl/ClearPlansCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ClearPlansCfg": {
    /*ro, req, object*/
        "ClearFlags": {
        /*ro, opt, object*/
            "doorStatusWeekPlan":  "true,false",
            /*ro, opt, string, whether to clear the week schedule of the door
control*/
            "cardReaderWeekPlan":  "true,false",
            /*ro, opt, string, whether to clear the week schedule of the card
reader authentication mode control*/
            "userRightWeekPlan":  "true,false",
```

```
            /*ro, opt, string, whether to clear the week schedule of the access
permission control*/
            "doorStatusHolidayPlan":  "true,false",
            /*ro, opt, string, whether to clear the holiday schedule of the door
control*/
            "cardReaderHolidayPlan":  "true,false",
            /*ro, opt, string, whether to clear the holiday schedule of the card
reader authentication mode control*/
            "userRightHolidayPlan":  "true,false",
            /*ro, opt, string, whether to clear the holiday schedule of the
access permission control*/
            "doorStatusHolidayGroup":  "true,false",
            /*ro, opt, string, whether to clear the holiday group of the door
control*/
            "cardReaderHolidayGroup":  "true,false",
            /*ro, opt, string, whether to clear the holiday group of the card
reader authentication mode control*/
            "userRightHolidayGroup":  "true,false",
            /*ro, opt, string, whether to clear the holiday group of the access
permission control*/
            "doorStatusTemplate":  "true,false",
            /*ro, opt, string, whether to clear the schedule template of the
door control*/
            "cardReaderTemplate":  "true,false",
            /*ro, opt, string, whether to clear the control schedule template of
the card reader authentication mode*/
            "userRightTemplate":  "true,false"
            /*ro, opt, string, whether to clear the schedule template of the
access permission control*/
        }
    }
}
```

13.5 Person and Credential Management

13.5.1 Add cards and link them with a person

URL

POST /ISAPI/AccessControl/CardInfo/Record?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "CardInfo": {
    /*req, object, card information*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID)*/
        "cardNo":  "1234567890",
        /*req, string, card No.*/
        "cardType":  "normalCard",
        /*opt, enum, card type, subType:string, desc:card type*/
        "leaderCard":  "1,3,5",
        /*opt, string, whether the access control points support first
fingerprint authentication function, desc:whether the access control points
support first fingerprint authentication function*/
```

```
        "checkCardNo":  true,
        /*opt, bool, whether to enable duplicated card verification,
desc:whether to enable duplicated card verification*/
        "checkEmployeeNo":  true,
        /*opt, bool, whether to check the existence of the employee No. (person
ID), desc:whether to check the existence of the employee No. (person ID)*/
        "operateType":  "byTerminal",
        /*opt, string, operation mode*/
        "terminalNoList": [1]
        /*opt, array, terminal ID list, subType:int, desc:terminal ID list*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.5.2 Get the total number of the added cards

URL

GET /ISAPI/AccessControl/CardInfo/Count?format=json&employeeNo=<employeeNo>&security=<security>&iv=<iv>

Query Parameter

Request Message

None

Response Message

```
{
    "CardInfoCount": {
    /*ro, req, object, number of cards*/
        "cardNumber":  100
        /*ro, opt, int, number of cards*/
    }
}
```

13.5.3 whether to delete card information

URL

PUT /ISAPI/AccessControl/CardInfo/Delete?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "CardInfoDelCond": {
    /*req, object*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object*/
            {
                "employeeNo":  "test"
                /*opt, string, employee No. (person ID)*/
            }
        ],
        "CardNoList": [
        /*opt, array, card No. list, subType:object, desc:(this node cannot
exist together with the EmployeeNoList, and if this node does not exist or is
set to NULL, it indicates deleting all cards)*/
            {
                "cardNo":  "1234567890"
                /*opt, string, card No.*/
            }
        ],
        "operateType":  "byTerminal",
        /*opt, enum, operation type, subType:string, desc:"byTerminal"-by
terminal*/
        "terminalNoList": [1]
        /*opt, array, terminal ID list, this node is required when operation
type is "byTerminal"; currently, only one terminal is supported, subType:int*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.5.4 Edit card information

URL

PUT /ISAPI/AccessControl/CardInfo/Modify?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "CardInfo": {
    /*req, object, card information*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID)*/
        "cardNo":  "1234567890",
        /*req, string, card No.*/
        "cardType":  "normalCard",
        /*opt, enum, card type, subType:string, desc:card type*/
        "leaderCard":  "1,3,5",
        /*opt, string, whether to support first card authentication function,
desc:e.g., the value "1, 3, 5" indicates that the access control points No.1,
No.3, and No.5 support first card authentication function*/
        "operateType":  "byTerminal",
        /*opt, string, operation type*/
        "terminalNoList": [1]
        /*opt, array, terminal ID list, subType:int, desc:this node is required
when operation type is "byTerminal"; currently, only one terminal is supported*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.5.5 Start deleting the fingerprint data

URL

PUT /ISAPI/AccessControl/FingerPrint/Delete?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "FingerPrintDelete": {
    /*req, object*/
        "mode":  "byEmployeeNo",
        /*req, enum, deleting mode, subType:string, desc:"byEmployeeNo" (delete
by employee No. (person ID)), "byCardReader" (delete by fingerprint module)*/
        "EmployeeNoDetail": {
        /*opt, object, delete by employee No. (person ID), dep:and,
{$.FingerPrintDelete.mode,eq,byEmployeeNo}, desc:this node is valid when mode is
"byEmployeeNo"*/
            "employeeNo":  "test",
            /*opt, string, employee No. (person ID) linked with the
fingerprint*/
            "enableCardReader": [1, 3, 5],
            /*opt, array, fingerprint module whose fingerprints should be
deleted, subType:int*/
            "fingerPrintID": [1, 3, 5]
            /*opt, array, No. of fingerprint to be deleted, subType:int*/
        },
        "CardReaderDetail": {
        /*opt, object, delete by fingerprint module, dep:and,
{$.FingerPrintDelete.mode,eq,byCardReader}, desc:this node is valid when mode is
"byCardReader"*/
            "cardReaderNo":  1,
            /*opt, int, fingerprint module No.*/
            "clearAllCard":  false,
            /*opt, bool, whether to delete the fingerprint information of all
cards, desc:"false" (no (delete by employee No.)), "true" (yes (delete the
fingerprint information of all employee No.))*/
            "employeeNo":  "test"
            /*opt, string, employee No. (person ID) linked with the
fingerprint*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.5.6 Delete person information

URL

PUT /ISAPI/AccessControl/UserInfo/Delete?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "UserInfoDelCond": {
    /*req, object, deleting conditions*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object*/
            {
                "employeeNo":  "test"
                /*req, string, employee No. (person ID), range:[1,32]*/
            }
        ],
        "operateType":  "byTerminal",
        /*opt, string, operation type*/
        "terminalNoList": [1]
        /*opt, array, terminal ID list, subType:int, desc:terminal ID list*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when
statusCode is not 1*/
}
```

13.5.7 Get the total number of fingerprints of a specific person

URL

GET /ISAPI/AccessControl/FingerPrint/Count?format=json&employeeNo=
<employeeNo>&security=<security>&iv=<iv>

Query Parameter

Request Message

None

Response Message

```
{
    "FingerPrintCountList": [
    /*ro, opt, array, subType:object*/
        {
            "cardReaderNo":  1,
            /*ro, opt, int, card reader No.*/
            "numberOfFP":  3,
            /*ro, req, int, number of fingerprints*/
            "fingerPrintIDs": []
            /*ro, opt, array, finger number, which is between 1 and 10,
subType:int*/
        }
    ]
}
```

13.5.8 Get the capability of deleting fingerprint data

URL

GET /ISAPI/AccessControl/FingerPrint/Delete/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FingerPrintDelete": {
    /*ro, opt, object*/
        "mode": {
        /*ro, req, object, mode*/
            "@opt":  "byEmployeeNo,byCardReader"
            /*ro, opt, string, deleting mode, desc:"byEmployeeNo"-delete by
employee No. (person ID), "byCardReader"-delete by fingerprint module*/
        },
        "EmployeeNoDetail": {
        /*ro, opt, object, delete by employee No. (person ID); this node is
valid when mode is "byEmployeeNo"*/
            "employeeNo": {
            /*ro, opt, object, employee No. (person ID) linked with the
fingerprint*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            },
            "enableCardReader": {
            /*ro, opt, object, fingerprint module whose fingerprints should be
deleted*/
                "@min":  1,
```

```
                /*ro, opt, int, the minimum value*/
                "@max":  512
                /*ro, opt, int, the maximum value*/
            },
            "fingerPrintID": {
            /*ro, opt, object, No. of fingerprint to be deleted*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  10
                /*ro, opt, int, the maximum value*/
            }
        },
        "CardReaderDetail": {
        /*ro, opt, object, delete by fingerprint module; this node is valid when
mode is "byCardReader"*/
            "cardReaderNo": {
            /*ro, opt, object, fingerprint module No.*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  512
                /*ro, opt, int, the maximum value*/
            },
            "clearAllCard":  "true,false",
            /*ro, opt, string, whether to delete the fingerprint information of
all cards: "false"-no (delete by employee No.), "true"-yes (delete the
fingerprint information of all employee No.)*/
            "employeeNo": {
            /*ro, opt, object, employee No. (person ID) linked with the
fingerprint*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            }
        }
    }
}
```

13.5.9 Set the fingerprint parameters

URL

POST /ISAPI/AccessControl/FingerPrint/SetUp?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "FingerPrintCfg": {
    /*opt, object, fingerprint parameters*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID) linked with the fingerprint,
range:[1,32]*/
        "enableCardReader": [1, 3, 5],
        /*req, array, fingerprint modules to apply fingerprint data to,
subType:int, desc:[1,3,5] indicates applying fingerprint data to fingerprint
modules No.1, No.3, and No.5*/
```

```
        "fingerPrintID":  1,
        /*req, int, fingerprint No., range:[1,10]*/
        "deleteFingerPrint":  true,
        /*opt, bool, whether to delete the fingerprint, desc:"true" (yes). This
field is required only when the fingerprint needs to be deleted; for adding or
editing fingerprint information, it can be empty*/
        "fingerType":  "normalFP",
        /*req, enum, fingerprint type, subType:string, desc:"normalFP" (normal
fingerprint), "hijackFP" (duress fingerprint), "patrolFP" (patrol fingerprint),
"superFP" (super fingerprint),"dismissingFP" (dismiss fingerprint)*/
        "fingerData":  "test",
        /*opt, string, fingerprint data, desc:it is encoded by Base64*/
        "leaderFP": [1, 3, 5],
        /*opt, array, whether the access control points support first
fingerprint authentication function, subType:int, range:[1,5], desc:[1,3,5]
indicates that access control points No.1, No.3, and No.5 support first
fingerprint authentication function*/
        "checkEmployeeNo":  true
        /*opt, bool, whether to check the existence of the employee No. (person
ID)*/
    }
}
```

Response Message

```
{
    "FingerPrintStatus": {
    /*ro, opt, object, fingerprint status*/
        "status":  "success",
        /*ro, opt, enum, status, subType:string, desc:"success", "failed". For
applying fingerprint data to the fingerprint module, this field will not be
returned. This field will be returned only when editing fingerprint parameters
or deleting fingerprints*/
        "StatusList": [
        /*ro, opt, array, status list, subType:object, desc:This field will be
returned only when applying fingerprint data to the fingerprint module. For
editing fingerprint parameters or deleting fingerprints, this filed will not be
returned*/
            {
                "id":  1,
                /*ro, opt, int, fingerprint module No.*/
                "cardReaderRecvStatus":  1,
                /*ro, opt, enum, fingerprint module status, subType:int, desc:0
(connecting failed), 1 (connected), 2 (the fingerprint module is offline), 3
(the fingerprint quality is poor, try again), 4 (the memory is full), 5 (the
fingerprint already exists), 6 (the fingerprint ID already exists), 7 (invalid
fingerprint ID), 8 (this fingerprint module is already configured), 10 (the
fingerprint module version is too old to support the employee No.)*/
                "errorMsg":  "test"
                /*ro, opt, string, error information, range:[1,32], desc:error
information*/
            }
        ]
    }
}
```

13.5.10 Edit fingerprint parameters

URL

POST /ISAPI/AccessControl/FingerPrintModify?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "FingerPrintModify": {
    /*opt, object*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID) linked with the fingerprint,
range:[1,32]*/
        "cardReaderNo":  1,
        /*req, int, fingerprint module No.*/
        "fingerPrintID":  1,
        /*req, int, fingerprint No., range:[1,10]*/
        "fingerType":  "normalFP",
        /*req, enum, fingerprint type, subType:string, desc:"normalFP" (normal
fingerprint), "hijackFP" (duress fingerprint), "patrolFP" (patrol fingerprint),
"superFP" (super fingerprint),"dismissingFP" (dismiss fingerprint)*/
        "leaderFP": [1, 3, 5]
        /*opt, array, whether the access control points support first
fingerprint authentication function, subType:int, desc:e.g., [1,3,5] indicates
that access control points No.1, No.3, and No.5 support first fingerprint
authentication function. If this node is not configured, the first fingerprint
authentication function will remain unchanged*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.5.11 Search the fingerprint information

URL

POST /ISAPI/AccessControl/FingerPrintUpload?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "FingerPrintCond": {
    /*opt, object, search condition*/
        "searchID":  "test",
        /*req, string, search ID, range:[1,32], desc:it is used to check whether
the current search requester is the same as the previous one. If they are the
same, the search record will be stored in the device to speed up the next
search*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID) linked with the fingerprint,
range:[1,32]*/
        "cardReaderNo":  1,
        /*opt, int, card reader No.*/
        "fingerPrintID":  1
        /*opt, int, finger ID, range:[1,10]*/
    }
}
```

Response Message

```
{
    "FingerPrintInfo": {
    /*ro, opt, object, fingerprint information*/
        "searchID":  "test",
        /*ro, req, string, search ID*/
        "status":  "OK",
        /*ro, req, enum, status, subType:string, desc:"OK" (the fingerprint
exists), "NoFP" (the fingerprint does not exist)*/
        "FingerPrintList": [
        /*ro, opt, array, subType:object*/
            {
                "cardReaderNo":  1,
                /*ro, req, int, card reader No., range:[1,10]*/
                "fingerPrintID":  1,
                /*ro, req, int, fingerprint No.*/
                "fingerType":  "normalFP",
                /*ro, req, enum, fingerprint type, subType:string,
desc:fingerprint type: "normalFP"-normal fingerprint,"hijackFP"-duress
fingerprint,"patrolFP"-patrol fingerprint,"superFP"-super
fingerprint,"dismissingFP"-dismiss fingerprint*/
                "fingerData":  "test",
                /*ro, req, string, fingerprint data encoded by Base64*/
                "leaderFP": [1, 3, 5]
                /*ro, opt, array, whether the access control points support
first fingerprint authentication function, subType:int, desc:e.g., [1,3,5]
indicates that access control points No.1, No.3, and No.5 support first
fingerprint authentication function*/
            }
        ]
    }
}
```

13.5.12 Search for cards

URL

POST /ISAPI/AccessControl/CardInfo/Search?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "CardInfoSearchCond": {
    /*req, object*/
        "searchID":  "test",
        /*req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list, desc:the start position of the search result in the result list. When there are multiple records and you cannot get all search results at a time,you can search for the records after the specified position next time. For example,if the maximum total number of matched results (totalMatches) supported by the device is M and the total number of matched results (totalMatches) stored in the device currently is N (here N is smaller than M),the valid range of this field is from 0 to N-1*/
        "maxResults":  30,
        /*req, int, the maximum number of search results this time by calling this URI, desc:if maxResults exceeds the range returned by the device capability, the device will return the maximum number of search results according to the device capability and will not return error message*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object*/
            {
                "employeeNo":  "test"
                /*opt, string, employee No. (person ID)*/
            }
        ],
        "CardNoList": [
        /*opt, array, card No. list, subType:object, desc:this node cannot exist together with EmployeeNoList*/
            {
                "cardNo":  "1234567890"
                /*opt, string, card No.*/
            }
        ]
    }
}
```

Response Message

```
{
    "CardInfoSearch": {
    /*ro, opt, object*/
        "searchID":  "test",
        /*ro, req, string, unique ID of search records*/
        "responseStatusStrg":  "OK",
```

```
        /*ro, req, enum, searching status description, subType:string, desc:"OK"
(searching completed), "MORE" (searching for more data), "NO MATCH" (no matched
data)*/
        "numOfMatches":  1,
        /*ro, req, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "CardInfo": [
        /*ro, opt, array, card information, subType:object*/
            {
                "employeeNo":  "test",
                /*ro, req, string, employee No. (person ID)*/
                "cardNo":  "test",
                /*ro, req, string, card No.*/
                "cardType":  "normalCard",
                /*ro, req, enum, card type, desc:card type: "normalCard"-normal
card,"patrolCard"-patrol card,"hijackCard"-duress card,"superCard"-super
card,"dismissingCard"-dismiss card,"emergencyCard"-emergency card (it is used to
assign permission to a temporary card,but it cannot open the door)*/
                "leaderCard":  "1,3,5"
                /*ro, opt, string, whether to support first card authentication
function, desc:the value "1, 3, 5" indicates that the access control points
No.1, No.3, and No.5 support first card authentication function*/
            }
        ]
    }
}
```

13.5.13 Local Person Change Event Confirmation

URL

PUT /ISAPI/AccessControl/UserAndRight/LocalUserChangeAcknowledgment?format=json

Query Parameter

None

Request Message

```
{
    "serialNo":  1
    /*req, int, Event Serial No., range:[1,150000], desc:Start from 1 and add 1
to the max. supported device.*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*opt, int, Status Code, desc:When 1 is not available, (one means succeeded
and no special status). It must be returned.*/
    "statusString":  "OK",
    /*opt, string, Status Description, range:[1,64], desc:When OK is not
available (OK means success and no special status). It must be returned.*/
    "subStatusCode":  "ok",
    /*opt, string, Child Status Code, range:[1,64], desc:It cannot be expressed
as OK. (OK means success and no special status.) It must be returned.*/
    "errorCode":  1,
    /*opt, int, Error Code, desc:When statusCode is not 1, it corresponds to
subStatusCode.*/
    "errorMsg":  "ok"
    /*opt, string, Error Information, desc:When statusCode is not 1, it must be
returned. The interpretation information is in protocol limit, and the device
can be optimized and enhanced in subsequent version iterations (without
restriction).*/
}
```

13.5.14 Get local person change event confirmation parameter capability.

URL

GET /ISAPI/AccessControl/UserAndRight/LocalUserChangeAcknowledgment/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "LocalUserChangeAcknowledgmentCap": {
    /*opt, object, Local Person Change Event Confirmation Capability,
desc:Corresponding Function: PUT
/ISAPI/AccessControl/UserAndRight/LocalUserChangeAcknowledgment ?format=json
Input Parameter Capability*/
        "serialNo": {
        /*req, object, Event Stream No. Range, desc:Start from 1 and add 1 to
the max. supported device.*/
            "@def":  1,
            /*opt, int, Default Value*/
            "@min":  1,
            /*opt, int, Min. Value*/
            "@max":  150000
            /*opt, int, Max. Value*/
        }
    }
}
```

13.5.15 Get the progress of applying fingerprint data

URL

GET /ISAPI/AccessControl/FingerPrintProgress?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FingerPrintStatus": {
    /*ro, opt, object*/
        "StatusList": [
        /*ro, opt, array, status list, subType:object*/
            {
                "id": 1,
                /*ro, opt, int, fingerprint module No.*/
                "cardReaderRecvStatus": 1,
                /*ro, opt, enum, fingerprint module status, subType:int, desc:0
(connecting failed), 1 (connected), 2 (the fingerprint module is offline), 3
(the fingerprint quality is poor, try again), 4 (the memory is full), 5 (the
fingerprint already exists), 6 (the fingerprint ID already exists), 7 (invalid
fingerprint ID), 8 (this fingerprint module is already configured), 10 (the
fingerprint module version is too old to support the employee No.)*/
                "errorMsg": "test"
                /*ro, opt, string, error information, range:[1,32], desc:error
information*/
            }
        ],
        "totalStatus": 1
        /*ro, req, enum, applying status, subType:int, desc:0 (applying), 1
(applied)*/
    }
}
```

13.5.16 Set card information

URL

PUT /ISAPI/AccessControl/CardInfo/SetUp?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "CardInfo": {
    /*opt, object, card information*/
        "employeeNo": "test",
        /*req, string, employee No. (person ID)*/
        "cardNo": "1234567890",
        /*req, string, card No.*/
        "deleteCard": true,
```

```
        /*opt, bool, whether to delete the card, desc:true (yes). This node is
required only when the card needs to be deleted; for adding or editing card
information, this node can be set to NULL*/
        "cardType":  "normalCard",
        /*req, enum, card type, subType:string, desc:normalCard (normal card),
patrolCard (patrol card), hijackCard (duress card), superCard (super card),
dismissingCard (dismiss card), emergencyCard (emergency card: it is used to
assign permission to a temporary card, but it cannot open the door)*/
        "leaderCard":  "1,3,5",
        /*opt, string, whether to support first card authentication function,
desc:e.g., the value "1, 3, 5" indicates that the access control points No.1,
No.3, and No.5 support first card authentication function*/
        "checkCardNo":  true,
        /*opt, bool, whether to enable duplicated card verification, desc:false
(disable), true (enable). If this node is not configured, the device will verify
the duplicated card by default. When there is no card information, you can set
checkCardNo to "false" to speed up data applying; otherwise, it is not
recommended to configure this node*/
        "checkEmployeeNo":  true
        /*opt, bool, whether to check the existence of the employee No. (person
ID), desc:"false"-no, "true"-yes. If this node is not configured, the device
will check the existence of the employee No. (person ID) by default. This node
is used for speeding up applying card information (by setting the value to
"false"). It is not recommended to configure this node*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.5.17 Get Local Person Change Record Parameters

URL

GET /ISAPI/AccessControl/UserAndRight/LocalUserChangeRecordParam?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "enabled":  "true",
    /*req, boolean, Enable, desc:By default, false means that the device local person (including certificate) needs to be recorded when it is configured as true.*/
    "uploadChannelType":  "ISAPIListen",
    /*opt, enum, Local Person Change Event Upload Channel Type, subType:string, dep:and,{enabled,eq,true}, desc:When enabling local person sync function, the device will upload local person sync event LocalUserChange to the channel configured in this field only. If it is the ISAPI listening channel, the message will be directly removed from the 200OK message returned from the platform (server). ; ISAPIListen: ISAPI listening channel; ISAPI: ISAPI client arming channel; ISUP: ISUP (formerly EHome) channel; EZVIZ: EZVIZ channel; PrivateSDK:Private SDK Client Arming Channel*/
    "pwQueryEnabled":  "true",
    /*opt, boolean, Available Password Search Enable, dep:and,{enabled,eq,true}, desc:true-enable, false-not enable*/
    "pwUploadChannelType":  "ISAPIListen"
    /*opt, enum, Available Password Event Upload Channel Type, subType:string, dep:and,{pwQueryEnabled,eq,true}, desc:; ISAPIListen: ISAPI listening channel; ISAPI: ISAPI client arming channel; ISUP: ISUP (formerly EHome) channel; EZVIZ: EZVIZ channel; PrivateSDK:Private SDK Client Arming Channel*/
}
```

13.5.18 Set Local Person Change Record Parameters

URL

PUT /ISAPI/AccessControl/UserAndRight/LocalUserChangeRecordParam?format=json

Query Parameter

None

Request Message

```
{
    "enabled":  "true",
    /*req, boolean, Enable, desc:By default, false means that the device local
person (including certificate) needs to be recorded when it is configured as
true.*/
    "uploadChannelType":  "ISAPIListen",
    /*opt, enum, Local Person Change Event Upload Channel Type, subType:string,
dep:and,{enabled,eq,true}, desc:When enabling local person sync function, the
device will upload local person sync event LocalUserChange to the channel
configured in this field only. If it is the ISAPI listening channel, the message
will be directly removed from the 200OK message returned from the platform
(server). ; ISAPIListen: ISAPI listening channel; ISAPI: ISAPI client arming
channel; ISUP: ISUP (formerly EHome) channel; EZVIZ: EZVIZ channel;
PrivateSDK:Private SDK Client Arming Channel*/
    "pwQueryEnabled":  "true",
    /*opt, boolean, Available Password Search Enable, dep:and,{enabled,eq,true},
desc:true-enable, false-not enable*/
    "pwUploadChannelType":  "ISAPIListen"
    /*opt, enum, Available Password Event Upload Channel Type, subType:string,
dep:and,{pwQueryEnabled,eq,true}, desc:; ISAPIListen: ISAPI listening channel;
ISAPI: ISAPI client arming channel; ISUP: ISUP (formerly EHome) channel; EZVIZ:
EZVIZ channel; PrivateSDK:Private SDK Client Arming Channel*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*opt, int, Status Code, desc:When 1 is not available, (one means succeeded
and no special status). It must be returned.*/
    "statusString":  "OK",
    /*opt, string, Status Description, range:[1,64], desc:When OK is not
available (OK means success and no special status). It must be returned.*/
    "subStatusCode":  "ok",
    /*opt, string, Child Status Code, range:[1,64], desc:It cannot be expressed
as OK. (OK means success and no special status.) It must be returned.*/
    "errorCode":  1,
    /*opt, int, Error Code, desc:When statusCode is not 1, it corresponds to
subStatusCode.*/
    "errorMsg":  "ok"
    /*opt, string, Error Information, desc:When statusCode is not 1, it must be
returned. The interpretation information is in protocol limit, and the device
can be optimized and enhanced in subsequent version iterations (without
restriction).*/
}
```

13.5.19 Obtained local person change record parameter capability.

URL

GET /ISAPI/AccessControl/UserAndRight/LocalUserChangeRecordParam/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "LocalUserChangeRecordParamCap": {
    /*opt, object, Obtained local person change record parameter capability.,
desc:Corresponding Function:
GET/PUT/ISAPI/AccessControl/UserAndRight/LocalUserChangeRecordParam ?
format=JSON*/
        "enabled": {
        /*req, object, Enable Range, desc:By default, false means that the
device local person (including certificate) needs to be recorded when it is
configured as true.*/
            "@opt": [],
            /*opt, array, Enable Enumeration List, subType:boolean, range:
[1,2]*/
            "@def":  "true"
            /*opt, boolean, Default Value*/
        },
        "uploadChannelType": {
        /*opt, object, Local person change event upload channel type range.,
desc:When enabling local person sync function, the device will upload local
person sync event LocalUserChange to the channel configured in this field only.
If it is the ISAPI listening channel, the message will be directly removed from
the 200OK message returned by the platform (server).*/
            "@opt": [],
            /*opt, array, Local Person Change Event Upload Channel Type
Enumeration List, subType:string, range:[1,5]*/
            "@def":  "ISAPIListen"
            /*opt, string, Default Value*/
        },
        "pwQueryEnabled": {
        /*opt, object, Available Password Search Enable Range, desc:true-enable,
false-disable*/
            "@opt": [],
            /*opt, array, Enumeration List of Available Passwords,
subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, Default Value*/
        },
        "pwUploadChannelType": {
        /*opt, object, Available Password Event Upload Channel Type Range*/
            "@opt": [],
            /*opt, array, Available Password Event Upload Channel Type
Enumeration List, subType:string, range:[1,5]*/
            "@def":  "ISAPIListen"
            /*opt, string, Default Value*/
        }
    }
}
```

13.5.20 Get the card management capability

URL

GET /ISAPI/AccessControl/CardInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardInfo": {
    /*ro, opt, object, card information*/
        "supportFunction": {
        /*ro, req, object, supported functions, desc:"post"-add, "delete",
"put"-edit, "get"-search, "setUp"-set*/
            "@opt":  "post, delete, put, get, setUp"
            /*ro, req, string*/
        },
        "CardInfoSearchCond": {
        /*ro, opt, object, search condition*/
            "searchID": {
            /*ro, opt, object, search ID*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  36
                /*ro, req, int, the maximum value*/
            },
            "maxResults": {
            /*ro, req, object, the maximum number of search results this time by
calling this URI*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  30
                /*ro, req, int, the maximum value*/
            },
            "EmployeeNoList": {
            /*ro, opt, object, person ID list*/
                "maxSize":  56,
                /*ro, req, int, the maximum value*/
                "employeeNo": {
                /*ro, opt, object, employee No. (person ID)*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  32
                    /*ro, req, int, the maximum value*/
                }
            },
            "CardNoList": {
            /*ro, opt, object, card No. list*/
                "maxSize":  56,
                /*ro, req, int, the maximum value*/
                "cardNo": {
                /*ro, opt, object, card No.*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  32
                    /*ro, req, int, the maximum value*/
```

```
                }
            }
        },
        "CardInfoDelCond": {
        /*ro, opt, object, deleting condition*/
            "EmployeeNoList": {
            /*ro, opt, object, person ID list*/
                "maxSize":  56,
                /*ro, req, int, the maximum value*/
                "employeeNo": {
                /*ro, opt, object, employee No. (person ID)*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  32
                    /*ro, req, int, the maximum value*/
                }
            },
            "CardNoList": {
            /*ro, opt, object, card No. list*/
                "maxSize":  56,
                /*ro, req, int, the maximum value*/
                "cardNo": {
                /*ro, opt, object, card No.*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  32
                    /*ro, req, int, the maximum value*/
                }
            }
        },
        "cardNo": {
        /*ro, req, object, card No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  32
            /*ro, req, int, the maximum value*/
        },
        "employeeNo": {
        /*ro, req, object, employee No. (person ID)*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  32
            /*ro, req, int, the maximum value*/
        },
        "cardType": {
        /*ro, req, object, card type*/
            "@opt":  "normalCard, patrolCard, hijackCard, superCard,
dismissingCard, emergencyCard"
            /*ro, req, string*/
        },
        "leaderCard": {
        /*ro, opt, object, whether it supports first card authentication
function*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  32
            /*ro, req, int, the maximum value*/
        },
```

```
        "checkCardNo":  "true,false",
        /*ro, opt, string, whether to enable duplicated card verification,
desc:"false"-disable, "true"-enable. If this node is not configured, the device
will verify the duplicated card by default this node is used for speeding up
applying card information (by setting the value to "false"). It is not
recommended to configure this node*/
        "checkEmployeeNo":  "true,false",
        /*ro, opt, string, whether to check the existence of the employee No.
(person ID), desc:"false"-no, "true"-yes. If this node is not configured, the
device will check the existence of the employee No. (person ID) by default. This
node is used for speeding up applying card information (by setting the value to
"false"). It is not recommended to configure this node*/
        "maxRecordNum":  10000,
        /*ro, opt, int, maximum number of records (cards) allowed,
desc:supported maximum number of records (card records)*/
        "numberPerPerson":  50
        /*ro, opt, enum, subType:int*/
    }
}
```

13.5.21 Get the configuration capability of fingerprint parameters

URL

GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FingerPrintCfg": {
    /*ro, opt, object*/
        "employeeNo": {
        /*ro, req, object, employee No. (person ID) linked with the
fingerprint*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "enableCardReader": {
        /*ro, req, object, fingerprint module to apply fingerprint data to*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  512
            /*ro, opt, int, the maximum value*/
        },
        "fingerPrintID": {
        /*ro, req, object, fingerprint No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  10
```

```
            /*ro, opt, int, the maximum value*/
        },
        "fingerType": {
        /*ro, req, object, fingerprint type*/
            "@opt":  "normalFP"
            /*ro, opt, enum, fingerprint type, subType:string, desc:"normalFP"
(normal fingerprint), "hijackFP" (duress fingerprint), "patrolFP" (patrol
fingerprint), "superFP" (super fingerprint), "dismissingFP" (dismiss
fingerprint)*/
        },
        "leaderFP": {
        /*ro, opt, object, whether to support first time authentication
function*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "checkEmployeeNo":  "true,false",
        /*ro, opt, string, whether to judge the existence of the employee No.
(person ID)*/
        "StatusList": {
        /*ro, opt, object, status list*/
            "id": {
            /*ro, opt, object, fingerprint module No.*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  512
                /*ro, opt, int, the maximum value*/
            },
            "cardReaderRecvStatus": {
            /*ro, opt, object, fingerprint module status*/
                "@opt":  "0,1,2,3,4,5,6,7,8,10"
                /*ro, opt, string, fingerprint module status*/
            },
            "errorMsg": {
            /*ro, opt, object, error information*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            }
        },
        "totalStatus": {
        /*ro, req, object, applying status*/
            "@opt":  "0,1"
            /*ro, opt, string, 0-applying, 1-applied*/
        },
        "isSupportFingerCover":  true,
        /*ro, opt, bool, whether to support overwriting the original fingerprint
when applying new fingerprint linked with the same person ID or employee No. If
it is supported, this node will be set to "true"; otherwise, this node will not
be returned*/
        "isSupportSetUp":  true,
        /*ro, opt, bool, whether to support setting fingerprint parameters. If
it is supported, this node will be set to "true"; otherwise, this node will not
be returned*/
        "maxRecordNum":  1,
```

```
        /*ro, opt, int, range:[0,10000]*/
        "isSupportCount":  true
        /*ro, opt, bool*/
    }
}
```

13.5.22 Get the total number of the added persons

URL

GET /ISAPI/AccessControl/UserInfo/Count?format=json&type=<type>

Query Parameter

Request Message

None

Response Message

```
{
    "UserInfoCount": {
    /*ro, req, object*/
        "userNumber":  100,
        /*ro, req, int, number of persons*/
        "bindFaceUserNumber":  100,
        /*ro, opt, int*/
        "bindFingerprintUserNumber":  100,
        /*ro, opt, int*/
        "bindCardUserNumber":  100,
        /*ro, opt, int*/
        "bindIrisUserNumber":  100
        /*ro, opt, int*/
    }
}
```

13.5.23 Set person information

URL

PUT /ISAPI/AccessControl/UserInfo/SetUp?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "UserInfo": {
    /*req, object, person information*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID)*/
        "deleteUser":  true,
        /*opt, bool, whether to delete the person*/
        "name":  "test",
        /*opt, string, person name*/
        "userType":  "normal",
```

```
        /*req, enum, person type, subType:string, dep:and,
{$.UserInfo.deleteUser,eq,false},{$.UserInfo.deleteUser,eq,不存在},
desc:"normal"-normal person (resident), "visitor"-visitor, "blacklist"-person in
the blocklist. "patient", "maintenance"; the maintenance can enter the room at
any time*/
        "closeDelayEnabled":  true,
        /*opt, bool, whether to enable door close delay, desc:true (yes), false
(no)*/
        "Valid": {
        /*req, object, the parameters of the effective period, dep:and,
{$.UserInfo.deleteUser,eq,false},{$.UserInfo.deleteUser,eq,不存在}, desc:If this
node is set to "false", the effective period is permanent. The effective period
can be a period of time between 1970-01-01 00:00:00 and 2037-12-31 23:59:59*/
            "enable":  true,
            /*req, bool, whether to enable the effective period, desc:true
(yes), false (no)*/
            "beginTime":  "test",
            /*req, string, start date of the effective period, desc:start time
of the effective period (if timeType does not exist or is "local",the beginTime
is the device local time,e.g.,2017-08-01T17:30:08; if timeType is "UTC",the
beginTime is UTC time,e.g.,2017-08-01T17:30:08+08:00)*/
            "endTime":  "test",
            /*req, string, end time of effective period, desc:end time of the
effective period (if timeType does not exist or is "local",the endTime is the
device local time,e.g.,2017-08-01T17:30:08; if timeType is "UTC",the endTime is
UTC time,e.g.,2017-08-01T17:30:08+08:00)*/
            "timeType":  "local"
            /*opt, enum, time type, subType:string, desc:"local" (device local
time), "UTC" (UTC time)*/
        },
        "belongGroup":  "1,3,5",
        /*opt, string, group*/
        "password":  "123456",
        /*opt, string, password*/
        "doorRight":  "1,3",
        /*opt, string, No. of the door or lock that has access permission,
desc:e.g., "1,3" indicates having permission to access door (lock) No. 1 and No.
3*/
        "RightPlan": [
        /*opt, array, door permission schedule (lock permission schedule),
subType:object*/
            {
                "doorNo":  1,
                /*opt, int, door No. (lock ID)*/
                "planTemplateNo":  "1,3,5"
                /*opt, string, schedule template No., desc:schedule template No.
"65535"-7*24h, "65534"-5*24h (Mon. to Fri.), "65533"-2*24h (Sat. to Sun.)*/
            }
        ],
        "maxOpenDoorTime":  0,
        /*opt, int, the maximum authentication attempts*/
        "roomNumber":  123,
        /*opt, int, room No.*/
        "floorNumber":  1,
        /*opt, int, floor No.*/
        "doubleLockRight":  true,
        /*opt, bool, whether to have the permission to open the double-locked
door, desc:true (yes), false (no)*/
```

```
        "localUIRight":  true,
        /*opt, bool, whether to have the permission to access the device local
UI: "true"-yes,"false"-no, desc:whether to have the permission to access the
device local UI: "true"-yes,"false"-no*/
        "userVerifyMode":  "card",
        /*opt, enum, person authentication mode, subType:string,
desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password),
"fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris); the priority of the
person authentication mode is higher than that of the card reader authentication
mode*/
        "checkUser":  true,
        /*opt, bool, whether to verify the duplicated person information,
desc:false (no), true (yes). If checkUser is not configured, the device will
verify the duplicated person information by default. When there is no person
information, you can set checkUser to "false" to speed up data applying;
otherwise, it is not recommended to configure this node*/
        "gender":  "male",
        /*opt, enum, gender of the person in the face picture, subType:string,
desc:"male", "female", "unknown"*/
        "PersonInfoExtends": [
        /*opt, array, extended fields for the additional person information,
subType:object, desc:this node is used to configure the extended person
information displayed on the device's UI. For MinMoe series facial recognition
terminals, currently only one value node can be supported for displaying the
employee No. and the node id is not supported*/
            {
                "id":  1,
                /*opt, int, extended ID of the additional person information,
range:[1,32], desc:It corresponds to the id in the message of the request URI
/ISAPI/AccessControl/personInfoExtendName?format=json and is used to link the
value of the node value and its name (the node name in the message of the
request URI /ISAPI/AccessControl/personInfoExtendName?format=json). If the node
id does not exist, the ID will start from 1 by default according to the array
order*/
```

```
                "value":  "test"
                /*opt, string, extended content of the additional person
information*/
            }
        ],
        "dynamicCode":  "123456",
        /*opt, string, dynamic permission code*/
        "callNumbers": ["1-1-1-401"],
        /*opt, array, room No. list to be called, subType:string, desc:this node
is the extension of roomNumber; when the node roomNumber and callNumber both
exist, the node callNumbers will be parsed first; domestic rule: 1-1-1-floor
number + room number (range of the room number: 1~999), e.g., 1-1-1-101, among
101, 1 is the floor number, 01 is the room number; 1-1-1-22204, among 22204, 222
is the floor number, 04 is the room number. Overseas rule: 1-1-1-room number
(range of the room number: 1~999), e.g., 1-1-1-111, 111 is the room number*/
        "floorNumbers": [1, 2],
        /*opt, array, floor No. list, subType:int, desc:this node is the
extension of floorNumber; when the node floorNumber and callNumber both exist,
the node callNumbers will be parsed first; domestic rule: if the callNumbers is
["1-1-1-101","1-1-1-302"], the floorNumbers is [1,3]; if the callNumber is ["1-
1-1-101","1-1-1-102"], the floorNumbers is [1,1]. Overseas rule: if the
callNumbers is ["1-1-1-1","1-1-1-99"], the number is values of floorNumbers
should be the same as that of the callNumbers. If the floorNumbers is [A, B], it
shows the number 1 room in A floor and the number 99 room in B floor*/
        "groupId":  1,
        /*opt, int, No. of the organization of local time and attendance*/
        "localAtndPlanTemplateId":  1,
        /*opt, int, local attendance schedule template*/
        "groupName":  "test",
        /*opt, string, organization name, range:[1,64]*/
        "ESDType":  "handAndFoot",
        /*opt, enum, ESD type, subType:string, desc:"handAndFoot"-detetct hands
and foot, "no", "hand"-detetc hands, "foot"-detect foot; the default value is
no*/
        "userLevel":  "Employee",
        /*opt, enum, user level, subType:string, desc:"Employee",
"DepartmentManager"-department manager*/
        "userName":  "test",
        /*opt, string, Account, range:[1,32], desc:digits, lower case letters,
upper case letters, and special characters (encrypted)); the node "account" and
"password" should be applied simultaneously*/
        "loginPassword":  "test"
        /*opt, string, password, range:[8,16], desc:including at least 2 types
of the following characters: digits, lower case letters, upper case letters, and
special characters (encrypted)); the node "account" and "password" should be
applied simultaneously*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this node is required when statusCode
is not 1*/
}
```

13.5.24 Search for person information

URL

POST /ISAPI/AccessControl/UserInfo/Search?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "UserInfoSearchCond": {
    /*req, object, search condition*/
        "searchID":  "123",
        /*req, string, search ID, desc:it is used to confirm the upper-level
platform or system. If the platform or the system is the same one during two
searching, the search history will be saved in the memory to speed up next
searching*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list,
desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked
position in the next search*/
        "maxResults":  30,
        /*req, int, the maximum number of search results, desc:if maxResults
exceeds the range returned by the device capability, the device will return the
maximum number of search results according to the device capability and will not
return error message*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object*/
            {
                "employeeNo":  "test"
                /*req, string, employee ID*/
            }
        ],
        "fuzzySearch":  "test",
        /*opt, string, fuzzy search by key words*/
        "groupIdList": [1, 2, 3, 4],
```

```
        /*opt, array, search range of number of the department of local time and
attendance, subType:int*/
        "groupNameList": ["name1", "name2"],
        /*opt, array, search range of name of the department of local time and
attendance, subType:string*/
        "arrangeType":  "personal",
        /*opt, enum, shift schedule type, subType:string, desc:"personal"-person
schedule, "group"-group schedule, "all"-all persons are arranged to shift
schedules, "none"-all persons are not arranged to shift schedules; this node is
used for searching persons' shift schedule type*/
        "userType":  "normal",
        /*opt, enum, person type, subType:string, desc:"normal"-normal person
(resident), "visitor"-visitor, "blacklist"-person in the blocklist. "patient",
"maintenance"; the maintenance can enter the room at any time*/
        "deviceIDList": [1, 2],
        /*opt, array, device No. (bed No.) list, subType:int*/
        "hasFace":  true,
        /*opt, bool, whether the face picture data is added to the device*/
        "hasCard":  true,
        /*opt, bool, whether the card is added to the device*/
        "hasFingerprint":  true,
        /*opt, bool, whether the fingerprint is added to the device*/
        "hasIris":  true,
        /*opt, bool, whether the iris data is added to the device*/
        "userLevel":  "Employee",
        /*opt, enum, user level, subType:string, desc:"Employee",
"DepartmentManager"-department manager*/
        "meetingID":  "test",
        /*opt, string, conference UUID, range:[1,32], desc:the device can filter
the person information in the conference by check-in status for displaying
information of person who checked in and checked late.*/
        "sortByNameFlag":  "A"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "UserInfoSearch": {
    /*ro, req, object, search results*/
        "searchID":  "test",
        /*ro, req, string, search ID*/
        "responseStatusStrg":  "OK",
        /*ro, req, enum, search status, subType:string, desc:"OK" (searching
completed), "NO MATCH" (no matched results), "MORE" (searching for more
results)*/
        "numOfMatches":  1,
        /*ro, req, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "UserInfo": [
        /*ro, opt, array, person information, subType:object*/
            {
                "employeeNo":  "test",
                /*ro, req, string, employee ID (person ID)*/
                "name":  "test",
```

```
                /*ro, opt, string, name*/
                "userType":  "normal",
                /*ro, req, enum, person type, subType:string, desc:"normal"
(normal person (house owner)), "visitor", "blockList" (person in blocklist),
"patient", "maintenance" (maintenance person)*/
                "sortByNamePosition":  1,
                /*ro, opt, int, range:[0,99999]*/
                "sortByNameFlag":  "A",
                /*ro, opt, enum, subType:string*/
                "closeDelayEnabled":  true,
                /*ro, opt, bool, whether to enable door close delay*/
                "Valid": {
                /*ro, opt, object, effective period parameters*/
                    "enable":  true,
                    /*ro, req, bool, whether to enable effective period*/
                    "beginTime":  "test",
                    /*ro, req, string, start time of the effective period,
desc:start time of the effective period (if timeType does not exist or is
"local",the beginTime is the device local time,e.g.,2017-08-01T17:30:08; if
timeType is "UTC",the beginTime is UTC time,e.g.,2017-08-01T17:30:08+08:00)*/
                    "endTime":  "test",
                    /*ro, req, string, end time of the effective period,
desc:end time of the effective period (if timeType does not exist or is
"local",the endTime is the device local time,e.g.,2017-08-01T17:30:08; if
timeType is "UTC",the endTime is UTC time,e.g.,2017-08-01T17:30:08+08:00)*/
                    "timeType":  "test"
                    /*ro, opt, string, time type*/
                },
                "belongGroup":  "1,3,5",
                /*ro, opt, string, group*/
                "password":  "123456",
                /*ro, opt, string, password*/
                "doorRight":  "1,3",
                /*ro, opt, string, No. of door or lock that has access
permission, desc:"1,3" indicates having permission to access door (lock) No. 1
and No. 3*/
                "RightPlan": [
                /*ro, opt, array, access permission schedule of the door or
lock, subType:object*/
                    {
                        "doorNo":  1,
                        /*ro, opt, int, door No. (lock ID)*/
                        "planTemplateNo":  "1,3,5"
                        /*ro, opt, string, schedule template No., desc:schedule
template No.; "65535"-7*24h, "65534"-5*24h (Mon. to Fri.), "65533"-2*24h (Sat.
to Sun.)*/
                    }
                ],
                "maxOpenDoorTime":  0,
                /*ro, opt, int, the maximum number of authentication attempts*/
                "openDoorTime":  0,
                /*ro, opt, int, number of authenticated attempts*/
                "roomNumber":  123,
                /*ro, opt, int, room No.*/
                "floorNumber":  1,
                /*ro, opt, int, floor No.*/
                "doubleLockRight":  true,
```

```
                /*ro, opt, bool, whether to have the permission to open the
double-locked door, desc:whether to have the permission to open the double-
locked door: "true"-yes,"false"-no*/
                "localUIRight":  true,
                /*ro, opt, bool, whether to have the permission to access the
device local UI, desc:whether to have the permission to access the device local
UI: "true"-yes,"false"-no*/
                "userVerifyMode":  "card",
                /*ro, opt, enum, person authentication mode, subType:string,
desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password),
"fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris); the priority of the
person authentication mode is higher than that of the card reader authentication
mode*/
                "dynamicCode":  "123456",
                /*ro, opt, string, dynamic permission code*/
                "callNumbers": ["1-1-1-401"],
                /*ro, opt, array, list of call No., subType:string, desc:this
node is the extension of roomNumber; when the node roomNumber and callNumber
both exist, the node callNumbers will be parsed first; domestic rule: 1-1-1-
floor number + room number (range of the room number: 1~999), e.g., 1-1-1-101,
among 101, 1 is the floor number, 01 is the room number; 1-1-1-22204, among
22204, 222 is the floor number, 04 is the room number. Overseas rule: 1-1-1-room
number (range of the room number: 1~999), e.g., 1-1-1-111, 111 is the room
number*/
                "floorNumbers": [1, 2],
                /*ro, opt, array, list of floor No., subType:int, desc:this node
is the extension of floorNumber; when the node floorNumber and callNumber both
exist, the node callNumbers will be parsed first; domestic rule: if the
callNumbers is ["1-1-1-101","1-1-1-302"], the floorNumbers is [1,3]; if the
callNumber is ["1-1-1-101","1-1-1-102"], the floorNumbers is [1,1]. Overseas
rule: if the callNumbers is ["1-1-1-1","1-1-1-99"], the number is values of
floorNumbers should be the same as that of the callNumbers. If the floorNumbers
is [A, B], it shows the number 1 room in A floor and the number 99 room in B
floor*/
```

```
                "groupId":  1,
                /*ro, opt, int, No. of the department of local time and
attendance*/
                "localAtndPlanTemplateId":  1,
                /*ro, opt, int, local attendance schedule template*/
                "numOfFace":  0,
                /*ro, opt, int, number of linked face pictures, desc:if this
node is not returned, it indicates that device does not support this function*/
                "numOfFP":  0,
                /*ro, opt, int, number of linked fingerprints, desc:if this node
is not returned, it indicates that device does not support this function*/
                "numOfCard":  0,
                /*ro, opt, int, number of linked cards, desc:if this node is not
returned, it indicates that device does not support this function*/
                "numOfIris":  0,
                /*ro, opt, int, number of linked iris pictures, desc:if this
node is not returned, it indicates that device does not support this function*/
                "numOfVP":  0,
                /*ro, opt, int*/
                "gender":  "male",
                /*ro, opt, enum, gender of the person in the face picture,
subType:string, desc:"male", "female", "unknown"*/
                "PersonInfoExtends": [
                /*ro, opt, array, person extension information, subType:object,
desc:this node is used to configure the extended person information displayed on
the device's UI. For MinMoe series facial recognition terminals, currently only
one value node can be supported for displaying the employee No. and the node id
is not supported*/
                    {
                        "id":  1,
                        /*ro, opt, int, serial No. of person extension
information, range:[1,32], desc:it corresponds to the id in the message of the
request URI /ISAPI/AccessControl/personInfoExtendName?format=json and is used to
link the value of the node value and its name. If the node id does not exist,
the ID will start from 1 by default according to the array order*/
                        "value":  "test"
                        /*ro, opt, string, extended content of the additional
person information*/
                    }
                ],
                "groupName":  "test",
                /*ro, opt, string, organization name, range:[1,64]*/
                "age":  0,
                /*ro, opt, int, age, range:[0,120]*/
                "PatientInfos": {
                /*ro, opt, object, patient information, desc:the node is
required and valid when the value of userType is patient*/
                    "deviceID":  "test",
                    /*ro, opt, string, device No.*/
                    "admissionTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, admission time*/
                    "chargeNurse":  "test",
                    /*ro, opt, string, primary nurse, range:[0,32]*/
                    "chargeDoctor":  "test",
                    /*ro, opt, string, primary doctor, range:[0,32]*/
                    "nursingLevel":  "tertiary",
```

```
                    /*ro, opt, enum, nursing level, subType:string,
desc:"unknow"-unknown, "tertiary"- tertiary nursing, "secondary"-secondary
nursing, "primary"-primary nursing, "special"-special nursing*/
                    "doctorsAdvice":  "test",
                    /*ro, opt, string, doctor's advise, range:[0,128]*/
                    "allergicHistory":  "test"
                    /*ro, opt, string, allergy, range:[0,128]*/
                },
                "TromboneRule": {
                /*ro, opt, object, trombone rule*/
                    "industryType":  "builidings",
                    /*ro, opt, enum, scene, subType:string, desc:the node
indicates the applied scene of the device linked to a person; the trombone rule
is in HI#KD#V10#001001001001003 format; HI is the scene, KD is device type, V10
is the version of private SIP; "builidings", "prison", "medicalTreatment"-
medical treatment, "broadcasting"*/
                    "unitType":  "indoor",
                    /*ro, opt, enum, device type, subType:string, desc:the node
indicates the type of the device linked to a person; the device trombone rule is
in HI#KD#V10#001001001001003 format; HI is the scene, KD is device type, V10 is
the version of private SIP (VOIP); "indoor"-indoor station, "villa"-villa door
station, "confirm"-doorphone, "outdoor"-door station, "fence" (outer door
station, "doorbell"-doorbell, "manage"-master station, "acs"-access control
device, "wardStation"-visit extension, "bedheadExtension"-beadhead extension,
"bedsideExtension"-bedside extension, "terminal"-terminal, "netAudio"-network
sound box, "interactive"-interaction terminal, "amplifier"*/
                    "SIPVersion":  "V10"
                    /*ro, opt, string, private SIP version, range:[0,32],
desc:the node indicates the SIP version of the device linked to a person, the
device trombone rule is in HI#KD#V10#001001001001003 format; HI is the scene, KD
is device type, V10 is the private SIP version;*/
                },
                "ESDType":  "handAndFoot",
                /*ro, opt, enum, ESD type, subType:string, desc:"handAndFoot"-
detetct hands and foot, "no", "hand"-detetc hands, "foot"-detect foot; the
default value is no*/
                "userLevel":  "Employee",
                /*ro, opt, enum, user level, subType:string, desc:"Employee",
"DepartmentManager"-department manager*/
                "userName":  "test",
                /*ro, opt, string, account, range:[1,32], desc:digits, lower
case letters, upper case letters, and special characters (encrypted)*/
                "faceURL":  "test",
                /*ro, opt, string, face picture URL, desc:when 80 port transmit
via 8000 port, the local URL of NVR or DVR is returned; ISUP and EZ transmit via
URL of NVR or DVR*/
                "LocalAttendance": {
                /*ro, opt, object, local time and attendance information*/
                    "templateName":  "test"
                    /*ro, opt, string, template / shift schedule name, desc:the
shift schedule of the person*/
                },
                "meetingSignInStatus":  "normal",
                /*ro, opt, enum, conference check-in status, subType:string,
desc:"normal"-checked-in, "late", "notSignIn"-not checked-in*/
                "deviceID":  "test"
                /*ro, opt, string*/
            }
```

```
        ]
    }
}
```

13.5.25 Add a person

URL

POST /ISAPI/AccessControl/UserInfo/Record?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "UserInfo": {
    /*req, object, person information*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID)*/
        "name":  "test",
        /*opt, string, person name*/
        "userType":  "normal",
        /*req, enum, person type, subType:string, desc:"normal" (normal person (household)), "visitor", "blackList" (person in blocklist), "maintenance" (maintenance persons including cleaners, repairmen, etc. )*/
        "closeDelayEnabled":  true,
        /*opt, bool, whether to enable door close delay*/
        "Valid": {
        /*req, object, parameters of the effective period,the effective period can be a period of time between 1970-01-01 00:00:00 and 2037-12-31 23:59:59*/
            "enable":  true,
            /*req, bool, whether to enable validity period, desc:if the node returns "false", it indicates the function is permanent*/
            "beginTime":  "2017-08-01T17:30:08+08:00",
            /*req, datetime, start time of the effective period, desc:if timeType does not exist or is "local", the beginTime is the device local time, e.g., 2017-08-01T17:30:08; if timeType is "UTC", the beginTime is UTC time, e.g., 2017-08-01T17:30:08+08:00*/
            "endTime":  "2017-08-01T17:30:08+08:00",
            /*req, datetime, end time of the effective period, desc:if timeType does not exist or is "local", the endTime is the device local time, e.g., 2017-08-01T17:30:08; if timeType is "UTC", the endTime is UTC time, e.g., 2017-08-01T17:30:08+08:00*/
            "timeType":  "local"
            /*opt, enum, time type, subType:string, desc:"local" (device local time), "UTC" (UTC time)*/
        },
        "belongGroup":  "1,3,5",
        /*opt, string, group*/
        "password":  "123456",
        /*opt, string, password*/
        "doorRight":  "1,3",
        /*opt, string, door permission*/
        "RightPlan": [
        /*opt, array, door permission schedule (lock permission schedule), subType:object*/
            {
                "doorNo":  1,
```

```
                /*opt, int, door No. (lock ID)*/
                "planTemplateNo":  "1,3,5"
                /*opt, string, schedule template No., desc:schedule template No.
"65535"-7*24h, "65534"-5*24h (Mon. to Fri.), "65533"-2*24h (Sat. to Sun.)*/
            }
        ],
        "maxOpenDoorTime":  0,
        /*opt, int, maximum authentication attempts, desc:0-unlimited*/
        "roomNumber":  123,
        /*opt, int, room No.*/
        "floorNumber":  1,
        /*opt, int, floor No.*/
        "doubleLockRight":  true,
        /*opt, bool, whether to have the permission to open the double-locked
door*/
        "localUIRight":  true,
        /*opt, bool, whether to have the permission to access the device local
UI*/
        "userVerifyMode":  "card",
        /*opt, enum, person authentication mode, subType:string,
desc:f"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password),
"fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris); the person
authentication mode is prior to the card reader authentication mode*/
        "checkUser":  true,
        /*opt, bool, whether to verify the duplicated person information,
desc:"false"-no, "true"-yes. If checkUser is not configured, the device will
verify the duplicated person information by default. When there is no person
information, you can set checkUser to "false" to speed up data applying;
otherwise, it is not recommended to configure this node*/
        "gender":  "male",
        /*opt, enum, gender of the person in the face picture, subType:string,
desc:"male", "female", "unknown"*/
        "PersonInfoExtends": [
```

```
        /*opt, array, extended fields for the additional person information,
subType:object, desc:This node is used to configure the extended person
information displayed on the device's UI. For MinMoe series facial recognition
terminals, currently only one value node can be supported for displaying the
employee No. and the node id is not supported*/
            {
                "id":  1,
                /*opt, int, extended ID of the additional person information,
range:[1,32], desc:it corresponds to the id in the message of the request URI
/ISAPI/AccessControl/personInfoExtendName?format=json and is used to link the
value of the node value and its name (the node name in the message of the
request URI /ISAPI/AccessControl/personInfoExtendName?format=json). If the node
id does not exist, the ID will start from 1 by default according to the array
order*/
                "value":  "test"
                /*opt, string, extended content of the additional person
information*/
            }
        ],
        "operateType":  "byTerminal",
        /*opt, string, operation type, desc:"byTerminal"-by terminal*/
        "terminalNoList": [1],
        /*opt, array, terminal ID list, subType:int, desc:this node is required
when operation type is "byTerminal"; currently, only one terminal is supported*/
        "dynamicCode":  "123456",
        /*opt, string, dynamic permission code*/
        "callNumbers": ["1-1-1-401"],
        /*opt, array, room No. list to be called, subType:string, desc:this node
is the extension of roomNumber; when the node roomNumber and callNumber both
exist, the node callNumbers will be parsed first; domestic rule: 1-1-1-floor
number + room number (range of the room number: 1~999), e.g., 1-1-1-101, among
101, 1 is the floor number, 01 is the room number; 1-1-1-22204, among 22204, 222
is the floor number, 04 is the room number. Overseas rule: 1-1-1-room number
(range of the room number: 1~999), e.g., 1-1-1-111, 111 is the room number*/
        "floorNumbers": [1, 2],
        /*opt, array, floor No. list, subType:int, desc:this node is the
extension of floorNumber; when the node floorNumber and callNumber both exist,
the node callNumbers will be parsed first; domestic rule: if the callNumbers is
["1-1-1-101","1-1-1-302"], the floorNumbers is [1,3]; if the callNumber is ["1-
1-1-101","1-1-1-102"], the floorNumbers is [1,1]. Overseas rule: if the
callNumbers is ["1-1-1-1","1-1-1-99"], the number is values of floorNumbers
should be the same as that of the callNumbers. If the floorNumbers is [A, B], it
shows the number 1 room in A floor and the number 99 room in B floor*/
        "groupId":  1,
        /*opt, int, No. of the organization of local time and attendance*/
        "localAtndPlanTemplateId":  1,
        /*opt, int, local attendance schedule template*/
        "groupName":  "test",
        /*opt, string, organization name, range:[1,64]*/
        "age":  0,
        /*opt, int, age, range:[0,120]*/
        "PatientInfos": {
        /*opt, object, patient information, desc:the node is required and valid
when the value of userType is patient*/
            "deviceID":  "test",
            /*opt, string, device No.*/
            "admissionTime":  "1970-01-01T00:00:00+08:00",
            /*opt, datetime, admission time*/
```

```
            "chargeNurse":  "test",
            /*opt, string, primary nurse, range:[0,32]*/
            "chargeDoctor":  "test",
            /*opt, string, primary doctor, range:[0,32]*/
            "nursingLevel":  "tertiary",
            /*opt, enum, nursing level, subType:string, desc:"unknow"-unknown,
"tertiary"- tertiary nursing, "secondary"-secondary nursing, "primary"-primary
nursing, "special"-special nursing*/
            "doctorsAdvice":  "test",
            /*opt, string, doctor's advise, range:[0,128]*/
            "allergicHistory":  "test"
            /*opt, string, allergy, range:[0,128]*/
        },
        "ESDType":  "handAndFoot",
        /*opt, enum, ESD type, subType:string, desc:"handAndFoot"-detetct hands
and foot, "no", "hand"-detetc hands, "foot"-detect foot; the default value is
no*/
        "userLevel":  "Employee",
        /*opt, enum, user level, subType:string, desc:"Employee",
"DepartmentManager"-department manager*/
        "userName":  "test",
        /*opt, string, account, range:[1,32], desc:digits, lower case letters,
upper case letters, and special characters (encrypted)); the node "account" and
"password" should be applied simultaneously*/
        "loginPassword":  "test",
        /*opt, string, password, range:[8,16], desc:including at least 2 types
of the following characters: digits, lower case letters, upper case letters, and
special characters (encrypted)); the node "account" and "password" should be
applied simultaneously*/
        "deviceID":  "test"
        /*opt, string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.5.26 Edit person information

URL

PUT /ISAPI/AccessControl/UserInfo/Modify?format=json&security=<security>&iv=<iv>

Query Parameter

Request Message

```
{
    "UserInfo": {
    /*req, object, person information*/
        "employeeNo":  "test",
        /*req, string, employee No. (person ID)*/
        "name":  "test",
        /*opt, string, name*/
        "userType":  "normal",
        /*opt, enum, person type, subType:string, desc:"normal"-normal person (resident), "visitor"-visitor, "blacklist"-person in the blocklist. "patient", "maintenance"; the maintenance can enter the room at any time*/
        "closeDelayEnabled":  true,
        /*opt, bool, whether to enable door close delay*/
        "Valid": {
        /*opt, object, parameters of the effective period,the effective period can be a period of time between 1970-01-01 00:00:00 and 2037-12-31 23:59:59*/
            "enable":  true,
            /*opt, bool, whether to enable validity period, desc:if the node returns "false", it indicates the function is permanent*/
            "beginTime":  "2017-08-01T17:30:08+08:00",
            /*opt, datetime, start time of the effective period, desc:if timeType does not exist or is "local", the beginTime is the device local time, e.g., 2017-08-01T17:30:08; if timeType is "UTC", the beginTime is UTC time, e.g., 2017-08-01T17:30:08+08:00*/
            "endTime":  "2017-08-01T17:30:08+08:00",
            /*opt, datetime, end time of the effective period, desc:if timeType does not exist or is "local", the endTime is the device local time, e.g., 2017-08-01T17:30:08; if timeType is "UTC", the endTime is UTC time, e.g., 2017-08-01T17:30:08+08:00*/
            "timeType":  "local"
            /*opt, enum, time type, subType:string, desc:"local" (device local time), "UTC" (UTC time)*/
        },
        "belongGroup":  "1,3,5",
        /*opt, string, group*/
        "password":  "123456",
        /*opt, string, password*/
        "doorRight":  "1,3",
        /*opt, string, door permission*/
        "RightPlan": [
        /*opt, array, door permission schedule (lock permission schedule), subType:object*/
            {
                "doorNo":  1,
                /*opt, int, door No. (lock ID)*/
                "planTemplateNo":  "1,3,5"
                /*opt, string, schedule template No., desc:schedule template No.; "65535"-7*24h, "65534"-5*24h (Mon. to Fri.), "65533"-2*24h (Sat. to Sun.)*/
            }
        ],
        "maxOpenDoorTime":  0,
        /*opt, int, maximum authentication attempts, desc:0-unlimited*/
```

```
        "roomNumber":  123,
        /*opt, int, room No.*/
        "floorNumber":  1,
        /*opt, int, floor No.*/
        "doubleLockRight":  true,
        /*opt, bool, whether to have the permission to open the double-locked
door*/
        "localUIRight":  true,
        /*opt, bool, whether to have the permission to access the device local
UI*/
        "userVerifyMode":  "card",
        /*opt, enum, person authentication mode, subType:string, desc:the person
authentication mode is prior to the card reader authentication mode; "cardAndPw"
(card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
        "checkUser":  true,
        /*opt, bool, whether to verify the duplicated person information,
desc:"false"-no, "true"-yes. If checkUser is not configured, the device will
verify the duplicated person information by default. When there is no person
information, you can set checkUser to "false" to speed up data applying;
otherwise, it is not recommended to configure this node*/
        "gender":  "male",
        /*opt, enum, gender of the person in the face picture, subType:string,
desc:"male", "female", "unknown"*/
        "PersonInfoExtends": [
        /*opt, array, extended fields for the additional person information,
subType:object, desc:this node is used to configure the extended person
information displayed on the device's UI. For MinMoe series facial recognition
terminals, currently only one value node can be supported for displaying the
employee No. and the node id is not supported*/
            {
                "id":  1,
```

```
                /*opt, int, extended ID of the additional person information,
range:[1,32], desc:it corresponds to the id in the message of the request URI
/ISAPI/AccessControl/personInfoExtendName?format=json and is used to link the
value of the node value and its name (the node name in the message of the
request URI /ISAPI/AccessControl/personInfoExtendName?format=json). If the node
id does not exist, the ID will start from 1 by default according to the array
order*/
                "value":  "test"
                /*opt, string, extended content of the additional person
information*/
            }
        ],
        "dynamicCode":  "123456",
        /*opt, string, dynamic permission code*/
        "callNumbers": ["1-1-1-401"],
        /*opt, array, room No. list to be called, subType:string, desc:this node
is the extension of roomNumber; when the node roomNumber and callNumber both
exist, the node callNumbers will be parsed first; domestic rule: 1-1-1-floor
number + room number (range of the room number: 1~999), e.g., 1-1-1-101, among
101, 1 is the floor number, 01 is the room number; 1-1-1-22204, among 22204, 222
is the floor number, 04 is the room number. Overseas rule: 1-1-1-room number
(range of the room number: 1~999), e.g., 1-1-1-111, 111 is the room number*/
        "floorNumbers": [1, 2],
        /*opt, array, floor No. list, subType:int, desc:this node is the
extension of floorNumber; when the node floorNumber and callNumber both exist,
the node callNumbers will be parsed first; domestic rule: if the callNumbers is
["1-1-1-101","1-1-1-302"], the floorNumbers is [1,3]; if the callNumber is ["1-
1-1-101","1-1-1-102"], the floorNumbers is [1,1]. Overseas rule: if the
callNumbers is ["1-1-1-1","1-1-1-99"], the number is values of floorNumbers
should be the same as that of the callNumbers. If the floorNumbers is [A, B], it
shows the number 1 room in A floor and the number 99 room in B floor*/
        "groupId":  1,
        /*opt, int, No. of the organization of local time and attendance*/
        "localAtndPlanTemplateId":  1,
        /*opt, int, local attendance schedule template*/
        "operateType":  "byTerminal",
        /*opt, string, operation type, desc:"byTerminal"-by terminal*/
        "terminalNoList": [1],
        /*opt, array, terminal ID list, subType:int, desc:this node is required
when operation type is "byTerminal"; currently, only one terminal is supported*/
        "groupName":  "test",
        /*opt, string, organization name, range:[1,64]*/
        "age":  0,
        /*opt, int, age, range:[0,120]*/
        "PatientInfos": {
        /*opt, object, patient information, desc:the node is required and valid
when the value of userType is patient*/
            "deviceID":  "test",
            /*opt, string, device No.*/
            "admissionTime":  "1970-01-01T00:00:00+08:00",
            /*opt, datetime, admission time*/
            "chargeNurse":  "test",
            /*opt, string, primary nurse, range:[0,32]*/
            "chargeDoctor":  "test",
            /*opt, string, primary doctor, range:[0,32]*/
            "nursingLevel":  "tertiary",
```

```
            /*opt, enum, nursing level, subType:string, desc:"unknow"-unknown,
"tertiary"- tertiary nursing, "secondary"-secondary nursing, "primary"-primary
nursing, "special"-special nursing*/
            "doctorsAdvice":  "test",
            /*opt, string, doctor's advise, range:[0,128]*/
            "allergicHistory":  "test"
            /*opt, string, allergy, range:[0,128]*/
        },
        "ESDType":  "handAndFoot",
        /*opt, enum, ESD type, subType:string, desc:"handAndFoot"-detetct hands
and foot, "no", "hand"-detetc hands, "foot"-detect foot; the default value is
no*/
        "userLevel":  "Employee",
        /*opt, enum, user level, subType:string, desc:"Employee",
"DepartmentManager"-department manager*/
        "userName":  "test",
        /*opt, string, account, range:[1,32], desc:digits, lower case letters,
upper case letters, and special characters*/
        "loginPassword":  "test",
        /*opt, string, password, range:[8,16], desc:including at least 2 types
of the following characters: digits, lower case letters, upper case letters, and
special characters (encrypted)); the node "account" and "password" should be
applied simultaneously*/
        "deviceID":  "test"
        /*opt, string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the
value of statusCode is not 1*/
}
```

13.5.27 Get the person management capability

URL

GET /ISAPI/AccessControl/UserInfo/capabilities?format=json&terminalNo=<terminalNo>

Query Parameter

Request Message

None

Response Message

```
{
    "UserInfo": {
    /*ro, req, object, person information*/
        "supportFunction": {
        /*ro, req, object, supported functions include adding, deleting,
editing, searching for person information, and getting total number of the added
persons*/
            "@opt":  "post,delete,put,get,setUp"
            /*ro, req, string, operation, desc:"post" (add), "delete", "put"
(edit), "get" (search), "setUp" (set)*/
        },
        "UserInfoSearchCond": {
        /*ro, opt, object, search conditions*/
            "maxResults": {
            /*ro, req, object, the maximum number of search results this time by
calling this URI*/
                "@min":  1,
                /*ro, req, int, the minimum value among elements of the array,
range:[1,100]*/
                "@max":  100
                /*ro, req, int, the maximum value among elements of the array,
range:[1,100]*/
            },
            "EmployeeNoList": {
            /*ro, opt, object, person ID list*/
                "maxSize":  56,
                /*ro, req, int, range of members, range:[1,56]*/
                "employeeNo": {
                /*ro, req, object, employee No. (person ID)*/
                    "@min":  1,
                    /*ro, req, int, the minimum value, range:[1,32]*/
                    "@max":  32
                    /*ro, req, int, the maximum value, range:[1,32]*/
                }
            },
            "fuzzySearch": {
            /*ro, opt, object, fuzzy search by key words*/
                "@min":  1,
                /*ro, req, int, the minimum value, range:[1,32]*/
                "@max":  32
                /*ro, req, int, the maximum value, range:[1,32]*/
            },
            "isSupportNumOfFace":  true,
            /*ro, opt, bool, whether it supports number of linked face pictures
when searching*/
            "isSupportNumOfFP":  true,
            /*ro, opt, bool, whether it supports number of linked fingerprints
when searching*/
            "isSupportNumOfCard":  true,
            /*ro, opt, bool, whether it supports number of linked cards when
searching*/
            "isSupportNumOfIris":  true,
            /*ro, opt, bool*/
```

```
            "isSupportNumOfVP":  true,
            /*ro, opt, bool*/
            "groupIdList": {
            /*ro, opt, object, list of the organization No. of local time and
attendance*/
                "@size":  1,
                /*ro, req, int, value range*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the minimum value*/
            },
            "groupNameList": {
            /*ro, opt, object, list of the organization name of local time and
attendance*/
                "@size":  1,
                /*ro, req, int, value range*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the minimum value*/
            },
            "arrangeType": {
            /*ro, opt, object, shift schedule type*/
                "@opt": ["personal", "group", "all", "none"]
                /*ro, req, array, options, subType:string*/
            },
            "userType": {
            /*ro, opt, object, person type, desc:"normal"-normal person
(resident), "visitor"-visitor, "blacklist"-person in the blocklist. "patient",
"maintenance"-maintenance person, cleaner, etc.;*/
                "@opt": ["normal", "visitor", "blackList", "patient",
"maintenance"]
                /*ro, opt, array, person type, subType:string*/
            },
            "deviceIDList": {
            /*ro, opt, object, number range of device ID*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  512
                /*ro, opt, int, the maximum value*/
            },
            "hasFace": {
            /*ro, opt, object, whether the face picture data is added to the
device*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "hasCard": {
            /*ro, opt, object, whether the card is added to the device*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "hasFingerprint": {
            /*ro, opt, object, whether the fingerprint is added to the device*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
```

```
            "hasIris": {
            /*ro, opt, object, whether the iris data is added to the device*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "userLevel": {
            /*ro, opt, object, user level*/
                "@opt": ["Employee", "DepartmentManager"]
                /*ro, req, array, user level, subType:string*/
            },
            "sortByNameFlag": {
            /*ro, opt, object*/
                "@opt": ["!", "A", "B", "C", "D", "E", "F", "G", "H", "I", "J",
"K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z"]
                /*ro, req, array, subType:string*/
            }
        },
        "UserInfoDelCond": {
        /*ro, opt, object, deleting conditions*/
            "EmployeeNoList": {
            /*ro, opt, object, person ID list*/
                "maxSize":  56,
                /*ro, opt, int, range of members*/
                "employeeNo": {
                /*ro, opt, object, employee No. (person ID)*/
                    "@min":  1,
                    /*ro, opt, int, the minimum value*/
                    "@max":  32
                    /*ro, opt, int, the maximum value*/
                }
            }
        },
        "employeeNo": {
        /*ro, req, object, employee No. (person ID)*/
            "@min":  1,
            /*ro, req, int, the minimum value, range:[1,32]*/
            "@max":  32
            /*ro, req, int, the maximum value, range:[1,32]*/
        },
        "name": {
        /*ro, opt, object, name*/
            "@min":  0,
            /*ro, req, int, the minimum value, range:[0,128]*/
            "@max":  128
            /*ro, req, int, the maximum value, range:[0,128]*/
        },
        "userType": {
        /*ro, req, object, person type*/
            "@opt":  "normal,visitor,blackList,patient,maintenance"
            /*ro, req, string, options*/
        },
        "closeDelayEnabled":  "true,false",
        /*ro, opt, string, whether to enable door close delay*/
        "Valid": {
        /*ro, req, object, parameters of the validity period*/
            "enable":  "true,false",
            /*ro, req, string, whether to enable validity period*/
            "beginTime": {
```

```
            /*ro, req, object, start time of the effective period (if timeType
does not exist or is "local", the beginTime is the device local time, e.g., :
2017-08-01T17:30:08; if timeType is "UTC", the beginTime is UTC time, e.g.,:
2017-08-01T17:30:08+08:00)*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            },
            "endTime": {
            /*ro, req, object, end time of the effective period (if timeType
does not exist or is "local", the endTime is the device local time, e.g.,: 2017-
08-01T17:30:08; if timeType is "UTC", the endTime is UTC time, e.g.,: 2017-08-
01T17:30:08+08:00)*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            },
            "timeRangeBegin":  "1970-01-01T00:00:00",
            /*ro, opt, string, start time that can be configured for beginTime,
desc:if the device does not return this node, the default start time that can be
configured for beginTime is "1970-01-01T00:00:00"*/
            "timeRangeEnd":  "2037-12-31T23:59:59",
            /*ro, opt, string, end time that can be configured for endTime,
desc:if the device does not return this node, the default start time that can be
configured for beginTime is "1970-01-01T00:00:00"*/
            "timeType": {
            /*ro, opt, object, time type*/
                "@opt":  "local,UTC"
                /*ro, opt, string, options*/
            }
        },
        "maxBelongGroup":  4,
        /*ro, opt, int, maximum number of groups that a person can belong to*/
        "belongGroup": {
        /*ro, opt, object, group*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "password": {
        /*ro, opt, object, password*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "doorRight": {
        /*ro, opt, object, No. of door or lock that has access permission*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "RightPlan": {
        /*ro, opt, object, door permission schedule (lock permission schedule)*/
```

```
            "maxSize":  32,
            /*ro, opt, int, range of members in the array*/
            "doorNo": {
            /*ro, opt, object, door No. (lock ID)*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            },
            "maxPlanTemplate":  4,
            /*ro, opt, int, maximum number of schedule templates that can be
configured for one door*/
            "planTemplateNo": {
            /*ro, opt, object, schedule template No.*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32,
                /*ro, opt, int, the maximum value*/
                "@opt": [65535, 65534, 65533]
                /*ro, opt, array, user level, subType:int, desc:"65535"-7*24h,
"65534"-5*24h (Mon. to Fri.), "65533"-2*24h (Sat. to Sun.)*/
            }
        },
        "maxOpenDoorTime": {
        /*ro, opt, object, maximum authentication attempts*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  100
            /*ro, opt, int, the maximum value*/
        },
        "openDoorTime": {
        /*ro, opt, object, authenticated attempts*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  100
            /*ro, opt, int, the maximum value*/
        },
        "roomNumber": {
        /*ro, opt, object, room No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  100
            /*ro, opt, int, the maximum value*/
        },
        "floorNumber": {
        /*ro, opt, object, floor No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  100
            /*ro, opt, int, the maximum value*/
        },
        "doubleLockRight":  "true,false",
        /*ro, opt, string, whether to have the permission to open the double-
locked door*/
        "localUIRight":  "true,false",
        /*ro, opt, string, whether to have the permission to access the device
local UI*/
        "userVerifyMode": {
```

```
        /*ro, opt, object, person authentication mode*/
            "@opt":
 "cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpO
rCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNo
AndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,fac
eOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw,employ
eeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCar
dOrPwOrIris,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,cardOrFpOrIrisOrPw,cardOrIri
sOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw,faceAndIrisAn
dPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw,cardAndVp,faceAndVp,fpAndVp,
pwAndVp"
            /*ro, opt, string, options, desc:"cardAndPw" (card+password), "card"
(card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw"
(fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard"
(fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password),
"faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp"
(face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card),
"face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint
or password), "employeeNoAndFp" (employee No.+fingerprint),
"employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard"
(face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint),
"employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or
face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or
password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or
fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "iris" (iris),
"faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris),
"faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or
face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or
password), "employeeNoAndFaceAndPw" (employee No.+face+password),
"cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or
fingerprint or password), "cardOrFpOrFaceOrIris" (card or fingerpriont or face
or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
        },
        "checkUser":  "true,false",
        /*ro, opt, string, whether to verify the duplicated person information*/
        "maxRecordNum":  0,
        /*ro, opt, int, supported maximum number of records (person records)*/
        "gender": {
        /*ro, opt, object, gender of the person in the face picture*/
            "@opt":  "male,female,unknown"
            /*ro, opt, string, options*/
        },
        "PersonInfoExtends": {
        /*ro, opt, object, extended fields for the additional person
information*/
            "maxSize":  3,
            /*ro, req, int, range of members in the array*/
            "id": {
            /*ro, opt, object, extended ID of the additional person
information*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
```

```
                /*ro, opt, int, the maximum value*/
            },
            "value": {
            /*ro, opt, object, extended content of the additional person
information*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  100
                /*ro, opt, int, the maximum value*/
            }
        },
        "dynamicCode": {
        /*ro, opt, object, dynamic code*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "callNumbers": {
        /*ro, opt, object, room No. list to be called,which is extended from
roomNumber and it is in higher priority; by default,the No. format is X-X-X-
X,e.g.,1-1-1-401,and for standard SIP,it can be the SIP number; this node must
be configured together with roomNumber*/
            "maxSize":  0,
            /*ro, opt, int, range of members in the array*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  100
            /*ro, opt, int, the maximum value*/
        },
        "floorNumbers": {
        /*ro, opt, object, floor No. list*/
            "maxSize":  0,
            /*ro, opt, int, range of members*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  100
            /*ro, opt, int, the maximum value*/
        },
        "groupId": {
        /*ro, opt, object, No. of the organization of local time and
attendance*/
            "@min":  0,
            /*ro, req, int, options*/
            "@max":  0
            /*ro, req, int, options*/
        },
        "localAtndPlanTemplateId": {
        /*ro, opt, object, local attendance schedule template*/
            "@min":  0,
            /*ro, req, int, options*/
            "@max":  0
            /*ro, req, int, options*/
        },
        "purePwdVerifyEnable":  true,
        /*ro, opt, bool, whether the device supports opening the door only by
password, desc:true (yes), this node is not returned (no). The password used to
open the door is the value of the node password in the message JSON_UserInfo*/
```

```
        "groupName": {
        /*ro, opt, object, organization name*/
            "@min":  1,
            /*ro, opt, int, the minimum length, range:[1,256]*/
            "@max":  256
            /*ro, opt, int, the maximum length, range:[1,256]*/
        },
        "age": {
        /*ro, opt, object, value range of age*/
            "@min":  0,
            /*ro, opt, int, the minimum value of age*/
            "@max":  120
            /*ro, opt, int, the maximum value of age*/
        },
        "PatientInfos": {
        /*ro, opt, object, patient information*/
            "admissionTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, admission time, desc:this node indicates the
format of the value*/
            "chargeNurse": {
            /*ro, opt, object, the value range of primary nurse*/
                "@min":  0,
                /*ro, opt, int, the minimum value, unit:Byte*/
                "@max":  64
                /*ro, opt, int, the maximum value, unit:Byte*/
            },
            "chargeDoctor": {
            /*ro, opt, object, the value range of primary doctor*/
                "@min":  0,
                /*ro, opt, int, the minimum value, unit:Byte*/
                "@max":  64
                /*ro, opt, int, the maximum value*/
            },
            "nursingLevel": {
            /*ro, opt, object, the value range and enumeration of nursing
level*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0,
                /*ro, opt, int, the maximum value*/
                "@opt": ["unknow", "tertiary", "secondary", "primary",
"special"]
                /*ro, opt, array, nursing level, subType:string*/
            },
            "doctorsAdvice": {
            /*ro, opt, object, the value range of doctor's advise*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  256
                /*ro, opt, int, the maximum value*/
            },
            "allergicHistory": {
            /*ro, opt, object, the value range of allergy*/
                "@min":  0,
                /*ro, opt, int, the minimum value of allergy*/
                "@max":  128
                /*ro, opt, int, the maximum value of allergy*/
            }
```

```
        },
        "ESDType": {
        /*ro, opt, object, ESD type*/
            "@opt": ["handAndFoot", "no", "hand", "foot"]
            /*ro, req, array, options, subType:string, desc:"handAndFoot"-
detetct hands and foot, "no", "hand"-detetc hands, "foot"-detect foot*/
        },
        "userLevel": {
        /*ro, opt, object, user level*/
            "@opt": ["Employee", "DepartmentManager"]
            /*ro, req, array, options, subType:string*/
        },
        "userName": {
        /*ro, opt, object, account*/
            "@min":  1,
            /*ro, req, int, options*/
            "@max":  32
            /*ro, req, int, options*/
        },
        "loginPassword": {
        /*ro, opt, object, password*/
            "@min":  8,
            /*ro, req, int, options*/
            "@max":  16
            /*ro, req, int, options*/
        },
        "meetingID": {
        /*ro, opt, object, conference UUID*/
            "@min":  1,
            /*ro, req, int, the maximum length*/
            "@max":  32
            /*ro, req, int, the maximum length*/
        },
        "meetingSignInStatus": {
        /*ro, opt, object, conference check-in status*/
            "@opt": ["normal", "late", "notSignIn"]
            /*ro, req, array, options, subType:string*/
        },
        "addressBookSearchResult": {
        /*ro, opt, object*/
            "employeeNo": {
            /*ro, req, object*/
                "@min":  1,
                /*ro, req, int, range:[1,32]*/
                "@max":  32
                /*ro, req, int, range:[1,32]*/
            },
            "name": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, range:[0,128]*/
                "@max":  128
                /*ro, req, int, range:[0,128]*/
            },
            "userType": {
            /*ro, req, object*/
                "@opt":  "normal,visitor,blackList"
                /*ro, req, string*/
```

```
            },
            "numOfFace": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, range:[0,1]*/
                "@max":  1
                /*ro, req, int, range:[0,1]*/
            },
            "numOfCard": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, range:[0,50]*/
                "@max":  50
                /*ro, req, int, range:[0,50]*/
            },
            "numOfFP": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, range:[0,10]*/
                "@max":  10
                /*ro, req, int, range:[0,10]*/
            },
            "numOfIris": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, range:[0,2]*/
                "@max":  2
                /*ro, req, int, range:[0,2]*/
            },
            "sortByNamePosition": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, range:[0,99999]*/
                "@max":  99999
                /*ro, req, int, range:[0,99999]*/
            },
            "sortByNameFlag": {
            /*ro, opt, object*/
                "@opt": ["!", "A", "B", "C", "D", "E", "F", "G", "H", "I", "J",
"K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z"]
                /*ro, req, array, subType:string*/
            }
        }
    }
}
```

13.5.28 Searched local person update event.

URL

POST /ISAPI/AccessControl/UserAndRight/SearchLocalUserChange?format=json&security=<security>&iv=<iv>

Query Parameter

Parameter Name	Parameter Type	Description
security	string	No security means data is not encrypted.Security=1, which means AES128 encryption of employeeNo field in the message.Security=2, which means AES256 encryption of employeeNo field in the message.
iv	string	iv refers to initialization vector. It is required when security=1 or 2.

Request Message

```
{
    "searchID":  "test",
    /*req, string, Search ID, range:[1,32], desc:1. It is generated by platform. It is unique and can only contain letters and digits. 2. SearchID content will be the same when searching condition is not changed. 3. If the searchID of the device is the same as the searchID of the task already executing, the error will be returned.*/
    "searchResultPosition":  1,
    /*req, int, The start position of the search result in the result list., range:[1,200000], desc:Start from 1*/
    "maxResults":  100,
    /*req, int, Max. Number of Results for Current Interaction, range:[1,100]*/
    "timeRange": {
    /*opt, object, Time Information*/
        "startTime":  "1970-01-01T00:00:00.000+08:00",
        /*req, datetime, Start Time*/
        "endTime":  "2022-01-01T00:00:00.000+08:00"
        /*req, datetime, End Time, desc:EndTime should be less than startTime.*/
    },
    "serialNoRange": {
    /*opt, object, Event Serial No. Information*/
        "startSerialNo":  1,
        /*req, int, Start Streaming No., range:[1,150000]*/
        "endSerialNo":  1
        /*req, int, End Streaming No., range:[1,150000]*/
    },
    "employeeNo":  "123"
    /*opt, string, Employee No., range:[1,64], desc:This field is sensitive information and needs to be encrypted (url parameter security = 1 or 2 to encrypt this field value, and security = 3 or 4 to encrypt the whole message).*/
}
```

Response Message

```
{
    "responseStatus":  "OK",
    /*req, enum, Search Status String Description, subType:string, desc:; OK:Search ended. MORE:Data waiting to be searched. NO MOTCH:No matched data. industrial:Exception occurred when searching.*/
```

```
    "numOfMatches":  1,
    /*req, int, Number of returned records., range:[0,100]*/
    "totalMatches":  1,
    /*req, int, Total Number of Qualified Records, range:[0,200000]*/
    "matchResults": [
    /*opt, array, Search Result, subType:object, range:[0,100]*/
        {
            "serialNo":  1,
            /*req, int, Event Serial No., range:[1,150000], desc:Start from 1
and add 1 to the max. supported device.*/
            "frontSerialNo":  1,
            /*opt, int, Previous Event Serial No., range:[1,150000], desc:If the
field is not returned, the platform will judge if the event is lost according to
serialNo. If yes, the platform will judge if the event is lost according to the
field and serialNo field (mainly used to solve the problem of serialNo
discontinuity caused by alarm subscription).*/
            "localUserChangeType":  "add",
            /*req, enum, Local Person Change Type, subType:string, desc:1. Add
refers to adding person. Only person employee No. can be uploaded to the device.
The platform will actively pull the person's full information (person +
certificate). 2. The delete refers to deleting person. The device only needs to
upload person employee No., and the platform has deleted the person's full
information (person + certificate). 3. Modify refers to the person (including
person information, adding certificate information, editing certificate
information, deleting certificate information). See userAndPermissionList for
details. ; add:Added. Modify:Edit; delete:Delete*/
            "employeeNo":  "123",
            /*req, string, Employee No., range:[1,64], desc:This field is
sensitive information and needs to be encrypted (url parameter security = 1 or 2
to encrypt this field value, and security = 3 or 4 to encrypt the whole
message).*/
            "userAndPermissionList": []
            /*opt, array, Changed person and certificate list., subType:string,
range:[1,6], dep:and,{localUserChangeType,eq,modify}, desc:Required when
localUserChangeType is modified.*/
        }
    ]
}
```

13.5.29 Get the ability to search local person to update event parameters.

URL

GET /ISAPI/AccessControl/UserAndRight/SearchLocalUserChange/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "SearchLocalUserChangeCap": {
```

```
    /*opt, object, Searched local person to update event capability.,
desc:Corresponding Function: POST
/ISAPI/AccessControl/UserAndRight/SearchLocalUserChange ?format=JSON &security=
<security>&iv=<iv>Input Parameter Capability*/
        "searchID": {
        /*req, object, Search ID Range, desc:1. It is generated by platform. It
is unique and can only contain letters and digits. 2. If the search condition is
unchanged, the searchID content will be unchanged. 3. If the searchID of the
device is the same as the searchID of the task already in progress, error will
be returned.*/
            "@def":  "test",
            /*opt, string, Default Value*/
            "@min":  1,
            /*opt, int, Min. Length*/
            "@max":  32
            /*opt, int, Max. Length*/
        },
        "searchResultPosition": {
        /*req, object, The search result should be in the value range of the
start position in the result list., desc:Start from 1*/
            "@def":  1,
            /*opt, int, Default Value*/
            "@min":  1,
            /*opt, int, Min. Value*/
            "@max":  200000
            /*opt, int, Max. Value*/
        },
        "maxResults": {
        /*req, object, Max. number of results of this interaction range.*/
            "@def":  100,
            /*opt, int, Default Value*/
            "@min":  1,
            /*opt, int, Min. Value*/
            "@max":  100
            /*opt, int, Max. Value*/
        },
        "timeRange": {
        /*opt, object, Time Information*/
            "startTime":  "1970-01-01T00:00:00.000+08:00",
            /*req, string, Start Time Value Example*/
            "endTime":  "2022-01-01T00:00:00.000+08:00"
            /*req, string, End Time Value Example, desc:EndTime should be
greater than or equal to startTime.*/
        },
        "serialNoRange": {
        /*opt, object, Event Serial No. Information*/
            "startSerialNo": {
            /*req, object, Start Streaming No. Range*/
                "@def":  1,
                /*opt, int, Default Value*/
                "@min":  1,
                /*opt, int, Min. Value*/
                "@max":  150000
                /*opt, int, Max. Value*/
            },
            "endSerialNo": {
            /*req, object, End Streaming No. Range*/
                "@def":  1,
```

```
                /*opt, int, Default Value*/
                "@min":  1,
                /*opt, int, Min. Value*/
                "@max":  150000
                /*opt, int, Max. Value*/
            }
        },
        "employeeNo": {
        /*opt, object, Employee ID Range*/
            "@def":  "123",
            /*opt, string, Default Value*/
            "@min":  1,
            /*opt, int, Min. Length*/
            "@max":  64
            /*opt, int, Max. Length*/
        }
    }
}
```

13.5.30 Get the progress of deleting fingerprint data

URL

GET /ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FingerPrintDeleteProcess": {
    /*ro, req, object*/
        "status":  "processing"
        /*ro, req, enum, deleting status, subType:string, desc:"processing",
"success", "failed"*/
    }
}
```

13.5.31 Get the total number of the added cards

URL

GET /ISAPI/AccessControl/CardInfo/Count?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardInfoCount": {
    /*ro, req, object, number of cards*/
        "cardNumber":  100
        /*ro, req, int, number of cards*/
    }
}
```

13.5.32 Get the total number of all persons' fingerprints

URL

GET /ISAPI/AccessControl/FingerPrint/Count?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FingerPrintCountList": [
    /*ro, req, array, subType:object, range:[0,1]*/
        {
            "numberOfFP":  3
            /*ro, req, int, number of fingerprints, desc:the total number of all persons' fingerprints*/
        }
    ]
}
```

13.5.33 Local Person Change Event

EventType:LocalUserChange

```
{
    "ipAddress":  "172.6.64.7",
    /*req, ipv4, Alarm Device IPv4 Address*/
    "ipv6Address":  "null",
    /*opt, ipv6, Alarm Device IPv6 Address*/
    "portNo":  80,
    /*opt, int, Alarm Device Port No., range:[1,65535]*/
    "protocol":  "HTTP",
    /*opt, string, Protocol Type, range:[0,64], desc:[HTTP #HTTP ,HTTPS#HTTPS,EHome #EHome ,OTAP#OTAP], the field has no practical significance.*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*opt, string, MAC Address, range:[1,32]*/
    "channelID":  1,
```

```
    /*opt, int, Device Channel No. Triggering Alarm, range:[1,65535],
desc:Triggered video channel No. (1. Uploaded video channel No. corresponding to
private protocol when SDK transparently transmitted ISAPI protocol; (2. Uploaded
video channel No. corresponding to Hik-Connect protocol when Hik-Connect
transparently transmitted ISAPI protocol; (3. Uploaded video channel No.
corresponding to ISUP protocol when ISUP transparently transmitted ISAPI
protocol);*/
    "channelName":  "ABC",
    /*opt, string, Channel Name (Camera Name), range:[0,64], desc:<channelID>
<channelName>Consistent with /ISAPI/Streaming/channels/*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*req, datetime, Alarm Triggering Time*/
    "activePostCount":  1,
    /*opt, int, Number of Uploaded Alarms, range:[1,2147483647], desc:Event
Triggering Frequency Pulse Event Definition: Event is triggered continuously (by
device detection frequency). For example: Motion Detection Instantaneous Event
Definition: Differentiate target. One target will be triggered once. Example:
ActivePostCount is used to distinguish between events triggered by the same
triggering source when the pulse event type is triggered. For example: Mobile
detection will be uploaded according to device detection frequency. If the
triggering source changes, the counting will start again. This can be integrated
as event triggering frequency.*/
    "eventType":  "LocalUserChange",
    /*req, string, Event Type, range:[1,64], desc:Local Person Change Event*/
    "eventState":  "active",
    /*req, enum, Event Status, subType:string, desc:Active for persistent event
- indicates valid event (the field is used for start or no process status);
inactive - indicates invalid event (end); remark:Under Heartbeat Type, the field
is assigned (representing heartbeat data, which is uploaded once in 10s);
Active:Valid Event; Inactive:Invalid event.*/
    "eventDescription":  "LocalUserChange",
    /*req, string, Event Description, range:[1,128], desc:Local Person Change
Event*/
    "deviceID":  "test0123",
    /*opt, string, Device ID, range:[1,64], desc:When ISUP protocol is accessed,
deviceID is the ISUP device ID registered by the device. When OTAP protocol is
accessed locally, deviceID is the OTAP device ID registered by the device. When
OTAP Hik-Connect is accessed, deviceID is the 9-digit short serial No. of Hik-
Connect registered by the device.*/
    "deviceUUID":  "12345",
    /*opt, string, Device No., range:[1,128], desc:Unique device ID,
<deviceID>obtained through field in protocol/ISAPI/System/deviceInfo (some
devices can configure deviceID parameters).*/
    "isDataRetransmission":  "true",
    /*opt, boolean, Retransmission Data Mark, desc:Uploading real-time detection
data failed. After restoring device exception, upload the data again.*/
    "UUID":  "079f23cd-0988-459f-96f5-fa1c507dd07c",
    /*opt, string, Upload Event ID, range:[0,64]*/
    "LocalUserChange": {
    /*req, object, Event Content*/
        "serialNo":  1,
        /*req, int, Event Serial No., range:[1,150000], desc:Start from 1 and
add 1 to the max. supported device.*/
        "frontSerialNo":  1,
```

```
        /*opt, int, Previous Event Serial No., range:[1,150000], desc:If the
field is not returned by the device, the platform will judge whether the event
is lost according to serialNo. If yes, the platform will judge whether the event
is lost according to the field and serialNo field (mainly used to solve the
problem of serialNo discontinuity caused by alarm subscription).*/
        "localUserChangeType":  "add",
        /*req, enum, Local Person Change Type, subType:string, desc:1. Add
refers to adding person. The device only needs to upload person employee No.,
and the platform actively pulls the person's full information (person +
certificate). 2. The delete refers to deleting person. The device only needs to
upload person employee No., and the platform has deleted the person's full
information (person + certificate). 3. Modify refers to editing person
(including editing person information, adding certificate information, editing
certificate information, and deleting certificate information). See
userAndPermissionList for details. ; add:Added. Modify:Edit; delete:Delete*/
        "employeeNo":  "123",
        /*req, string, Employee No., range:[1,64]*/
        "userAndPermissionList": [],
        /*opt, array, Changed person and certificate list., subType:string,
range:[1,6], dep:and,{localUserChangeType,eq,modify}, desc:Required when
localUserChangeType is modified.*/
        "UserInfo": {
        /*opt, object, Person Information, desc:Only custom items are supported.
Baseline is not supported.*/
            "employeeNo":  "test",
            /*req, string, Employee No., range:[1,32]*/
            "name":  "test",
            /*opt, string, Name, range:[0,128]*/
            "userType":  "normal",
            /*req, enum, Person Type, subType:string, desc:Maintenance personnel
should be able to enter the room at any time. normal:Normal Person (Owner);
visitor:Visitor (Visitor); blackList:Non-authorized list person;
maintenance:Maintenance personnel (including cleaning personnel, maintenance
personnel, etc.); patient:Patient; custom 1:Custom Person Type 1;
Customer2:Custom Person Type 2; Customer3:Custom person type 3; custom4:Custom
person type 4; custom 5:Custom Person Type 5*/
            "closeDelayEnabled":  "true",
            /*opt, boolean, Close Door Delay*/
            "Valid": {
            /*req, object, Person Validity Period*/
                "enable":  "true",
                /*req, boolean, Enable Validity Period, desc:1. Enable is true,
and the person is not valid for a long time, and the bininTime should be less
than the endTme. 2. Enable is false, and the person is valid for a long time.
The upper layer still needs to apply beninTime and endTime. The device also
needs to verify the validity of the bininTime and endTime.*/
                "beginTime":  "2017-08-01T17:30:08.000+08:00",
                /*req, datetime, Validity Period Start Time, desc:When the
timeType field does not exist or is local, the groupTime is the device local
time. For example: 2017-08-01T17:30:08; When timeType field is UTC, beginTime is
UTC time, e.g.: 2017-08-01T17:30:08+08:00*/
                "endTime":  "2017-08-01T17:30:08.000+08:00",
                /*req, datetime, End Time of Validity, desc:When the timeType
field does not exist or is local, the groupTime is the device local time. For
example: 2017-08-01T17:30:08; When timeType field is UTC, beginTime is UTC time,
e.g.: 2017-08-01T17:30:08+08:00*/
                "timeType":  "local"
```

```
                /*opt, enum, Time Type, subType:string, desc:; local:Device
Local Time; UTC: UTC Time*/
            },
            "belongGroup":  "1,3,5",
            /*opt, string, Group, range:[0,32]*/
            "password":  "123456",
            /*opt, string, Credential Password, range:[4,8], desc:Pure Digital
Composition*/
            "doorRight":  "1,3",
            /*opt, string, Door Permission, range:[0,256]*/
            "RightPlan": [
            /*opt, array, Door Permission Schedule (Lock Permission Schedule),
subType:object, range:[0,64]*/
                {
                    "doorNo":  1,
                    /*opt, int, Door No. (lock ID), range:[1,64]*/
                    "planTemplateNo":  "1,3,5"
                    /*opt, string, Schedule Template No., range:[1,64], desc:The
same door and schedule template is handled by permission or . No schedule
template No. by default. Effective 24 hours from 65535-7*, 24 hours from Monday
to Friday, and 24 hours from Saturday.*/
                }
            ],
            "maxOpenDoorTime":  0,
            /*opt, int, Max. Authentication Times, range:[0,65535], desc:No
limit.*/
            "doubleLockRight":  "true",
            /*opt, boolean, Unlock Door Opening Permission*/
            "localUIRight":  "true",
            /*opt, boolean, Local UI access permission of device.*/
            "userVerifyMode":  "card"
```

```
            /*opt, enum, Person Verification Mode, subType:string, desc:Person
verification mode is higher than card reader verification mode. If it is empty,
person verification mode is invalid. ; cardAndPw:Card + Password; card:Card
Swiping; cardOrPw:Card Swiping or Password; fp:Fingerprint; fpAndPw:Fingerprint
+ Password; fpOrCard:Fingerprint or Card Swiping; fpAndCard:Fingerprint + Card
Swiping; fpAndCardAndPw:Fingerprint + swiping card + password;
faceOrFpOrCardOrPw:Face picture or fingerprint or swiping card or password;
faceAndFp:Face picture + fingerprint; faceAndPw:Face picture + password;
faceAndCard:Face picture + swiping card. Face:Face picture;
employeeNoAndPw:Employee ID + Password; fpOrPw:Fingerprint or password;
employeeNoAndFp:Employee No. + Fingerprint; employeeNoAndFpAndPw:Employee No. +
Fingerprint + Password; faceAndFpAndCard:Face picture + fingerprint + swiping
card; faceAndPwAndFp:Face picture + password + fingerprint;
employeeNoAndFace:Employee No. + Face Picture; faceOrfaceAndCard:Face picture or
face picture + swiping card. fpOrface:Fingerprint or face picture;
cardOrfaceOrPw:Card swiping or face picture or password; cardOrFace:Card swiping
or face picture; cardOrFaceOrFp:Card swiping or face picture or fingerprint;
faceOrPw:Face picture or password; employeeNoAndFaceAndPw:Employee ID + Face
Picture + Password; cardOrFaceOrFaceAndCard:Card swiping or face picture or face
picture + swiping card; Iris:Iris; faceOrFpOrCardOrPwOrIris:Face picture or
fingerprint or swiping card or password or iris; faceOrCardOrPwOrIris:Face
picture or swiping card or password or iris; cardOrFpOrFaceOrIris:Card swiping
or fingerprinting or face picture or iris; fpOrFaceOrIrisOrPw:Fingerprint or
face picture or iris or password; cardOrFpOrIrisOrPw:Card swiping or fingerprint
or iris or password; cardOrIrisOrPw:Card swiping or iris or password;
cardAndIris:Card + Iris; fpAndIris:Fingerprint + iris; faceAndIris:Face picture
+ iris; irisAndPw:Iris + Password; cardAndIrisAndPw:Card + iris + password;
faceAndIrisAndPw:Face picture + iris + password; cardAndFaceAndIris:Card + Face
Picture + Iris; Pw:Password; cardOrFpOrPw:Card swiping or fingerprint or
password; faceOrFpOrPw:Face picture or fingerprint or password; cardAndVp:Card +
Sound; faceAndVp:Face picture + sound; fpAndVp:Fingerprint + Sound;
pwAndVp:Password + Sound; remoteControl:Remote Control;
remoteControlAndFace:Remote Control + Face Picture; remoteControlOrFace:Remote
control or face picture; remoteControlOrCard:Remote Control or Card;
remoteControlOrFp:Remote Control or Fingerprint; remoteControlOrPw:Remote
Control or Password; remoteControlOrCardOrFp:Remote Control or Card or
Fingerprint; remoteControlOrCardOrFace:Remote control or card or face picture;
remoteControlOrCardOrPw:Remote control or card or password;
remoteControlOrFpOrFace:Remote control or fingerprint or face picture;
remoteControlOrFpOrPw:Remote Control or Fingerprint or Password;
remoteControlOrFaceOrPw:Remote control or face picture or password;
remoteControlOrCardOrFpOrFace:Remote control or card or fingerprint or face
picture; remoteControlOrCardOrFpOrPw:Remote control or card or fingerprint or
password; remoteControlOrFpOrFaceOrPw:Remote Control or Fingerprint or Face
Picture or Password; remoteControlOrCardOrFaceOrPw:Remote control or card or
face picture or password; remoteControlOrCardOrFpOrFaceOrPw:Remote Control or
Card or Fingerprint or Face Picture or Password*/
        },
        "FaceInfo": {
        /*opt, object, Face Picture Information, desc:Only custom items are
supported. Baseline is not supported.*/
            "List": [
            /*opt, array, Face Picture Information List, subType:object, range:
[1,2]*/
                {
                    "FDID":  "1",
                    /*opt, string, Face Picture Library ID, range:[1,64]*/
                    "faceID":  1,
```

```
                    /*opt, int, Face Picture ID, range:[1,32]*/
                    "modelData":  "test",
                    /*opt, string, Face Picture Modeling Data, range:[1,512],
desc:Adopt Base64 Encoding*/
                    "faceName":  "test",
                    /*opt, string, Face Picture Name, range:[1,64], desc:Match
to name in the boundary.*/
                    "facePicURL":  "test"
                    /*opt, string, Face Picture URL, range:[1,1024]*/
                }
            ]
        },
        "CardInfo": {
        /*opt, object, Card Information, desc:Only custom items are supported.
Baseline is not supported.*/
            "List": [
            /*opt, array, Card Information List, subType:object, range:[1,32]*/
                {
                    "cardNo":  "test",
                    /*opt, string, Card No., range:[1,32]*/
                    "cardType":  "normalCard"
                    /*opt, enum, Card Type, subType:string, desc:Added card. The
field is not entered. The card type is normal. The field is not entered.
NormalCard:Normal Card; PatrolCard:Patrol Card; hijackCard:duress card;
superCard:Super Card; DismissingCard:Cancel the card; mergeCard:Emergency
Management Card (for authorizing temporary card permission).*/
                }
            ]
        },
        "FPInfo": {
        /*opt, object, Fingerprint Information, desc:Only custom items are
supported. Baseline is not supported.*/
            "List": [
            /*opt, array, Fingerprint Information List, subType:object, range:
[1,10]*/
                {
                    "fingerID":  1,
                    /*opt, int, Finger No., range:[1,10]*/
                    "fingerType":  "normalFP",
                    /*opt, enum, Fingerprint Type, subType:string, desc:Added
fingerprint. The field is not entered. The fingerprint type is normal by
default. The field is not entered. NormalFP:Normal Fingerprint; hijackFP:duress
fingerprint; patrolFP:Patrol Fingerprint; superFP:Super Fingerprint;
DismissingFP:Cancel Fingerprint*/
                    "fingerData":  "test"
                    /*opt, string, Fingerprint Data, range:[1,512], desc:Adopt
Base64 Code*/
                }
            ]
        }
    }
}
```

13.6 Face Picture Library Management (To be Optimized)

13.6.1 Get the information of all face picture libraries

URL

GET /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL": "test",
    /*ro, opt, string, request URL*/
    "statusCode": 1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString": "ok",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error details, desc:this node is required when the value
of statusCode is not 1*/
    "FDLib": [
    /*ro, opt, array, face picture library information, subType:object*/
        {
            "faceLibType": "blackFD",
            /*ro, opt, enum, face picture library type, subType:string,
desc:"blackFD" (list library), "staticFD" (static library)*/
            "name": "test",
            /*ro, opt, string, name of the face picture library, range:[0,48]*/
            "customInfo": "test",
            /*ro, opt, string, custom information, range:[0,192]*/
            "FDID": "test"
            /*ro, opt, string, face picture library ID*/
        }
    ]
}
```

13.6.2 Create a face picture library

URL

POST /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

```
{
    "faceLibType":  "blackFD,staticFD",
    /*req, string, face picture library type, desc:face picture library type*/
    "name":  "test",
    /*req, string, name of the face picture library, range:[0,48], desc:name of
the face picture library*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information*/
    "libArmingType":  "armingLib",
    /*opt, enum, subType:string*/
    "libAttribute":  "blackList",
    /*opt, enum, subType:string*/
    "FDID":  "test"
    /*opt, string, face picture library ID, desc:face picture library ID*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value
of statusCode is not 1*/
    "FDID":  "test"
    /*ro, opt, string, face picture library ID, desc:face picture library ID*/
}
```

13.6.3 Get face picture library capability

URL

GET /ISAPI/Intelligent/FDLib/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "test",
```

```
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, this field is required when the value of
statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, detailed error description, this field is required when
the value of statusCode is not 1*/
    "FDNameMaxLen":  64,
    /*ro, req, int, maximum length of face picture library name*/
    "customInfoMaxLen":  192,
    /*ro, req, int, maximum length of custom information*/
    "FDMaxNum":  3,
    /*ro, req, int, maximum number of face picture libraries*/
    "FDRecordDataMaxNum":  12345,
    /*ro, req, int, maximum face records supported by face picture library*/
    "supportFDFunction":  "post,delete,put,get,setUp",
    /*ro, req, string, the supported operations on face picture library,
desc:"post" (create), "delete" (delete), "put" (edit), "get" (search), "setUp"
(set)*/
    "isSuportFDSearch":  true,
    /*ro, req, bool, whether supports searching in face picture library*/
    "isSupportFDSearchDataPackage":  true,
    /*ro, req, bool, whether supports packaging the found data in the face
picture library*/
    "isSuportFSsearchByPic":  true,
    /*ro, req, bool, whether supports searching by picture in the face picture
library*/
    "isSuportFSsearchByPicGenerate":  true,
    /*ro, req, bool, whether supports exporting search by picture results from
the face picture library*/
    "isSuportFDSearchDuplicate":  true,
    /*ro, req, bool, whether supports duplication checking*/
    "isSuportFDSearchDuplicateGenerate":  true,
    /*ro, req, bool, whether supports exporting the duplication checking
results*/
    "isSuportFCSearch":  true,
    /*ro, req, bool, whether supports searching face picture comparison alarms*/
    "isSupportFCSearchDataPackage":  true,
    /*ro, req, bool, whether supports packaging the search results of face
picture comparison alarms*/
    "isSupportFDExecuteControl":  true,
    /*ro, req, bool, whether supports creating relation between face picture
libraries and cameras*/
    "generateMaxNum":  1234,
    /*ro, req, int, maximum face records can be exported from face picture
library*/
    "faceLibType":  "blackFD,staticFD",
    /*ro, opt, string, face picture library type*/
    "modelMaxNum":  1000,
    /*ro, opt, int, the maximum number of search results*/
    "isSupportModelData":  true,
    /*ro, opt, bool, whether it supports applying model data*/
```

```
    "isSuportFDLibArmingType":  true,
    /*ro, opt, bool, whether it supports face picture library arming type*/
    "isSuportFDLibSearch":  true,
    /*ro, opt, bool, whether it supports searching face picture library*/
    "FDArmingRecordDataMaxNum":  12345,
    /*ro, opt, int, the supported maximum number of face records in the face
picture arming library*/
    "isSupportControlPersonRecordByHumanId":  true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record
by humanId*/
    "isSupportControlPersonRecordByRowKey":  true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record
by rowKey*/
    "isSupportFaceLibRebuildCfg":  true,
    /*ro, opt, bool, whether it supports recreating face picture library
information and configuration*/
    "isSupportFDMove":  true,
    /*ro, opt, bool, whether it supports moving face data in the face picture
library in a batch, desc:the related URI is
/ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json*/
    "faceURLLen":  1024,
    /*ro, opt, int, the maximum size of the face picture URL, desc:if this node
is not returned, the default size of the face picture URL supported by the
device is 256 bytes; otherwise, the device should support that the value of this
node is greater than or equal to 256*/
    "isSupportArmingLibCfg":  true,
    /*ro, opt, bool, whether it supports configuring parameters of the armed
face picture library, desc:related URI:
/ISAPI/Intelligent/FDLib/armingLibCfg/capabilities?format=json*/
    "isSupportModelTransformation":  true,
    /*ro, opt, bool, whether it supports converting face picture models in the
face picture list library, desc:related URI:
/ISAPI/Intelligent/FDLib/model/transformation/capabilities?format=json*/
    "featurePointTypeList": ["face", "leftEye", "rightEye", "leftMouthCorner",
"rightMouthCorner", "nose"],
    /*ro, opt, array, feature point types of face pictures supported by the
device, subType:string*/
    "isSupportPresort":  true,
    /*ro, opt, bool, whether it supports configuring pre-categorization,
desc:related URI: /ISAPI/Intelligent/FDLib/presort/capabilities?format=json*/
    "libAttribute": {
    /*ro, opt, object, library attribute type, desc:"general" (normal library),
"blockList" (blocklist library), "VIP" (VIP library), "passerby" (allowlist
library, which cannot be deleted)*/
        "@opt": ["general", "blackList", "VIP", "passerby"]
        /*ro, opt, array, options, subType:string*/
    },
    "faceType": {
    /*ro, opt, object, face picture type*/
        "@opt": ["normalFace", "patrolFace", "hijackFace", "superFace"]
        /*ro, opt, array, options, subType:string*/
    },
    "saveFacePic": {
    /*ro, opt, object, whether to save the registered face picture*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "leaderPermission": {
```

```
    /*ro, opt, object, first authentication permission*/
        "@size":  4,
        /*ro, opt, int, the maximum number of array members, range:[1,4]*/
        "@min":  1,
        /*ro, opt, int, member value range, range:[1,4]*/
        "@max":  4
        /*ro, opt, int, member value range, range:[1,4]*/
    },
    "facePicFormat": {
    /*ro, opt, object*/
        "@opt": ["jpg", "png", "bmp"]
        /*ro, req, array, subType:string*/
    }
}
```

13.7 Permission Schedules for Persons and Access Points

13.7.1 Get the schedule template configuration capability of the access permission control

URL

GET /ISAPI/AccessControl/UserRightPlanTemplate/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserRightPlanTemplate": {
    /*ro, opt, object*/
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false"
(disabled)*/
        "templateName": {
        /*ro, opt, object, template name*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "weekPlanNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
```

```
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "holidayGroupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

13.7.2 Set the holiday group parameters of the access permission control schedule

URL

PUT /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Request Message

```
{
    "UserRightHolidayGroupCfg": {
    /*req, object, the holiday group parameters of the access permission control schedule*/
        "enable":  true,
        /*req, bool, whether to enable, desc:true (yes), false (no)*/
        "groupName":  "test",
        /*req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5",
        /*req, string, holiday group schedule No., desc:holiday group schedule No.*/
        "operateType":  "byTerminal",
        /*opt, enum, operation type, subType:string, desc:"byTerminal" (by terminal), "byOrg" (by organization), "byTerminalOrg" (by terminal organization)*/
        "terminalNoList": [1, 2, 3, 4],
        /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal" or "byTerminalOrg"*/
        "orgNoList": [1, 2, 3, 4]
        /*opt, array, organization ID list, subType:int, desc:this node is required when operation type is "byOrg" or "byTerminalOrg"*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
```

```
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required
when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required
when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.7.3 Set the schedule template parameters of the access permission control

URL

PUT /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json

Query Parameter

Request Message

```
{
    "UserRightPlanTemplate": {
    /*req, object, the schedule template of the access permission control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:true (yes), false (no)*/
        "templateName":  "test",
        /*req, string, template name*/
        "weekPlanNo":  1,
        /*req, int, week schedule No.*/
        "holidayGroupNo":  "1,3,5",
        /*req, string, holiday group No., desc:holiday group No.*/
        "operateType":  "byTerminal",
        /*opt, enum, operation type, subType:string, desc:"byTerminal" (by
terminal), "byOrg" (by organization), "byTerminalOrg" (by terminal
organization)*/
        "terminalNoList": [1, 2, 3, 4],
        /*opt, array, terminal ID list, subType:int, desc:this node is required
when operation type is "byTerminal" or "byTerminalOrg"*/
        "orgNoList": [1, 2, 3, 4]
        /*opt, array, organization ID list, subType:int, desc:this node is
required when operation type is "byOrg" or "byTerminalOrg"*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "test",
```

```
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required
when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required
when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.7.4 Set the week schedule parameters of the access permission control

URL

PUT /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Request Message

```
{
    "UserRightWeekPlanCfg": {
    /*opt, object, the week schedule parameters of the access permission
control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true" (enable), "false"
(disable)*/
        "WeekPlanCfg": [
        /*req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*req, enum, days of the week, subType:string, desc:"Monday",
"Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*req, int, time period No., range:[1,8], desc:it is between 1
and 8*/
                "enable":  true,
                /*req, bool, whether to enable, desc:"true" (enable), "false"
(disable)*/
                "TimeSegment": {
                /*req, object, time period*/
                    "beginTime":  "10:10:00",
                    /*req, string, start time of the time period, desc:(device
local time)*/
                    "endTime":  "12:10:00"
                    /*req, string, end time of the time period, desc:(device
local time)*/
                },
                "authenticationTimesEnabled":  true,
                /*opt, bool*/
                "authenticationTimes":  10
                /*opt, int, range:[1,255], step:1*/
            }
        ]
```

```
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required
when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required
when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.7.5 Get weekly schedule configuration parameters

URL

GET /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "UserRightWeekPlanCfg": {
    /*ro, opt, object, weekly schedule configuration parameters*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
        "WeekPlanCfg": [
        /*ro, req, array, weekly schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*ro, req, enum, day of the week, subType:string, desc:"Monday",
"Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*ro, req, int, time period No., range:[1,8], desc:it is between
1 and 8*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
                "TimeSegment": {
```

```
                /*ro, req, object, time period*/
                    "beginTime":  "10:10:00",
                    /*ro, req, string, start time of the time period,
desc:device local time*/
                    "endTime":  "12:10:00"
                    /*ro, req, string, end time of the time period, desc:device
local time*/
                },
                "authenticationTimesEnabled":  true,
                /*ro, opt, bool*/
                "authenticationTimes":  10
                /*ro, opt, int, range:[1,255], step:1*/
            }
        ]
    }
}
```

13.7.6 Get the weekly schedule configuration capability of the access permission control

URL

GET /ISAPI/AccessControl/UserRightWeekPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserRightWeekPlanCfg": {
    /*ro, opt, object*/
        "planNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false"
(disabled)*/
        "WeekPlanCfg": {
        /*ro, opt, object, weekly schedule parameters*/
            "maxSize":  56,
            /*ro, opt, int, the maximum value*/
            "week": {
            /*ro, opt, object, week*/
                "@opt":  "Monday"
                /*ro, opt, enum, days of the week, subType:string,
desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday",
"Sunday"*/
            },
            "id": {
```

```
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  8
                /*ro, opt, int, the maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether it is enabled, desc:"true" (enabled),
"false" (disabled)*/
            "TimeSegment": {
            /*ro, opt, object, time period*/
                "beginTime":  "test",
                /*ro, opt, string, start time, desc:(device local time)*/
                "endTime":  "test",
                /*ro, opt, string, end time, desc:(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour",
"minute", "second". If this node is not returned, the default time accuracy is
"minute"*/
            },
            "authenticationTimesEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            },
            "authenticationTimes": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, range:[1,255], step:1*/
                "@max":  255
                /*ro, req, int, range:[1,255], step:1*/
            }
        }
    }
}
```

13.7.7 Get holiday schedule configuration parameters

URL

GET /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "UserRightHolidayPlanCfg": {
    /*ro, req, object, holiday schedule configuration parameters*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
        "beginDate":  "1970-01-01",
        /*ro, req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-01",
```

```
        /*ro, req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8], desc:it is between
1 and 8*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
                "TimeSegment": {
                /*ro, req, object, time period*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "00:00:00"
                    /*ro, req, time, end time of the time period, desc:device
local time*/
                },
                "authenticationTimesEnabled":  true,
                /*ro, opt, bool*/
                "authenticationTimes":  10
                /*ro, opt, int, range:[1,255], step:1*/
            }
        ]
    }
}
```

13.7.8 Set the holiday schedule parameters of the access permission control

URL

PUT /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Request Message

```
{
    "UserRightHolidayPlanCfg": {
    /*req, object, the holiday schedule parameters of the access permission
control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true" (enable), "false"
(disable)*/
        "beginDate":  "1970-01-01",
        /*req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-01",
        /*req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*req, array, subType:object*/
            {
                "id":  1,
                /*req, int, time period No., range:[1,8], desc:it is between 1
and 8*/
                "enable":  true,
```

```
                /*req, bool, whether to enable, desc:"true" (enable), "false"
(disable)*/
                "TimeSegment": {
                /*opt, object, time period*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time of the time period, desc:device
local time*/
                    "endTime":  "00:00:00"
                    /*req, time, end time of the time period, desc:device local
time*/
                },
                "authenticationTimesEnabled":  true,
                /*opt, bool*/
                "authenticationTimes":  10
                /*opt, int, range:[1,255], step:1*/
            }
        ]
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required
when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required
when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.7.9 Get the holiday schedule configuration capability of the access permission control

URL

GET /ISAPI/AccessControl/UserRightHolidayPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserRightHolidayPlanCfg": {
    /*ro, req, object*/
        "planNo": {
        /*ro, opt, object, holiday schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false"
(disabled)*/
        "beginDate":  "1970-01-01",
        /*ro, opt, date, start date of the holiday, desc:(device local time)*/
        "endDate":  "1970-01-01",
        /*ro, opt, date, end date of the holiday, desc:(device local time)*/
        "HolidayPlanCfg": {
        /*ro, opt, object, holiday schedule parameter*/
            "maxSize":  8,
            /*ro, opt, int, the maximum value*/
            "id": {
            /*ro, opt, object, time period No.*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  8
                /*ro, opt, int, the maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether it is enabled, desc:"true" (enabled),
"false" (disabled)*/
            "TimeSegment": {
            /*ro, opt, object, time period*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time, desc:(device local time)*/
                "endTime":  "00:00:00",
                /*ro, opt, time, end time, desc:(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour",
"minute", "second". If this node is not returned, the default time accuracy is
"minute"*/
            },
            "authenticationTimesEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            },
            "authenticationTimes": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, range:[1,255], step:1*/
                "@max":  255
                /*ro, req, int, range:[1,255], step:1*/
            }
        }
    }
}
```

13.7.10 Get the holiday group configuration parameters of the access permission control schedule

URL

GET /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "UserRightHolidayGroupCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (yes), false (no)*/
        "groupName":  "test",
        /*ro, req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*ro, req, string, holiday group schedule No., desc:holiday group
schedule No.*/
    }
}
```

13.7.11 Get the holiday group configuration capability of the access permission control

URL

GET /ISAPI/AccessControl/UserRightHolidayGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserRightHolidayGroupCfg": {
    /*ro, req, object*/
        "groupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false"
(disabled)*/
        "groupName": {
        /*ro, opt, object, holiday group name*/
```

```
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "holidayPlanNo": {
        /*ro, opt, object, holiday group schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

13.7.12 Get the schedule template configuration parameters of the access permission control

URL

GET /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "UserRightPlanTemplate": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (yes), false (no)*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, week schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*ro, req, string, holiday group No., desc:holiday group No.*/
    }
}
```

13.8 Access Control Module Management

13.8.1 Get the door configuration capability

URL

GET /ISAPI/AccessControl/Door/param/<doorID>/capabilities

Query Parameter

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DoorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <doorNo min="1" max="10">
    <!--ro, opt, int, door No., attr:min{req, int},max{req, int}-->1
  </doorNo>
  <doorName min="1" max="32">
    <!--ro, opt, string, door name, attr:min{req, int},max{req, int}-->test
  </doorName>
  <magneticType opt="alwaysClose,alwaysOpen">
    <!--ro, opt, enum, magnetic contact type, subType:string, attr:opt{req,
string}, desc:"alwaysClose” (remain locked), "alwaysOpen” (remain unlocked)--
>alwaysClose
  </magneticType>
  <openButtonType opt="alwaysClose,alwaysOpen">
    <!--ro, opt, enum, door button type, subType:string, attr:opt{req, string},
desc:"alwaysClose” (remain locked), "alwaysOpen” (remain unlocked)-->alwaysClose
  </openButtonType>
  <openDuration min="1" max="255">
    <!--ro, opt, int, door open duration (floor relay action time),
attr:min{req, int},max{req, int}-->1
  </openDuration>
  <disabledOpenDuration min="1" max="255">
    <!--ro, opt, int, door open duration by disability card (delay duration of
closing the door), attr:min{req, int},max{req, int}-->1
  </disabledOpenDuration>
  <magneticAlarmTimeout min="0" max="255">
    <!--ro, opt, int, alarm time of magnetic contact detection timeout, range:
[0,255], unit:s, attr:min{req, int},max{req, int}, desc:alarm time of magnetic
contact detection timeout,which is between 0 and 255,0 refers to not triggering
alarm,unit: second-->1
  </magneticAlarmTimeout>
  <enableDoorLock opt="true,false">
    <!--ro, opt, bool, whether to enable locking door when the door is closed,
attr:opt{req, string}-->true
  </enableDoorLock>
  <enableLeaderCard opt="true,false">
    <!--ro, opt, bool, whether to enable remaining open with first card,
attr:opt{req, string}-->true
  </enableLeaderCard>
  <leaderCardMode opt="disable,alwaysOpen,authorize">
    <!--ro, opt, enum, first card mode, subType:string, attr:opt{req, string},
desc:"disable", "alwaysOpen” (remain open with first card), "authorize” (first
card authentication)-->disable
  </leaderCardMode>
  <leaderCardOpenDuration min="1" max="1440">
    <!--ro, opt, int, duration of remaining open with first card, attr:min{req,
int},max{req, int}-->1
  </leaderCardOpenDuration>
  <stressPassword min="1" max="8">
    <!--ro, opt, string, duress password, attr:min{req, int},max{req, int},
desc:the maximum length is 8 bytes, and the duress password should be encoded by
Base64 for transmission-->test
  </stressPassword>
  <superPassword min="1" max="8">
```

```
    <!--ro, opt, string, super password, attr:min{req, int},max{req, int},
desc:the maximum length is 8 bytes, and the duress password should be encoded by
Base64 for transmission-->test
  </superPassword>
  <unlockPassword min="1" max="8">
    <!--ro, opt, string, dismiss password,t, attr:min{req, int},max{req, int},
desc:the maximum length is 8 bytes, and the duress password should be encoded by
Base64 for transmission-->test
  </unlockPassword>
  <useLocalController opt="true,false">
    <!--ro, opt, bool, whether it is connected to the distributed controller,
attr:opt{req, string}-->true
  </useLocalController>
  <localControllerID min="0" max="64">
    <!--ro, opt, int, distributed controller No., range:[0,64], attr:min{req,
int},max{req, int}, desc:ro,distributed controller No.,which is between 1 and
64,0-unregistered-->1
  </localControllerID>
  <localControllerDoorNumber min="0" max="4">
    <!--ro, opt, int, distributed controller door No., range:[0,4],
attr:min{req, int},max{req, int}, desc:ro,distributed controller door No.,which
is between 1 and 4,0-unregistered-->1
  </localControllerDoorNumber>
  <localControllerStatus opt="0,1,2,3,4,5,6,7,8,9">
    <!--ro, opt, enum, online status of the distributed controller, subType:int,
attr:opt{req, string}, desc:0-offline, 1-network online, 2-RS-485 serial port 1
on loop circuit 1, 3-RS-485 serial port 2 on loop circuit 1, 4-RS-485 serial
port 1 on loop circuit 2, 5-RS-485 serial port 2 on loop circuit 2, 6-RS-485
serial port 1 on loop circuit 3, 7-RS-485 serial port 2 on loop circuit 3,8-RS-
485 serial port 1 on loop circuit 4,  9-RS-485 serial port 2 on loop circuit 4--
>1
  </localControllerStatus>
  <lockInputCheck opt="true,false">
    <!--ro, opt, bool, whether to enable door lock input detection,
attr:opt{req, string}-->true
  </lockInputCheck>
  <lockInputType opt="alwaysClose,alwaysOpen">
    <!--ro, opt, enum, door lock input type, subType:string, attr:opt{req,
string}, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked)--
>alwaysClose
  </lockInputType>
  <doorTerminalMode opt="preventCutAndShort,preventCutAndShort,common">
    <!--ro, opt, enum, working mode of door terminal, subType:string,
attr:opt{req, string}, desc:working mode of door terminal: "preventCutAndShort"-
prevent from broken-circuit and short-circuit (default),"common"--
>preventCutAndShort
  </doorTerminalMode>
  <openButton opt="true,false">
    <!--ro, opt, bool, whether to enable door button, attr:opt{req, string},
desc:whether to enable door button: "true"-yes (default),"false"-no-->true
  </openButton>
  <ladderControlDelayTime min="1" max="255">
    <!--ro, opt, int, elevator control delay time (for visitor), attr:min{req,
int},max{req, int}-->1
  </ladderControlDelayTime>
  <Leader>
    <!--ro, opt, object-->
    <continuousVerificationTimes min="1" max="10">
```

```
      <!--ro, opt, int, range:[1,10], attr:min{req, int},max{req, int}-->1
    </continuousVerificationTimes>
    <continuousVerificationDuration min="1" max="10">
      <!--ro, opt, int, range:[5,60], unit:s, attr:min{req, int},max{req, int}--
>20
    </continuousVerificationDuration>
    <effectiveTimeEnabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </effectiveTimeEnabled>
    <dayBeginTime>
      <!--ro, opt, time, dep:and,
{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </dayBeginTime>
    <beginEffectiveTime>
      <!--ro, opt, time, dep:and,
{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </beginEffectiveTime>
    <endEffectiveTime>
      <!--ro, opt, time, dep:and,
{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </endEffectiveTime>
  </Leader>
  <verificationPassOpenDoor opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </verificationPassOpenDoor>
  <relayReverseEnabled opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </relayReverseEnabled>
</DoorParam>
```

13.8.2 Set the door (floor) parameters

URL

PUT /ISAPI/AccessControl/Door/param/<doorID>?security=<security>&iv=<iv>

Query Parameter

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DoorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, door parameter, attr:version{req, string, protocolVersion}-->
  <doorName>
    <!--opt, string, door name-->test
  </doorName>
  <magneticType>
    <!--opt, enum, door magnetic sensor type, subType:string, desc:"alwaysClose"
(remain locked), "alwaysOpen" (remain unlocked)-->alwaysClose
  </magneticType>
  <openButtonType>
    <!--opt, enum, open door button type, subType:string, desc:"alwaysClose"
(remain locked), "alwaysOpen" (remain unlocked)-->alwaysClose
  </openButtonType>
  <openDuration>
    <!--opt, int, door open duration, range:[1,255], unit:s-->1
```

```
  </openDuration>
  <disabledOpenDuration>
    <!--opt, int, door open duration by disability card (delay duration of
closing the door), range:[1,255], unit:s-->1
  </disabledOpenDuration>
  <magneticAlarmTimeout>
    <!--opt, int, alarm time of magnetic contact detection timeout, range:
[0,255], unit:s, desc:0 refers to not triggering alarm-->1
  </magneticAlarmTimeout>
  <enableDoorLock>
    <!--opt, bool, lock door when door closed-->true
  </enableDoorLock>
  <enableLeaderCard>
    <!--opt, bool, whether to enable remaining open with first card-->true
  </enableLeaderCard>
  <leaderCardMode>
    <!--opt, enum, first card mode, subType:string, desc:first card mode:
"disable","alwaysOpen"-remain open with first card,"authorize"-first card
authentication. If this node is configured,the node <enableLeaderCard> is
invalid-->disable
  </leaderCardMode>
  <leaderCardOpenDuration>
    <!--opt, int, duration of remaining open with first card, range:[0,1440],
unit:min, dep:and,{$.DoorParam.leaderCardMode,eq,alwaysOpen}-->1
  </leaderCardOpenDuration>
  <stressPassword>
    <!--opt, string, duress password, desc:the maximum length is 8 bytes, and
the duress password should be encoded by Base64 for transmission-->test
  </stressPassword>
  <superPassword>
    <!--opt, string, super password, desc:the maximum length is 8 bytes, and the
duress password should be encoded by Base64 for transmission-->test
  </superPassword>
  <unlockPassword>
    <!--opt, string, unlock password, desc:the maximum length is 8 bytes, and
the duress password should be encoded by Base64 for transmission-->test
  </unlockPassword>
  <useLocalController>
    <!--opt, bool, whether it is connected to the distributed controller-->true
  </useLocalController>
  <localControllerID>
    <!--opt, int, distributed controller No., range:[0,64], desc:ro,distributed
controller No.,which is between 1 and 64,0-unregistered-->1
  </localControllerID>
  <localControllerDoorNumber>
    <!--opt, int, distributed controller door No., range:[0,4],
desc:ro,distributed controller door No.,which is between 1 and 4,0-unregistered-
->1
  </localControllerDoorNumber>
  <localControllerStatus>
    <!--opt, enum, online status of the distributed controller, subType:int,
desc:ro,online status of the distributed controller: 0-offline,1-network
online,2-RS-485 serial port 1 on loop circuit 1,3-RS-485 serial port 2 on loop
circuit 1,4-RS-485 serial port 1 on loop circuit 2,5-RS-485 serial port 2 on
loop circuit 2,6-RS-485 serial port 1 on loop circuit 3,7-RS-485 serial port 2
on loop circuit 3,8-RS-485 serial port 1 on loop circuit 4,9-RS-485 serial port
2 on loop circuit 4-->1
  </localControllerStatus>
```

```
  <lockInputCheck>
    <!--opt, bool, whether to enable door lock input detection-->true
  </lockInputCheck>
  <lockInputType>
    <!--opt, enum, door lock input type, subType:string, desc:"alwaysClose”
(remain locked), "alwaysOpen” (remain unlocked), lt remains locked by default--
>alwaysClose
  </lockInputType>
  <doorTerminalMode>
    <!--opt, enum, working mode of door terminal, subType:string,
desc:"preventCutAndShort” (prevent from broken-circuit and short-circuit
(default)),"common"-->preventCutAndShort
  </doorTerminalMode>
  <openButton>
    <!--opt, bool, whether to enable door button-->true
  </openButton>
  <ladderControlDelayTime>
    <!--opt, int, elevator control delay time (for visitor), range:[1,255],
unit:min-->1
  </ladderControlDelayTime>
  <Leader>
    <!--opt, object-->
    <continuousVerificationTimes>
      <!--opt, int, range:[1,10]-->1
    </continuousVerificationTimes>
    <continuousVerificationDuration>
      <!--opt, int, range:[5,60], unit:s-->20
    </continuousVerificationDuration>
    <effectiveTimeEnabled>
      <!--req, bool-->true
    </effectiveTimeEnabled>
    <dayBeginTime>
      <!--opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}--
>00:00:00
    </dayBeginTime>
    <beginEffectiveTime>
      <!--opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}--
>00:00:00
    </beginEffectiveTime>
    <endEffectiveTime>
      <!--opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}--
>00:00:00
    </endEffectiveTime>
  </Leader>
  <verificationPassOpenDoor>
    <!--opt, bool-->true
  </verificationPassOpenDoor>
  <relayReverseEnabled>
    <!--opt, bool-->true
  </relayReverseEnabled>
</DoorParam>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string,
protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device
Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK”
(succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML
Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description--
>OK
  </subStatusCode>
</ResponseStatus>
```

13.8.3 Get the door (floor) configuration parameters

URL

GET /ISAPI/AccessControl/Door/param/<doorID>?security=<security>&iv=<iv>

Query Parameter

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DoorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <doorName>
    <!--ro, opt, string, door name-->test
  </doorName>
  <magneticType>
    <!--ro, opt, enum, magnetic contact type, subType:string,
desc:"alwaysClose"-remain locked, "alwaysOpen"-remain unlocked-->alwaysClose
  </magneticType>
  <openButtonType>
    <!--ro, opt, enum, door button type, subType:string, desc:"alwaysClose"-
remain locked, "alwaysOpen"-remain unlocked-->alwaysClose
  </openButtonType>
  <openDuration>
    <!--ro, opt, int, door open duration, range:[1,255], unit:s-->1
  </openDuration>
  <disabledOpenDuration>
    <!--ro, opt, int, door open duration by disability card (delay duration of
closing the door), range:[1,255], unit:s-->1
  </disabledOpenDuration>
```

```
  <magneticAlarmTimeout>
    <!--ro, opt, int, alarm time of magnetic contact detection timeout, range:
[0,255], unit:s, desc:0 refers to not triggering alarm-->1
  </magneticAlarmTimeout>
  <enableDoorLock>
    <!--ro, opt, bool, whether to enable locking door when the door is closed--
>true
  </enableDoorLock>
  <enableLeaderCard>
    <!--ro, opt, bool, whether to enable remaining open with first card. This
node is invalid when leaderCardMode is configured-->true
  </enableLeaderCard>
  <leaderCardMode>
    <!--ro, opt, enum, first card mode, subType:string,
desc:"disable","alwaysOpen"-remain open with first card,"authorize"-first card
authentication. If this node is configured,the node <enableLeaderCard> is
invalid-->disable
  </leaderCardMode>
  <leaderCardOpenDuration>
    <!--ro, opt, int, duration of remaining open with first card, range:
[0,1440], unit:min, dep:and,{$.DoorParam.leaderCardMode,eq,alwaysOpen}-->1
  </leaderCardOpenDuration>
  <stressPassword>
    <!--ro, opt, string, duress password, desc:the maximum length is 8 bytes,
and the duress password should be encoded by Base64 for transmission-->test
  </stressPassword>
  <superPassword>
    <!--ro, opt, string, super password, desc:wo,super password,the maximum
length is 8 bytes,and the super password should be encoded by Base64 for
transmission-->test
  </superPassword>
  <unlockPassword>
    <!--ro, opt, string, dismiss password, desc:the maximum length is 8 bytes,
and the dismiss password should be encoded by Base64 for transmission-->test
  </unlockPassword>
  <useLocalController>
    <!--ro, opt, bool, whether it is connected to the distributed controller--
>true
  </useLocalController>
  <localControllerID>
    <!--ro, opt, int, distributed controller No., which is between 1 and 64,
range:[0,64], desc:0-unregistered-->1
  </localControllerID>
  <localControllerDoorNumber>
    <!--ro, opt, int, distributed controller door No., range:[0,4], desc:0-
unregistered-->1
  </localControllerDoorNumber>
  <localControllerStatus>
    <!--ro, opt, enum, online status of the distributed controller, subType:int,
desc:0-offline,1-network online,2-RS-485 serial port 1 on loop circuit 1,3-RS-
485 serial port 2 on loop circuit 1,4-RS-485 serial port 1 on loop circuit 2,5-
RS-485 serial port 2 on loop circuit 2,6-RS-485 serial port 1 on loop circuit
3,7-RS-485 serial port 2 on loop circuit 3,8-RS-485 serial port 1 on loop
circuit 4,9-RS-485 serial port 2 on loop circuit 4-->1
  </localControllerStatus>
  <lockInputCheck>
    <!--ro, opt, bool, whether to enable door lock input detection-->true
  </lockInputCheck>
```

```
  <lockInputType>
    <!--ro, opt, enum, door lock input type, subType:string, desc:"alwaysClose"-
remain locked (default), "alwaysOpen"-remain unlocked-->alwaysClose
  </lockInputType>
  <doorTerminalMode>
    <!--ro, opt, enum, working mode of door terminal, subType:string,
desc:"preventCutAndShort"-prevent from broken-circuit and short-circuit
(default),"common"-->preventCutAndShort
  </doorTerminalMode>
  <openButton>
    <!--ro, opt, bool, whether to enable door button: "true"-yes (default),
"false"-no-->true
  </openButton>
  <ladderControlDelayTime>
    <!--ro, opt, int, elevator control delay time (for visitor), range:[1,255],
unit:min-->1
  </ladderControlDelayTime>
  <Leader>
    <!--ro, opt, object-->
    <continuousVerificationTimes>
      <!--ro, opt, int, range:[1,10]-->1
    </continuousVerificationTimes>
    <continuousVerificationDuration>
      <!--ro, opt, int, range:[5,60], unit:s-->20
    </continuousVerificationDuration>
    <effectiveTimeEnabled>
      <!--ro, req, bool-->true
    </effectiveTimeEnabled>
    <dayBeginTime>
      <!--ro, opt, time, dep:and,
{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </dayBeginTime>
    <beginEffectiveTime>
      <!--ro, opt, time, dep:and,
{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </beginEffectiveTime>
    <endEffectiveTime>
      <!--ro, opt, time, dep:and,
{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </endEffectiveTime>
  </Leader>
  <verificationPassOpenDoor>
    <!--ro, opt, bool-->true
  </verificationPassOpenDoor>
  <relayReverseEnabled>
    <!--ro, opt, bool-->true
  </relayReverseEnabled>
</DoorParam>
```

13.9 Credential Recognition Module Management

13.9.1 Get the capability of configuring card No. authentication rule

URL

GET /ISAPI/AccessControl/CardVerificationRule/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardVerificationRuleCap": {
    /*ro, req, object*/
        "cardNoLenMode": {
        /*ro, opt, object, length mode of card No. authentication (comparison), desc:length mode of card No. authentication (comparison)*/
            "@opt": ["full", "4Bytes", "3Bytes", "wiegand27", "wiegand35", "Corporate1000_35", "Corporate1000_48", "H10302_37", "H10304_37", "wiegand_26CSN", "H10313O_32CSN", "wiegand_56CSN", "wiegand_58"]
            /*ro, opt, array, options, subType:string*/
        },
        "CardVerificationRuleRes": {
        /*ro, opt, object*/
            "checkStatus": {
            /*ro, opt, object, status of switching card No. authentication (comparison) mode, desc:"continue" (switching result can be searched for later), "ok" (switching completed), "duplicate" (duplicate data exist and switching failed)*/
                "@opt": ["continue", "ok", "duplicate"]
                /*ro, opt, array, options, subType:string*/
            },
            "progress": {
            /*ro, opt, object, switching progress in percentage, desc:which is between 0 and 100,and 100 indicates that the card No. authentication (comparison) mode is switched*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            }
        },
        "reverseCardNoEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, options, subType:bool, range:[1,2]*/
        }
    }
}
```

13.9.2 Get the capability of configuring parameters of the facial recognition mode.

URL

GET /ISAPI/AccessControl/FaceRecognizeMode/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FaceRecognizeMode": {
    /*ro, opt, object, facial recognition mode*/
        "mode": {
        /*ro, req, object, mode*/
            "@opt":  "normalMode"
            /*ro, opt, enum, mode options, subType:string, desc:"normalMode"
(normal mode), "deepMode" (deep mode)*/
        }
    }
}
```

13.9.3 Set the card reader parameters

URL

PUT /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json

Query Parameter

Request Message

```
{
    "CardReaderCfg": {
    /*req, object, card reader information*/
        "enable":  true,
        /*req, bool, whether to enable, desc:true-yes, false-no*/
        "okLedPolarity":  "cathode",
        /*opt, enum, OK LED polarity, subType:string, desc:"cathode", "anode"*/
        "errorLedPolarity":  "cathode",
        /*opt, enum, Error LED polarity, subType:string, desc:"cathode",
"anode"*/
        "buzzerPolarity":  "cathode",
        /*opt, enum, buzzer polarity, subType:string, desc:"cathode", "anode"*/
        "swipeInterval":  1,
        /*opt, int, time interval of repeated authentication, unit:s, desc:it is
valid for authentication modes such as fingerprint, card, face, etc.*/
        "pressTimeout":  1,
        /*opt, int, timeout to reset entry on keypad, unit:s*/
        "enableFailAlarm":  true,
        /*opt, bool, whether to enable excessive failed authentication attempt
alarm*/
        "maxReadCardFailNum":  1,
        /*opt, int, maximum number of failed authentication attempts*/
        "enableTamperCheck":  true,
        /*opt, bool, whether to enable tampering detection*/
        "offlineCheckTime":  1,
        /*opt, int, time to detect after the card reader is offline, unit:s*/
        "fingerPrintCheckLevel":  1,
```

```
        /*opt, enum, fingerprint recognition level, subType:int, desc:1-1/10
false acceptance rate (FAR), 2-1/100 false acceptance rate (FAR), 3-1/1000 false
acceptance rate (FAR), 4-1/10000 false acceptance rate (FAR), 5-1/100000 false
acceptance rate (FAR), 6-1/1000000 false acceptance rate (FAR), 7-1/10000000
false acceptance rate (FAR), 8-1/100000000 false acceptance rate (FAR), 9-3/100
false acceptance rate (FAR), 10-3/1000 false acceptance rate (FAR), 11-3/10000
false acceptance rate (FAR), 12-3/100000 false acceptance rate (FAR), 13-
3/1000000 false acceptance rate (FAR), 14-3/10000000 false acceptance rate
(FAR), 15-3/100000000 false acceptance rate (FAR), 16-Automatic Normal, 17-
Automatic Secure, 18-Automatic More Secure (currently not support)*/
        "useLocalController":  true,
        /*opt, bool, whether it is connected to the distributed controller*/
        "localControllerID":  1,
        /*opt, int, distributed controller No., range:[0,64], dep:and,
{$.CardReaderCfg.localControllerID,eq,true}, desc:0-unregistered. This field is
valid only when useLocalController is "true"*/
        "localControllerReaderID":  1,
        /*opt, int, card reader ID of the distributed controller, dep:and,
{$.CardReaderCfg.localControllerID,eq,true}, desc:0-unregistered. This field is
valid only when useLocalController is "true"*/
        "cardReaderChannel":  1,
        /*opt, enum, communication channel No. of the card reader, subType:int,
dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-Wiegand or offline,
1-RS-485A, 2-RS-485B. This field is valid only when useLocalController is
"true"*/
        "fingerPrintImageQuality":  1,
        /*opt, enum, fingerprint image quality, subType:int, desc:1-low quality
(V1), 2-medium quality (V1), 3-high quality (V1), 4-highest quality (V1), 5-low
quality (V2), 6-medium quality (V2), 7-high quality (V2), 8-highest quality
(V2)*/
        "fingerPrintContrastTimeOut":  1,
        /*opt, enum, fingerprint comparison timeout, subType:int, desc:it is
between 1 and 20, unit: second, 255-infinite*/
        "fingerPrintRecogizeInterval":  1,
        /*opt, enum, fingerprint scanning interval, subType:int, desc:it is
between 1 and 10, unit: second, 255-no delay*/
        "fingerPrintMatchFastMode":  1,
        /*opt, enum, fingerprint matching quick mode, subType:int, desc:1-quick
mode 1, 2-quick mode 2, 3-quick mode 3, 4-quick mode 4, 5-quick mode 5, 255-
automatic*/
        "fingerPrintModuleSensitive":  1,
        /*opt, enum, fingerprint module sensitivity, subType:int,
desc:fingerprint module sensitivity,which is between 1 and 8*/
        "fingerPrintModuleLightCondition":  "outdoor",
        /*opt, enum, fingerprint module light condition, subType:string,
desc:"outdoor", "indoor"*/
        "faceMatchThresholdN":  1,
        /*opt, int, threshold of face picture 1:N comparison, range:[0,100],
desc:threshold of face picture 1:N comparison,which is between 0 and 100*/
        "faceQuality":  1,
        /*opt, int, face picture quality, range:[0,100]*/
        "faceRecogizeTimeOut":  1,
        /*opt, enum, face recognition timeout, subType:int, desc:it is between 1
and 20, unit: second, 255-infinite*/
        "faceRecogizeInterval":  1,
        /*opt, enum, face recognition interval, subType:int, desc:it is between
1 and 10, unit: second, 255-no delay*/
```

```
        "cardReaderFunction":  ["fingerPrint", "face", "fingerVein", "iris",
"card"],
        /*opt, enumarray, card reader type, subType:string, desc:"fingerPrint"-
fingerprint, "face", "fingerVein"-finger vein, “iris”. For example,
["fingerPrint","face"] indicates that the card reader supports both fingerprint
and face*/
        "cardReaderDescription":  "Wiegand\u000485Offline",
        /*opt, string, card reader description, desc:if the card reader is the
Wiegand card reader or if offline, this field will be set to "Wiegand" or
"485Offline”*/
        "faceImageSensitometry":  1,
        /*opt, int, face picture exposure, range:[0,655535]*/
        "livingBodyDetect":  true,
        /*opt, bool, whether to enable human detection*/
        "faceMatchThreshold1":  1,
        /*opt, int, threshold of face picture 1:1 comparison, range:[0,100],
desc:threshold of face picture 1:1 comparison,which is between 0 and 100*/
        "buzzerTime":  1,
        /*opt, int, buzzing duration, range:[0,59999], unit:s, desc:0-long
buzzing*/
        "faceMatch1SecurityLevel":  1,
        /*opt, enum, security level of face 1:1 recognition, subType:int,
desc:1-normal, 2-high, 3-higher*/
        "faceMatchNSecurityLevel":  1,
        /*opt, enum, security level of face 1:N recognition, subType:int,
desc:1-normal, 2-high, 3-higher*/
        "envirMode":  "indoor",
        /*opt, enum, environment mode of face recognition, subType:string,
desc:"indoor", "other”*/
        "liveDetLevelSet":  "low",
        /*opt, enum, threshold level of liveness detection, subType:string,
desc:"low", "middle"-medium, "high"*/
        "liveDetThreshold":  1,
        /*opt, int, range:[0,100]*/
        "liveDetAntiAttackCntLimit":  1,
        /*opt, int, number of anti-attacks of liveness detection,, range:
[1,255], desc:this value should be configured as the same one on both client and
device*/
        "enableLiveDetAntiAttack":  true,
        /*opt, bool, whether to enable anti-attack for liveness detection*/
        "liveDetAntiAttackLockedTime":  1,
        /*opt, int, range:[0,300], unit:s*/
        "supportDelFPByID":  true,
        /*opt, bool, whether the card reader supports deleting fingerprint by
fingerprint ID, desc:true-yes, false-no*/
        "fingerPrintCapacity":  1,
        /*opt, int, fingerprint capacity*/
        "fingerPrintNum":  1,
        /*opt, int, number of added fingerprints*/
        "defaultVerifyMode":  "cardAndPw",
```

```
        /*opt, enum, default authentication mode of the fingerprint and card
reader (factory defaults):, subType:string, desc:factory defaults; "cardAndPw"
(card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
        "faceRecogizeEnable":  1,
        /*opt, enum, whether to enable facial recognition, subType:int, desc:1-
enable, 2-disable, 3-attendence checked in/out by recognition of multiple
faces*/
        "FPAlgorithmVersion":  "test",
        /*opt, string, fingerprint algorithm library version, range:[1,32]*/
        "cardReaderVersion":  "test",
        /*opt, string, card reader version, range:[1,32]*/
        "enableReverseCardNo":  true,
        /*opt, bool, whether to enable reversing the card No.*/
        "independSwipeIntervals":  0,
        /*opt, int, time interval of person authentication, unit: second. This
time interval is calculated for each person separately and is different from
swipeInterval, desc:time interval of person authentication,unit: second. This
time interval is calculated for each person separately and is different from
swipeInterval*/
        "maskFaceMatchThresholdN":  1,
        /*opt, int, 1:N face picture (face with mask and normal background)
comparison threshold, range:[0,100], desc:1:N face picture (face with mask and
normal background) comparison threshold,value range: [0,100]*/
        "maskFaceMatchThreshold1":  1,
        /*opt, int, 1:1 face picture (face with mask and normal background)
comparison threshold, range:[0,100], desc:1:1 face picture (face with mask and
normal background) comparison threshold,value range: [0,100]*/
        "faceMotionDetLevel":  "low",
        /*opt, enum, face motion detection level, subType:string, desc:"high",
"medium", "low"*/
        "showMode":  "normal",
```

```
        /*opt, enum, display mode, subType:string, desc:this node is not valid;
simple mode indicates that the device displays authentication results exclude
employee No., name, etc.; the device applies normal mode by default;
advertisement mode indicates that the device displays both advertisement and
authentication results; meeting mode indicates that the device displays check-in
page of the conference; custom mode indicates that the device displays layout of
the interface customized by users; "concise"-simple mode, "normal"-normal mode
(default), "advertising"-advertisement mode, "meeting"-meeting mode,
"selfDefine"-custom mode*/
        "enableScreenOff":  true,
        /*opt, bool, whether to enable auto locking the screen*/
        "screenOffTimeout":  1
        /*opt, int, time, step:1, unit:s*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.9.4 Get the card reader configuration parameters

URL

GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json

Query Parameter

Request Message

None

Response Message

```
{
    "CardReaderCfg": {
    /*ro, req, object, card reader information*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false
(disable)*/
        "okLedPolarity":  "cathode",
```

```
        /*ro, opt, enum, OK LED polarity, subType:string, desc:"cathode",
"anode"*/
        "errorLedPolarity":  "cathode",
        /*ro, opt, enum, Error LED polarity, subType:string, desc:"cathode",
"anode"*/
        "buzzerPolarity":  "cathode",
        /*ro, opt, enum, buzzer polarity, subType:string, desc:“cathode",
"anode"*/
        "swipeInterval":  1,
        /*ro, opt, int, time interval of repeated authentication, unit:s,
desc:which is valid for authentication modes such as fingerprint, card, face,
etc.*/
        "pressTimeout":  1,
        /*ro, opt, int, timeout to reset entry on keypad, unit:s*/
        "enableFailAlarm":  true,
        /*ro, opt, bool, whether to enable excessive failed authentication
attempts alarm*/
        "maxReadCardFailNum":  1,
        /*ro, opt, int, maximum number of failed authentication attempts*/
        "enableTamperCheck":  true,
        /*ro, opt, bool, whether to enable tampering detection*/
        "offlineCheckTime":  1,
        /*ro, opt, int, time to detect after the card reader is offline,
unit:s*/
        "fingerPrintCheckLevel":  1,
        /*ro, opt, enum, fingerprint recognition level, subType:int, desc:1-1/10
false acceptance rate (FAR), 2-1/100 false acceptance rate (FAR), 3-1/1000 false
acceptance rate (FAR), 4-1/10000 false acceptance rate (FAR), 5-1/100000 false
acceptance rate (FAR), 6-1/1000000 false acceptance rate (FAR), 7-1/10000000
false acceptance rate (FAR), 8-1/100000000 false acceptance rate (FAR), 9-3/100
false acceptance rate (FAR),  10-3/1000 false acceptance rate (FAR), 11-3/10000
false acceptance rate (FAR), 12-3/100000 false acceptance rate (FAR), 13-
3/1000000 false acceptance rate (FAR), 14-3/10000000 false acceptance rate
(FAR), 15-3/100000000 false acceptance rate (FAR), 16-Automatic Normal, 17-
Automatic Secure, 18-Automatic More Secure (currently not support)*/
        "useLocalController":  true,
        /*ro, opt, bool, whether it is connected to the distributed controller*/
        "localControllerID":  1,
        /*ro, opt, int, distributed controller No., range:[0,64], dep:and,
{$.CardReaderCfg.localControllerID,eq,true}, desc:which is between 1 and 64, 0-
unregistered. This field is valid only when useLocalController is "true”*/
        "localControllerReaderID":  1,
        /*ro, opt, int, card reader ID of the distributed controller, 0-
unregistered, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:this
field is valid only when useLocalController is "true”*/
        "cardReaderChannel":  1,
        /*ro, opt, enum, communication channel No. of the card reader,
subType:int, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-Wiegand
or offline, 1-RS-485A, 2-RS-485B. This field is valid only when
useLocalController is "true”*/
        "fingerPrintImageQuality":  1,
        /*ro, opt, enum, fingerprint image quality, subType:int, desc:1-low
quality (V1), 2-medium quality (V1), 3-high quality (V1), 4-highest quality
(V1), 5-low quality (V2), 6-medium quality (V2), 7-high quality (V2), 8-highest
quality (V2)*/
        "fingerPrintContrastTimeOut":  1,
```

```
        /*ro, opt, enum, fingerprint comparison timeout, subType:int,
desc:fingerprint comparison timeout,which is between 1 and 20,unit: second,255-
infinite*/
        "fingerPrintRecogizeInterval":  1,
        /*ro, opt, enum, fingerprint scanning interval, subType:int,
desc:fingerprint scanning interval,which is between 1 and 10,unit: second,255-no
delay*/
        "fingerPrintMatchFastMode":  1,
        /*ro, opt, enum, fingerprint matching quick mode, subType:int, desc:1-
quick mode 1, 2-quick mode 2, 3-quick mode 3, 4-quick mode 4, 5-quick mode 5,
255-automatic*/
        "fingerPrintModuleSensitive":  1,
        /*ro, opt, enum, fingerprint module sensitivity, subType:int,
desc:fingerprint module sensitivity,which is between 1 and 8*/
        "fingerPrintModuleLightCondition":  "outdoor",
        /*ro, opt, enum, fingerprint module light condition, subType:string,
desc:"outdoor", "indoor"*/
        "faceMatchThresholdN":  1,
        /*ro, opt, int, threshold of face picture 1:N comparison,which is
between 0 and 100, range:[0,100]*/
        "faceQuality":  1,
        /*ro, opt, int, face picture quality, range:[0,100]*/
        "faceRecogizeTimeOut":  1,
        /*ro, opt, enum, face recognition timeout, subType:int, desc:face
recognition timeout,which is between 1 and 20,unit: second,255-infinite*/
        "faceRecogizeInterval":  1,
        /*ro, opt, enum, face recognition interval, subType:int, desc:face
recognition interval,which is between 1 and 10,unit: second,255-no delay*/
        "cardReaderFunction":  ["fingerPrint", "face", "card"],
        /*ro, opt, enumarray, card reader type, subType:string,
desc:"fingerPrint", "face", "fingerVein". For example, ["fingerPrint", "face"]
indicates that the card reader supports both fingerprint and face*/
        "cardReaderDescription":  "Wiegand\u000485Offline",
        /*ro, opt, string, card reader description, desc:if the card reader is
the Wiegand card reader or if offline, this field will be set to "Wiegand" or
"485Offline"*/
        "faceImageSensitometry":  1,
        /*ro, opt, int, face picture exposure, range:[0,655535]*/
        "livingBodyDetect":  true,
        /*ro, opt, bool, whether to enable human detection*/
        "faceMatchThreshold1":  1,
        /*ro, opt, int, threshold of face picture 1:1 comparison, range:
[0,100]*/
        "buzzerTime":  1,
        /*ro, opt, int, buzzing duration, range:[0,59999], unit:s, desc:buzzing
duration,which is between 0 and 5999,unit: second,0-long buzzing*/
        "faceMatch1SecurityLevel":  1,
        /*ro, opt, enum, security level of face 1:1 recognition, subType:int,
desc:1 (normal), 2 (high), 3 (higher)*/
        "faceMatchNSecurityLevel":  1,
        /*ro, opt, enum, security level of face 1:N recognition: 1-normal,2-
high,3-higher, subType:int, desc:1 (normal), 2 (high), 3 (higher)*/
        "envirMode":  "other",
        /*ro, opt, enum, environment mode of face recognition, subType:string,
desc:"indoor", "other"*/
        "liveDetLevelSet":  "low",
        /*ro, opt, enum, threshold level of liveness detection, subType:string,
desc:"low", "middle", "high"*/
```

```
        "liveDetThreshold":  1,
        /*ro, opt, int, range:[0,100]*/
        "liveDetAntiAttackCntLimit":  1,
        /*ro, opt, int, number of anti-attacks of liveness detection, range:
[1,255], desc:this value should be configured as the same one on both client and
device*/
        "enableLiveDetAntiAttack":  true,
        /*ro, opt, bool, whether to enable anti-attack for liveness detection*/
        "liveDetAntiAttackLockedTime":  1,
        /*ro, opt, int, range:[0,300], unit:s*/
        "supportDelFPByID":  true,
        /*ro, opt, bool, whether the card reader supports deleting fingerprint
by fingerprint ID, desc:"true"-yes, "false"-no*/
        "fingerPrintCapacity":  1,
        /*ro, opt, int, fingerprint capacity*/
        "fingerPrintNum":  1,
        /*ro, opt, int, number of added fingerprints*/
        "defaultVerifyMode":  "cardAndPw",
        /*ro, opt, enum, default authentication mode of the fingerprint and card
reader (factory defaults), subType:string, desc:factory defaults; "cardAndPw"
(card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card
or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee
No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password),
"faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face),
"faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face),
"cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or
face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or
face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris),
"fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw"
(card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password),
"cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
        "faceRecogizeEnable":  1,
        /*ro, opt, enum, whether to enable facial recognition, subType:int,
desc:1 (enable), 2 (disable), 3 (attendence checked in/out by recognition of
multiple faces)*/
        "FPAlgorithmVersion":  "test",
        /*ro, opt, string, fingerprint algorithm library version, range:[1,32]*/
        "cardReaderVersion":  "test",
        /*ro, opt, string, card reader version, range:[1,32]*/
        "enableReverseCardNo":  true,
        /*ro, opt, bool, whether to enable reversing the card No.*/
        "independSwipeIntervals":  0,
```

```
        /*ro, opt, int, time interval of person authentication, desc:unit:
second. This time interval is calculated for each person separately and is
different from swipeInterval*/
        "maskFaceMatchThresholdN":  1,
        /*ro, opt, int, 1:N face picture (face with mask and normal background)
comparison threshold, range:[0,100]*/
        "maskFaceMatchThreshold1":  1,
        /*ro, opt, int, 1:1 face picture (face with mask and normal background)
comparison threshold, range:[0,100]*/
        "faceMotionDetLevel":  "low",
        /*ro, opt, enum, face motion detection level, subType:string,
desc:"high", "medium", "low"*/
        "showMode":  "normal",
        /*ro, opt, enum, display mode, subType:string, desc:this node is not
valid; simple mode indicates that the device displays authentication results
exclude employee No., name, etc.; the device applies normal mode by default;
advertisement mode indicates that the device displays both advertisement and
authentication results; meeting mode indicates that the device displays check-in
page of the conference; custom mode indicates that the device displays layout of
the interface customized by users; "concise"-simple mode, "normal"-normal mode
(default), "advertising"-advertisement mode, "meeting"-meeting mode,
"selfDefine"-custom mode*/
        "enableScreenOff":  true,
        /*ro, opt, bool, whether to enable auto locking the screen*/
        "screenOffTimeout":  1
        /*ro, opt, int, time, step:1, unit:s*/
    }
}
```

13.9.5 Set the parameters of intelligent identity recognition terminal

URL

PUT /ISAPI/AccessControl/IdentityTerminal

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IdentityTerminal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, the parameters of intelligent identity recognition terminal,
attr:version{req, string, protocolVersion}-->
  <terminalMode>
    <!--opt, enum, terminal mode, subType:string, desc:"authMode”
(authentication mode), "registerMode” (registration mode)-->authMode
  </terminalMode>
  <idCardReader>
    <!--opt, enum, ID card reader, subType:string, desc:“iDR210”, “DS-K1F110-I”,
“DS-K1F110-B”, “DS-K1F110-AB”, “none”, “DS-K1F1001-I (USB)”, “DS-K1F1002-I
(USB)”-->iDR210
  </idCardReader>
  <camera>
    <!--opt, enum, camera, subType:string, desc:"C270", "DS-2CS5432B-S"-->C270
  </camera>
```

```
  <fingerPrintModule>
    <!--opt, enum, fingerprint module, subType:string, desc:fingerprint module
type: ALIWARD,HikModule-->ALIWARD
  </fingerPrintModule>
  <videoStorageTime>
    <!--opt, int, time for saving video, range:[0,10], desc:unit: second-->1
  </videoStorageTime>
  <faceContrastThreshold>
    <!--opt, int, face picture comparison threshold, range:[0,100]-->1
  </faceContrastThreshold>
  <twoDimensionCode>
    <!--opt, enum, whether to enable QR code recognition, subType:string,
desc:“enable”, “disable”-->enable
  </twoDimensionCode>
  <blackListCheck>
    <!--opt, enum, whether to enable blocklist verification, subType:string,
desc:“enable”, “disable”-->enable
  </blackListCheck>
  <idCardCheckCenter>
    <!--opt, enum, ID card comparison mode, subType:string, desc:“local”
(compare with ID card of local storage), “server” (compare with ID card of
remote server storage)-->local
  </idCardCheckCenter>
  <faceAlgorithm>
    <!--opt, enum, face picture algorithm, subType:string, desc:"DeepLearn"
(deep learning algorithm), "Tradition" (third-party algorithm)-->DeepLearn
  </faceAlgorithm>
  <comNo>
    <!--opt, int, COM No., range:[1,9]-->1
  </comNo>
  <memoryLearning>
    <!--opt, enum, whether to enable learning and memory function,
subType:string, desc:“enable”, “disable”-->enable
  </memoryLearning>
  <saveCertifiedImage>
    <!--opt, enum, whether to enable saving authenticated picture,
subType:string, desc:“enable”, “disable”-->enable
  </saveCertifiedImage>
  <MCUVersion>
    <!--opt, string, MCU Version-->test
  </MCUVersion>
  <usbOutput>
    <!--opt, enum, whether to enable USB output of ID card reader,
subType:string, desc:“enable”, “disable”-->enable
  </usbOutput>
  <serialOutput>
    <!--opt, enum, whether to enable serial port output of ID card reade,
subType:string, desc:“enable”, “disable”-->enable
  </serialOutput>
  <readInfoOfCard>
    <!--opt, enum, set content to be read from CPU card, subType:string,
desc:“serialNo” (read serial No.), “file” (read file)-->serialNo
  </readInfoOfCard>
  <workMode>
    <!--opt, enum, authentication mode, subType:string, desc:“passMode”,
“accessControlMode”-->passMode
  </workMode>
  <ecoMode>
```

```
    <!--opt, object, ECO mode-->
    <eco>
      <!--opt, enum, whether to enable ECO mode, subType:string, desc:“enable”, “disable”-->enable
    </eco>
    <faceMatchThreshold1>
      <!--opt, int, 1V1 face picture comparison threshold of ECO mode, range: [0,100]-->1
    </faceMatchThreshold1>
    <faceMatchThresholdN>
      <!--opt, int, 1VN face picture comparison threshold of ECO mode, range: [0,100]-->1
    </faceMatchThresholdN>
    <changeThreshold>
      <!--opt, int, ECO mode threshold, desc:switching threshold of ECO mode,which is between 0 and 8-->0
    </changeThreshold>
    <maskFaceMatchThresholdN>
      <!--opt, int, 1:N face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100]-->1
    </maskFaceMatchThresholdN>
    <maskFaceMatchThreshold1>
      <!--opt, int, 1:1 face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100]-->1
    </maskFaceMatchThreshold1>
    <alwaysInfrared>
      <!--opt, bool, whether to enable infrared recognition, desc:if this node exists and changeThreshold is 8, it indicates that the device will not always enable the infrared recognition-->true
    </alwaysInfrared>
    <ageFaceMatchList>
      <!--opt, array, list of different age groups in ECO mode, subType:object, range:[0,5], desc:before set this node, age group matching should be enabled, see URL: PUT /ISAPI/AccessControl/FaceCompareCond, and ageFaceMatchEnabled should be true-->
      <ageFaceMatch>
        <!--opt, object, age group matching in ECO mode-->
        <ageLevel>
          <!--opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40 years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
        </ageLevel>
        <ageFaceMatchThreshold1>
          <!--opt, int, matching threshold when authenticating via age groups in ECO mode (1:1), range:[0,100]-->1
        </ageFaceMatchThreshold1>
        <ageFaceMatchThresholdN>
          <!--opt, int, matching threshold when authenticating via age groups in ECO mode (1:N), range:[0,100]-->1
        </ageFaceMatchThresholdN>
      </ageFaceMatch>
    </ageFaceMatchList>
  </ecoMode>
  <readCardRule>
    <!--opt, enum, card No. setting rule, subType:string, desc:"wiegand26", "wiegand34"-->wiegand26
  </readCardRule>
```

```
  <enableScreenOff>
    <!--opt, bool, whether the device enters the sleep mode when there is no
operation after the configured sleep time-->true
  </enableScreenOff>
  <screenOffTimeout>
    <!--opt, int, sleep time, desc:unit: second-->1
  </screenOffTimeout>
  <enableScreensaver>
    <!--opt, bool, whether to enable the screen saver function-->true
  </enableScreensaver>
  <faceModuleVersion>
    <!--opt, string, face recognition module version, range:[1,32]-->test
  </faceModuleVersion>
  <showMode>
    <!--opt, enum, display mode, subType:string, desc:simple mode indicates that
the device displays authentication results exclude employee No., name, etc.; the
device applies normal mode by default; advertisement mode indicates that the
device displays both advertisement and authentication results; meeting mode
indicates that the device displays check-in page of the conference; custom mode
indicates that the device displays layout of the interface customized by users;
"concise"-simple mode, "normal"-normal mode (default), "advertising"-
advertisement mode, "meeting"-meeting mode, "selfDefine"-custom mode-->concise
  </showMode>
  <popUpPreviewWindow>
    <!--opt, bool, whether to pop up live view window, dep:or,
{$.IdentityTerminal.showMode,eq,advertising}-->true
  </popUpPreviewWindow>
  <needDeviceCheck>
    <!--opt, bool, whether it need device check in permission free mode, dep:or,
{$.IdentityTerminal.workMode,eq,passMode}-->true
  </needDeviceCheck>
  <previewShowTime>
    <!--opt, int, display duration in live view, range:[1,99], unit:s, dep:or,
{$.IdentityTerminal.popUpPreviewWindow,eq,true}-->1
  </previewShowTime>
</IdentityTerminal>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response status, attr:version{req, string,
protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, desc:request URL-->test
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:read-only,status code:
0,1-OK,2-Device Busy,3-Device Error,4-Invalid Operation,5-Invalid XML Format,6-
Invalid XML Content,7-Reboot Required,9-Additional Error-->1
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description, subType:string, desc:“OK”
(succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML
Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
```

```
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:read-only,describe the error
reason in detail-->test
  </subStatusCode>
</ResponseStatus>
```

13.9.6 Get the parameters of intelligent identity recognition terminal

URL

GET /ISAPI/AccessControl/IdentityTerminal

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IdentityTerminal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of intelligent identity recognition terminal,
attr:version{req, string, protocolVersion}-->
  <terminalMode>
    <!--ro, opt, enum, terminal mode, subType:string, desc:"authMode"
(authentication mode), "registerMode" (registration mode)-->authMode
  </terminalMode>
  <idCardReader>
    <!--ro, opt, enum, ID card reader model, subType:string, desc:“iDR210”, “DS-
K1F110-I”, “DS-K1F1110-B”, “DS-K1F1110-AB”, “none”, “DS-K1F1001-I(USB)”, “DS-
K1F1002-I(USB)”-->iDR210
  </idCardReader>
  <camera>
    <!--ro, opt, enum, camera, subType:string, desc:camera model: C270,DS-
2CS5432B-S-->C270
  </camera>
  <fingerPrintModule>
    <!--ro, opt, enum, fingerprint module type, subType:string-->ALIWARD
  </fingerPrintModule>
  <videoStorageTime>
    <!--ro, opt, int, time for saving video, range:[0,10], desc:unit: second-->1
  </videoStorageTime>
  <faceContrastThreshold>
    <!--ro, opt, int, face picture comparison threshold, range:[0,100]-->1
  </faceContrastThreshold>
  <twoDimensionCode>
    <!--ro, opt, enum, whether to enable QR code recognition, subType:string,
desc:“enable”, “disable”-->enable
  </twoDimensionCode>
  <blackListCheck>
    <!--ro, opt, enum, whether to enable blocklist verification, subType:string,
desc:“enable”, “disable”-->enable
  </blackListCheck>
```

```
  <idCardCheckCenter>
    <!--ro, opt, enum, ID card comparison mode, subType:string, desc:“local” (compare with ID card of local storage), “server” (compare with ID card of remote server storage)-->local
  </idCardCheckCenter>
  <faceAlgorithm>
    <!--ro, opt, enum, face picture algorithm, subType:string, desc:"DeepLearn" (deep learning algorithm), "Tradition" (third-party algorithm)-->DeepLearn
  </faceAlgorithm>
  <comNo>
    <!--ro, opt, int, COM No., range:[1,9]-->1
  </comNo>
  <memoryLearning>
    <!--ro, opt, enum, whether to enable learning and memory function, subType:string, desc:“enable”, “disable”-->enable
  </memoryLearning>
  <saveCertifiedImage>
    <!--ro, opt, enum, whether to enable saving authenticated picture, subType:string, desc:“enable”, “disable”-->enable
  </saveCertifiedImage>
  <MCUVersion>
    <!--ro, opt, string, MCU version information-->test
  </MCUVersion>
  <usbOutput>
    <!--ro, opt, enum, whether to enable USB output of ID card reader, subType:string, desc:“enable”, “disable”-->enable
  </usbOutput>
  <serialOutput>
    <!--ro, opt, enum, whether to enable serial port output of ID card reader, subType:string, desc:“enable”, “disable”-->enable
  </serialOutput>
  <readInfoOfCard>
    <!--ro, opt, enum, set content to be read from CPU card, subType:string, desc:“serialNo” (read serial No.), “file” (read file)-->serialNo
  </readInfoOfCard>
  <workMode>
    <!--ro, opt, enum, authentication mode, subType:string, desc:“passMode”, “accessControlMode”-->passMode
  </workMode>
  <ecoMode>
    <!--ro, opt, object, ECO mode-->
    <eco>
      <!--ro, opt, enum, whether to enable ECO mode, subType:string, desc:“enable”, “disable”-->enable
    </eco>
    <faceMatchThreshold1>
      <!--ro, req, int, 1V1 face picture comparison threshold of ECO mode, range:[0,100]-->1
    </faceMatchThreshold1>
    <faceMatchThresholdN>
      <!--ro, req, int, 1:N face picture comparison threshold of ECO mode, range:[0,100]-->1
    </faceMatchThresholdN>
    <changeThreshold>
      <!--ro, opt, int, switching threshold of ECO mode, desc:switching threshold of ECO mode,which is between 0 and 8-->0
    </changeThreshold>
    <maskFaceMatchThresholdN>
```

```
      <!--ro, opt, int, 1:N face picture (face with mask and normal background
picture) comparison threshold of ECO mode, range:[0,100]-->0
    </maskFaceMatchThresholdN>
    <maskFaceMatchThreshold1>
      <!--ro, opt, int, 1:1 face picture (face with mask and normal background
picture) comparison threshold of ECO mode, range:[0,100]-->0
    </maskFaceMatchThreshold1>
    <alwaysInfrared>
      <!--ro, opt, bool, whether to enable infrared recognition, desc:if this
node exists and changeThreshold is 8, it indicates that the device will not
always enable the infrared recognition-->true
    </alwaysInfrared>
    <ageFaceMatchList>
      <!--ro, opt, array, list of different age groups in ECO mode,
subType:object, range:[0,5], desc:before set this node, age group matching
should be enabled, see URL: PUT /ISAPI/AccessControl/FaceCompareCond, and
ageFaceMatchEnabled should be true-->
      <ageFaceMatch>
        <!--ro, opt, object, age group matching in ECO mode-->
        <ageLevel>
          <!--ro, opt, enum, age groups, subType:int, desc:0 (child who are 0~6
years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40
years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than
66 years old)-->1
        </ageLevel>
        <ageFaceMatchThreshold1>
          <!--ro, opt, int, matching threshold when authenticating via age
groups in ECO mode (1:1), range:[0,100]-->1
        </ageFaceMatchThreshold1>
        <ageFaceMatchThresholdN>
          <!--ro, opt, int, matching threshold when authenticating via age
groups in ECO mode (1:N), range:[0,100]-->1
        </ageFaceMatchThresholdN>
      </ageFaceMatch>
    </ageFaceMatchList>
  </ecoMode>
  <readCardRule>
    <!--ro, opt, enum, card No. setting rule, subType:string, desc:"wiegand26",
"wiegand34"-->wiegand26
  </readCardRule>
  <enableScreenOff>
    <!--ro, opt, bool, whether the device enters the sleep mode when there is no
operation after the configured sleep time-->true
  </enableScreenOff>
  <screenOffTimeout>
    <!--ro, opt, int, sleep time, desc:unit: second-->1
  </screenOffTimeout>
  <enableScreensaver>
    <!--ro, opt, bool, whether to enable the screen saver function-->true
  </enableScreensaver>
  <faceModuleVersion>
    <!--ro, opt, string, face recognition module version, range:[1,32]-->test
  </faceModuleVersion>
  <showMode>
    <!--ro, opt, enum, display mode, subType:string, desc:"concise" (simple
mode,only the authentication result will be displayed), "normal" (normal mode).
The default mode is normal mode. If this node does not exist, the default mode
is normal mode-->concise
```

```
  </showMode>
  <popUpPreviewWindow>
    <!--ro, opt, bool, whether to pop up live view window, dep:or,
{$.IdentityTerminal.showMode,eq,advertising}-->true
  </popUpPreviewWindow>
  <needDeviceCheck>
    <!--ro, opt, bool, whether it need device check in permission free mode,
dep:or,{$.IdentityTerminal.workMode,eq,passMode}-->true
  </needDeviceCheck>
  <previewShowTime>
    <!--ro, opt, int, display duration in live view, range:[1,99], unit:s,
dep:or,{$.IdentityTerminal.popUpPreviewWindow,eq,true}-->1
  </previewShowTime>
</IdentityTerminal>
```

13.9.7 Get configuration capability of intelligent identity recognition terminal

URL

GET /ISAPI/AccessControl/IdentityTerminal/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IdentityTerminal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of intelligent identity recognition terminal,
attr:version{req, string, protocolVersion}-->
  <terminalMode opt="authMode,registerMode">
    <!--ro, opt, enum, terminal mode, subType:string, attr:opt{req, string},
desc:"authMode" (authentication mode), "registerMode" (registration mode)--
>authMode
  </terminalMode>
  <idCardReader opt="iDR210,DS-K1F110-I,DS-K1F1110-B,DS-K1F1110-AB,none ">
    <!--ro, opt, enum, ID card reader model, subType:string, attr:opt{req,
string}, desc:"iDR210", "DS-K1F110-I", "DS-K1F1110-B", "DS-K1F1110-AB", "none",
"DS-K1F1001-I (USB)", "DS-K1F1002-I (USB)"-->iDR210
  </idCardReader>
  <camera opt="C270,DS-2CS5432B-S">
    <!--ro, opt, enum, camera, subType:string, attr:opt{req, string},
desc:"C270", "DS-2CS5432B-S"-->C270
  </camera>
  <fingerPrintModule opt="ALIWARD,HikModule">
    <!--ro, opt, enum, fingerprint module, subType:string, attr:opt{req,
string}, desc:fingerprint module-->ALIWARD
  </fingerPrintModule>
  <videoStorageTime min="0" max="10">
    <!--ro, opt, int, time for saving video (unit: second), range:[0,10],
attr:min{req, int},max{req, int}, desc:unit: second-->1
```

```
  </videoStorageTime>
  <faceContrastThreshold min="0" max="100">
    <!--ro, opt, int, face picture comparison threshold, range:[0,100],
attr:min{req, int},max{req, int}-->1
  </faceContrastThreshold>
  <twoDimensionCode opt="enable,disable">
    <!--ro, opt, enum, whether to enable QR code recognition, subType:string,
attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </twoDimensionCode>
  <blackListCheck opt="enable,disable">
    <!--ro, opt, enum, whether to enable blocklist verification, subType:object,
attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </blackListCheck>
  <idCardCheckCenter opt="local,server">
    <!--ro, opt, enum, ID card comparison mode, subType:string, attr:opt{req,
string}, desc:“local” (compare with ID card of local storage), “server” (compare
with ID card of remote server storage)-->local
  </idCardCheckCenter>
  <faceAlgorithm opt="DeepLearn,Tradition">
    <!--ro, opt, enum, face picture algorithm, subType:string, attr:opt{req,
string}, desc:"DeepLearn" (deep learning algorithm), "Tradition" (third-party
algorithm)-->DeepLearn
  </faceAlgorithm>
  <comNo min="1" max="9">
    <!--ro, opt, int, COM No., range:[1,9], attr:min{req, int},max{req, int}-->1
  </comNo>
  <memoryLearning opt="enable,disable">
    <!--ro, opt, enum, whether to enable learning and memory function,
subType:object, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </memoryLearning>
  <saveCertifiedImage opt="enable,disable">
    <!--ro, opt, enum, whether to enable saving authenticated picture,
subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </saveCertifiedImage>
  <MCUVersion min="1" max="10">
    <!--ro, opt, string, MCU version information, attr:min{req, int},max{req,
int}-->test
  </MCUVersion>
  <usbOutput opt="enable,disable">
    <!--ro, opt, enum, whether to enable USB output of ID card reader,
subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </usbOutput>
  <serialOutput opt="enable,disable">
    <!--ro, opt, enum, whether to enable serial port output of ID card reader,
subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </serialOutput>
  <readInfoOfCard opt="serialNo,file">
    <!--ro, opt, enum, set content to be read from CPU card, subType:string,
attr:opt{req, string}, desc:"serialNo" (read the serial No.), "file" (read the
file)-->serialNo
  </readInfoOfCard>
  <workMode opt="passMode,accessControlMode">
    <!--ro, opt, enum, authentication mode, subType:string, attr:opt{req,
string}, desc:authentication mode-->passMode
  </workMode>
  <ecoMode>
    <!--ro, opt, object, ECO mode-->
    <eco opt="enable,disable">
```

```
      <!--ro, opt, enum, whether to enable ECO mode, subType:string,
attr:opt{req, string}, desc:“enable”, “disable”-->enable
    </eco>
    <faceMatchThreshold1 min="0" max="100">
      <!--ro, opt, int, 1V1 face picture comparison threshold of ECO mod, range:
[0,100], attr:min{req, int},max{req, int}-->1
    </faceMatchThreshold1>
    <faceMatchThresholdN min="0" max="100">
      <!--ro, opt, int, 1:N face picture comparison threshold of ECO mode,
range:[0,100], attr:min{req, int},max{req, int}-->1
    </faceMatchThresholdN>
    <changeThreshold min="1" max="10">
      <!--ro, opt, int, switching threshold of ECO mode, attr:min{req,
int},max{req, int}, desc:switching threshold of ECO mode,which is between 0 and
8-->0
    </changeThreshold>
    <maskFaceMatchThresholdN min="0" max="100">
      <!--ro, opt, int, 1:N face picture (face with mask and normal background
picture) comparison threshold of ECO mode, range:[0,100], attr:min{req,
int},max{req, int}-->1
    </maskFaceMatchThresholdN>
    <maskFaceMatchThreshold1 min="0" max="100">
      <!--ro, opt, int, 1:1 face picture (face with mask and normal background
picture) comparison threshold of ECO mode, range:[0,100], attr:min{req,
int},max{req, int}-->1
    </maskFaceMatchThreshold1>
    <alwaysInfrared opt="true,false">
      <!--ro, opt, bool, whether to enable infrared recognition, attr:opt{req,
string}, desc:if this node exists and changeThreshold is 8, it indicates that
the device will not always enable the infrared recognition-->true
    </alwaysInfrared>
    <ageFaceMatchList size="5">
      <!--ro, opt, array, list of different age groups in ECO mode,
subType:object, range:[0,5], attr:size{req, int}-->
      <ageFaceMatch>
        <!--ro, opt, object, age group matching in ECO mode-->
        <ageLevel opt="0,1,2,3,4">
          <!--ro, opt, enum, age groups, subType:int, attr:opt{req, string},
desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2
(youth and prime who are 18~40 years old), 3 (middle age who are 41~65 years
old), 4 (elderly who are more than 66 years old)-->1
        </ageLevel>
        <ageFaceMatchThreshold1 min="0" max="100">
          <!--ro, opt, int, matching threshold when authenticating via age
groups in ECO mode (1:1), range:[0,100], attr:min{req, int},max{req, int}-->1
        </ageFaceMatchThreshold1>
        <ageFaceMatchThresholdN min="0" max="100">
          <!--ro, opt, int, matching threshold when authenticating via age
groups in ECO mode (1:N), range:[0,100], attr:min{req, int},max{req, int}-->1
        </ageFaceMatchThresholdN>
      </ageFaceMatch>
    </ageFaceMatchList>
  </ecoMode>
  <readCardRule opt="wiegand26,wiegand34">
    <!--ro, opt, enum, card No. setting rule, subType:string, attr:opt{req,
string}, desc:card No. setting rule: "wiegand26","wiegand34"-->wiegand26
  </readCardRule>
  <enableScreenOff opt="true,false">
```

```
    <!--ro, opt, bool, whether the device enters the sleep mode when there is no
operation after the configured sleep time, attr:opt{req, string}-->true
  </enableScreenOff>
  <screenOffTimeout min="0" max="10">
    <!--ro, opt, int, sleep time, attr:min{req, int},max{req, int}, desc:unit:
second-->1
  </screenOffTimeout>
  <enableScreensaver opt="true,false">
    <!--ro, opt, bool, whether to enable the screen saver function,
attr:opt{req, string}-->true
  </enableScreensaver>
  <faceModuleVersion min="0" max="10">
    <!--ro, opt, string, face recognition module version, attr:min{req,
int},max{req, int}-->test
  </faceModuleVersion>
  <showMode opt="concise,normal,advertising,meeting,selfDefine,boxStatus">
    <!--ro, opt, enum, display mode, subType:string, attr:opt{req, string},
desc:"concise" (simple mode,only the authentication result will be displayed),
"normal" (normal mode). The default mode is normal mode. If this node does not
exist, the default mode is normal mode-->concise
  </showMode>
  <popUpPreviewWindow opt="true,false">
    <!--ro, opt, bool, whether to pop up live view window, dep:or,
{$.IdentityTerminal.showMode,eq,advertising}, attr:opt{req, string}-->true
  </popUpPreviewWindow>
  <needDeviceCheck opt="true,false">
    <!--ro, opt, bool, whether it need device check in permission free mode,
dep:or,{$.IdentityTerminal.workMode,eq,passMode}, attr:opt{req, string}-->true
  </needDeviceCheck>
  <previewShowTime min="1" max="99">
    <!--ro, opt, int, display duration in live view, range:[1,99], attr:min{req,
int},max{req, int}-->1
  </previewShowTime>
</IdentityTerminal>
```

13.9.8 Get the configuration capability of the card reader

URL

GET /ISAPI/AccessControl/CardReaderCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardReaderCfg": {
    /*ro, req, object, Set Card Reader Parameters*/
        "cardReaderNo": {
        /*ro, opt, object, card reader No., desc:card reader No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
```

```
            "@max":  512,
            /*ro, opt, int, the maximum value*/
            "@opt": [1, 4]
            /*ro, opt, array, value range, subType:int, desc:turnstiles only
support No.1 and No.4; returning the node "@min" and "@max" are not required*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:"true"-yes,"false"-no*/
        "okLedPolarity": {
        /*ro, opt, object, OK LED polarity*/
            "@opt":  "cathode,anode"
            /*ro, req, string, options, desc:"cathode", "anode"*/
        },
        "errorLedPolarity": {
        /*ro, opt, object, error LED polarity*/
            "@opt":  "cathode,anode"
            /*ro, req, string, options, desc:"cathode", "anode"*/
        },
        "buzzerPolarity": {
        /*ro, opt, object, buzzer polarity, desc:buzzer polarity:
"cathode","anode"*/
            "@opt":  "cathode,anode"
            /*ro, req, string, options, desc:"cathode", "anode"*/
        },
        "swipeInterval": {
        /*ro, opt, object, time interval of repeated authentication, desc:it is
valid for authentication modes such as fingerprint, card, face, etc., unit:
second*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "pressTimeout": {
        /*ro, opt, object, timeout to reset entry on keypad, desc:unit: second*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "enableFailAlarm":  "true,false",
        /*ro, opt, string, whether to enable excessive failed authentication
attempts alarm*/
        "maxReadCardFailNum": {
        /*ro, opt, object, maximum number of failed authentication attempts*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "enableTamperCheck":  "true,false",
        /*ro, opt, string, whether to enable tampering detection*/
        "offlineCheckTime": {
        /*ro, opt, object, time to detect after the card reader is offline,
desc:unit: second*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
```

```
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintCheckLevel": {
        /*ro, opt, object, fingerprint recognition level*/
            "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18"
            /*ro, req, string, options*/
        },
        "useLocalController":  "true,false",
        /*ro, opt, string, whether it is connected to the distributed
controller*/
        "localControllerID": {
        /*ro, opt, object, distributed controller No., desc:it is between 1 and
64, 0 (unregistered). This field is valid only when useLocalController is
"true"*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  64
            /*ro, req, int, the maximum value*/
        },
        "localControllerReaderID": {
        /*ro, opt, object, card reader ID of the distributed controller, desc:0
(unregistered). This field is valid only when useLocalController is "true"*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  4
            /*ro, req, int, the maximum value*/
        },
        "cardReaderChannel": {
        /*ro, opt, object, communication channel No. of the card reader,
desc:this field is valid only when useLocalController is "true"*/
            "@opt":  "0,1,2"
            /*ro, req, string, options*/
        },
        "fingerPrintImageQuality": {
        /*ro, opt, object, fingerprint image quality, desc:fingerprint image
quality: 1 (low quality (V1)), 2 (medium quality (V1)), 3 (high quality (V1)), 4
(highest quality (V1)), 5 (low quality (V2)), 6 (medium quality (V2)), 7 (high
quality (V2)), 8 (highest quality (V2))*/
            "@opt":  "1,2,3,4,5,6,7,8"
            /*ro, req, string, options*/
        },
        "fingerPrintContrastTimeOut": {
        /*ro, opt, object, fingerprint comparison timeout, desc:it is between 1
and 20, unit: second, 0 (infinite)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  20
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintRecogizeInterval": {
        /*ro, opt, object, fingerprint scanning interval, desc:it is between 1
and 10, unit: second, 0 (no delay)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  10
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintMatchFastMode": {
```

```
        /*ro, opt, object, fingerprint matching quick mode, desc:1 (quick mode
1), 2 (quick mode 2), 3 (quick mode 3), 4 (quick mode 4), 5 (quick mode 5), 0
(automatic)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  5
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintModuleSensitive": {
        /*ro, opt, object, fingerprint module sensitivity, desc:it is between 1
and 8*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  8
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintModuleLightCondition": {
        /*ro, opt, object, fingerprint module light condition*/
            "@opt":  "outdoor,indoor"
            /*ro, req, string, options, desc:"outdoor", "indoor"*/
        },
        "faceMatchThresholdN": {
        /*ro, opt, object, threshold of face picture 1:N comparison, desc:it is
between 0 and 100*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "faceQuality": {
        /*ro, opt, object, face picture quality, desc:it is between 0 and 100*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "faceRecogizeTimeOut": {
        /*ro, opt, object, face recognition timeout, desc:it is between 1 and
20, unit: second, 0 (infinite)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  20
            /*ro, req, int, the maximum value*/
        },
        "faceRecogizeInterval": {
        /*ro, opt, object, face recognition interval, desc:it is between 1 and
10, unit: second, 0 (no delay)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  10
            /*ro, req, int, the maximum value*/
        },
        "cardReaderFunction": {
        /*ro, opt, object, card reader type*/
            "@opt":  "fingerPrint,face,fingerVein,iris,card"
            /*ro, req, string, options, desc:"fingerPrint" (fingerprint),
"face", "fingerVein" (finger vein), “iris”*/
        },
```

```
        "cardReaderDescription": {
        /*ro, opt, object, card reader description, desc:if the card reader is
the Wiegand card reader or if offline, this field will be set to "Wiegand" or
"485Offline"*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "isSupportOfflineSendPermission":  true,
        /*ro, opt, bool, whether to enable applying permissions offline on the
card reader, desc:AIOT (Artificial Intelligence & Internet of Things) supports
apply person's permissions when the card reader is offline*/
        "faceImageSensitometry": {
        /*ro, opt, object, face picture exposure, desc:it is between 0 and
65535*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  65535
            /*ro, req, int, the maximum value*/
        },
        "livingBodyDetect":  "true,false",
        /*ro, opt, string, whether to enable human detection*/
        "faceMatchThreshold1": {
        /*ro, opt, object, threshold of face picture 1:1 comparison, desc:it is
between 0 and 100*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "buzzerTime": {
        /*ro, opt, object, buzzing duration, desc:it is between 0 and 5999,
unit: second, 0 (long buzzing)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  5999
            /*ro, req, int, the maximum value*/
        },
        "faceMatch1SecurityLevel": {
        /*ro, opt, object, security level of face 1:1 recognition*/
            "@opt":  "1,2,3"
            /*ro, req, string, options, desc:1 (normal), 2 (high), 3 (higher)*/
        },
        "faceMatchNSecurityLevel": {
        /*ro, opt, object, security level of face 1:N recognition*/
            "@opt":  "1,2,3"
            /*ro, req, string, options, desc:1 (normal), 2 (high), 3 (higher)*/
        },
        "envirMode": {
        /*ro, opt, object, environment mode of face recognition*/
            "@opt":  "indoor,other"
            /*ro, req, string, options, desc:"indoor", "other"*/
        },
        "liveDetLevelSet": {
        /*ro, opt, object, threshold level of liveness detection*/
            "@opt":  "low,middle,high,normal,professional"
            /*ro, req, string, options, desc:"low", "middle" (medium), "high"*/
```

```
        },
        "liveDetThreshold": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, range:[0,100]*/
            "@max":  100
            /*ro, req, int, range:[0,100]*/
        },
        "liveDetAntiAttackCntLimit": {
        /*ro, opt, object, number of anti-attacks of liveness detection, desc:it
is between 1 and 255. This value should be configured as the same one on both
client and device*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "enableLiveDetAntiAttack":  "true,false",
        /*ro, opt, string, whether to enable anti-attack for liveness detection,
desc:whether to enable anti-attack for liveness detection*/
        "liveDetAntiAttackLockedTime": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, range:[0,300]*/
            "@max":  300
            /*ro, req, int, range:[0,300]*/
        },
        "supportDelFPByID":  "true,false",
        /*ro, opt, string, whether the card reader supports deleting the
fingerprint by fingerprint ID, desc:"true" (yes), "false" (no)*/
        "fingerPrintCapacity": {
        /*ro, opt, object, maximum number of fingerprints that can be added*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintNum": {
        /*ro, opt, object, number of added fingerprints*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "defaultVerifyMode": {
        /*ro, opt, object, default authentication mode of the card reader,
desc:default authentication mode of the card reader (factory defaults)*/
            "@opt":
 "cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpO
rCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNo
AndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,fac
eOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw,employ
eeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCar
dOrPwOrIris,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,cardOrFpOrIrisOrPw,cardOrIri
sOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw,faceAndIrisAn
dPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw,cardAndVp,faceAndVp,fpAndVp,
pwAndVp"
            /*ro, req, string, options*/
```

```
        },
        "FPAlgorithmVersion": {
        /*ro, opt, object, fingerprint algorithm library version*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "cardReaderVersion": {
        /*ro, opt, object, card reader version*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "faceRecogizeEnable": {
        /*ro, opt, object, whether to enable facial recognition*/
            "@opt":  "1,2,3"
            /*ro, req, string, options, desc:1 (enable), 2 (disable), 3
(attendence checked in/out by recognition of multiple faces)*/
        },
        "enableReverseCardNo":  "true,false",
        /*ro, opt, string, whether to enable reversing the card No.*/
        "independSwipeIntervals": {
        /*ro, opt, object, time interval of person authentication, desc:unit:
second. This time interval is calculated for each person separately and is
different from swipeInterval*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "maskFaceMatchThresholdN": {
        /*ro, opt, object, 1:N face picture (face with mask and normal
background) comparison threshold, desc:value range: [0,100]*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "maskFaceMatchThreshold1": {
        /*ro, opt, object, 1:1 face picture (face with mask and normal
background) comparison threshold, desc:value range: [0,100]*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "faceMotionDetLevel": {
        /*ro, opt, object, face motion detection level*/
            "@opt": ["low", "meduim", "height"]
            /*ro, req, array, options, subType:string*/
        },
        "showMode": {
        /*ro, opt, object, display mode*/
            "@opt": ["concise", "normal", "advertising", "meeting",
"selfDefine"]
            /*ro, req, array, options, subType:string*/
```

```
        },
        "enableScreenOff": {
        /*ro, opt, object, whether to enable auto locking the screen*/
            "@opt": [true, false]
            /*ro, req, array, options, subType:bool*/
        },
        "screenOffTimeout": {
        /*ro, opt, object, time*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  10
            /*ro, req, int, the maximum value*/
        }
    }
}
```

13.9.9 Set the facial recognition mode parameters

URL

PUT /ISAPI/AccessControl/FaceRecognizeMode?format=json

Query Parameter

None

Request Message

```
{
    "FaceRecognizeMode": {
    /*opt, object, facial recognition mode*/
        "mode":  "normalMode"
        /*req, enum, facial recognition mode, subType:string, desc:"normalMode"
(normal mode), "deepMode" (deep mode)*/
    }
}
```

Response Message

```
{
    "requestURL":  "/ISAPI/Intelligent/FDLib/asyncImportDatas?format=json",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.9.10 Get the parameters of the facial recognition mode.

URL

GET /ISAPI/AccessControl/FaceRecognizeMode?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FaceRecognizeMode": {
    /*ro, opt, object, facial recognition mode*/
        "mode":  "normalMode"
        /*ro, req, enum, facial recognition mode, subType:string,
desc:"normalMode" (normal mode), "deepMode" (deep mode)*/
    }
}
```

13.9.11 Set the parameters of card No. authentication mode

URL

PUT /ISAPI/AccessControl/CardVerificationRule?format=json

Query Parameter

None

Request Message

```
{
    "CardVerificationRule": {
    /*req, object*/
        "cardNoLenMode":  "full",
        /*req, enum, length mode of card No. authentication (comparison),
subType:string, desc:"full", "3Bytes", "4Bytes". After the card No.
authentication (comparison) mode is switched, the device should check the card
No. compatibility*/
        "reverseCardNoEnabled":  true
        /*opt, bool*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.9.12 Get the parameters of card No. authentication mode

URL

GET /ISAPI/AccessControl/CardVerificationRule?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardVerificationRule": {
    /*ro, req, object*/
        "cardNoLenMode":  "full",
        /*ro, req, enum, length mode of card No. authentication (comparison),
subType:string, desc:"full", "3Bytes", "4Bytes". After the card No.
authentication (comparison) mode is switched, the device should check the card
No. compatibility*/
        "reverseCardNoEnabled":  true
        /*ro, opt, bool*/
    }
}
```

13.10 Time Management

13.10.1 Set device time synchronization management parameters

URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}--
>
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP
time synchronization), “manual” (manual time synchronization), “satellite”
(satellite time synchronization), “platform” (platform time synchronization),
“NONE” (time synchronization is not allowed or no time synchronization source),
“GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,
{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:00
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,
{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}-->CST-
8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min,
desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--opt, bool, whether the time returned by the current device is that in
the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,
{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as
platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,
{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via
platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-
->1
  </platformNo>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string,
protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
```

```
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device
Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK”
(succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML
Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual
time synchronization of central analysis cluster, this field is returned if time
synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time,
range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
</ResponseStatus>
```

13.10.2 Get device time synchronization management parameters

URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string,
protocolVersion}-->
  <timeMode>
```

```
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP”
(NTP time synchronization), “manual” (manual time synchronization), “satellite”
(satellite time synchronization), “platform” (platform time synchronization),
“NONE” (time synchronization is not allowed or no time synchronization source),
“GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,
{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:00
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,
{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}-->CST-
8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min,
desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST
(Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,
{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as
platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,
{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via
platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-
->1
  </platformNo>
</Time>
```

13.10.3 Get the capability of device time synchronization management

URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string,
protocolVersion}-->
```

```
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and, {$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and, {$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI: /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
```

```
</Time>
```

13.11 Access Control Device Management

13.11.1 Get the capability of getting the working status of

the access controller

URL

GET /ISAPI/AccessControl/AcsWorkStatus/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsWorkStatus": {
    /*ro, req, object*/
        "doorLockStatus": {
        /*ro, opt, object, door lock status (relay status): 0-normally close, 1-normally open, 2-short-circuit alarm, 3-broken-circuit alarm, 4-exception alarm*/
            "@opt":  "0,1,2,3,4"
            /*ro, opt, string*/
        },
        "doorStatus": {
        /*ro, opt, object, door (floor) status: 1-sleep, 2-remain unlocked (free), 3-remain locked (disabled), 4-normal status (controlled)*/
            "@opt":  "1,2,3,4"
            /*ro, opt, string*/
        },
        "magneticStatus": {
        /*ro, opt, object, magnetic contact status: 0-normally close,1-normally open, 2-short-circuit alarm, 3-broken-circuit alarm, 4-exception alarm*/
            "@opt":  "0,1,2,3,4"
            /*ro, opt, string, magnetic contact status*/
        },
        "caseStatus": {
        /*ro, opt, object, event trigger status*/
            "@min":  1,
            /*ro, opt, int, event trigger status*/
            "@max":  1
            /*ro, opt, int, event trigger status*/
        },
        "batteryVoltage": {
        /*ro, opt, object, storage battery power voltage, the actual value will be 10 times of this value*/
            "@min":  1,
            /*ro, opt, int, storage battery power voltage*/
            "@max":  1
            /*ro, opt, int, storage battery power voltage*/
        },
```

```
        "batteryLowVoltage":  "true,false",
        /*ro, opt, string, whether the storage battery is in low voltage status:
"true"-yes,"false"-no*/
        "powerSupplyStatus": {
        /*ro, opt, object, device power supply status: "ACPowerSupply"-
alternative current, "BatteryPowerSupply"-storage battery power supply*/
            "@opt":  "ACPowerSupply,BatteryPowerSupply"
            /*ro, opt, string*/
        },
        "multiDoorInterlockStatus": {
        /*ro, opt, object, multi-door interlocking status: "close"-disabled,
"open"-enabled*/
            "@opt":  "close,open"
            /*ro, opt, string*/
        },
        "antiSneakStatus": {
        /*ro, opt, object, anti-passback status: "close"-disabled,"open"-
enabled*/
            "@opt":  "close,open"
            /*ro, opt, string*/
        },
        "hostAntiDismantleStatus": {
        /*ro, opt, object, tampering status of the access control device:
"close"-disabled, "open"-enabled*/
            "@opt":  "close,open"
            /*ro, opt, string*/
        },
        "indicatorLightStatus": {
        /*ro, opt, object, indicator status: "offLine"-offline, "onLine"-
online*/
            "@opt":  "offLine,onLine"
            /*ro, opt, string*/
        },
        "cardReaderOnlineStatus": {
        /*ro, opt, object, online status of the authentication unit*/
            "@min":  1,
            /*ro, opt, int, the minimum value, range:[1,8]*/
            "@max":  1
            /*ro, opt, int, the maximum value, range:[1,8]*/
        },
        "netReaderOnlineStatus": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int, the minimum value, range:[1,8]*/
            "@max":  1
            /*ro, opt, int, the maximum value, range:[1,8]*/
        },
        "POEPortList": {
        /*ro, opt, object*/
            "@size":  8,
            /*ro, opt, int, range:[1,8]*/
            "port": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int, the minimum value, range:[1,8]*/
                "@max":  8
                /*ro, opt, int, the maximum value, range:[1,8]*/
            },
```

```
            "readerID": {
            /*ro, opt, object*/
                "@size":  8,
                /*ro, opt, int, range:[1,8]*/
                "@min":  1,
                /*ro, opt, int, the minimum value, range:[1,8]*/
                "@max":  8
                /*ro, opt, int, the maximum value, range:[1,8]*/
            }
        },
        "cardReaderAntiDismantleStatus": {
        /*ro, opt, object, tampering status of the authentication unit*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "cardReaderVerifyMode": {
        /*ro, opt, object, current authentication mode of the authentication
unit, desc:1-sleep,2-card+password,3-card,4-card or password,5-fingerprint,6-
fingerprint+password,7-fingerprint or card,8-fingerprint+card,9-
fingerprint+card+password,10-face or fingerprint or card or password,11-
face+fingerprint,12-face+password,13-face+card,14-face,15-employee
No.+password,16-fingerprint or password,17-employee No.+fingerprint,18-employee
No.+fingerprint+password,19-face+fingerprint+card,20-
face+password+fingerprint,21-employee No.+face,22-face or face+card,23-
fingerprint or face,24-card or face or password,25-card or face,26-card or face
or fingerprint,27-card or fingerprint or password*/
            "@opt":
 "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,
30,31,32,33,34"
            /*ro, opt, string*/
        },
        "setupAlarmStatus": {
        /*ro, opt, object, No. of armed input port*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "alarmInStatus": {
        /*ro, opt, object, No. of input port with alarms*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "alarmOutStatus": {
        /*ro, opt, object, No. of output port with alarms*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "cardNum": {
        /*ro, opt, object, number of added cards*/
            "@min":  1,
            /*ro, opt, int*/
```

```
            "@max":  1
            /*ro, opt, int*/
        },
        "fireAlarmStatus": {
        /*ro, opt, object, fire alarm status*/
            "@opt":  "normal,shortCircuit,brokenCircuit"
            /*ro, opt, string, "normal","shortCircuit"-short-circuit alarm,
"brokenCircuit"-broken-circuit alarm*/
        },
        "batteryChargeStatus": {
        /*ro, opt, object, battery charging status*/
            "@opt":  "charging,uncharged"
            /*ro, opt, string, "charging","uncharged”*/
        },
        "masterChannelControllerStatus": {
        /*ro, opt, object, online status of the main lane controller*/
            "@opt":  "offLine,onLine"
            /*ro, opt, string, "offLine"-offline, "onLine"-online*/
        },
        "slaveChannelControllerStatus": {
        /*ro, opt, object, online status of the sub lane controller*/
            "@opt":  "offLine,onLine"
            /*ro, opt, string, "offLine"-offline,"onLine"-online*/
        },
        "antiSneakServerStatus": {
        /*ro, opt, object, anti-passback server status: "disable"-disabled,
"normal", "disconnect"-disconnected*/
            "@opt":  "disable,normal,disconnect"
            /*ro, opt, string*/
        },
        "netStatus": {
        /*ro, opt, object, network connection status*/
            "@opt": ["connect", "disconnect", "ipConflict"]
            /*ro, opt, array, subType:string*/
        },
        "InterfaceStatusList": {
        /*ro, opt, object*/
            "@size":  2,
            /*ro, opt, int*/
            "id": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  2
                /*ro, opt, int*/
            },
            "netStatus": {
            /*ro, opt, object*/
                "@opt": ["connect", "disconnect", "ipConflict"]
                /*ro, opt, array, subType:string*/
            }
        },
        "signalStatus": {
        /*ro, opt, object, signal status*/
            "@opt": ["noSignal", "2G", "3G", "4G", "5G"]
            /*ro, opt, array, subType:string*/
        },
        "sipStatus": {
```

```
        /*ro, opt, object*/
            "@opt": ["connect", "disconnect", "unregistered"]
            /*ro, opt, array, subType:string*/
        },
        "ezvizStatus": {
        /*ro, opt, object*/
            "@opt": ["unregistered", "notAdd", "connect"]
            /*ro, opt, array, subType:string*/
        },
        "voipStatus": {
        /*ro, opt, object*/
            "@opt": ["unregistered", "connect"]
            /*ro, opt, array, subType:string*/
        },
        "wifiStatus": {
        /*ro, opt, object*/
            "@opt": ["connect", " disconnect", "connecting", "noModule"]
            /*ro, opt, array, subType:string*/
        },
        "TFCardStatus": {
        /*ro, opt, object*/
            "@opt": ["mounted", "unmounted"]
            /*ro, opt, array, subType:string*/
        },
        "acrossHostInterlockStatus": {
        /*ro, opt, object*/
            "@opt": ["disable", "normal", "disconnect"]
            /*ro, opt, array, subType:string*/
        },
        "faceReceivingStatus": {
        /*ro, opt, object*/
            "@opt": ["receiving", "received"]
            /*ro, opt, array, subType:string*/
        },
        "QRCodeReaderStatusList": {
        /*ro, opt, object*/
            "id": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  2
                /*ro, opt, int*/
            },
            "QRCodeReaderStatus": {
            /*ro, opt, object*/
                "@opt": ["onLine", "offLine"]
                /*ro, opt, array, subType:string*/
            }
        }
    }
}
```

13.11.2 Get the working status of the access controller

URL

GET /ISAPI/AccessControl/AcsWorkStatus?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsWorkStatus": {
    /*ro, req, object*/
        "doorLockStatus":  [1, 2, 1, 2],
        /*ro, opt, enumarray, door lock status (relay status), subType:int,
desc:door lock status (relay status): 0 (normally close), 1 (normally open), 2
(short-circuit alarm), 3 (broken-circuit alarm), 4 (exception alarm). For
example, [1,2,1,2] indicates that door lock 1 is normally open, door lock 2
triggers short-circuit alarm, door lock 3 is normally open, and door lock 4
triggers short-circuit alarm*/
        "doorStatus":  [1, 2, 1, 2],
        /*ro, opt, enumarray, door (floor) status, subType:int, desc:door
(floor) status: 1 (sleep), 2 (remain unlocked (free)), 3 (remain locked
(disabled)), 4 (normal status (controlled)). For example, [1,2,1,2] indicates
that door 1 is sleeping, door 2 remains unlocked, door 3 is sleeping, and door 4
remains unlocked*/
        "magneticStatus":  [1, 2, 1, 2],
        /*ro, opt, enumarray, magnetic contact status, subType:int,
desc:magnetic contact status: 0 (normally close), 1 (normally open), 2 (short-
circuit alarm), 3 (broken-circuit alarm), 4 (exception alarm). For example,
[1,2,1,2] indicates that magnetic contact No.1 is normally open, magnetic
contact No.2 triggers short-circuit alarm, magnetic contact No.3 is normally
open, and magnetic contact No.4 triggers short-circuit alarm*/
        "caseStatus": [1, 3, 5],
        /*ro, opt, array, event trigger status, subType:int, desc:event trigger
status, e.g., [1,3,5] indicates that event trigger No.1, No.3, and No.5 have
input*/
        "batteryVoltage":  50,
        /*ro, opt, int, storage battery power voltage, desc:storage battery
power voltage, the actual value will be 10 times of this value, unit: Volt*/
        "batteryLowVoltage":  false,
        /*ro, opt, bool, whether the storage battery is in low voltage status,
desc:"true" (yes), "false" (no)*/
        "powerSupplyStatus":  "ACPowerSupply",
        /*ro, opt, enum, device power supply status, subType:string,
desc:"ACPowerSupply" (alternative current), "BatteryPowerSupply" (storage
battery power supply)*/
        "multiDoorInterlockStatus":  "close",
        /*ro, opt, enum, multi-door interlocking status, subType:string,
desc:"close" (disabled), "open" (enabled)*/
        "antiSneakStatus":  "open",
        /*ro, opt, enum, anti-passback status, subType:string, desc:"close"
(disabled), "open" (enabled)*/
        "hostAntiDismantleStatus":  "open",
        /*ro, opt, enum, tampering status of the access control device,
subType:string, desc:"close" (disabled), "open" (enabled)*/
        "indicatorLightStatus":  "onLine",
```

```
        /*ro, opt, enum, indicator status, subType:string, desc:"offLine" (offline), "onLine" (online)*/
        "cardReaderOnlineStatus": [1, 3, 5],
        /*ro, opt, array, online status of the authentication unit, subType:int, desc:online status of the authentication unit, e.g., [1,3,5] indicates that authentication unit No.1, No.3, and No.5 are online*/
        "netReaderOnlineStatus": [1, 3, 5],
        /*ro, opt, array, subType:int*/
        "POEPortList": [
        /*ro, opt, array, subType:object, range:[1,8]*/
            {
                "port":  1,
                /*ro, req, int, range:[1,8]*/
                "readerID": [1, 2, 3]
                /*ro, opt, array, subType:int*/
            }
        ],
        "cardReaderAntiDismantleStatus": [1, 3, 5],
        /*ro, opt, array, tampering status of the authentication unit, subType:int, desc:tampering status of the authentication unit, e.g., [1,3,5] indicates that the tampering function of authentication unit No.1, No.3, and No.5 is enabled*/
        "cardReaderVerifyMode":  [3, 5, 3, 5],
        /*ro, opt, enumarray, current authentication mode of the authentication unit, subType:int, desc:1 (sleep), 2 (card + password), 3 (card), 4 (card or password), 5 (fingerprint), 6 (fingerprint + password), 7 (fingerprint or card), 8 (fingerprint + card), 9 (fingerprint + card + password),10 (face or fingerprint or card or password), 11 (face + fingerprint), 12 (face + password), 13 (face + card), 14 (face), 15 (employee No. + password), 16 (fingerprint or password), 17 (employee No. + fingerprint), 18 (employee No. + fingerprint + password), 19 (face + fingerprint + card), 20 (face + password + fingerprint), 21 (employee No. + face), 22 (face or face + card), 23 (fingerprint or face), 24 (card or face or password), 25 (card or face), 26 (card or face or fingerprint), 27 (card or fingerprint or password). For example, [3,5,3,5] indicates that the authentication mode of authentication unit 1 is "card", the authentication mode of authentication unit 2 is "fingerprint", the authentication mode of authentication unit 3 is "card", and the authentication mode of authentication unit 4 is "fingerprint"*/
        "setupAlarmStatus": [1, 3, 5],
        /*ro, opt, array, No. of armed input port, subType:int, desc:No. of armed input port, e.g., [1,3,5] indicates that input port No.1, No.3, and No.5 are armed*/
        "alarmInStatus": [1, 3, 5],
        /*ro, opt, array, No. of input port with alarms, subType:int, desc:No. of input port with alarms, e.g., [1,3,5] indicates that input port No.1, No.3, and No.5 trigger alarms*/
        "alarmOutStatus": [1, 3, 5],
        /*ro, opt, array, No. of output port with alarms, subType:int, desc:No. of output port with alarms, e.g., [1,3,5] indicates that output port No.1, No.3, and No.5 trigger alarms*/
        "cardNum":  3,
        /*ro, opt, int, number of added cards, desc:number of added cards*/
        "fireAlarmStatus":  "normal",
        /*ro, opt, enum, fire alarm status, subType:string, desc:fire alarm status: "normal", "shortCircuit" (short-circuit alarm), "brokenCircuit" (broken-circuit alarm)*/
        "batteryChargeStatus":  "charging",
```

```
        /*ro, opt, enum, battery charging status, subType:string, desc:battery
charging status: "charging", "uncharged"*/
        "masterChannelControllerStatus":  "onLine",
        /*ro, opt, enum, online status of the main-lane controller,
subType:string, desc:online status of the main lane controller: "offLine"
(offline), "onLine" (online)*/
        "slaveChannelControllerStatus":  "onLine",
        /*ro, opt, enum, online status of the sub lane controller,
subType:string, desc:online status of the sub lane controller: "offLine"
(offline), "onLine" (online)*/
        "antiSneakServerStatus":  "normal",
        /*ro, opt, enum, anti-passback server status, subType:string, desc:anti-
passback server status: "disable" (disabled), "normal", "disconnect"
(disconnected)*/
        "netStatus":  "connect",
        /*ro, opt, enum, subType:string*/
        "InterfaceStatusList": [
        /*ro, opt, array, subType:object*/
            {
                "id":  1,
                /*ro, opt, int*/
                "netStatus":  "connect"
                /*ro, opt, enum, subType:string*/
            }
        ],
        "signalStatus":  "noSignal",
        /*ro, opt, enum, subType:string*/
        "sipStatus":  "connect",
        /*ro, opt, enum, subType:string*/
        "ezvizStatus":  "unregistered",
        /*ro, opt, enum, subType:string*/
        "voipStatus":  "unregistered",
        /*ro, opt, enum, subType:string*/
        "wifiStatus":  "connect",
        /*ro, opt, enum, subType:string*/
        "TFCardStatus":  "mounted",
        /*ro, opt, enum, subType:string*/
        "acrossHostInterlockStatus":  "disable",
        /*ro, opt, enum, subType:string*/
        "faceReceivingStatus":  "receiving",
        /*ro, opt, enum, subType:string*/
        "QRCodeReaderStatusList": [
        /*ro, opt, array, subType:object*/
            {
                "id":  1,
                /*ro, opt, int*/
                "QRCodeReaderStatus":  "onLine"
                /*ro, opt, enum, subType:string*/
            }
        ]
    }
}
```

13.11.3 Get the configuration capability of the access controller

URL

GET /ISAPI/AccessControl/AcsCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsCfg": {
    /*ro, req, object*/
        "RS485Backup":  "true,false",
        /*ro, opt, string, whether to enable downstream RS-485 communication
redundancy, desc:"true" (yes), "false" (no)*/
        "showCapPic":  "true,false",
        /*ro, opt, string, whether to display the captured picture, desc:"true"
(yes), "false" (no)*/
        "showUserInfo":  "true,false",
        /*ro, opt, string, whether to display user information, desc:"true"
(yes), "false" (no)*/
        "overlayUserInfo":  "true,false",
        /*ro, opt, string, whether to overlay user information, desc:"true"
(yes), "false" (no)*/
        "voicePrompt":  "true,false",
        /*ro, opt, string, whether to enable voice prompt, desc:"true" (yes),
"false" (no)*/
        "uploadCapPic":  "true,false",
        /*ro, opt, string, whether to upload the picture from linked capture,
desc:"true" (yes), "false" (no)*/
        "saveCapPic":  "true,false",
        /*ro, opt, string, whether to save the captured picture, desc:"true"
(yes), "false" (no)*/
        "inputCardNo":  "true,false",
        /*ro, opt, string, whether to allow inputting card No. on keypad,
desc:"true" (yes), "false" (no)*/
        "enableWifiDetect":  "true,false",
        /*ro, opt, string, whether to enable Wi-Fi probe, desc:"true" (yes),
"false" (no)*/
        "enable3G4G":  "true,false",
        /*ro, opt, string, whether to enable 3G/4G, desc:"true" (yes), "false"
(no)*/
        "protocol": {
        /*ro, opt, object, communication protocol type of the card reader,
desc:"Private" (private protocol), "OSDP" (OSDP protocol)*/
            "@opt":  "Private,OSDP"
            /*ro, opt, string, optional item*/
        },
        "enableCaptureCertificate":  "true,false",
        /*ro, opt, string, whether to enable capturing the ID picture, desc:true
(yes), false (no). If this node does not exist, it indicates that this function
is not supported*/
        "showPicture":  "true,false",
        /*ro, opt, string, whether to display the authenticated picture,
desc:"true" (yes), "false" (no)*/
        "showEmployeeNo":  "true,false",
```

```
        /*ro, opt, string, whether to display the authenticated employee ID,
desc:"true" (yes), "false" (no)*/
        "showName":  "true,false",
        /*ro, opt, string, whether to display the authenticated name,
desc:"true" (yes), "false" (no)*/
        "desensitiseEmployeeNo": {
        /*ro, opt, object, whether to enable employee No. de-identification for
local UI display*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "desensitiseName": {
        /*ro, opt, object, whether to enable name de-identification for local UI
display*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalEnabled": {
        /*ro, opt, object, whether to enable temperature measurement*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalMode": {
        /*ro, opt, object, whether to enable temperature measurement only mode:
true-enable (only for temperature measurement),false-disable (default)*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalPictureEnabled": {
        /*ro, opt, object, whether to enable uploading visible light pictures in
temperature measurement only mode: true-enable,false-disable (default). This
field is used to control uploading captured pictures and visible light pictures,
desc:whether to enable uploading visible light pictures in temperature
measurement only mode: true-enable,false-disable (default). This field is used
to control uploading captured pictures and visible light pictures*/
            "@opt": [true, false]
            /*ro, opt, array, whether to enable uploading visible light pictures
in temperature measurement only mode, subType:bool*/
        },
        "isSupportThermalIp":  true,
        /*ro, opt, bool, whether it supports configuring IP address of the
thermography device, desc:whether it supports configuring IP address of the
thermography device: true-yes,this field is not returned-no*/
        "highestThermalThreshold": {
        /*ro, opt, object, maximum value of the temperature threshold,
desc:float,upper limit of the temperature threshold which is accurate to one
decimal place,unit: Celsius*/
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  2.0
            /*ro, opt, float*/
        },
        "lowestThermalThreshold": {
        /*ro, opt, object, minimum value of the temperature threshold,
desc:float,lower limit of the temperature threshold which is accurate to one
decimal place,unit: Celsius*/
            "@min":  1.0,
            /*ro, opt, float*/
```

```
            "@max":  2.0
            /*ro, opt, float*/
        },
        "thermalDoorEnabled": {
        /*ro, opt, object, whether to open the door according to the temperature
threshold, desc:whether to open the door when the temperature is above the upper
limit (highestThermalThreshold) or below the lower limit
(lowestThermalThreshold) of the threshold: true (open the door), false (not open
the door (default))*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "QRCodeEnabled": {
        /*ro, opt, object, whether to enable QR code function, desc:whether to
enable QR code function: true-enable,false-disable (default)*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "remoteCheckDoorEnabled": {
        /*ro, opt, object, whether to enable controlling the door by remote
verification, desc:whether to enable controlling the door by remote
verification: true-control,false-not control (default)*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "checkChannelType": {
        /*ro, opt, object, verification channel type, desc:verification channel
type*/
            "@opt": ["Ezviz", "ISUP", "ISAPI", "PrivateSDK", "ISAPIListen"]
            /*ro, opt, array, subType:string*/
        },
        "isSupportChannelIp":  true,
        /*ro, opt, bool, whether it supports configuring IP address of the
verification channel, desc:whether it supports configuring IP address of the
verification channel: true-yes,this field is not returned-no*/
        "needDeviceCheck": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "uploadVerificationPic":  "true,false",
        /*ro, opt, string, whether to upload the authenticated picture,
desc:"true"-yes, "false"-no*/
        "saveVerificationPic":  "true,false",
        /*ro, opt, string, whether to save the authenticated picture,
desc:"true" (yes), "false" (no)*/
        "saveFacePic":  "true,false",
        /*ro, opt, string, whether to save the registered face picture,
desc:"true" (yes), "false" (no)*/
        "thermalUnit": {
        /*ro, opt, object, temperature unit, desc:object,temperature unit:
"celsius" (default),"fahrenheit"*/
            "@opt": ["celsius", "fahrenheit"]
            /*ro, opt, array, subType:string*/
        },
        "highestThermalThresholdF": {
        /*ro, opt, object, maximum value of the temperature threshold, desc:the
value is accurate to one decimal place, and the unit is Fahrenheit*/
```

```
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  1.0
            /*ro, opt, float*/
        },
        "lowestThermalThresholdF": {
        /*ro, opt, object, minimum value of the temperature threshold, desc:the
value is accurate to one decimal place, and the unit is Fahrenheit*/
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  1.0
            /*ro, opt, float*/
        },
        "enable5G": {
        /*ro, opt, object, whether to enable 5G*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalCompensation": {
        /*ro, opt, object, temperature compensation*/
            "@min":  -99.9,
            /*ro, opt, float*/
            "@max":  99.9
            /*ro, opt, float*/
        },
        "externalCardReaderEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "combinationAuthenticationTimeout": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,20], step:1, unit:s*/
            "@max":  20
            /*ro, req, int, range:[1,20], step:1, unit:s*/
        },
        "combinationAuthenticationLimitOrder": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "passwordEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "showGender": {
        /*ro, opt, object, whether to display gender*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "showSignInTime": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "showsCustomInfo": {
```

```
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "showMobileWebQRCode": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "fireAlarmInputType": {
        /*ro, opt, object*/
            "@opt": ["alwaysClose", "alwaysOpen"]
            /*ro, req, array, subType:string*/
        },
        "buzzerEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        }
    }
}
```

13.11.4 Set the parameters of the access controller

URL

PUT /ISAPI/AccessControl/AcsCfg?format=json

Query Parameter

None

Request Message

```
{
    "AcsCfg": {
    /*req, object, parameters of the access controller*/
        "RS485Backup":  true,
        /*opt, bool, whether to enable downstream RS-485 communication
redundancy*/
        "showCapPic":  true,
        /*opt, bool, whether to display the captured picture*/
        "showUserInfo":  true,
        /*opt, bool, whether to display user information*/
        "overlayUserInfo":  true,
        /*opt, bool, whether to overlay user information*/
        "voicePrompt":  true,
        /*opt, bool, whether to enable voice prompt*/
        "uploadCapPic":  true,
        /*opt, bool, whether to upload the picture from linked capture*/
        "saveCapPic":  true,
        /*opt, bool, whether to save the captured picture*/
        "inputCardNo":  true,
        /*opt, bool, whether to allow inputting card No. on keypad*/
        "enableWifiDetect":  true,
        /*opt, bool, whether to enable Wi-Fi probe*/
        "enable3G4G":  true,
        /*opt, bool, whether to enable 3G/4G*/
```

```
        "protocol":  "Private",
        /*opt, enum, communication protocol type of the card reader,
subType:string, desc:"Private" (private protocol), "OSDP" (OSDP protocol)*/
        "enableCaptureCertificate":  true,
        /*opt, bool, whether to enable capturing the ID picture, desc:true
(yes), false (no). If this node does not exist, it indicates that this function
is not supported*/
        "showPicture":  true,
        /*opt, bool, whether to display the authenticated picture*/
        "showEmployeeNo":  true,
        /*opt, bool, whether to display the authenticated employee ID*/
        "showName":  true,
        /*opt, bool, whether to display the authenticated name*/
        "desensitiseEmployeeNo":  true,
        /*opt, bool, whether to enable employee No. de-identification for local
UI display, dep:or,{$.AcsCfg.showEmployeeNo,eq,true}*/
        "desensitiseName":  true,
        /*opt, bool, whether to enable name de-identification for local UI
display, dep:or,{$.AcsCfg.showName,eq,true}*/
        "thermalEnabled":  true,
        /*opt, bool, whether to enable temperature measurement*/
        "thermalMode":  true,
        /*opt, bool, whether to enable temperature measurement only mode*/
        "thermalPictureEnabled":  true,
        /*opt, bool, whether to enable uploading visible light pictures in
temperature measurement only mode*/
        "thermalIp":  "192.168.1.1",
        /*opt, string, IP address of the thermography device, desc:for access
control devices, each device only requires one IP address; for metal detector
doors, this field does not need to be configured*/
        "highestThermalThreshold":  37.3,
        /*opt, float, maximum value of the temperature threshold*/
        "lowestThermalThreshold":  38.5,
        /*opt, float, minimum value of the temperature threshold*/
        "thermalDoorEnabled":  false,
        /*opt, bool, whether to open the door when the temperature is above the
upper limit (highestThermalThreshold) or below the lower limit
(lowestThermalThreshold) of the threshold: true-open the door,false-not open the
door (default), desc:whether to open the door when the temperature is above the
upper limit (highestThermalThreshold) or below the lower limit
(lowestThermalThreshold) of the threshold: true (open the door), false (not open
the door (default))*/
        "QRCodeEnabled":  false,
        /*opt, bool, whether to enable QR code function*/
        "remoteCheckDoorEnabled":  false,
        /*opt, bool, whether to enable controlling the door by remote
verification*/
        "checkChannelType":  "Ezviz",
        /*opt, enum, verification channel type, subType:string,
desc:verification channel type: "Ezviz"-EZVIZ channel,"ISUP"-ISUP
channel,"ISAPI"-ISAPI channel,"PrivateSDK"-private SDK channel,"ISAPIListen"-
ISAPI listening channel. This field is valid when remoteCheckDoorEnabled is
true*/
        "channelIp":  "test",
        /*opt, string, IP address of the verification channel, desc:this field
is valid when checkChannelType is "PrivateSDK"*/
        "needDeviceCheck":  true,
        /*opt, bool, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}*/
```

```
        "uploadVerificationPic":  true,
        /*opt, bool, whether to upload the authenticated picture*/
        "saveVerificationPic":  true,
        /*opt, bool, whether to save the authenticated picture*/
        "saveFacePic":  true,
        /*opt, bool, whether to save the registered face picture*/
        "thermalUnit":  "celsius",
        /*opt, enum, temperature unit, subType:string, desc:"celsius",
"fahrenheit"*/
        "highestThermalThresholdF":  1.0,
        /*opt, float, the maximum value of the temperature threshold, desc:the
value is accurate to one decimal place, and the unit is Fahrenheit this node is
used to check whether to open the door when the temperature is higher than the
threshold*/
        "lowestThermalThresholdF":  1.0,
        /*opt, float, the minimum value of the temperature threshold, desc:the
value is accurate to one decimal place, and the unit is Fahrenheit this node is
used to check whether to open the door when the temperature is higher than the
threshold*/
        "enable5G":  true,
        /*opt, bool, whether to enable 5G*/
        "thermalCompensation":  1.0,
        /*opt, float, temperature compensation, desc:float,temperature
compensation,the value is accurate to one decimal place. The unit depends on the
node thermalUnit. If the node thermalUnit does not exist,the default unit is
Celsius*/
        "externalCardReaderEnabled":  true,
        /*opt, bool*/
        "combinationAuthenticationTimeout":  1,
        /*opt, int, range:[1,20], step:1, unit:s*/
        "combinationAuthenticationLimitOrder":  true
        /*opt, bool*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error description*/
}
```

13.11.5 Get the configuration parameters of the access controller

URL

GET /ISAPI/AccessControl/AcsCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsCfg": {
    /*ro, req, object*/
        "RS485Backup":  true,
        /*ro, opt, bool, whether to enable downstream RS-485 communication
redundancy*/
        "showCapPic":  true,
        /*ro, opt, bool, whether to display the captured picture*/
        "showUserInfo":  true,
        /*ro, opt, bool, whether to display user information*/
        "overlayUserInfo":  true,
        /*ro, opt, bool, whether to overlay user information*/
        "voicePrompt":  true,
        /*ro, opt, bool, whether to enable voice prompt*/
        "uploadCapPic":  true,
        /*ro, opt, bool, whether to upload the picture from linked capture*/
        "saveCapPic":  true,
        /*ro, opt, bool, whether to save the captured picture*/
        "inputCardNo":  true,
        /*ro, opt, bool, whether to allow inputting card No. on keypad*/
        "enableWifiDetect":  true,
        /*ro, opt, bool, whether to enable Wi-Fi probe*/
        "enable3G4G":  true,
        /*ro, opt, bool, whether to enable 3G/4G*/
        "protocol":  "Private",
        /*ro, opt, enum, communication protocol type of the card reader,
subType:string, desc:"Private” (private protocol), "OSDP” (OSDP protocol)*/
        "enableCaptureCertificate":  true,
        /*ro, opt, bool, whether to enable capturing the ID picture, desc:true
(yes), false (no). If this node does not exist, it indicates that this function
is not supported*/
        "showPicture":  true,
        /*ro, opt, bool, whether to display the authenticated picture*/
        "showEmployeeNo":  true,
        /*ro, opt, bool, whether to display the authenticated employee ID*/
        "showName":  true,
        /*ro, opt, bool, whether to display the authenticated name*/
        "desensitiseEmployeeNo":  true,
        /*ro, opt, bool, whether to enable employee No. de-identification for
local UI display, dep:or,{$.AcsCfg.showEmployeeNo,eq,true}*/
        "desensitiseName":  true,
        /*ro, opt, bool, whether to enable name de-identification for local UI
display, dep:or,{$.AcsCfg.showName,eq,true}*/
        "thermalEnabled":  true,
        /*ro, opt, bool, whether to enable temperature measurement*/
        "thermalMode":  true,
        /*ro, opt, bool, whether to enable temperature measurement only mode*/
        "thermalPictureEnabled":  true,
```

```
        /*ro, opt, bool, whether to enable uploading visible light pictures in
temperature measurement only mode: true-enable,false-disable (default). This
field is used to control uploading captured pictures and visible light
pictures*/
        "thermalIp":  "192.168.1.1",
        /*ro, opt, string, IP address of the thermography device, desc:for
access control devices, each device only requires one IP address; for metal
detector doors, this field does not need to be configured*/
        "highestThermalThreshold":  37.3,
        /*ro, opt, float, upper limit of the temperature threshold*/
        "lowestThermalThreshold":  38.5,
        /*ro, opt, float, lower limit of the temperature threshold*/
        "thermalDoorEnabled":  false,
        /*ro, opt, bool, whether to open the door according to the temperature
threshold, desc:whether to open the door when the temperature is above the upper
limit (highestThermalThreshold) or below the lower limit
(lowestThermalThreshold) of the threshold: true (open the door), false (not open
the door (default))*/
        "QRCodeEnabled":  false,
        /*ro, opt, bool, whether to enable QR code function*/
        "remoteCheckDoorEnabled":  false,
        /*ro, opt, bool, whether to enable controlling the door by remote
verification*/
        "checkChannelType":  "Ezviz",
        /*ro, opt, enum, verification channel type, subType:string,
desc:verification channel type*/
        "channelIp":  "test",
        /*ro, opt, string, IP address of the verification channel, desc:this
field is valid when checkChannelType is "PrivateSDK"*/
        "needDeviceCheck":  true,
        /*ro, opt, bool, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}*/
        "uploadVerificationPic":  true,
        /*ro, opt, bool, whether to upload the authenticated picture*/
        "saveVerificationPic":  true,
        /*ro, opt, bool, whether to save the authenticated picture*/
        "saveFacePic":  true,
        /*ro, opt, bool, whether to save the registered face picture*/
        "thermalUnit":  "celsius",
        /*ro, opt, enum, temperature unit, subType:string, desc:"celsius",
"fahrenheit"*/
        "highestThermalThresholdF":  1.0,
        /*ro, opt, float, the maximum value of the temperature threshold,
desc:the value is accurate to one decimal place, and the unit is Fahrenheit this
node is used to check whether to open the door when the temperature is higher
than the threshold*/
        "lowestThermalThresholdF":  1.0,
        /*ro, opt, float, the minimum value of the temperature threshold,
desc:the value is accurate to one decimal place, and the unit is Fahrenheit this
node is used to check whether to open the door when the temperature is higher
than the threshold*/
        "enable5G":  true,
        /*ro, opt, bool, whether to enable 5G*/
        "thermalCompensation":  1.0,
        /*ro, opt, float, temperature compensation, desc:float,temperature
compensation,the value is accurate to one decimal place. The unit depends on the
node thermalUnit. If the node thermalUnit does not exist,the default unit is
Celsius*/
        "externalCardReaderEnabled":  true,
```

```
        /*ro, opt, bool*/
        "combinationAuthenticationTimeout":  1,
        /*ro, opt, int, range:[1,20], step:1, unit:s*/
        "combinationAuthenticationLimitOrder":  true,
        /*ro, opt, bool*/
        "passwordEnabled":  true,
        /*ro, opt, bool*/
        "showGender":  true,
        /*ro, opt, bool, whether to display gender*/
        "showSignInTime":  true,
        /*ro, opt, bool*/
        "showsCustomInfo":  true,
        /*ro, opt, bool*/
        "showMobileWebQRCode":  true,
        /*ro, opt, bool*/
        "fireAlarmInputType":  "alwaysOpen",
        /*ro, opt, enum, subType:string*/
        "buzzerEnabled":  true
        /*ro, opt, bool*/
    }
}
```

13.12 Access Control Event Management

13.12.1 Get the total number of access control events by specific conditions

URL

POST /ISAPI/AccessControl/AcsEventTotalNum?format=json

Query Parameter

None

Request Message

```
{
    "AcsEventTotalNumCond": {
    /*req, object*/
        "major":  1,
        /*req, int, major alarm type, desc:(the type value should be transformed
to the decimal number), refer to Access Control Event Types for details*/
        "minor":  1024,
        /*req, int, sub type, step:1, desc:(the type value should be transformed
to the decimal number),refer to Access Control Event Types for details*/
        "startTime":  "1970-01-01+08:00",
        /*opt, date, start time (UTC time)*/
        "endTime":  "1970-01-01+08:00",
        /*opt, date, end time (UTC time)*/
        "cardNo":  "test",
        /*opt, string, card No.*/
        "name":  "test",
        /*opt, string, name of the card holder*/
        "picEnable":  true,
        /*opt, bool, whether to upload the picture along with the event
information, desc:whether to contain pictures: "true"-yes,"false"-no*/
        "beginSerialNo":  1,
```

```
        /*opt, int, start serial No.*/
        "endSerialNo":  100,
        /*opt, int, end serial No.*/
        "employeeNoString":  "test"
        /*opt, string, employee No. (person ID), range:[1,32]*/
    }
}
```

Response Message

```
{
    "AcsEventTotalNum": {
    /*ro, req, object*/
        "totalNum":  1,
        /*ro, req, int, total number of events that match the search
conditions*/
        "existedEventNum":  1
        /*ro, opt, int*/
    }
}
```

13.12.2 Get the capability of searching for access control events

URL

GET /ISAPI/AccessControl/AcsEvent/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsEvent": {
    /*ro, req, object, access control events*/
        "AcsEventCond": {
        /*ro, opt, object, search conditions*/
            "searchID": {
            /*ro, req, object, search ID, it is used to check whether the
current search requester is the same as the previous one. If they are the same,
the search record will be stored in the device to speed up the next search*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "searchResultPosition": {
            /*ro, req, object, the start position of the search result in the
result list*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
```

```
            },
            "maxResults": {
            /*ro, req, object, the maximum number of search results that can be
obtained by calling this URL*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "major": {
            /*ro, opt, object, major alarm type (the type value should be
transformed to the decimal number)*/
                "@opt":  "0,1,2,3,5"
                /*ro, req, string, major type*/
            },
            "minorAlarm": {
            /*ro, opt, object, minor alarm type (the type value should be
transformed to the decimal number)*/
                "@opt":  "1024,1025,1026,1027…"
                /*ro, req, string, minor alarm type*/
            },
            "minorException": {
            /*ro, opt, object, minor exception type (the type value should be
transformed to the decimal number)*/
                "@opt":  "39,58,59,1024…"
                /*ro, req, string, minor exception type*/
            },
            "minorOperation": {
            /*ro, opt, object, minor operation type (the type value should be
transformed to the decimal number)*/
                "@opt":  "80,90,112,113…"
                /*ro, req, string, minor operation type*/
            },
            "minorEvent": {
            /*ro, opt, object, minor event type (the type value should be
transformed to the decimal number)*/
                "@opt":  "1,2,3,4…"
                /*ro, req, string, minor event type*/
            },
            "startTime": {
            /*ro, opt, object, start time*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,32]*/
                "@max":  32
                /*ro, req, int, the maximum value, range:[0,32]*/
            },
            "endTime": {
            /*ro, opt, object, end time*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,32]*/
                "@max":  32
                /*ro, req, int, the maximum value, range:[0,32]*/
            },
            "cardNo": {
            /*ro, opt, object, card No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
```

```
                /*ro, req, int, the maximum value*/
            },
            "name": {
            /*ro, opt, object, name of the card holder*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "videoChannel": {
            /*ro, opt, object, video channel No. (this node is newly added to
DeepinMind devices for attendance)*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "picEnable":  "true,false",
            /*ro, opt, string, whether to include pictures*/
            "beginSerialNo": {
            /*ro, opt, object, start serial No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "endSerialNo": {
            /*ro, opt, object, end serial No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "employeeNoString": {
            /*ro, opt, object, employee No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "isAbnomalTemperature": {
            /*ro, opt, object, whether the skin-surface temperature is
abnormal*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool, desc:true (yes), false
(no)*/
            },
            "temperatureSearchCond": {
            /*ro, opt, object, temperature search condition*/
                "@opt": ["all", "normal", "abnormal"]
                /*ro, req, array, options, subType:string*/
            },
            "isAttendanceInfo": {
            /*ro, opt, object, whether it contains attendance records*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool, desc:true (yes), false
(no)*/
            },
```

```
            "hasRecordInfo": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            }
        },
        "InfoList": {
        /*ro, opt, object, information list*/
            "maxSize":  10,
            /*ro, opt, int, the maximum value*/
            "time": {
            /*ro, opt, object, time (UTC time)*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,32]*/
                "@max":  32
                /*ro, req, int, the maximum value, range:[0,32]*/
            },
            "netUser": {
            /*ro, opt, object, user name*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "remoteHostAddr": {
            /*ro, opt, object, remote host address*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "cardNo": {
            /*ro, opt, object, card No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "cardType": {
            /*ro, opt, object, card type*/
                "@opt":  "1,2,3,4,5,6,7,8"
                /*ro, req, string, 1 (normal card), 2 (disability card), 3
(blocklist card), 4 (patrol card), 5 (duress card), 6 (super card), 7 (visitor
card), 8 (dismiss card)*/
            },
            "whiteListNo": {
            /*ro, opt, object, allowlist No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "reportChannel": {
            /*ro, opt, object, channel type for uploading alarms/events*/
                "@opt":  "1,2,3"
                /*ro, req, string, "1" (for uploading arming information), "2"
(for uploading by central group 1), "3" (for uploading by central group 2)*/
            },
```

```
            "cardReaderKind": {
            /*ro, opt, object, card reader type*/
                "@opt":  "1,2,3,4"
                /*ro, req, string, "1" (IC card reader), "2" (ID card reader),
"3" (QR code scanner), "4" (fingerprint module)*/
            },
            "cardReaderNo": {
            /*ro, opt, object, card reader No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "doorNo": {
            /*ro, opt, object, door (floor) No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "verifyNo": {
            /*ro, opt, object, multiple authentication No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "alarmInNo": {
            /*ro, opt, object, alarm input No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "alarmOutNo": {
            /*ro, opt, object, alarm output No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "caseSensorNo": {
            /*ro, opt, object, event trigger No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "RS485No": {
            /*ro, opt, object, RS-485 channel No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "multiCardGroupNo": {
            /*ro, opt, object, group No.*/
                "@min":  1,
```

```
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "accessChannel": {
            /*ro, opt, object, turnstile No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "deviceNo": {
            /*ro, opt, object, device No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "distractControlNo": {
            /*ro, opt, object, distributed access controller No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "employeeNo": {
            /*ro, opt, object, employee No. (person ID)*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "localControllerID": {
            /*ro, opt, object, distributed access controller No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "InternetAccess": {
            /*ro, opt, object, network interface No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "type": {
            /*ro, opt, object, zone type*/
                "@opt":  "0,1,2,3,4,5,6,7,8,9,10,11,255"
                /*ro, req, string, 0 (instant alarm zone), 1 (24-hour zone), 2
(delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm zone), 6
(perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-
hour shock zone), 10 (emergency door open zone), 11 (emergency door closed
zone), 255 (none)*/
            },
            "MACAddr": {
            /*ro, opt, object, MAC Address*/
                "@min":  1,
```

```
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "swipeCardType": {
            /*ro, opt, object, card swiping type*/
                "@opt":  "0,1"
                /*ro, req, string, "0" (invalid), "1" (QR code)*/
            },
            "serialNo": {
            /*ro, opt, object, event serial No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "channelControllerID": {
            /*ro, opt, object, lane controller No.*/
                "@opt":  "0,1"
                /*ro, req, string, "1" (main lane controller), "2" (sub lane
controller)*/
            },
            "channelControllerLampID": {
            /*ro, opt, object, light board ID of lane controller*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "channelControllerIRAdaptorID": {
            /*ro, opt, object, IR adapter No. of lane controller*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "channelControllerIREmitterID": {
            /*ro, opt, object, active infrared intrusion detector No. of lane
controller*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "userType": {
            /*ro, opt, object, person types*/
                "@opt":  "normal,visitor,blackList,administrators"
                /*ro, req, string, "normal" (normal person (household)),
"visitor" (visitor), "blacklist" (person in blocklist), "administrators"
(administrator)*/
            },
            "currentVerifyMode": {
            /*ro, opt, object, current authentication mode of the card reader*/
```

```
                "@opt":
 "cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpO
rCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNo
AndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,fac
eOrfaceAndCard,fpOrface,cardOrfaceOrPw,iris,faceOrFpOrCardOrPwOrIris,faceOrCardO
rPwOrIris"
                /*ro, req, string, "cardAndPw"-card+password, "card",
"cardOrPw"-card or password, "fp"-fingerprint, "fpAndPw"-fingerprint+password,
"fpOrCard"-fingerprint or card, "fpAndCard"-fingerprint+card, "fpAndCardAndPw"-
fingerprint+card+password, "faceOrFpOrCardOrPw"-face or fingerprint or card or
password, "faceAndFp"-face+fingerprint, "faceAndPw"-face+password,
"faceAndCard"-face+card, "face", "employeeNoAndPw"-employee No.+password,
"fpOrPw"-fingerprint or password, "employeeNoAndFp"-employee No.+fingerprint,
"employeeNoAndFpAndPw"-employee No.+fingerprint+password, "faceAndFpAndCard"-
face+fingerprint+card, "faceAndPwAndFp"-face+password+fingerprint,
"employeeNoAndFace"-employee No.+face, "faceOrfaceAndCard"-face or face+card,
"fpOrface"-fingerprint or face, "cardOrfaceOrPw"-card or face or password,
"cardOrFpOrPw"-card or fingerprint or password*/
            },
            "QRCodeInfo": {
            /*ro, opt, object, QR code information*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "thermometryUnit": {
            /*ro, opt, object, temperature unit*/
                "@opt": ["celsius", "fahrenheit", "kelvin"]
                /*ro, req, array, "celsius" (Celsius (default)), "fahrenheit"
(Fahrenheit), "kelvin" (Kelvin), subType:string*/
            },
            "currTemperature": {
            /*ro, opt, object, skin-surface temperature*/
                "@min":  1,
                /*ro, req, int, skin-surface temperature, which is accurate to
one decimal place*/
                "@max":  1
                /*ro, req, int, skin-surface temperature, which is accurate to
one decimal place*/
            },
            "isAbnomalTemperature": {
            /*ro, opt, object, whether the skin-surface temperature is
abnormal*/
                "@opt": [true, false]
                /*ro, req, array, whether the skin-surface temperature is
abnormal (true-yes), subType:bool*/
            },
            "RegionCoordinates": {
            /*ro, opt, object, coordinates of the skin-surface temperature*/
                "positionX": {
                /*ro, opt, object, X-coordinate*/
                    "@min":  0,
                    /*ro, req, int, the minimum value, normalized X-coordinate
which is between 0 and 1000*/
                    "@max":  1000
                    /*ro, req, int, the maximum value, normalized X-coordinate
which is between 0 and 1000*/
```

```
                },
                "positionY": {
                /*ro, opt, object, Y-coordinate*/
                    "@min":  0,
                    /*ro, req, int, the minimum value, normalized Y-coordinate
which is between 0 and 1000*/
                    "@max":  1000
                    /*ro, req, int, the maximum value, normalized Y-coordinate
which is between 0 and 1000*/
                }
            },
            "mask": {
            /*ro, opt, object, whether the person is wearing mask*/
                "@opt": ["unknown", "yes", "no"]
                /*ro, req, array, "unknown", "yes", "no", subType:string*/
            },
            "pictureURL": {
            /*ro, opt, object, URL of the captured picture*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "filename": {
            /*ro, opt, object, file name, desc:if multiple pictures are
returned, the file name of each picture should be unique*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "attendanceStatus": {
            /*ro, opt, object, attendance status, desc:"undefined", "checkIn"
(check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn" (end
of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
                "@opt":
 "undefined,checkIn,checkOut,breakOut,breakIn,overtimeIn,overtimeOut"
                /*ro, req, string, options*/
            },
            "label": {
            /*ro, opt, object, self-defined attendance name*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "statusValue": {
            /*ro, opt, object, status value*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "helmet": {
            /*ro, opt, object, whether the person is wearing hard hat,
desc:"unknown", "yes", "no"*/
                "@opt":  "unknown,yes,no"
                /*ro, req, string, options*/
```

```
            },
            "visibleLightPicUrl": {
            /*ro, opt, object, URL of the visible light picture*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "thermalPicUrl": {
            /*ro, opt, object, URL of the thermal imaging picture*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "HealthInfo": {
            /*ro, opt, object, health information*/
                "healthCode": {
                /*ro, opt, object, health code status*/
                    "@opt": [0, 1, 2, 3, 4, 5, 6, 7]
                    /*ro, req, array, options, subType:int, desc:0 (no request),
1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR code), 5
(no such person), 6 (other error, e.g., searching failed due to API exception),
7 (searching for the health code timed out)*/
                },
                "NADCode": {
                /*ro, opt, object, nucleic acid test result, desc:0 (no result),
1 (negative, which means normal), 2 (positive, which means diagnosed), 3 (the
result has expired)*/
                    "@opt": [0, 1, 2, 3, 4]
                    /*ro, req, array, options, subType:int*/
                },
                "travelCode": {
                /*ro, opt, object, trip code, desc:0 (no trip in the past 14
days), 1 (has left the current area left in the past 14 days), 2 (has been to
the high-risk area in the past 14 days), 3 (other)*/
                    "@opt": [0, 1, 2, 3, 4]
                    /*ro, req, array, options, subType:int*/
                },
                "travelInfo": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int, step:1*/
                    "@max":  15
                    /*ro, req, int, step:1*/
                },
                "vaccineStatus": {
                /*ro, opt, object, whether the person is vaccinated, desc:0 (not
vaccinated), 1 (vaccinated)*/
                    "@opt": [0, 1, 2, 3, 4]
                    /*ro, req, array, options, subType:int*/
                },
                "vaccineNum": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int, step:1*/
                    "@max":  3
                    /*ro, req, int, step:1*/
```

```
                }
            },
            "meetingID": {
            /*ro, opt, object, meeting ID*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  32
                /*ro, req, int, the maximum value*/
            },
            "PersonInfoExtends": {
            /*ro, opt, object, extended nodes for the additional person
information*/
                "maxSize":  3,
                /*ro, req, int, range of members in the array*/
                "id": {
                /*ro, opt, object, extended ID of the additional person
information*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  32
                    /*ro, req, int, the maximum value*/
                },
                "value": {
                /*ro, opt, object, extended content of the additional person
information*/
                    "@min":  0,
                    /*ro, req, int, the minimum value*/
                    "@max":  100
                    /*ro, req, int, the maximum value*/
                }
            },
            "name": {
            /*ro, opt, object, name*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  128
                /*ro, req, int, the maximum value*/
            },
            "FaceRect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin
is the upper-left corner of the screen*/
                "height": {
                /*ro, req, object, height*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                },
                "width": {
                /*ro, req, object, width*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                },
                "x": {
                /*ro, req, object, X-coordinate of the upper-left corner of the
frame*/
```

```
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                },
                "y": {
                /*ro, req, object, Y-coordinate of the upper-left corner of the
frame*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                }
            },
            "RecordInfo": {
            /*ro, opt, object*/
                "startTime":  "1970-01-01T00:00:00+08:00",
                /*ro, opt, datetime*/
                "endTime":  "1970-01-01T00:00:00+08:00",
                /*ro, opt, datetime*/
                "playbackURL": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int, the minimum length*/
                    "@max":  256
                    /*ro, req, int, the maximum length*/
                }
            },
            "currentAuthenticationTimes": {
            /*ro, req, object*/
                "@min":  0,
                /*ro, req, int*/
                "@max":  255
                /*ro, req, int*/
            },
            "allowAuthenticationTimes": {
            /*ro, req, object*/
                "@min":  0,
                /*ro, req, int*/
                "@max":  255
                /*ro, req, int*/
            }
        }
    }
}
```

13.12.3 Search for access control events

URL

POST /ISAPI/AccessControl/AcsEvent?format=json

Query Parameter

None

Request Message

```
{
    "AcsEventCond": {
    /*req, object, access control events*/
        "searchID":  "test",
        /*req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list:, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search. If the maximum number of totalMatches supported by the device is M and the number of totalMatches stored in the device now is N (N<=M), the valid range of this node is 0 to N-1*/
        "maxResults":  30,
        /*req, int, the maximum number of search results, which is defined by the device capability, will be returned if the value of maxResults reaches the limit, desc:if maxResults exceeds the range returned by the device capability, the device will return the maximum number of search results according to the device capability and will not return error message*/
        "major":  1,
        /*req, int, major type, desc:the type value should be transformed to the decimal number; see Access Control Event Types for details*/
        "minor":  1024,
        /*req, int, minor type, desc:the type value should be transformed to the decimal number; see Access Control Event Types for details*/
        "startTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, start time (UTC time)*/
        "endTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, end time (UTC time)*/
        "cardNo":  "test",
        /*opt, string, card No.*/
        "name":  "test",
        /*opt, string, name of the card holder*/
        "videoChannel":  1,
        /*opt, int, video channel No., range:[1,86400], desc:this node is newly added to DeepinMind devices for attendance*/
        "picEnable":  true,
        /*opt, bool, whether to upload the picture along with the event information, desc:false (no), true (yes, default value); (1. all matched events will be uploaded without pictures; 2. all matched events will be uploaded with pictures if there are any; 3. if this node is not configured, the default value is true)*/
        "beginSerialNo":  1,
        /*opt, int, start serial No.*/
        "endSerialNo":  1,
        /*opt, int, end serial No.*/
        "employeeNoString":  "test",
        /*opt, string, employee No. (person ID)*/
        "timeReverseOrder":  true,
        /*opt, bool, whether to return events in descending order of time (later events will be returned first), desc:true (yes), false or this node is not returned (no)*/
        "isAbnomalTemperature":  true,
        /*opt, bool, whether the skin-surface temperature is abnormal*/
        "temperatureSearchCond":  "all",
```

```
        /*opt, enum, temperature search condition, subType:string, desc:when
this node and isAbnormalTemperature both exist, isAbnormalTemperature is
invalid; "all" (event with temperature), "normal" (event with normal
temperature), "abnormal" (event with abnormal temperature)*/
        "isAttendanceInfo":  true,
        /*opt, bool, whether it contains attendance records, desc:this node is
newly added to HEOP protocol; if this node is true, main type, minor type,
employee No., name, and time will be returned*/
        "hasRecordInfo":  true
        /*opt, bool*/
    }
}
```

Response Message

```
{
    "AcsEvent": {
    /*ro, req, object, access control events*/
        "searchID":  "test",
        /*ro, req, string, search ID, it is used to check whether the current
search requester is the same as the previous one. If they are the same, the
search record will be stored in the device to speed up the next search*/
        "responseStatusStrg":  "OK",
        /*ro, req, string, searching status description*/
        "numOfMatches":  1,
        /*ro, req, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "InfoList": [
        /*ro, opt, array, information list, subType:object*/
            {
                "major":  1,
                /*ro, req, int, major alarm type*/
                "minor":  1,
                /*ro, req, int, minor alarm type*/
                "time":  "2016-12-12T17:30:08+08:00",
                /*ro, req, string, time (UTC time)*/
                "netUser":  "test",
                /*ro, opt, string, user name*/
                "remoteHostAddr":  "test",
                /*ro, opt, string, remote host address*/
                "videoChannel":  1,
                /*ro, opt, int, video channel No., range:[1,86400], desc:this
node is newly added to DeepinMind devices for attendance*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  1,
                /*ro, opt, enum, card type, subType:int, desc:1 (normal card), 2
(disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6
(super card), 7 (visitor card), 8 (dismiss card)*/
                "whiteListNo":  1,
                /*ro, opt, int, allowlist No.*/
                "reportChannel":  1,
                /*ro, opt, int, channel type for uploading alarm/event*/
                "cardReaderKind":  1,
                /*ro, opt, int, card reader type: 1 (IC card reader)*/
                "cardReaderNo":  1,
```

```
                /*ro, opt, int, card reader No.*/
                "doorNo": 1,
                /*ro, opt, int, door or floor No.*/
                "verifyNo": 1,
                /*ro, opt, int, multi-factor authentication No.*/
                "alarmInNo": 1,
                /*ro, opt, int, alarm input No.*/
                "alarmOutNo": 1,
                /*ro, opt, int, alarm output No.*/
                "caseSensorNo": 1,
                /*ro, opt, int, event trigger No.*/
                "RS485No": 1,
                /*ro, opt, int, RS-485 channel No.*/
                "multiCardGroupNo": 1,
                /*ro, opt, int, group No.*/
                "accessChannel": 1,
                /*ro, opt, int, RS-485 channel No.*/
                "deviceNo": 1,
                /*ro, opt, int, device No.*/
                "distractControlNo": 1,
                /*ro, opt, int, distributed controller No.*/
                "employeeNoString": "test",
                /*ro, opt, string, employee No. (person ID)*/
                "localControllerID": 1,
                /*ro, opt, int, distributed controller No.*/
                "InternetAccess": 1,
                /*ro, opt, int, network interface No.*/
                "type": 1,
                /*ro, opt, int, zone type, desc:0 (instant alarm zone), 1 (24-
hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm
zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone),
9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door closed
zone), 255 (none)*/
                "MACAddr": "test",
                /*ro, opt, string, MAC address*/
                "swipeCardType": 1,
                /*ro, opt, enum, card swiping type, subType:int, desc:0
(invalid), 1 (QR code)*/
                "serialNo": 1,
                /*ro, opt, int, event serial No.*/
                "channelControllerID": 1,
                /*ro, opt, int, lane controller ID*/
                "channelControllerLampID": 1,
                /*ro, opt, int, light board ID of lane controller, range:
[1,255]*/
                "channelControllerIRAdaptorID": 1,
                /*ro, opt, int, IR adapter ID of lane controller, range:
[1,255]*/
                "channelControllerIREmitterID": 1,
                /*ro, opt, int, active infrared intrusion detector No. of lane
controller, range:[1,255]*/
                "userType": "normal",
                /*ro, opt, string, person type*/
                "currentVerifyMode": "cardAndPw",
```

```
                /*ro, opt, enum, current authentication mode of the card reader,
subType:string, desc:"cardAndPw" (card + password); "card", "cardOrPw" (card or
password),  "fp" (fingerprint), "fpAndPw" (fingerprint + password), "fpOrCard"
"fingerprint or card", "fpAndCard" (fingerprint + card), "fpAndCardAndPw"
(fingerprint + card + password), "faceOrFpOrCardOrPw" (face or fingerprint or
card or password), "faceAndFp" (face + fingerprint), "faceAndPw" (face +
password), "faceAndCard" (face + card), "face", "employeeNoAndPw" (emplyee No.
+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.
+ fingerprint), "employeeNoAndFpAndPw" (employee No. + fingerprint + password),
"faceAndFpAndCard" (face + fingerprint + card), "faceAndPwAndFp" (face +
password + fingerprint), "employeeNoAndFace" (employee No. + face),
"faceOrfaceAndCard" (face or face + card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "faceOrFpOrCardOrPwOrIris" (face or
fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "sleep", "invalid"*/
                "QRCodeInfo":  "test",
                /*ro, opt, string, QR code information*/
                "thermometryUnit":  "celsius",
                /*ro, opt, enum, temperature unit, subType:string,
desc:"celsius" (Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin"
(Kelvin)*/
                "currTemperature":  36.5,
                /*ro, opt, float, skin-surface temperature, which is accurate to
one decimal place*/
                "isAbnomalTemperature":  true,
                /*ro, opt, bool, whether the skin-surface temperature is
abnormal (true-yes)*/
                "RegionCoordinates": {
                /*ro, opt, object, coordinates of the skin-surface temperature*/
                    "positionX":  254,
                    /*ro, opt, int, normalized X-coordinate which is between 0
and 1000*/
                    "positionY":  133
                    /*ro, opt, int, normalized Y-coordinate which is between 0
and 1000*/
                },
                "mask":  "unknown",
                /*ro, opt, enum, whether the person wears a mask,
subType:string, desc:"unknown"*/
                "pictureURL":  "test",
                /*ro, opt, string, picture URL*/
                "filename":  "picture1",
                /*ro, opt, string, file name, desc:if multiple pictures are
returned at a time, filename of each picture should be unique*/
                "attendanceStatus":  "undefined",
                /*ro, opt, enum, attendance status, subType:string,
desc:"undefined", "checkIn" (check-in), "checkOut" (check-out), "breakOut"
(start of break), "breakIn" (end of break), "overtimeIn" (start of overtime),
"overTimeOut" (end of overtime)*/
                "label":  "test",
                /*ro, opt, string, custom attendance name*/
                "statusValue":  1,
                /*ro, opt, int, status value*/
                "helmet":  "unknown",
                /*ro, opt, enum, whether the person wears a hard hat,
subType:string, desc:"unknown", "yes", "no"*/
                "visibleLightPicUrl":  "test",
                /*ro, opt, string, visible light picture URL*/
```

```
                "thermalPicUrl":  "test",
                /*ro, opt, string, URL of the thermal imaging picture*/
                "appType":  "attendance",
                /*ro, opt, enum, application type, subType:string,
desc:"attendance" (Time & Attendance module), "signIn" (Check-In module, which
is only used for FocSign products)*/
                "HealthInfo": {
                /*ro, opt, object, health information*/
                    "healthCode":  1,
                    /*ro, opt, enum, health code status, subType:int, desc:0 (no
request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR
code), 5 (no such person), 6 (other error, e.g., searching failed due to API
exception), 7 (searching for the health code timed out)*/
                    "NADCode":  1,
                    /*ro, opt, enum, nucleic acid test result, subType:int,
desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means
diagnosed), 3 (the result has expired)*/
                    "travelCode":  1,
                    /*ro, opt, enum, trip code, subType:int, desc:0 (no trip in
the past 14 days), 1 (has left the current area in the past 14 days), 2 (has
been to the high-risk area in the past 14 days), 3 (other)*/
                    "travelInfo":  "test",
                    /*ro, opt, string*/
                    "vaccineStatus":  1,
                    /*ro, opt, enum, whether the person is vaccinated,
subType:int, desc:0 (not vaccinated), 1 (vaccinated)*/
                    "vaccineNum":  1
                    /*ro, opt, int, step:1*/
                },
                "meetingID":  "test",
                /*ro, opt, string, meeting ID*/
                "PersonInfoExtends": [
                /*ro, opt, array, additional person information, subType:object,
desc:this node displays additional person information on the device*/
                    {
                        "id":  1,
                        /*ro, opt, int, extended ID of the additional person
information, range:[1,32], desc:related URL:
/ISAPI/AccessControl/personInfoExtendName?format=json; this node is used for
displaying the name of value; if ID does not exists, it starts from 1*/
                        "value":  "test"
                        /*ro, opt, string, extended content of the additional
person information*/
                    }
                ],
                "name":  "test",
                /*ro, opt, string, name, desc:person name*/
                "FaceRect": {
                /*ro, opt, object, rectangle frame for human face, desc:the
origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate of the upper-left corner of
the frame, range:[0.000,1.000]*/
                    "y":  0.000
```

```
                    /*ro, req, float, Y-coordinate of the upper-left corner of
the frame, range:[0.000,1.000]*/
                },
                "RecordInfo": {
                /*ro, opt, object*/
                    "startTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, recording start time*/
                    "endTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, recording end time*/
                    "playbackURL":
 "rtsp://10.65.130.168:554/ISAPI/Streaming/tracks/201/?
starttime=20190213T091134Z&amp;endtime=20190213T092116Z"
                    /*ro, opt, string, range:[0,256]*/
                },
                "currentAuthenticationTimes":  1,
                /*ro, opt, int, range:[0,255], step:1*/
                "allowAuthenticationTimes":  1
                /*ro, opt, int, range:[0,255], step:1*/
            }
        ]
    }
}
```

13.12.4 Get the capability of getting total number of access control events by specific conditions

URL

GET /ISAPI/AccessControl/AcsEventTotalNum/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsEvent": {
    /*ro, opt, object*/
        "AcsEventTotalNumCond": {
        /*ro, opt, object, search conditions*/
            "major": {
            /*ro, req, object, major alarm type*/
                "@opt":  "0,1,2,3,5"
                /*ro, opt, string, major alarm type*/
            },
            "minorAlarm": {
            /*ro, req, object, minor alarm type*/
                "@opt":  "1024,1025,1026,1027"
                /*ro, opt, string, minor alarm type*/
            },
            "minorException": {
            /*ro, req, object, minor exception type*/
                "@opt":  "39,58,59,1024"
                /*ro, opt, string, minor exception type*/
```

```
        },
        "minorOperation": {
        /*ro, req, object, minor operation type*/
            "@opt":  "80,90,112,113"
            /*ro, opt, string, minor operation type*/
        },
        "minorEvent": {
        /*ro, opt, object, minor event type*/
            "@opt":  "1,2,3,4"
            /*ro, opt, string, minor event type*/
        },
        "startTime": {
        /*ro, opt, object, start time*/
            "@min":  1,
            /*ro, opt, int, start time (UTC time)*/
            "@max":  1
            /*ro, opt, int, end time (UTC time)*/
        },
        "endTime": {
        /*ro, opt, object, end time*/
            "@min":  1,
            /*ro, opt, int, start time (UTC time)*/
            "@max":  1
            /*ro, opt, int, end time (UTC time)*/
        },
        "cardNo": {
        /*ro, opt, object, card No.*/
            "@min":  1,
            /*ro, opt, int, card No.*/
            "@max":  32
            /*ro, opt, int, card No.*/
        },
        "name": {
        /*ro, opt, object, name of the card holder*/
            "@min":  1,
            /*ro, opt, int, name of the card holder*/
            "@max":  32
            /*ro, opt, int, name of the card holder*/
        },
        "picEnable":  "true,false",
        /*ro, opt, string*/
        "beginSerialNo": {
        /*ro, opt, object, start serial No.*/
            "@min":  1,
            /*ro, opt, int, start serial No.*/
            "@max":  1
            /*ro, opt, int, start serial No.*/
        },
        "endSerialNo": {
        /*ro, opt, object, end serial No.*/
            "@min":  1,
            /*ro, opt, int, end serial No.*/
            "@max":  1
            /*ro, opt, int, end serial No.*/
        },
        "employeeNoString": {
        /*ro, opt, object, employee No. (person ID)*/
            "@min":  1,
```

```
                /*ro, opt, int, employee No. (person ID)*/
                "@max":  32
                /*ro, opt, int, employee No. (person ID)*/
            }
        },
        "totalNum": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        }
    }
}
```

13.12.5 Access control event

EventType:AccessControllerEvent

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the
alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string,
desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the
video channel No. of private protocol. When ISAPI protocol is transmitted via EZ
protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of
ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when
ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel
No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the
channel No. is the video channel No. of EZ protocol. When ISAPI protocol is
transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded, desc:times that
the same alarm has been uploaded*/
    "eventType":  "AccessControllerEvent",
    /*ro, req, string, event type, desc:"AccessControllerEvent" (access control
event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event:
"active" (valid event or event starts), "inactive" (invalid event or the event
ends). For the heartbeat, the node value indicates the heartbeat data, and it is
uploaded every 10 seconds*/
    "eventDescription":  "AccessControllerEvent",
```

```
    /*ro, req, string, event description, desc:"AccessControllerEvent" (access
control event)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID), desc:this node must be returned when
ISAPI event information is transmitted via ISUP*/
    "AccessControllerEvent": {
    /*ro, req, object, access control event information*/
        "deviceName":  "test",
        /*ro, opt, string, device name*/
        "majorEventType":  1,
        /*ro, req, int, major alarm type, desc:the type value should be
transformed to the decimal number; see Access Control Alarm Types for details*/
        "subEventType":  1,
        /*ro, req, int, minor alarm type, desc:the type value should be
transformed to the decimal number; see Access Control Alarm Types for details*/
        "inductiveEventType":  "authenticated",
        /*ro, opt, enum, inductive event type, subType:string, desc:this node is
used by storage devices; for access control devices, this node is invalid;
"authenticated", "authenticationFailed", "openingDoor", "closingDoor",
"doorException", "remoteOperation", "timeSynchronization", "deviceException",
"deviceRecovered", "alarmTriggered", "alarmRecovered" (arming restoring event),
"callCenter"*/
        "netUser":  "test",
        /*ro, opt, string, user name for network operations*/
        "remoteHostAddr":  "test",
        /*ro, opt, string, remote host address*/
        "cardNo":  "test",
        /*ro, opt, string, card No.*/
        "cardType":  1,
        /*ro, opt, enum, card type, subType:int, desc:1 (normal card), 2
(disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6
(super card), 7 (visitor card), 8 (dismiss card)*/
        "name":  "test",
        /*ro, opt, string, person name*/
        "sex":  "male",
        /*ro, opt, enum, subType:string, desc:"male", "female"*/
        "whiteListNo":  1,
        /*ro, opt, int, allowlist No.*/
        "reportChannel":  1,
        /*ro, opt, enum, channel type for uploading alarms/events, subType:int,
desc:1 (uploading in arming mode), 2 (uploading by central group 1), 3
(uploading by central group 2)*/
        "cardReaderKind":  1,
        /*ro, opt, enum, card reader type, subType:int, desc:1 (IC card reader),
2 (ID card reader), 3 (QR code scanner), 4 (fingerprint module)*/
        "cardReaderNo":  1,
        /*ro, opt, int, card reader No., step:1, desc:card reader No.*/
        "doorNo":  1,
        /*ro, opt, int, door (floor) No.*/
        "verifyNo":  1,
        /*ro, opt, int, multiple authentication No.*/
        "alarmInNo":  1,
        /*ro, opt, int, alarm input No.*/
        "alarmOutNo":  1,
        /*ro, opt, int, alarm output No.*/
        "caseSensorNo":  1,
        /*ro, opt, int, event trigger No.*/
        "RS485No":  1,
```

```
        /*ro, opt, int, RS-485 channel No.*/
        "multiCardGroupNo":  1,
        /*ro, opt, int, group No.*/
        "accessChannel":  1,
        /*ro, opt, int, turnstile No.*/
        "deviceNo":  1,
        /*ro, opt, int, device No.*/
        "distractControlNo":  1,
        /*ro, opt, int, distributed access controller No.*/
        "employeeNo":  1,
        /*ro, opt, int, employee No. (person ID)*/
        "employeeNoString":  "test",
        /*ro, opt, string, employee No. (person ID), desc:if the node employeeNo
exists or the value of employeeNoString can be converted to that of employeeNo,
this node is required. For the upper-layer platform or client software, the node
employeeNoString will be parsed in priority; if employeeNoString is not
configured, the node employeeNo will be parsed*/
        "employeeName":  "test",
        /*ro, opt, string, person name, desc:this node is only used for FocSign
products*/
        "localControllerID":  1,
        /*ro, opt, int, distributed access controller No., desc:0 (access
controller), 1 to 64 (distributed access controller No. 1 to distributed access
controller No. 64)*/
        "InternetAccess":  1,
        /*ro, opt, enum, network interface No., subType:int, desc:1 (upstream
network interface No. 1), 2 (upstream network interface No. 2), 3 (downstream
network interface No. 1)*/
        "type":  1,
        /*ro, opt, enum, zone type, subType:int, desc:0 (instant zone), 1 (24-
hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm
zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone),
9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door closed
zone), 255 (none)*/
        "MACAddr":  "test",
        /*ro, opt, string, MAC address*/
        "swipeCardType":  1,
        /*ro, opt, enum, card swiping types, subType:int, desc:0 (invalid), 1
(QR code)*/
        "serialNo":  1,
        /*ro, opt, int, event serial No., range:[1,100000], desc:it starts at 1
and each record increases by 1. It will be overwritten repeatedly when reaching
the maximum value supported by the device*/
        "channelControllerID":  1,
        /*ro, opt, enum, lane controller ID, subType:int, desc:1 (main lane
controller), 2 (sub-lane controller)*/
        "channelControllerLampID":  1,
        /*ro, opt, int, light board ID of lane controller, range:[1,255]*/
        "channelControllerIRAdaptorID":  1,
        /*ro, opt, int, IR adaptor ID of the lane controller, range:[1,255]*/
        "channelControllerIREmitterID":  1,
        /*ro, opt, int, active infrared intrusion detector No. of the lane
controller, range:[1,255]*/
        "userType":  "normal",
        /*ro, opt, enum, person type, subType:string, desc:"normal" (normal
person (resident)), "visitor" (visitor), "blacklist" (person in the blocklist),
"administrators" (administrator)*/
        "currentVerifyMode":  "cardAndPw",
```

```
        /*ro, opt, enum, current authentication mode of the card reader,
subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw"
(card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password),
"fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card),
"fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or
fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw"
(face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw"
(employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp"
(employee No.+fingerprint), "employeeNoAndFpAndPw" (employee
No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card),
"faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee
No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or
face), "cardOrfaceOrPw" (card or face or password), "iris" (iris),
"faceOrFpOrCardOrPwOrIris" (face, fingerprint, card, password, or iris),
"faceOrCardOrPwOrIris" (face, card, password, or iris)*/
        "currentEvent":  true,
        /*ro, opt, bool, whether it is a real-time event*/
        "QRCodeInfo":  "test",
        /*ro, opt, string, QR code information*/
        "thermometryResult":  "success",
        /*ro, opt, enum, temperature screening result, subType:string,
desc:"success", "fail"*/
        "thermometryUnit":  "celsius",
        /*ro, opt, enum, temperature unit, subType:string, desc:"celsius"
(Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
        "currTemperature":  36.1,
        /*ro, opt, float, skin-surface temperature, which is accurate to one
decimal place*/
        "isAbnomalTemperature":  true,
        /*ro, opt, bool, whether the skin-surface temperature is abnormal*/
        "RegionCoordinates": {
        /*ro, opt, object, coordinates of the skin-surface temperature*/
            "positionX":  0,
            /*ro, opt, int, normalized X-coordinate which is between 0 and 1000,
range:[0,1000]*/
            "positionY":  0
            /*ro, opt, int, normalized Y-coordinate which is between 0 and 1000,
range:[0,1000]*/
        },
        "remoteCheck":  true,
        /*ro, opt, bool, whether remote verification is required: true-yes,
false-no (default)*/
        "mask":  "unknown",
        /*ro, opt, enum, whether the person wears a mask, subType:string,
desc:"unknown", "yes", "no"*/
        "frontSerialNo":  1,
        /*ro, opt, int, serial No. of the previous event, desc:if this node does
not exist, the platform will check whether the event loss occurred according to
the node serialNo. If both the serialNo and frontSerialNo are returned, the
platform will check whether the event loss occurred according to both nodes. It
is mainly used to solve the problem that the serialNo is inconsistent after
subscribing events or alarms*/
        "attendanceStatus":  "checkIn",
        /*ro, opt, enum, attendance status, subType:string, desc:"checkIn"
(check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn" (end
of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
        "label":  "test",
        /*ro, opt, string, self-defined attendance name*/
```

```
        "statusValue":  1,
        /*ro, opt, int, status value*/
        "pictureURL":  "test",
        /*ro, opt, string, URL of the captured picture, range:[0,256]*/
        "visibleLightURL":  "test",
        /*ro, opt, string, visible light picture URL of the thermal imaging
camera, range:[0,256]*/
        "thermalURL":  "test",
        /*ro, opt, string, URL of the thermal picture, range:[0,256]*/
        "faceBasemapURL":  "test",
        /*ro, opt, string, range:[0,256]*/
        "picturesNumber":  1,
        /*ro, opt, int, number of captured pictures*/
        "unlockType":  "password",
        /*ro, opt, enum, unlocking type, subType:string, desc:this node is
returned when the minor type is MINOR_UNCLOCK_RECORD; "password" (unlock by
password), "hijcking" (unlock by duress), "card" (unlock by card), "householder"
(unlock by householder), "centerplatform" (unlock by management center),
"bluetooth" (unlock by bluetooth), "qrcode" (unlocked via QR code), "face"
(unlock by recognizing face), "fingerprint" (unlock by fingerprint)*/
        "classroomId":  "test",
        /*ro, opt, string, class ID*/
        "classroomName":  "test",
        /*ro, opt, string, class name*/
        "analysisModule":  "signageApp",
        /*ro, opt, enum, analysis module, subType:string, desc:this node is not
returned, and the value is report via signage App; "signageApp" (signage App),
"faceSDK" (face picture SDK)*/
        "customInfo":  "test",
        /*ro, opt, string, custom information*/
        "helmet":  "unknown",
        /*ro, opt, enum, whether the person is wearing hard hat, subType:string,
desc:"unknown", "yes", "no"*/
        "purePwdVerifyEnable":  true,
        /*ro, opt, bool, whether the device supports opening the door only by
password,
        desc:opening the door only by password:
        the password in authentication method is person password; checking the
repetition of person password is not supported by the device, it should be
performed by the upper-layer platform; adding, deleting, editing, and searching
for person password locally is not supported by the device*/
        "appType":  "attendance",
        /*ro, opt, enum, application type (for FocSign products),
subType:string, desc:"attendance" (Time & Attendance module), "signIn" (Check-In
module)*/
        "HealthInfo": {
        /*ro, opt, object, health information*/
            "healthCode":  1,
            /*ro, opt, enum, health code status, subType:int, desc:0 (no
request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR
code), 5 (no such person), 6 (other error, e.g., searching failed due to API
exception), 7 (searching for the health code timed out)*/
            "NADCode":  1,
            /*ro, opt, enum, nucleic acid test result, subType:int, desc:0 (no
result), 1 (negative, which means normal), 2 (positive, which means diagnosed),
3 (the result has expired)*/
            "NADMsg":  "test",
            /*ro, opt, string, range:[0,64]*/
```

```
            "NADTime":  1,
            /*ro, opt, enum, subType:int*/
            "travelCode":  1,
            /*ro, opt, enum, trip code, subType:int, desc:0 (no trip in the past
14 days), 1 (has left the current area left in the past 14 days), 2 (has been to
the high-risk area in the past 14 days), 3 (other)*/
            "travelInfo":  "test",
            /*ro, opt, string, trip information, desc:the empty string indicates
that searching trip failed*/
            "vaccineStatus":  1,
            /*ro, opt, enum, whether the person is vaccinated, subType:int,
desc:0 (not vaccinated), 1 (vaccinated)*/
            "vaccineNum":  1,
            /*ro, opt, int, step:1*/
            "vaccineMsg":  "test",
            /*ro, opt, string, range:[0,64]*/
            "ANTCode":  1,
            /*ro, opt, enum, subType:int*/
            "ANTMsg":  "test"
            /*ro, opt, string, range:[0,64]*/
        },
        "PhysicalInfo": {
        /*ro, opt, object, BMI information, desc:this node is obtained after
authentication by BMI scales which is connected to MinMoe terminals*/
            "weight":  7000,
            /*ro, opt, int, weight, unit:kg*/
            "height":  18000
            /*ro, opt, int, height, unit:cm*/
        },
        "meetingID":  "test",
        /*ro, opt, string, meeting ID, range:[1,32]*/
        "PersonInfoExtends": [
        /*ro, opt, array, additional person information, subType:object,
desc:this node displays additional person information on the device*/
            {
                "id":  1,
                /*ro, opt, int, extended ID of the additional person
information, range:[1,32], desc:related URL:
/ISAPI/AccessControl/personInfoExtendName?format=json; this node is used for
displaying the name of value; if ID does not exists, it starts from 1*/
                "value":  "test"
                /*ro, opt, string, extended content of the additional person
information*/
            }
        ],
        "customPrompt":  "test",
        /*ro, opt, string, custom prompt message, range:[1,128], desc:this node
is displayed when the authentication result is authenticated, authentication
failed, or stranger*/
        "FaceRect": {
        /*ro, opt, object, rectangle frame for human face, desc:the origin is
the upper-left corner of the screen*/
            "height":  1.000,
            /*ro, req, float, height, range:[0.000,1.000]*/
            "width":  1.000,
            /*ro, req, float, width, range:[0.000,1.000]*/
            "x":  0.000,
```

```
            /*ro, req, float, X-coordinate of the upper-left corner of the
frame, range:[0.000,1.000]*/
            "y":  0.000
            /*ro, req, float, Y-coordinate of the upper-left corner of the
frame, range:[0.000,1.000]*/
        },
        "faceSimilarity":  90,
        /*ro, opt, int, Similarity, range:[0,100]*/
        "faceRecognitionDistance":  0.1,
        /*ro, opt, float, unit:m*/
        "eyesDistance":  20,
        /*ro, opt, int, range:[0,100], step:1*/
        "faceRecognitionFailedReason":  "attackBlacklist",
        /*ro, opt, enum, subType:string*/
        "currentAuthenticationTimes":  1,
        /*ro, opt, int, range:[0,255], step:1*/
        "allowAuthenticationTimes":  1,
        /*ro, opt, int, range:[0,255], step:1*/
        "LocalAttendanceData": {
        /*ro, opt, object*/
            "attendanceResult": [
            /*ro, opt, array, subType:object*/
                {
                    "date":  "1970-01-01",
                    /*ro, opt, date*/
                    "week":  1,
                    /*ro, opt, enum, subType:int, desc:1 (Monday), 2 (Tuesday),
3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
                    "personalAttendanceStatus":  "normal"
                    /*ro, opt, enum, subType:string*/
                }
            ]
        },
        "hasRecord":  true
        /*ro, opt, bool*/
    },
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string,
desc:"no" (no authentication, it is used for the cloud protocol), "digest"
(digest authentication, it is used for local picture URL returned by NVR or
DVR)*/
}
```

13.13 Face Picture Library Record Management

13.13.1 Set the face picture data in the face picture library

URL

PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json

Query Parameter

None

Request Message

```
{
```

```
    "faceURL":  "http",
    /*opt, string, picture storage URL, which is inputted when uploading the face picture by URL*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:"blackFD" (list library), "staticFD" (static library), the maximum length is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
    "FPID":  " ",
    /*opt, string, face record ID, desc:it can be generated by the device or inputted. If it is inputted, it should be the unique ID with the combination of letters and digits, and the maximum length is 63 bytes; if it is generated by the device automatically, it is the same as the employee No. (person ID)*/
    "deleteFP":  true,
    /*opt, bool, whether to delete the face record, desc:true (yes). This field required when the face record needs to be deleted; for adding or editing the face record, it should not be configured*/
    "modelData":  "test",
    /*opt, string, target model data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method*/
    "PicFeaturePoints": [
    /*opt, array, feature picture coordinates, subType:object, desc:if the device only supports three types of feature points, when the platform applies more than three types of feature points, the device will not return error information*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized coordinates*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized coordinates*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:it is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:it is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic":  true,
    /*opt, bool, whether to save the registered face picture*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:first authentication permission*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the
value of statusCode is not 1*/
}
```

13.13.2 Get the total number of face records in a face picture library

URL

GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error description, desc:this node is required when the
value of statusCode is not 1*/
    "FDID":  "test",
    /*ro, opt, string, face picture library ID*/
    "faceLibType":  "blackFD",
```

```
    /*ro, opt, enum, face picture library type, subType:string, desc:face
picture library type*/
    "name":  "test",
    /*ro, opt, string, name of the face picture library, desc:the maximum length
is 48*/
    "recordDataNumber":  123
    /*ro, opt, int, total number of records*/
}
```

13.13.3 Get the total number of face records in all face picture libraries

URL

GET /ISAPI/Intelligent/FDLib/Count?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":
 "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode
is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, see the description of this node and above nodes in the
message of JSON_ResponseStatus, desc:this node is required when the value of
statusCode is not 1*/
    "FDRecordDataInfo": [
    /*ro, opt, array, information of face records in face picture library,
subType:object, desc:this node is valid when errorCode is 1 and errorMsg is
"ok"*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID, desc:the maximum size is
63 bytes*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, subType:string,
desc:face picture library type "blackFD" list library,"staticFD" static library,
the maximum size is 32 bytes*/
            "name":  "test",
```

```
            /*ro, opt, string, face picture library name, desc:the maximum size
is 48 bytes*/
            "recordDataNumber":  123,
            /*ro, opt, int, number of records*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:arming type*/
            "libAttribute":  "general",
            /*ro, opt, enum, subType:string*/
            "personnelFileEnabled":  true
            /*ro, opt, bool*/
        }
    ],
    "totalRecordDataNumber":  500000,
    /*ro, opt, int, total number of records, desc:total number of records*/
    "FDCapacity": {
    /*ro, opt, object*/
        "total":  0.000,
        /*ro, req, float*/
        "use":  0.000,
        /*ro, req, float*/
        "remain":  0.000,
        /*ro, req, float, remaining space, desc:remaining space*/
        "maxRecordDataNumber":  0,
        /*ro, opt, int*/
        "useRecordDataNumber":  0,
        /*ro, opt, int*/
        "remainRecordDataNumber":  0
        /*ro, opt, int*/
    }
}
```

13.13.4 Edit face records in the face picture library in a batch

URL

PUT /ISAPI/Intelligent/FDLib/FDModify?format=json

Query Parameter

None

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture
by URL, the maximum length is 256 bytes*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture
library type: "blackFD"-list library,"staticFD"-static library,the maximum
length is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the
maximum length is 63 bytes,multiple face picture libraries should be separated
by commas*/
    "FPID":  " ",
    /*opt, string, face record ID*/
    "name":  "test",
```

```
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which
is used in search), "male", “female”, “unknown"*/
    "bornTime":  "2004-05-03",
    /*req, time, date of birth of the person in the face picture in ISO8601 time
format, the maximum length is 20 bytes, desc:date of birth of the person in the
face picture in ISO8601 time format,the maximum length is 20 bytes*/
    "city":  "130100",
    /*opt, string, code of the city of birth for the person in the face picture,
the maximum length is 32 bytes, desc:code of the city of birth for the person in
the face picture,the maximum length is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string*/
    "certificateNumber":  "test",
    /*opt, string, ID No., the maximum length is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, it is valid when faceLibType is "blackFD",
range:[0,192]*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag*/
    "address":  "test",
    /*opt, string, person address, it is valid when faceLibType is "staticFD”*/
    "customInfo":  "test",
    /*opt, string, custom information, it is valid when faceLibType is
"staticFD”*/
    "modelData":  "test",
    /*opt, string, target model data, non-modeled binary data needs to be
encrypted by base64 during transmission*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal"-by terminal*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when
operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied. If the device only supports
three types of feature points, when the platform applies more than three types
of feature points,the device will not return error information, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face",
"leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose”*/
            "coordinatePoint": {
            /*opt, object, point coordinates*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-
coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-
coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:width which is between 0
and 1000. This node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:height which is between
0 and 1000. This node is required when featurePointType is "face"*/
            }
```

```
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when
statusCode is not 1*/
}
```

13.13.5 Add a face record to the face picture library

URL

POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json

Query Parameter

None

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture
by URL*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture
library type*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:the maximum size is 63 bytes*/
    "FPID":  " test",
    /*opt, string, face record ID, desc:face record ID,it can be generated by
device or inputted. If it is inputted,it should be the unique ID with the
combination of letters and digits,and the maximum length is 63 bytes; if it is
generated by the device automatically,it is the same as the employee No. (person
ID)*/
    "name":  "张三",
```

```
    /*req, string, name of person in the face picture, desc:the maximum size is
96 bytes*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string,
desc:“male”, “female”, “unknown”; the maximum size is 32 bytes*/
    "bornTime":  "2004-05-03",
    /*req, string, birthday of person in the face picture, desc:the maximum size
is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture,
desc:the maximum size is 32 bytes*/
    "certificateType":  "officerID",
    /*opt, enum, certificate type, subType:string*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., desc:the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, desc:the maximum size is 192 bytes; it is
valid when faceLibType is "blackFD”*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, desc:up to 4 tags, which are separated by commas.
The maximum size is 195 bytes. It is valid when faceLibType is "blackFD”*/
    "address":  "test",
    /*opt, string, person address, desc:the maximum size is 192 bytes. It is
valid when faceLibType is "staticFD”*/
    "customInfo":  "test",
    /*opt, string, custom information, desc:the maximum size is 192 bytes. It is
valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:target model data,non-modeled
binary data needs to be encrypted by Base64 during transmission*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal” (operate by
terminal)*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when
operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face",
"leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point,
desc:object,coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-
coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-
coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:this node is required
when featurePointType is "face"*/
                "height":  1
```

```
                /*opt, int, height, range:[0,1000], desc:this node is required
when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:see the description of this node
and above nodes in the message of JSON_ResponseStatus*/
    "FPID":  "test",
    /*ro, opt, string, face record ID, desc:face record ID returned when the
face record is added,it is unique,and the maximum size is 63 bytes. This node is
valid when errorCode is "1" and errorMsg is "ok"*/
    "rowKey":  "test"
    /*ro, opt, string*/
}
```

13.13.6 Delete the face record(s) in the face picture library

URL

PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>

Query Parameter

Request Message

```
{
    "FPID": [
    /*opt, array, list of face record ID, subType:object*/
        {
            "value":  "test",
            /*req, string, face record ID, range:[0,63]*/
            "rowKey":  "test"
            /*opt, string, main key of face picture library, range:[0,64],
desc:main key of face picture library*/
```

```
        }
    ],
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal" (by
terminal)*/
    "terminalNoList": [1]
    /*opt, array, terminal ID list, subType:int, desc:terminal ID list*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an
error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded).
It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It
is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it
corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when
statusCode is not 1*/
}
```

13.13.7 Search for the face records in the a face picture library or multiple face picture libraries

URL

POST /ISAPI/Intelligent/FDLib/FDSearch?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list,
desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked
position in the next search*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture
library type: "blackFD"-list library,"staticFD"-static library,the maximum size
is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the
maximum size is 63*/
```

```
    "FPID":  " ",
    /*opt, string, face record ID*/
    "startTime":  "2004-05-03",
    /*opt, string, start date of birth*/
    "endTime":  "2004-05-03",
    /*opt, string, end time of birth*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which
is used in search), "male", "female", "unknown"*/
    "city":  "130100",
    /*opt, string, city code, desc:city code of birth for the person in the face
picture,the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string, desc:the maximum size is 10
bytes,certificate type: "officerID"-officer ID,"ID"-identify
card,passport,other*/
    "certificateNumber":  "test",
    /*opt, string, certificate No.*/
    "isInLibrary":  "yes",
    /*opt, enum, whether the picture is in library (whether modeling is
successful), subType:string, desc:whether modeling is successful: unknown, no,
yes*/
    "isDisplayCaptureNum":  true,
    /*opt, bool, whether to display number of captured pictures*/
    "rowKey":  "test",
    /*opt, string, main key of face picture library, desc:searching by rowKey
can be more efficient*/
    "transfer":  true
    /*opt, bool, Transferring, desc:whether to enable transfer*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "activeNumMax",
    /*ro, req, enum, sub status code, subType:string*/
    "errorCode":  1,
    /*ro, opt, int, detailed error description, this field is required when the
value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, detailed error description, this field is required when
the value of statusCode is not 1*/
    "responseStatusStrg":  "OK",
    /*ro, opt, enum, status search, subType:string, desc:"OK" (searching
completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
```

```
    "MatchList": [
    /*ro, opt, array, list of matched records, subType:object*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID*/
            "FDName":  "名单库A",
            /*ro, opt, string*/
            "FPID":  "test",
            /*ro, opt, string, face record ID*/
            "faceURL":  "test",
            /*ro, opt, string, face picture URL*/
            "name":  "test",
            /*ro, opt, string, name*/
            "gender":  "any",
            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited
condition which is used in search), "male", "female", "unknown"*/
            "bornTime":  "2004-05-03",
            /*ro, opt, string, birth date of the person in the face picture*/
            "city":  "130100",
            /*ro, opt, string*/
            "certificateType":  "ID",
            /*ro, req, enum, certificate type, subType:string*/
            "certificateNumber":  "test",
            /*ro, opt, string, certificate No.*/
            "caseInfo":  "test",
            /*ro, opt, string, remarks*/
            "tag":  "aa,bb,cc,dd",
            /*ro, opt, string, custom tag*/
            "address":  "test",
            /*ro, opt, string, person address*/
            "customInfo":  "test",
            /*ro, opt, string, custom information*/
            "isInLibrary":  "yes",
            /*ro, opt, enum, subType:string, desc:whether modeling is
successful: unknown, no, yes*/
            "captureNum":  12,
            /*ro, opt, int, number of captured pictures*/
            "rowKey":  "test",
            /*ro, opt, string, main key of face picture library, desc:searching
by rowKey can be more efficient*/
            "modelData":  "test",
            /*ro, opt, string, target modeling data, desc:during the process of
transmission, the non-modeling binary data will be encrypted with Base64
method*/
            "faceType":  "normalFace",
            /*ro, opt, enum, face picture library type, subType:string*/
            "saveFacePic":  true,
            /*ro, opt, bool*/
            "leaderPermission": [1, 2, 3, 4],
            /*ro, opt, array, subType:int, range:[1,4]*/
            "numberOfArrivalsAtSiteLastDays": [
            /*ro, opt, array, subType:object*/
                {
                    "day":  1,
                    /*ro, opt, int, range:[1,7]*/
                    "numberOfArrivalsAtSite":  3
                    /*ro, opt, int, range:[1,99]*/
                }
```

```
            ],
            "dwellTimeOnPreviousDay":  4,
            /*ro, opt, int, range:[0,86400], unit:min*/
            "totalNumberOfArrivalsAtSite":  4
            /*ro, opt, int, range:[1,99]*/
        }
    ]
}
```

13.14 Device Basic Information Management

13.14.1 Set the device information parameters

URL

PUT /ISAPI/System/deviceInfo

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string,
protocolVersion}-->
  <deviceName>
    <!--req, string, device name, range:[1,128]-->test
  </deviceName>
  <deviceID>
    <!--req, string, device No., range:[1,128]-->test
  </deviceID>
  <deviceDescription>
    <!--opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--opt, string, location, range:[1,128]-->hangzhou
  </deviceLocation>
  <deviceStatus>
    <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-
->normal
  </deviceStatus>
  <DetailAbnormalStatus>
    <!--ro, opt, object, error status details,it is valid only when deviceStatus
is "abnormal", desc:it is valid only when deviceStatus is abnormal-->
    <hardDiskFull>
      <!--ro, opt, bool, read-only,whether the error of "HDD full" occurred:
"true"-yes,"false"-no-->true
    </hardDiskFull>
    <hardDiskError>
      <!--ro, opt, bool, read-only,whether the error of "HDD error" occurred:
"true"-yes,"false"-no-->true
    </hardDiskError>
    <ethernetBroken>
      <!--ro, opt, bool, read-only,whether the error of "network disconnected"
occurred: "true"-yes,"false"-no-->true
```

```
    </ethernetBroken>
    <ipaddrConflict>
      <!--ro, opt, bool, IP address is conflicted.-->true
    </ipaddrConflict>
    <illegalAccess>
      <!--ro, opt, bool, illegal access-->true
    </illegalAccess>
    <recordError>
      <!--ro, opt, bool, read-only,whether the error of "recording exception"
occurred: "true"-yes,"false"-no-->true
    </recordError>
    <raidLogicDiskError>
      <!--ro, opt, bool, read-only,whether the error of "RAID exception"
occurred: "true"-yes,"false"-no-->true
    </raidLogicDiskError>
    <spareWorkDeviceError>
      <!--ro, opt, bool, read-only,whether the error of "working device
exception" occurred: "true"-yes,"false"-no-->true
    </spareWorkDeviceError>
  </DetailAbnormalStatus>
  <systemContact>
    <!--ro, opt, string, contact information of the device, range:[1,32]-->STD-
CGI
  </systemContact>
  <model>
    <!--ro, req, string, device model, range:[1,64]-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--ro, req, string, device serial No., range:[1,48]-->iDS-9632NX-
I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--ro, req, string, MAC Address, range:[1,64]-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate>
    <!--ro, opt, string, compile date of the device firmware version-->2019-11-
01
  </firmwareReleasedDate>
  <encoderVersion>
    <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
  </encoderVersion>
  <encoderReleasedDate>
    <!--ro, opt, string, compile date of the device encoder version-->2019-11-02
  </encoderReleasedDate>
  <bootVersion>
    <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
  </bootVersion>
  <bootReleasedDate>
    <!--ro, opt, string, release date of boot-->2019-11-03
  </bootReleasedDate>
  <panelVersion>
    <!--ro, opt, string, panel version, range:[1,32]-->V1.0
  </panelVersion>
  <hardwareVersion>
    <!--ro, opt, string, hardware version, range:[1,16]-->0x0
```

```
  </hardwareVersion>
  <decoderVersion>
    <!--ro, opt, string, device decoder version, range:[1,32]-->V1.0
  </decoderVersion>
  <decoderReleasedDate>
    <!--ro, opt, string, compile date of the device decoder version-->2019-01-01
  </decoderReleasedDate>
  <softwareVersion>
    <!--ro, opt, string, read-only,software version, range:[1,32]-->V1.23
  </softwareVersion>
  <capacity>
    <!--ro, opt, int, read-only,unit: MB,device capacity, range:[1,10240],
desc:unit: MB-->1
  </capacity>
  <usedCapacity>
    <!--ro, opt, int, read-only,unit: MB,capacity usage, range:[1,10240],
desc:unit: MB-->1
  </usedCapacity>
  <deviceType>
    <!--ro, req, enum, device type, subType:string, desc:"Blade", "BladePS",
"CVR", "DM", "DMbehavior", "DVR", "DVS", "DockStation", "FA", "FD", "FS",
"HAWK", "HMCP", "HybirdNVR", "IPCamera", "IPDome", "IPZoom", "InfoReleaseSys",
"InfoTerminal', "MCU", "NVR", "PHA", "PURE", "RFID", "SipServer", "Switch",
"lightFace", "Visitor", "PWA", "PHAPro", "ACS", "VIS", "FacePaymentTerminal",
"InteractiveTerminal", "Cabinet", "VoiceSpeaker", "PowerAmplifier",
"FireControlMatrix", "PagingMicrophone", "ElevatorControl", "conferencePlat"
(interactive tablet), "AIOT", "OPCA", "NetworkReceiver", "PanicAlarmPanel",
"bodyCamera", "videoMatrix", "EmbeddedCentralController", "networkMic", "IPA",
"PersonnelChannel", "MediaComponentGateway"-->Blade
  </deviceType>
  <telecontrolID>
    <!--opt, int, keyfob ID, range:[1,255]-->1
  </telecontrolID>
  <supportBeep>
    <!--ro, opt, bool, whether to support buzzer-->true
  </supportBeep>
  <supportVideoLoss>
    <!--ro, opt, bool, whether the device supports video loss-->true
  </supportVideoLoss>
  <firmwareVersionInfo>
    <!--ro, opt, string, firmware version information, range:[1,32]-->test
  </firmwareVersionInfo>
  <actualFloorNum>
    <!--ro, opt, int, actual number of floors,which is between 1 and 128, range:
[1,128]-->1
  </actualFloorNum>
  <localZoneNum>
    <!--ro, opt, int, number of zones (local), range:[0,16]-->1
  </localZoneNum>
  <alarmOutNum>
    <!--ro, opt, int, number of alarm outputs, range:[0,16]-->1
  </alarmOutNum>
  <distanceResolution>
    <!--ro, opt, float, distance resolution, range:[0.000,0.999]-->0.000
  </distanceResolution>
  <angleResolution>
    <!--ro, opt, float, angle resolution, range:[0.000,0.999]-->0.000
  </angleResolution>
```

```
  <speedResolution>
    <!--ro, opt, float, resolution of speed,unit: m/s, range:[0.000,0.999]--
>0.000
  </speedResolution>
  <detectDistance>
    <!--ro, opt, float, detection distance, range:[0.000,0.999]-->0.000
  </detectDistance>
  <relayNum>
    <!--ro, opt, int, number of relays (local), range:[0,16]-->1
  </relayNum>
  <electroLockNum>
    <!--ro, opt, int, number of locks (local), range:[0,16]-->1
  </electroLockNum>
  <sirenNum>
    <!--ro, opt, int, number of sirens, range:[0,16]-->1
  </sirenNum>
  <alarmLamp>
    <!--ro, opt, int, number of alarm lamps, range:[0,16]-->1
  </alarmLamp>
  <RS485Num>
    <!--ro, opt, int, number of 485, range:[0,16]-->1
  </RS485Num>
  <radarVersion>
    <!--ro, opt, string, radar version, range:[1,32]-->test
  </radarVersion>
  <cameraModuleVersion>
    <!--ro, opt, string, camera modulo version, range:[1,32]-->test
  </cameraModuleVersion>
  <mainversion>
    <!--ro, opt, int, main version No., range:[1,255]-->1
  </mainversion>
  <subversion>
    <!--ro, opt, int, sub version No., range:[1,255]-->1
  </subversion>
  <upgradeversion>
    <!--ro, opt, int, update version No., range:[1,255]-->1
  </upgradeversion>
  <customizeversion>
    <!--ro, opt, int, custom version No., range:[1,255]-->1
  </customizeversion>
  <companyName>
    <!--ro, opt, string, manufacturer name, range:[1,32]-->test
  </companyName>
  <copyright>
    <!--ro, opt, string, version information, range:[1,32]-->test
  </copyright>
  <systemName>
    <!--ro, opt, enum, storage system name, subType:string,
desc:"storageManagement" (storage management system),
"distributedStorageManagement" (distributed storage management system)--
>storageManagement
  </systemName>
  <systemStatus>
    <!--ro, opt, enum, system status, subType:string, desc:"configured",
"unConfigured" (not configured)-->configured
  </systemStatus>
  <isLeaderDevice>
```

```
    <!--ro, opt, bool, whether it is the corresponding device of the resource IP address-->true
  </isLeaderDevice>
  <clusterVersion>
    <!--ro, opt, string, cluster version, range:[1,32], desc:it is valid when isLeaderDevice returns true-->test
  </clusterVersion>
  <manufacturer>
    <!--ro, opt, enum, manufacturer information, subType:string, desc:it is used to distinguish OEM device-->test
  </manufacturer>
  <centralStorageVersion>
    <!--ro, opt, string, center storage version, range:[1,16]-->test
  </centralStorageVersion>
  <powerOnMode>
    <!--ro, opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
  </powerOnMode>
  <customizedInfo>
    <!--ro, opt, string, custom project No., range:[1,32], desc:no value will be returned if it is baseline device. custom project No. will be returned if is custom device-->test
  </customizedInfo>
  <verificationCode>
    <!--ro, opt, string, device verification code-->test
  </verificationCode>
  <supportUrl>
    <!--ro, opt, string, service portal-->test
  </supportUrl>
  <subSerialNumber>
    <!--ro, opt, string, sub serial No.-->test
  </subSerialNumber>
  <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese">
    <!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french", "russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch", "hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic", "traditionalChinese"-->chinese
  </languageType>
  <DockStation>
    <!--ro, opt, object, dock station configuration-->
    <Platform>
      <!--ro, opt, object, platform configuration-->
      <type>
        <!--ro, req, enum, platform type: none,9533,8618,ISAPI, subType:string, desc:"none", "9533", "8618", "ISAPI"-->none
      </type>
      <ip>
        <!--ro, opt, string, IP address, range:[1,32]-->test
      </ip>
      <port>
        <!--ro, opt, int, communication port, range:[1,65535]-->1
      </port>
      <userName>
```

```
        <!--ro, req, string, user name,which is used for the dock station to log in to platform, range:[1,32]-->test
      </userName>
      <password>
        <!--ro, req, string, password, range:[1,16]-->test
      </password>
    </Platform>
    <centralStorageBackupEnabled>
      <!--ro, opt, bool, whether to enable central storage backup-->true
    </centralStorageBackupEnabled>
  </DockStation>
  <webVersion>
    <!--ro, opt, string, Web version No., range:[1,32]-->test
  </webVersion>
  <deviceRFProgramVersion>
    <!--ro, opt, string, device RF program version, range:[1,32]-->test
  </deviceRFProgramVersion>
  <securityModuleSerialNo>
    <!--ro, opt, string, security module serial No., range:[1,32]-->test
  </securityModuleSerialNo>
  <securityModuleVersion>
    <!--ro, opt, string, security module version, range:[1,32]-->test
  </securityModuleVersion>
  <securityChipVersion>
    <!--ro, opt, string, security chip version, range:[1,32]-->test
  </securityChipVersion>
  <securityModuleKeyVersion>
    <!--ro, opt, string, security module key version, range:[1,32]-->test
  </securityModuleKeyVersion>
  <UIDLampRecognition>
    <!--ro, opt, object, recognize the device information by UID-->
    <enabled>
      <!--ro, opt, bool, whether to enable the function-->true
    </enabled>
  </UIDLampRecognition>
  <confDeviceIdPrefix>
    <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
  </confDeviceIdPrefix>
  <OEMCode>
    <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
  </OEMCode>
  <simpleAlgorithmVersion>
    <!--ro, opt, string, single algorithm version, desc:for fire control analyzer, it only supports one recognition algorithm if this node is returned to the device information-->test
  </simpleAlgorithmVersion>
  <bootTime>
    <!--ro, opt, datetime, boot time-->1970-01-01T00:00:00+08:00
  </bootTime>
  <intelligentAnalysisEngineModel>
    <!--ro, opt, string, engine model, range:[1,32]-->test
  </intelligentAnalysisEngineModel>
  <marketType>
    <!--ro, opt, enum, market type, subType:int, desc:0 (invalid), 1 (distribution type), 2 (industry type)-->0
  </marketType>
```

```
  <algorithmVersion>
    <!--ro, opt, string, HCNN algorithm version, desc:its Data Dictionary is
provided by research institute for matching the algorithm model-->test
  </algorithmVersion>
  <firmware>
    <!--ro, opt, string, firmware version, desc:its Data Dictionary is provided
by research institute for matching the algorithm model-->test
  </firmware>
  <engineList>
    <!--ro, opt, object, list of device computing power, desc:it returns engine
No. by number of GPU chips-->
    <engine>
      <!--ro, opt, int, returned engine No.-->1
    </engine>
  </engineList>
  <platform>
    <!--ro, opt, enum, platform type, subType:int, desc:1 (TX1), 2 (P4), 3
(3559), 4 (3519), 5 (3516)-->1
  </platform>
  <platformName>
    <!--ro, opt, string, the platform where the algorithm runs, desc:TX1/K81:
NVR, KT: data center,H5: front-end device, H7: front-end device, K82: NVR, G3:
front-end device-->test
  </platformName>
  <touchScreenVersionInfo>
    <!--ro, opt, string, touch screen module version-->test
  </touchScreenVersionInfo>
  <protocolFileURL>
    <!--ro, opt, string, protocol notice URI, range:[1,32]-->test
  </protocolFileURL>
  <recycleRecordEnabled>
    <!--ro, opt, bool, whether to enable overwritten recording-->false
  </recycleRecordEnabled>
  <decordChannelNums>
    <!--ro, opt, int, number of decoding channels-->0
  </decordChannelNums>
  <VGANums>
    <!--ro, opt, int, number of VGA ports-->0
  </VGANums>
  <USBNums>
    <!--ro, opt, int, number of USB ports-->0
  </USBNums>
  <auxoutNums>
    <!--ro, opt, int, number of auxiliary ports-->0
  </auxoutNums>
  <expansionBoardVersion>
    <!--ro, opt, string, extension board version information, range:[1,32]--
>test
  </expansionBoardVersion>
  <initWizzardDisplay>
    <!--ro, opt, bool, whether it displays initialization wizard, desc:it is
only used for displaying configuration of applications integrated to the wizard,
such as local GUI and Hik-Connect app by default, the value is true. it is false
after the local GUI configures the wizard-->true
  </initWizzardDisplay>
  <beaconID>
    <!--ro, opt, string, device RF program version No., range:[0,32], desc:by
default, it is the current value-->test
```

```
  </beaconID>
  <isResetDeviceLanguage>
    <!--ro, opt, bool, whether it supports resetting the device language (only
for Admin and Installer), desc:if the value is true, the Hik-Connect app and the
Web Client display device information in the target language; if the value is
false, the Hik-Connect app and the Web Client display the original device
information-->false
  </isResetDeviceLanguage>
  <dispalyNum>
    <!--ro, opt, int, number of windows-->0
  </dispalyNum>
  <bspVersion>
    <!--ro, opt, string, BSP software version-->test
  </bspVersion>
  <dspVersion>
    <!--ro, opt, string, DSP software version-->test
  </dspVersion>
  <localUIVersion>
    <!--ro, opt, string, local UI version-->test
  </localUIVersion>
  <detectorType>
    <!--ro, opt, enum, detector type, subType:string, desc:"PDA" (photo-diode
array)-->PDA
  </detectorType>
  <wiegandOutNum>
    <!--ro, opt, int, number of Wiegand outputs, range:[0,1]-->1
  </wiegandOutNum>
  <ChipVersionInfoList>
    <!--ro, opt, array, subType:object, range:[1,6]-->
    <ChipVersionInfo>
      <!--ro, opt, object-->
      <ID>
        <!--ro, req, int, range:[0,1]-->1
      </ID>
      <type>
        <!--ro, req, enum, subType:string-->MCU
      </type>
      <firmwareVersion>
        <!--ro, req, string, range:[0,64]-->1.0.0
      </firmwareVersion>
    </ChipVersionInfo>
  </ChipVersionInfoList>
  <personBagLinkAlgoEngineVersion>
    <!--ro, opt, string, engine version of the person & package linkage module,
range:[0,64], desc:this node is for analyzer (security inspection)-->1.0.0
  </personBagLinkAlgoEngineVersion>
  <BIOSVersion>
    <!--ro, opt, string, BIOS version, range:[0,16]-->test
  </BIOSVersion>
  <contactInformation>
    <!--ro, opt, string, contact information, range:[0,64]-->test
  </contactInformation>
  <materialScanAlgorithmVersion>
    <!--opt, string, material scan algorithm version, range:[0,64],
desc:algorithm version of imaging radar panel for material scan-->test
  </materialScanAlgorithmVersion>
  <audioVersion>
    <!--ro, opt, string, range:[1,32]-->test
```

```
  </audioVersion>
</DeviceInfo>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string,
protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device
Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK"
(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML
Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

13.14.2 Get configuration capability of the device information parameters

URL

GET /ISAPI/System/deviceInfo/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string,
protocolVersion}-->
  <deviceName min="1" max="128">
    <!--ro, req, string, device name, range:[1,128], attr:min{opt,
string},max{opt, string}-->test
```

```
  </deviceName>
  <deviceID min="1" max="128">
    <!--ro, opt, string, device ID, range:[1,128], attr:min{opt,
string},max{opt, string}-->123456
  </deviceID>
  <deviceDescription min="1" max="128">
    <!--ro, opt, string, device description, range:[1,128], attr:min{opt,
string},max{opt, string}-->test
  </deviceDescription>
  <deviceLocation opt="STD-CGI">
    <!--ro, opt, string, device location, range:[1,32], attr:opt{opt, string}--
>hangzhou
  </deviceLocation>
  <systemContact opt="STD-CGI">
    <!--ro, opt, string, manufacturer, range:[1,32], attr:opt{opt, string}--
>test
  </systemContact>
  <model min="1" max="64">
    <!--ro, opt, string, device model, range:[1,64], attr:min{opt,
string},max{opt, string}-->iDS-9632NX-I8/X
  </model>
  <serialNumber min="1" max="48">
    <!--ro, opt, string, device serial No., range:[1,48], attr:min{opt,
string},max{opt, string}-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress min="1" max="64">
    <!--ro, opt, string, MAC address, range:[1,64], attr:min{opt,
string},max{opt, string}-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion min="1" max="64">
    <!--ro, opt, string, device firmware version, range:[1,64], attr:min{opt,
string},max{opt, string}-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate min="1" max="64">
    <!--ro, opt, string, release date of the device firmware version, range:
[1,64], attr:min{opt, string},max{opt, string}-->build 181129
  </firmwareReleasedDate>
  <encoderVersion min="1" max="64">
    <!--ro, opt, string, device description, range:[1,64], attr:min{opt,
string},max{opt, string}-->V7.3
  </encoderVersion>
  <encoderReleasedDate min="1" max="64">
    <!--ro, opt, string, release date, range:[1,64], attr:min{opt,
string},max{opt, string}-->build 20161223
  </encoderReleasedDate>
  <bootVersion min="1" max="16">
    <!--ro, opt, string, BOOT version No., range:[1,16], attr:min{opt,
string},max{opt, string}-->V1.3.4
  </bootVersion>
  <bootReleasedDate min="1" max="16">
    <!--ro, opt, string, release date of boot, range:[1,16], attr:min{opt,
string},max{opt, string}-->100316
  </bootReleasedDate>
  <hardwareVersion min="1" max="16">
    <!--ro, opt, string, hardware version, range:[1,16], attr:min{opt,
string},max{opt, string}-->0x0
  </hardwareVersion>
```

```
  <deviceType
opt="IPCamera,IPDome,DVR,HybirdNVR,NVR,DVS,IPZoom,CVR,DockStation,RFID,InfoReleaseSys,MCU,SipServer,PHA,PWA,InfoTerminal,Switch,Visitor,PHAPro,ACS,VIS,FacePaymentTerminal,InteractiveTerminal,Cabinet,VoiceSpeaker,PowerAmplifier,FireControlMatrix,PagingMicrophone,ElevatorControl,conferencePlat,OPCA,NetworkReceiver,PanicAlarmPanel,videoMatrix,EmbeddedCentralController,networkMic,PersonnelChannel,MediaComponentGateway,ImagingRadarPanel,TrafficTerminalServer,securityRadar,TrafficTerminalServer,UPLC,GC">
    <!--ro, opt, string, device type, attr:opt{opt, string}, desc:PWA continues to be used in wireless security control panel (Ax Pro series)-->IPCamera
  </deviceType>
  <subDeviceType
opt="accessControlTerminal,attendanceCheckDevice,multiChannelAccessController,personAndIdCardDevice,doorStation,mainStation,swingBarrier,flapBarrier,tripodTurnstile">
    <!--ro, opt, string, attr:opt{req, string}-->accessControlTerminal
  </subDeviceType>
  <telecontrolID min="1" max="16">
    <!--ro, opt, int, remote control ID, range:[1,16], attr:min{opt, string},max{opt, string}-->8
  </telecontrolID>
  <subChannelEnabled>
    <!--ro, opt, bool, whether to enable two channels-->true
  </subChannelEnabled>
  <thrChannelEnabled>
    <!--ro, opt, bool, whether to enable three channels-->true
  </thrChannelEnabled>
  <powerOnMode opt="button,adapter" def="button">
    <!--ro, opt, enum, device startup mode, subType:string, attr:opt{opt, string},def{opt, string}, desc:"button", "adapter"-->button
  </powerOnMode>
  <customizedInfo min="0" max="128">
    <!--ro, opt, string, device custom No., range:[0,128], attr:min{req, int},max{req, int}, desc:for baseline devices, returing this node is not required-->test
  </customizedInfo>
  <languageType
opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese,slovenian,swedish,norwegian,romanian,danish,german,polish,dutch,hungarian,slovak,serbian,southAmericanSpanish,ukrainian,croatian,irish,bulgarian,hebrew,thai,indonesian,arabic,traditionalChinese,lithuanian,anglicism,estonian">
    <!--ro, opt, string, language type, range:[1,32], attr:opt{opt, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french", "russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch", "hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic", "traditionalChinese"-->chinese
  </languageType>
  <localZoneNum min="0" max="16">
    <!--ro, opt, int, number of local zones, range:[0,16], attr:min{opt, string},max{opt, string}-->1
  </localZoneNum>
  <alarmOutNum min="0" max="16">
    <!--ro, opt, int, number of alarm outputs, range:[0,16], attr:min{opt, string},max{opt, string}-->1
  </alarmOutNum>
```

```
  <relayNum min="0" max="16">
    <!--ro, opt, int, number of local relays, range:[0,16], attr:min{opt,
string},max{opt, string}-->1
  </relayNum>
  <electroLockNum min="0" max="16">
    <!--ro, opt, int, number of local electronic locks, range:[0,16],
attr:min{opt, string},max{opt, string}-->1
  </electroLockNum>
  <sirenNum min="0" max="16">
    <!--ro, opt, int, number of sirens, range:[0,16], attr:min{opt,
string},max{opt, string}-->1
  </sirenNum>
  <alarmLamp min="0" max="16">
    <!--ro, opt, int, number of strobe lights, range:[0,16], attr:min{opt,
string},max{opt, string}-->1
  </alarmLamp>
  <RS485Num min="0" max="16">
    <!--ro, opt, int, number of local RS-485, range:[0,16], attr:min{opt,
string},max{opt, string}-->1
  </RS485Num>
  <manufacturer opt="hikvision, ,Hostone">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->hikvision
  </manufacturer>
  <DockStation>
    <!--ro, opt, object, dock station information configuration-->
    <Platform>
      <!--ro, opt, object, platform dock station information configuration-->
      <type opt="none, 9533, 8618, ISAPI">
        <!--ro, req, enum, platform which is connected to the dock station,
subType:string, attr:opt{opt, string}, desc:"none", "9533", "8618", "ISAPI"--
>none
      </type>
      <ip>
        <!--ro, opt, string, IP address, range:[1,32]-->0.0.0.0
      </ip>
      <port>
        <!--ro, opt, int, port No., range:[1,65535]-->1
      </port>
      <userName>
        <!--ro, req, string, user name, range:[1,32]-->test
      </userName>
      <password>
        <!--ro, req, string, password, range:[1,16]-->test
      </password>
    </Platform>
    <centralStorageBackupEnabled opt="true, false">
      <!--ro, opt, bool, whether to enable center storage backup, attr:opt{opt,
string}, desc:when it is enabled, the device uploads all data to the center
storage for backup before uploading them to the platform; when it is disabled,
the device uploads data to the platform directly, and for those important data,
the platform will apply the file list to the device for center storage (related
URI: /ISAPI/ContentMgmt/record/control/centralStorageBackup), then the device
will upload the data with storage URI to the platform-->true
    </centralStorageBackupEnabled>
    <isSupportManufacturer>
```

```
      <!--ro, opt, bool, whether it supports configuring third-party
manufacturer for the dock station,
desc:/ISAPI/Traffic/dockStation/slot/Manufacturer/capabilities?format=json--
>true
    </isSupportManufacturer>
  </DockStation>
  <webVersion>
    <!--ro, opt, string, Web version No., range:[1,32]-->1.0
  </webVersion>
  <deviceRFProgramVersion>
    <!--ro, opt, string, device RF program version No., range:[1,32]-->1.0
  </deviceRFProgramVersion>
  <securityModuleSerialNo>
    <!--ro, opt, string, security module serial No., range:[1,48]-->1.0
  </securityModuleSerialNo>
  <securityModuleVersion>
    <!--ro, opt, string, security module version No., range:[1,32]-->1.0
  </securityModuleVersion>
  <securityChipVersion>
    <!--ro, opt, string, security chip version No., range:[1,32]-->1.0
  </securityChipVersion>
  <securityModuleKeyVersion>
    <!--ro, opt, string, security module key version No., range:[1,32]-->1.0
  </securityModuleKeyVersion>
  <UIDLampRecognition>
    <!--ro, opt, object, recognize the device information by UID-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{opt, string}-->true
    </enabled>
  </UIDLampRecognition>
  <confDeviceIdPrefix opt="true,false">
    <!--ro, opt, bool, whether the meeting uses the device name as its prefix,
attr:opt{opt, string}-->true
  </confDeviceIdPrefix>
  <simpleAlgorithmVersion min="0" max="10">
    <!--ro, opt, string, single algorithm version, attr:min{req, int},max{req,
int}-->test
  </simpleAlgorithmVersion>
  <bootTime>
    <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
  </bootTime>
  <isSupportNewVersionDevlanguageSwitch>
    <!--ro, opt, bool, whether it supports switching languages for the new
protocol version-->true
  </isSupportNewVersionDevlanguageSwitch>
  <intelligentAnalysisEngineModel>
    <!--ro, opt, string, engine model, range:[1,32]-->K81
  </intelligentAnalysisEngineModel>
  <marketType opt="0,1,2">
    <!--ro, opt, enum, market type, subType:int, attr:opt{req, string}, desc:0
(invalid), 1 (distribution type), 2 (industry type)-->0
  </marketType>
  <protocolFileURL min="1" max="32">
    <!--ro, opt, string, protocol notice URI, attr:min{opt, string},max{opt,
string}-->test
  </protocolFileURL>
  <recycleRecordEnabled opt="true,false">
```

```
    <!--ro, opt, bool, whether to enable overwritten recording, attr:opt{req,
string}-->false
  </recycleRecordEnabled>
  <decordChannelNums>
    <!--ro, opt, int, number of decoding channels-->0
  </decordChannelNums>
  <VGANums>
    <!--ro, opt, int, number of VGA ports-->0
  </VGANums>
  <USBNums>
    <!--ro, opt, int, number of USB ports-->0
  </USBNums>
  <auxoutNums>
    <!--ro, opt, int, number of auxiliary ports-->0
  </auxoutNums>
  <expansionBoardVersion min="0" max="32">
    <!--ro, opt, string, extension board version information, range:[0,32],
attr:min{req, int},max{req, int}-->test
  </expansionBoardVersion>
  <initWizzardDisplay opt="true,false">
    <!--ro, opt, bool, whether it displays initialization wizard, attr:opt{req,
string}, desc:it is only used for displaying configuration of applications
integrated to the wizard, such as local GUI and Hik-Connect app. By default, the
value is true; it is false after the wizard is configured on the local GUI--
>true
  </initWizzardDisplay>
  <beaconID min="0" max="32">
    <!--ro, opt, string, device RF program version No., range:[0,32],
attr:min{req, int},max{req, int}, desc:by default, it is the current value--
>test
  </beaconID>
  <isResetDeviceLanguage opt="true,false">
    <!--ro, opt, bool, whether to reset the language, attr:opt{req, string},
desc:only available for Admin/Installer: true-after the language reset, the
device information will be displayed in the target language on Hik-Connect
Mobile Client and Web Client; false-after the language reset, the device
information will be displayed in the original language on Hik-Connect Mobile
Client and Web Client-->false
  </isResetDeviceLanguage>
  <dispalyNum>
    <!--ro, opt, int, number of device screens-->2
  </dispalyNum>
  <bspVersion min="1" max="1">
    <!--ro, opt, string, BSP version, attr:min{req, int},max{req, int}-->test
  </bspVersion>
  <dspVersion min="1" max="1">
    <!--ro, opt, string, DSP version, attr:min{req, int},max{req, int}-->test
  </dspVersion>
  <localUIVersion min="1" max="1">
    <!--ro, opt, string, local UI version, attr:min{req, int},max{req, int}--
>test
  </localUIVersion>
  <OPCASubType opt="PDA,pump,column,autoSampler,SOC">
    <!--ro, opt, enum, the sub type of optical physical and chemical analyzer,
subType:string, attr:opt{req, string}, desc:"detector"; "pump"; "columnOven"
(column oven); "autoSampler" (auto sampler)-->detector
  </OPCASubType>
  <wiegandOutNum min="0" max="1">
```

```
    <!--ro, opt, int, number of Wiegand outputs, range:[0,1], attr:min{opt,
string},max{opt, string}-->1
  </wiegandOutNum>
  <ChipVersionInfoList size="6">
    <!--ro, opt, object, chip version information list, attr:size{req, int}-->
    <ChipVersionInfo>
      <!--ro, opt, object, chip version information-->
      <ID min="0" max="1">
        <!--ro, req, int, chip No., range:[0,1], attr:min{req, int},max{req,
int}, desc:the No.s of chips of different types can be duplicate. Users can
distinguish them by chip type and chip No. together-->1
      </ID>
      <type opt="MCU,FPGA,MSTAR">
        <!--ro, req, enum, chip type, subType:string, attr:opt{req, string},
desc:"MCU", "FPGA", "MSTAR"-->MCU
      </type>
      <firmwareVersion min="0" max="64">
        <!--ro, req, string, chip firmware version, range:[0,64], attr:min{req,
int},max{req, int}-->1.0.0
      </firmwareVersion>
      <chipName min="0" max="32">
        <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}--
>test
      </chipName>
    </ChipVersionInfo>
  </ChipVersionInfoList>
  <personBagLinkAlgoEngineVersion min="0" max="64">
    <!--ro, opt, string, engine version of the person & package linkage module,
range:[0,64], attr:min{req, int},max{req, int}, desc:this node is for analyzer
(security inspection)-->1.0.0
  </personBagLinkAlgoEngineVersion>
  <BIOSVersion min="0" max="16">
    <!--ro, opt, string, BIOS version, range:[0,16], attr:min{req, int},max{req,
int}-->V1.3.4
  </BIOSVersion>
  <contactInformation min="0" max="64">
    <!--ro, opt, string, contact information, range:[0,64], attr:min{req,
int},max{req, int}-->test
  </contactInformation>
  <PedestrianWarningModuleVersion>
    <!--ro, opt, object, pedestrian warning module information, desc:this node
is returned only for pedestrian warning devices-->
    <pedestrianWarningMCUVersion min="1" max="32">
      <!--ro, opt, string, MCU program version, range:[0,16], attr:min{req,
int},max{req, int}-->V1.3.4
    </pedestrianWarningMCUVersion>
    <pedestrianWarningRadarVersion min="1" max="32">
      <!--ro, opt, string, the program version of pedestrian warning radar,
range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningRadarVersion>
    <pedestrianRangingModuleVersion min="1" max="32">
      <!--ro, opt, string, the program version of the pedestrian distance
measurement module, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianRangingModuleVersion>
  </PedestrianWarningModuleVersion>
  <encryptionModel min="0" max="16">
```

```
    <!--ro, opt, string, device encryption model, range:[0,16], attr:min{req,
int},max{req, int}, desc:it cannot be modified and is a hexadecimal digit
currently-->test
  </encryptionModel>
  <UWBVersion min="1" max="32">
    <!--ro, opt, string, UWB version, range:[1,32], attr:min{req, int},max{req,
int}-->V1.3.4
  </UWBVersion>
  <audioBoard>
    <!--ro, opt, object, audio board information-->
    <audioBoardModel min="0" max="32">
      <!--ro, opt, string, audio board model, range:[0,32], attr:min{req,
int},max{req, int}, desc:currently only C8 and R8 are supported-->C8
    </audioBoardModel>
    <audioBoardVersion min="0" max="32">
      <!--ro, opt, string, audio board version, range:[0,32], attr:min{req,
int},max{req, int}-->V1.0.0 build 211210
    </audioBoardVersion>
  </audioBoard>
  <materialScanAlgorithmVersion min="0" max="64">
    <!--ro, opt, string, material scan algorithm version, range:[0,64],
attr:min{req, int},max{req, int}, desc:algorithm version of imaging radar panel
for material scan-->1.0.0
  </materialScanAlgorithmVersion>
  <productionDate>
    <!--ro, opt, date-->2022-04-02
  </productionDate>
  <wifiModuleMACAddress min="1" max="48">
    <!--ro, opt, string, range:[1,48], attr:min{req, int},max{req, int}--
>44:47:cc:c8:d9:e4
  </wifiModuleMACAddress>
  <isSupportExportDeviceFingerFile>
    <!--ro, opt, bool-->true
  </isSupportExportDeviceFingerFile>
  <isSupportPromptMessage>
    <!--ro, opt, bool-->true
  </isSupportPromptMessage>
  <shortSerialNumber min="1" max="9">
    <!--ro, opt, string, range:[1,9], attr:min{req, int},max{req, int}-->test
  </shortSerialNumber>
  <OSCoreVersionInfo>
    <!--ro, opt, object-->
    <OSCoreVersion min="1" max="32">
      <!--ro, req, string, range:[1,32], attr:min{req, int},max{req, int}--
>1.0.0
    </OSCoreVersion>
    <minisysVersion min="1" max="32">
      <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}--
>1.0.0
    </minisysVersion>
    <networkServiceVersion min="1" max="32">
      <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}--
>1.0.0
    </networkServiceVersion>
    <upgradeServiceVersion min="1" max="32">
      <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}--
>1.0.0
    </upgradeServiceVersion>
```

```
  </OSCoreVersionInfo>
  <isSupportExportDeviceIcon>
    <!--ro, opt, bool-->true
  </isSupportExportDeviceIcon>
  <xTransVersion min="1" max="32">
    <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}-->1.0.0
  </xTransVersion>
</DeviceInfo>
```

13.14.3 Get firmware code list

URL

GET /ISAPI/System/firmwareCodeV2?startIndex=<startIndex>&maxNumber=<maxNumber>

Query Parameter

Parameter Name	Parameter Type	Description
startIndex	string	To control the message size, the quantity of returned firmware IDs returned by device depends on the client request. For example, if there are 100 firmware IDs of a device, and up to 32 IDs are allowed in every request, then the device returns only 32 IDs according to the start index (startIndex). Examples: GET /ISAPI/System/firmwareCodeV2?startIndex=1&maxNumber=32
maxNumber	string	To control the message size, the quantity of returned firmware IDs returned by device depends on the client request. For example, if there are 100 firmware IDs of a device, and up to 32 IDs are allowed in every request, then the device returns only 32 IDs according to the start index (startIndex). Examples: GET /ISAPI/System/firmwareCodeV2?startIndex=1&maxNumber=32

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FirmwareCodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, firmware ID list, subType:object, attr:version{req,
string, protocolVersion}-->
  <FirmwareCode>
    <!--ro, opt, object, firmware ID-->
    <index>
      <!--ro, req, int, index, desc:it starts from 1-->1
    </index>
    <code>
```

```
      <!--ro, req, string, firmware ID, desc:N/A--
>00000002000001000000000100000001000000010000000000000001ffffffff0001020307e1060
100001000
    </code>
    <version>
      <!--ro, req, string, version-->V5.5.160 build 210416
    </version>
    <channel>
      <!--ro, opt, int, channel No., desc:this node is for IP channel-->1
    </channel>
    <devTypeName>
      <!--ro, opt, string, device model name which IP channel corresponds to,
desc:device model name which IP channel corresponds to-->test
    </devTypeName>
    <moduleType>
      <!--ro, opt, enum, device component type, subType:string,
desc:"cardReader" (card reader), "FPModule" (fingerprint module),
"securityModule" (security module), "extendModule" (I/O extended module),
"channelController" (lane controller), "IRModule" (infrared module),
"lampModule" (indicator module), "elevatorController" (sub elevator controller),
"FPAlgorithmProgram" (fingerprint algorithm programma of card reader),
"dockStationHost" (dock station host), "dockStationScreen" (dock station
screen)-->cardReader
    </moduleType>
    <moduleID>
      <!--ro, opt, int, device component No., desc:it depends on the value of
moduleType-->1
    </moduleID>
  </FirmwareCode>
</FirmwareCodeList>
```

13.14.4 Get device information parameters

URL

GET /ISAPI/System/deviceInfo

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string,
protocolVersion}-->
  <deviceName>
    <!--ro, req, string, device name, range:[1,128]-->test
  </deviceName>
  <deviceID>
    <!--ro, req, string, device No., range:[1,128]-->test
  </deviceID>
```

```
  <deviceDescription>
    <!--ro, opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--ro, opt, string, device location, range:[1,128]-->hangzhou
  </deviceLocation>
  <deviceStatus>
    <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-
->normal
  </deviceStatus>
  <DetailAbnormalStatus>
    <!--ro, opt, object, error status details, desc:it is valid only when
deviceStatus is "abnormal"-->
    <hardDiskFull>
      <!--ro, opt, bool, the disk is full-->true
    </hardDiskFull>
    <hardDiskError>
      <!--ro, opt, bool, the disk error-->true
    </hardDiskError>
    <ethernetBroken>
      <!--ro, opt, bool, network disconnected-->true
    </ethernetBroken>
    <ipaddrConflict>
      <!--ro, opt, bool, IP address is conflicted-->true
    </ipaddrConflict>
    <illegalAccess>
      <!--ro, opt, bool, illegal access-->true
    </illegalAccess>
    <recordError>
      <!--ro, opt, bool, recording exception-->true
    </recordError>
    <raidLogicDiskError>
      <!--ro, opt, bool, virtual disk exception in the array-->true
    </raidLogicDiskError>
    <spareWorkDeviceError>
      <!--ro, opt, bool, hot spare active device exception-->true
    </spareWorkDeviceError>
  </DetailAbnormalStatus>
  <systemContact>
    <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
  </systemContact>
  <model>
    <!--ro, req, string, device model, range:[1,64]-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--ro, req, string, device serial No., range:[1,48]-->iDS-9632NX-
I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--ro, req, string, MAC address, range:[1,64]-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate>
    <!--ro, opt, string, Firmware Version-->2019-11-01
  </firmwareReleasedDate>
  <encoderVersion>
```

```
    <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
  </encoderVersion>
  <encoderReleasedDate>
    <!--ro, opt, string, compile date of the device encoder version-->2019-11-02
  </encoderReleasedDate>
  <bootVersion>
    <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
  </bootVersion>
  <bootReleasedDate>
    <!--ro, opt, string, release date of boot-->2019-11-03
  </bootReleasedDate>
  <panelVersion>
    <!--ro, opt, string, panel version, range:[1,32]-->V1.0
  </panelVersion>
  <hardwareVersion>
    <!--ro, opt, string, hardware version, range:[1,16]-->0x0
  </hardwareVersion>
  <decoderVersion>
    <!--ro, opt, string, device decoder version, range:[1,32]-->V1.0
  </decoderVersion>
  <decoderReleasedDate>
    <!--ro, opt, string, compile date of the device decoder version-->2019-01-01
  </decoderReleasedDate>
  <softwareVersion>
    <!--ro, opt, string, software version, range:[1,32]-->V1.23
  </softwareVersion>
  <capacity>
    <!--ro, opt, int, device capacity, range:[1,10240], desc:unit), " MB-->1
  </capacity>
  <usedCapacity>
    <!--ro, opt, int, used capacity of the device, range:[1,10240], desc:unit:
MB-->1
  </usedCapacity>
  <deviceType>
```

```
    <!--ro, req, enum, device type, subType:string, desc:"Blade" (vehicle
analysis server), "BladePS" (security vehicle analysis server), "CVR","DM" (full
analysis server), "DMbehavior" (behavior analysis device), "DVR","DVS",
"DockStation" (dock station), "FA" (real-time face picture comparison and alarm
server), "FD" (capture detection server), "FS" (face picture static data
server), "HAWK" (video structural analysis server), "HMCP" (model comparison
server), "HybirdNVR" (hybrid DVR), "IPCamera" (network camera), "IPDome"
(network speed dome), "IPZoom" (integrated camera), "InfoReleaseSys"
(information release system), "InfoTerminal" (information release terminal),
"MCU", "NVR", "PHA" (security control panel), "PURE" (pure analysis server),
"RFID", "SipServer" (SIP server), "Switch", "lightFace" (lightweight facial
recognition server), "Visitor" (visitor terminal), "PWA", "PHAPro" (Ax Pro
security control panel), "ACS" (access control device), "VIS" (video intercom
device), "FacePaymentTerminal" (payment terminal), "InteractiveTerminal"
(intelligent interaction terminal), "Cabinet" (intelligent cabinet),
"VoiceSpeaker" (floor standing speaker), "PowerAmplifier"(power amplifier),
"FireControlMatrix" (fire protection matrix), "PagingMicrophone",
"ElevatorControl" (elevator controller), "conferencePlat" (interactive tablet),
"AIOT" (Artificial Intelligence & Internet of Things), "OPCA" (Optical physical
and chemical analyzer), "NetworkReceiver" (network alarm receiver),
"PanicAlarmPanel" (panic alarm station), "bodyCamera" (body camera),
"videoMatrix", "EmbeddedCentralController" (IoT control host), "networkMic"
(network microphone), "IPA", "PersonnelChannel", "MediaComponentGateway",
"ImagingRadarPanel", "TrafficTerminalServer", "DecodeStitchControl", "PDA",
"securityRadar", "TrafficTerminalServer", "VideoConferenceTerminal", "UPLC"
(ultra-high performance liquid chromatography), "GC" (gas chromatography),
"inquestDevice" (interrogation device), "simpleCourtTrialVideoRecording",
"standardCourtTrialVideoRecording", "educationSharingDevice" (education sharing
server), "captureCamera" (capture camera), "positioningSystem", "fisheye",
"panoVuSeriesCamera", "360PanoramicCamera", "panoramicCamera",
"UHFfixedCardReader", "codesplitter", "industrySecurityControlPanel",
"powerandEnvironmentDevice", "transcoder", "LCDscreen", "gatekeeper",
"codecombiner", "rearProjectionDisplayDevice", "screenSever", "captureServer",
"Y10Series", "IoTgateway", "networkDeviceManagement", "ARterminal",
"videoSecurityControlPanel", "networkSecurityControlPanel", "4GNetworkModule",
"domesticSecurityControlPanel", "UAV", "mobileDevice", "intelligentAnalyzer",
"intelligentTrafficServer" (intelligent traffic server), "matrixGateway",
"fiberOpticTransceiver" (optical transceiver), "intergratedVideoWallController"
(integrated multi-screen controller), "distributedVideoWallController"
(distributed multi-screen controller), "matrixSwitch", "fiberOpticalTransceiver"
(fiber converter), "wirelessDevices", "mobileSecurityProduct" (mobile security
product), "NVS", "facialRecognitionServerPureAnalysis" (pure analysis facial
recognition server), "smartLock" (smart lock), "radar", "fireProtectionGateway"
(fire protection gateway), "analyzer", "securityInspectionSystem" (security
inspection system), "walkThroughMetalDetector" (walk-through metal detector),
"recheckingArea" (rechecking area), "judicialPlat", "wirelessRouter" (wireless
router), "AC" (wireless controller), "fitAP" (thin access point), "fatAP" (fat
access point), "TrafficCenterEventServer" (traffic center event detection
server)-->Blade
  </deviceType>
  <subDeviceType>
```

```
    <!--ro, opt, enum, device sub type, subType:string,
desc:"accessControlTerminal" (access control terminal; it is valid when the
value of deviceType is "ACS"), "attendanceCheckDevice" (attendance check
devices; it is valid when the value of deviceType is "ACS"),
"multiChannelAccessController" (multi-channel access controller; it is valid
when the value of deviceType is "ACS"), "personAndIdCardDevice" (person and ID
card device; it is valid when the value of deviceType is "ACS"), "doorStation"
(door station; it is valid when the value of deviceType is "VIS"), "indoor"
(indoor station; it is valid when the value of deviceType is "VIS"),
"mainStation" (main station; it is valid when the value of deviceType is "VIS"),
"dropGate" (swing barrier; it is valid when the value of deviceType is
"PersonnelChannel"), "wingGate" (flap barrier; it is valid when the value of
deviceType is "PersonnelChannel"), "threeRollerGate" (tripod turnstile; it is
valid when the value of deviceType is "PersonnelChannel")--
>accessControlTerminal
  </subDeviceType>
  <telecontrolID>
    <!--ro, opt, int, remote control ID, range:[1,255]-->1
  </telecontrolID>
  <supportBeep>
    <!--ro, opt, bool, whether to support buzzer-->true
  </supportBeep>
  <supportVideoLoss>
    <!--ro, opt, bool, whether it supports video loss-->true
  </supportVideoLoss>
  <firmwareVersionInfo>
    <!--ro, opt, string, firmware version information, range:[1,32]-->B-R-H5-0
  </firmwareVersionInfo>
  <actualFloorNum>
    <!--ro, opt, int, actual number of floors, range:[1,128]-->1
  </actualFloorNum>
  <localZoneNum>
    <!--ro, opt, int, number of local zones, range:[0,16]-->1
  </localZoneNum>
  <alarmOutNum>
    <!--ro, opt, int, number of alarm outputs, range:[0,16]-->1
  </alarmOutNum>
  <alarmInNum>
    <!--ro, opt, int, number of alarm inputs, range:[0,16]-->1
  </alarmInNum>
  <distanceResolution>
    <!--ro, opt, float, distance resolution, range:[0.000,0.999]-->0.000
  </distanceResolution>
  <angleResolution>
    <!--ro, opt, float, angle resolution, range:[0.000,0.999]-->0.000
  </angleResolution>
  <speedResolution>
    <!--ro, opt, float, speed resolution, range:[0.000,0.999]-->0.000
  </speedResolution>
  <detectDistance>
    <!--ro, opt, float, detection distance, range:[0.000,0.999]-->0.000
  </detectDistance>
  <relayNum>
    <!--ro, opt, int, number of relays (local), range:[0,16]-->1
  </relayNum>
  <electroLockNum>
    <!--ro, opt, int, number of local electronic locks, range:[0,16]-->1
  </electroLockNum>
```

```
  <sirenNum>
    <!--ro, opt, int, number of sirens, range:[0,16]-->1
  </sirenNum>
  <alarmLamp>
    <!--ro, opt, int, number of alarm lamps, range:[0,16]-->1
  </alarmLamp>
  <RS485Num>
    <!--ro, opt, int, number of local RS-485, range:[0,16]-->1
  </RS485Num>
  <radarVersion>
    <!--ro, opt, string, radar version, range:[1,32]-->test
  </radarVersion>
  <cameraModuleVersion>
    <!--ro, opt, string, camera module version, range:[1,32]-->test
  </cameraModuleVersion>
  <mainversion>
    <!--ro, opt, int, main version No., range:[1,255]-->1
  </mainversion>
  <subversion>
    <!--ro, opt, int, sub version No., range:[1,255]-->1
  </subversion>
  <upgradeversion>
    <!--ro, opt, int, updated version No., range:[1,255]-->1
  </upgradeversion>
  <customizeversion>
    <!--ro, opt, int, customized version No., range:[1,255]-->1
  </customizeversion>
  <companyName>
    <!--ro, opt, string, manufacturer name, range:[1,32]-->test
  </companyName>
  <copyright>
    <!--ro, opt, string, copyright Information, range:[1,32]-->test
  </copyright>
  <systemName>
    <!--ro, opt, enum, storage system name, subType:string,
desc:"storageManagement" (storage management system),
"distributedStorageManagement" (distributed storage management system)--
>storageManagement
  </systemName>
  <systemStatus>
    <!--ro, opt, enum, system status, subType:string, desc:“configured”,
“unConfigured” (not configured)-->configured
  </systemStatus>
  <isLeaderDevice>
    <!--ro, opt, bool, whether it is the corresponding device of the resource IP
address-->true
  </isLeaderDevice>
  <clusterVersion>
    <!--ro, opt, string, system cluster version, range:[1,32], desc:this node is
valid when the value of isLeaderDevice is true-->test
  </clusterVersion>
  <manufacturer>
    <!--ro, opt, enum, manufacturer information, subType:string, desc:N/A--
>hikvision
  </manufacturer>
  <centralStorageVersion>
    <!--ro, opt, string, center storage version, range:[1,16]-->test
  </centralStorageVersion>
```

```
  <powerOnMode>
    <!--ro, opt, enum, startup mode, subType:string, desc:“button” (press button to power on), “adapter” (connect adapter to power on)-->button
  </powerOnMode>
  <customizedInfo>
    <!--ro, opt, string, customized project No., range:[1,32], desc:no value will be returned if it is baseline device. custom project No. will be returned if is custom device-->test
  </customizedInfo>
  <verificationCode>
    <!--ro, opt, string, device verification code-->test
  </verificationCode>
  <supportUrl>
    <!--ro, opt, string, service portal-->test
  </supportUrl>
  <subSerialNumber>
    <!--ro, opt, string, sub serial No.-->test
  </subSerialNumber>
  <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese">
    <!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french", "russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch", "hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic", "traditionalChinese", "lithuanian", "anglicism", "estonian"-->chinese
  </languageType>
  <DockStation>
    <!--ro, opt, object, dock station information configuration-->
    <Platform>
      <!--ro, opt, object, platform dock station information configuration-->
      <type>
        <!--ro, req, enum, platform which is connected to the dock station, subType:string, desc:"none", "9533", "8618", "ISAPI"-->none
      </type>
      <ip>
        <!--ro, opt, string, IP address, range:[1,32]-->test
      </ip>
      <port>
        <!--ro, opt, int, communication port, range:[1,65535]-->1
      </port>
      <userName>
        <!--ro, req, string, user name, range:[1,32]-->test
      </userName>
    </Platform>
    <centralStorageBackupEnabled>
      <!--ro, opt, bool, whether to enable center storage backup-->true
    </centralStorageBackupEnabled>
  </DockStation>
  <webVersion>
    <!--ro, opt, string, Web version No., range:[1,32]-->test
  </webVersion>
  <deviceRFProgramVersion>
    <!--ro, opt, string, device RF program version, range:[1,32]-->test
  </deviceRFProgramVersion>
  <securityModuleSerialNo>
```

```
    <!--ro, opt, string, security module serial No., range:[1,32]-->test
  </securityModuleSerialNo>
  <securityModuleVersion>
    <!--ro, opt, string, security module version, range:[1,32]-->test
  </securityModuleVersion>
  <securityChipVersion>
    <!--ro, opt, string, security chip version, range:[1,32]-->test
  </securityChipVersion>
  <securityModuleKeyVersion>
    <!--ro, opt, string, security module key version, range:[1,32]-->test
  </securityModuleKeyVersion>
  <UIDLampRecognition>
    <!--ro, opt, object, recognize the device information by UID-->
    <enabled>
      <!--ro, opt, bool, whether to enable the function-->true
    </enabled>
  </UIDLampRecognition>
  <confDeviceIdPrefix>
    <!--ro, opt, bool, whether the meeting uses the device name as its prefix--
>true
  </confDeviceIdPrefix>
  <OEMCode>
    <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard
device), 0 (neutral device)-->1
  </OEMCode>
  <simpleAlgorithmVersion>
    <!--ro, opt, string, single algorithm version, desc:for fire control
analyzer, it only supports one recognition algorithm if this node is returned to
the device information-->test
  </simpleAlgorithmVersion>
  <bootTime>
    <!--ro, opt, datetime, system boot time, desc:ISO8601 time (TD format, local
time and time difference)-->1970-01-01T08:00:00+08:00
  </bootTime>
  <intelligentAnalysisEngineModel>
    <!--ro, opt, string, engine model, range:[1,32]-->test
  </intelligentAnalysisEngineModel>
  <marketType>
    <!--ro, opt, enum, market type, subType:int, desc:0 (invalid), 1
(distribution type), 2 (industry type)-->0
  </marketType>
  <algorithmVersion>
    <!--ro, opt, string, HCNN algorithm version, desc:its Data Dictionary is
provided by research institute for matching the algorithm model-->test
  </algorithmVersion>
  <firmware>
    <!--ro, opt, string, firmware version, desc:its Data Dictionary is provided
by research institute for matching the algorithm model-->test
  </firmware>
  <engineList>
    <!--ro, opt, object, list of device computing power, desc:it returns engine
No. by number of GPU chips-->
    <engine>
      <!--ro, opt, int, returned engine No.-->1
    </engine>
  </engineList>
  <platform>
```

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    <!--ro, opt, enum, platform type, subType:int, desc:1 (TX1), 2 (P4), 3
(3559), 4 (3519), 5 (3516)-->1
  </platform>
  <platformName>
    <!--ro, opt, string, the platform where the algorithm runs, desc:"TX1/K81"
(NVR), "KT" (data center), "H5" (front-end device), "H7" (front-end device),
"K82" (NVR), "G3" (front-end device)-->test
  </platformName>
  <touchScreenVersionInfo>
    <!--ro, opt, string, touch screen module version-->test
  </touchScreenVersionInfo>
  <protocolFileURL>
    <!--ro, opt, string, protocol notice URI, range:[1,32]-->test
  </protocolFileURL>
  <recycleRecordEnabled>
    <!--ro, opt, bool, whether to enable overwritten recording-->false
  </recycleRecordEnabled>
  <decordChannelNums>
    <!--ro, opt, int, number of decoding channels-->0
  </decordChannelNums>
  <VGANums>
    <!--ro, opt, int, number of VGA ports-->0
  </VGANums>
  <USBNums>
    <!--ro, opt, int, number of USB ports-->0
  </USBNums>
  <auxoutNums>
    <!--ro, opt, int, number of auxiliary ports-->0
  </auxoutNums>
  <expansionBoardVersion>
    <!--ro, opt, string, extension board version information, range:[1,32]--
>test
  </expansionBoardVersion>
  <initWizzardDisplay>
    <!--ro, opt, bool, whether it displays initialization wizard, desc:by
default, the value is true. The status turns to false after the local GUI wizard
completed-->true
  </initWizzardDisplay>
  <beaconID>
    <!--ro, opt, string, device RF program version No., range:[0,32], desc:by
default, it is the current value-->test
  </beaconID>
  <isResetDeviceLanguage>
    <!--ro, opt, bool, whether to reset the language, desc:if the value is true,
the Hik-Connect app and the Web Client display device information in the target
language; if the value is false, the Hik-Connect app and the Web Client display
the original device information-->false
  </isResetDeviceLanguage>
  <dispalyNum>
    <!--ro, opt, int, number of windows-->0
  </dispalyNum>
  <bspVersion>
    <!--ro, opt, string, BSP software version-->test
  </bspVersion>
  <dspVersion>
    <!--ro, opt, string, DSP software version-->test
  </dspVersion>
  <localUIVersion>
```

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    <!--ro, opt, string, local UI version-->test
  </localUIVersion>
  <OPCASubType>
    <!--ro, opt, enum, sub type of optical physical and chemical analyzer,
subType:string, desc:"pump", "autoSampler" (auto sampler), "detector",
"columnOven" (column oven), "SOC" (gas chromatography)-->detector
  </OPCASubType>
  <wiegandOutNum>
    <!--ro, opt, int, number of Wiegand outputs, range:[0,1]-->1
  </wiegandOutNum>
  <ChipVersionInfoList>
    <!--ro, opt, array, chip version information list, subType:object, range:
[1,6]-->
    <ChipVersionInfo>
      <!--ro, opt, object, chip version information-->
      <ID>
        <!--ro, req, int, chip No., range:[0,1], desc:the No.s of chips of
different types can be duplicate. Users can distinguish them by chip type and
chip No. together-->1
      </ID>
      <type>
        <!--ro, req, enum, chip type, subType:string, desc:"MCU", "FPGA",
"MSTAR"-->MCU
      </type>
      <firmwareVersion>
        <!--ro, req, string, chip firmware version, range:[0,64]-->1.0.0
      </firmwareVersion>
      <chipName>
        <!--ro, opt, string, chip name, range:[0,32], desc:chip name-->test
      </chipName>
    </ChipVersionInfo>
  </ChipVersionInfoList>
  <personBagLinkAlgoEngineVersion>
    <!--ro, opt, string, engine version of the person & package linkage module,
range:[0,64], desc:this node is for analyzer (security inspection)-->1.0.0
  </personBagLinkAlgoEngineVersion>
  <BIOSVersion>
    <!--ro, opt, string, BIOS version, range:[0,16]-->V1.3.4
  </BIOSVersion>
  <contactInformation>
    <!--ro, opt, string, contact information, range:[0,64]-->test
  </contactInformation>
  <temperatureModuleVersionInfo>
    <!--ro, opt, string, temperature measurement component version-->test
  </temperatureModuleVersionInfo>
  <PedestrianWarningModuleVersion>
    <!--ro, opt, object, pedestrian warning module information, desc:this node
is returned only for pedestrian warning devices-->
    <pedestrianWarningMCUVersion>
      <!--ro, opt, string, MCU program version, range:[0,32]-->V1.3.4
    </pedestrianWarningMCUVersion>
    <pedestrianWarningRadarVersion>
      <!--ro, opt, string, the program version of pedestrian warning radar,
range:[0,32]-->V1.3.4
    </pedestrianWarningRadarVersion>
    <pedestrianRangingModuleVersion>
      <!--ro, opt, string, the program version of the pedestrian distance
measurement module, range:[0,32]-->V1.3.4
```

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    </pedestrianRangingModuleVersion>
  </PedestrianWarningModuleVersion>
  <encryptionModel>
    <!--ro, opt, string, device encryption model, range:[0,16], desc:it cannot
be modified and is a hexadecimal digit currently-->test
  </encryptionModel>
  <UWBVersion>
    <!--ro, opt, string, UWB version, range:[1,32]-->V1.3.0
  </UWBVersion>
  <audioBoard>
    <!--ro, opt, object, audio board information-->
    <audioBoardModel>
      <!--ro, opt, string, audio board model, range:[0,32], desc:currently only
C8 and R8 are supported-->C8
    </audioBoardModel>
    <audioBoardVersion>
      <!--ro, opt, string, audio board version, range:[0,32]-->V1.0.0 build
211210
    </audioBoardVersion>
  </audioBoard>
  <materialScanAlgorithmVersion>
    <!--ro, opt, string, material scanning algorithm version, range:[0,64],
desc:material scanning algorithm version-->test
  </materialScanAlgorithmVersion>
  <regionVersion>
    <!--ro, opt, enum, N/A, subType:string, desc:N/A-->basic
  </regionVersion>
  <productionDate>
    <!--ro, opt, date, date of manufacture-->2022-04-02
  </productionDate>
  <wifiModuleMACAddress>
    <!--ro, opt, string, Mac address of Wi-Fi module, range:[1,48]--
>44:47:cc:c8:d9:e4
  </wifiModuleMACAddress>
  <displayInterfaceSize>
    <!--ro, opt, float, display interface size, range:[0.0,16.0], unit:in,
desc:diagonal length of display interface for video intercom devices-->0
  </displayInterfaceSize>
  <releaseRegion>
    <!--ro, opt, enum, N/A, subType:string, desc:N/A-->China
  </releaseRegion>
  <shortSerialNumber>
    <!--ro, opt, string, short serial No., range:[1,9]-->test
  </shortSerialNumber>
  <audioVersion>
    <!--ro, opt, string, audio version, range:[1,32]-->test
  </audioVersion>
  <communicationFrequency>
    <!--ro, opt, enum, device wireless communication frequency, subType:string,
desc:"433MHz", "868MHz"-->433MHz
  </communicationFrequency>
  <OSCoreVersionInfo>
    <!--ro, opt, object, OS/core component version information-->
    <OSCoreVersion>
      <!--ro, req, string, OS/core component version No., range:[1,32]-->1.0.0
    </OSCoreVersion>
    <minisysVersion>
      <!--ro, opt, string, minisys version No., range:[1,32]-->1.0.0
```

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    </minisysVersion>
    <networkServiceVersion>
      <!--ro, opt, string, network service version No., range:[1,32]-->1.0.0
    </networkServiceVersion>
    <upgradeServiceVersion>
      <!--ro, opt, string, upgrade service version No., range:[1,32]-->1.0.0
    </upgradeServiceVersion>
  </OSCoreVersionInfo>
  <xTransVersion>
    <!--ro, opt, string, version No. of xtrans chip communication component,
range:[1,32], desc:this node is for analyzer-->1.0.0
  </xTransVersion>
</DeviceInfo>
```